

# Remainder Terms of the Riemann Zeta Function

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August 31, 2026

## Abstract

Starting from the Riemann–Siegel formula for  $\zeta(s)$  given in Siegel’s 1932 paper, we introduce a change of variable,  $t = I(T)$ , that replaces Siegel’s coupled pair “imaginary part  $t$  / summation cutoff  $m$ ” with one continuous cutoff parameter  $T$ . Five results follow. (1) We split Siegel’s single remainder integral  $R$  into two exact pieces and prove  $R = R_1 + R_2$ , so that  $\zeta = \Sigma_1 + R_1 + \Sigma_2 + R_2$ ; the algebra is formally verified in Lean in an appendix, as are the critical-line equalities that descend from it. (2) Each remainder is nothing more than *one additional, fractional partial summand* appended to its Dirichlet sum:  $R_1 = \hat{d}_1 (m+1)^{-s}$  and  $R_2 = \hat{d}_2 \chi(s) (m+1)^{s-1}$  with  $m = \lfloor T \rfloor$ ; we prove the weights  $\hat{d}_1, \hat{d}_2$  are real across the strip, equal and positive on the critical line, and positive off it outside two narrow windows of width  $O(1/T)$  per unit interval of  $T$ . (3) The weights are periodic in the index:  $d_1$  repeats one waveform in every unit interval of  $T$ , converging to the explicit tangent waveform  $\mathcal{W}_\infty$ , and at fixed fractional part  $\{T\} = x$  the normalized weight  $\sqrt{m+1} d_1$  converges to a closed-form limit profile  $d(x)$ . (4) We find thin loci in the strip, the equal-leg and folded-leg ovals, that are the only places off the critical line where a zero of  $\zeta$  could possibly lie. (5) We construct a counting function  $N_*(T)$  from critical-line information alone; unlike the Riemann–von Mangoldt count it sees only the zeros on the line, and it is continuous and increasing (the latter under the Riemann hypothesis), passing an integer exactly where  $Z$  changes sign.

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# 1 Introduction

Siegel published his study of Riemann’s unpublished papers in 1932 [21]. Working through the Nachlaß at Göttingen<sup>1</sup> he recovered, from calculations Riemann never published, an asymptotic evaluation of  $\zeta$  by the saddle-point method. The result is what is now called the Riemann–Siegel formula, and it is the starting point of this paper.

This paper does seven things: the five results announced in the abstract, together with the reparameterization they rest on and an account of the geometry from which they came.

1. **A reparameterization.** We introduce a change of variable  $t = I(T)$  that collapses Siegel’s two roles for  $t$  (the imaginary part of the input and the quantity determining the summation cutoff) into one continuous cutoff parameter  $T$  (§3). The map  $t = I(T)$  is exact, while our choice  $m = \lfloor T \rfloor$  differs from Siegel’s standard cutoff. The Riemann–Siegel identity remains exact with this new cutoff.
2. **Two new remainders.** The Riemann–Siegel formula states that  $\zeta = \Sigma_1 + \Sigma_2 + R$ , where  $\Sigma_1, \Sigma_2$  are finite Dirichlet sums and  $R$  is a remainder integral. We define two exact remainder terms,  $R_1$  and  $R_2$ , and prove

$$R = R_1 + R_2, \quad \text{hence} \quad \zeta = \Sigma_1 + R_1 + \Sigma_2 + R_2.$$

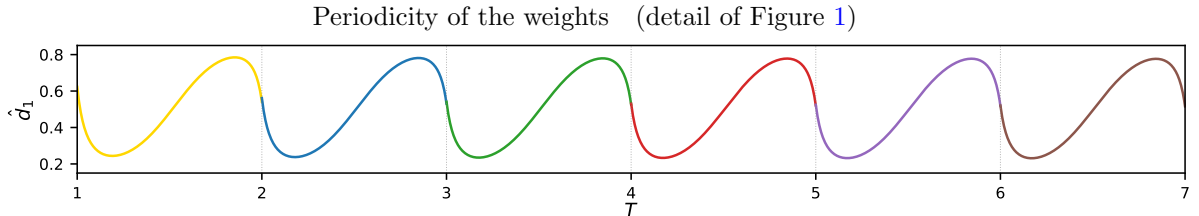
This is the paper’s central identity, stated and proved as Theorem 4.2 (§4). A formalization of its proof in Lean is given in Appendix B.

3. **Just a fractional summand.** The two new remainders are not opaque integrals: each is exactly *one more summand* of its Dirichlet sum, namely the  $(m + 1)$ st term, scaled by a real “fractional” weight  $\hat{d}_1$  or  $\hat{d}_2$  (§5). The exact zeta function thus reduces to the main sum and the dual sum, each carried one fractional summand past its cutoff, with no other remainder term:

$$\zeta(s) = \sum_{n=1}^m n^{-s} + \hat{d}_1 (m + 1)^{-s} + \chi(s) \sum_{n=1}^m n^{s-1} + \hat{d}_2 \chi(s) (m + 1)^{s-1},$$

which is (23) of §5. The weights are positive on the critical line, where they also coincide,  $d_1 = d_2$ , and positive off it outside narrow windows (Proposition 4.1, proved in Appendix A).

4. **Periodicity.** In the variable  $\{T\}$  the weights repeat one fixed waveform in every unit interval of  $T$  (§6):  $d_1$  converges to the explicit tangent waveform  $\mathcal{W}_\infty$  of (25), and at fixed fractional part  $\{T\} = x$  the normalized weight  $\sqrt{m + 1} d_1$  converges to a closed-form limit profile  $d(x)$ . The waveform and its limit were first observed geometrically, as the yin and yang curves of §6.4; only afterwards did we show they can be reached symbolically from a function in Siegel’s paper (§6.3).



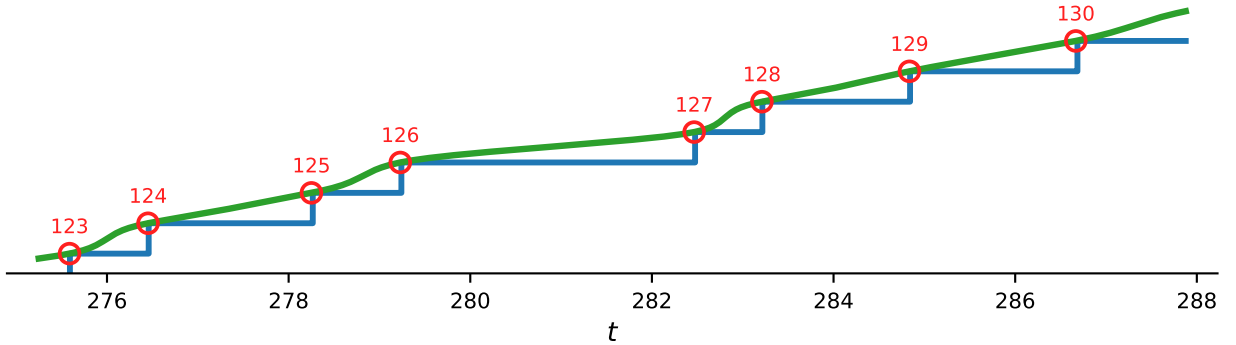
<sup>1</sup>The chronology pins the reading to Göttingen, where the Nachlaß is held: Siegel studied it there in the early 1930s (though he didn’t move to Göttingen until 1938) and published in 1932, before his first visit to the Institute for Advanced Study (January–June 1935) and his longer residence there (1940–1951).

5. **Where the insights came from: geometry.** None of this was first derived; it was first *seen*. Drawing the partial-sum chains of  $\zeta$  as links and joints in the complex plane, and sweeping  $\sigma$  and  $T$  continuously, is how the mapping  $I(T)$ , the yin-yang curves, the split of  $R$ , and the weights were found (§9, §10). The bisector point, the two legs it makes with the origin and  $\zeta$ , and the angles  $\vartheta_1, \vartheta_2$  organize everything that follows.
6. **Loci that confine the zeros.** A zero of  $\zeta$  requires the two legs to be equal in length *and* folded back onto one another. Off the critical line the equal-leg locus consists of thin ovals, and those ovals are the only places off the line where a zero could possibly lie. A zero also needs the fold angle to reach  $\pi$  there, so showing the two conditions never meet on an oval is yet another possible route to the Riemann hypothesis (§9.4, §9.5).
7. **A new zero-counting function.** From critical-line information alone we build

$$N_*(T) = \frac{1}{\pi} \arg\left(Z - \frac{iZ'}{\vartheta'}\right) + \frac{3}{2},$$

which is (163) of §12: unlike the Riemann–von Mangoldt count  $N(t)$ , which counts all zeros in the strip,  $N_*$  counts only the zeros on the critical line; it is continuous, increasing under the Riemann hypothesis, and passes an integer exactly where  $Z$  changes sign (§12).  $N(t)$  is a step function and tells nothing about  $|Z|$ , while the vector behind  $N_*$  carries  $|Z|$  along with the count: its winding gives the zeros and its modulus gives the envelope of  $|Z|$ .

**Riemann–von Mangoldt count  $N(t)$ , and  $N_*(T)$**



## 2 Siegel's 1932 formula

We begin with the Riemann–Siegel formula, which expresses  $\zeta(s)$  as two finite Dirichlet sums plus a remainder integral, as it appears in Siegel's 1932 paper [21] (his equation 13):

$$\zeta(s) = \sum_{n=1}^m n^{-s} + \frac{(2\pi)^s}{2\Gamma(s) \cos(\frac{\pi s}{2})} \sum_{n=1}^m n^{s-1} + \frac{(2\pi)^s e^{\frac{\pi i s}{2}}}{\Gamma(s) (e^{2\pi i s} - 1)} \int_{C_2} g(x) dx, \quad (1)$$

where  $m$  is an integer cutoff, fixed below in (8), and in Siegel's notation  $\epsilon = e^{-\pi i/4} = (1-i)/\sqrt{2}$ . The oriented path  $C_2$  consists of two half-lines meeting at  $\eta(1 - \frac{\epsilon}{2})$ , with  $\eta = +\sqrt{t/2\pi}$  the saddle point; one half-line passes through  $\eta$ , the other through  $-(m + \frac{1}{2})$ , and  $g$  is defined below.

It is convenient to name the three pieces. Recall the functional-equation factor

$$\chi(s) := 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) = \frac{(2\pi)^s}{2\Gamma(s) \cos(\frac{\pi s}{2})}, \quad (2)$$

so that, writing  $s = \sigma + it$  and taking the first  $m$  terms,

$$\Sigma_1(s) := \sum_{n=1}^m n^{-s} \quad (\text{the main sum}), \quad (3)$$

$$\Sigma_2(s) := \frac{(2\pi)^s}{2\Gamma(s) \cos(\frac{\pi s}{2})} \sum_{n=1}^m n^{s-1} = \chi(s) \sum_{n=1}^m n^{s-1} \quad (\text{the dual sum}), \quad (4)$$

$$R(s) := \frac{(2\pi)^s e^{\frac{\pi i s}{2}}}{\Gamma(s) (e^{2\pi i s} - 1)} \int_{C_2} g(x) dx, \quad (5)$$

where

$$g(x) := x^{s-1} \frac{e^{-2\pi i m x}}{e^{2\pi i x} - 1}. \quad (6)$$

With these names, (1) reads

$$\zeta(s) = \Sigma_1 + \Sigma_2 + R. \quad (7)$$

In Siegel's formulation  $m$  is the summation cutoff, and to minimize the remainder he takes

$$m = \left\lfloor \sqrt{\frac{t}{2\pi}} \right\rfloor. \quad (8)$$

Thus  $t$  plays two roles at once: it is the imaginary part of the input  $s$ , and, through (8), it also fixes the number of terms  $m$ . In this way  $m$  depends on  $t$ . The next section reverses the dependency.

### 3 Reparameterization and cutoff choice: The $I(T)$ mapping

Unless a statement explicitly gives a larger domain, assume throughout that  $T > 0$ ,  $0 < \sigma < 1$ ,  $t = I(T)$ , and  $s = \sigma + it$ .

We introduce a change-of-variable mapping for  $t$ , and with it fix the variables used for the rest of the paper:

$$t = \frac{\pi(2T+1)}{\log(\frac{1}{T}+1)} = I(T), \quad s = \sigma + it, \quad m = \lfloor T \rfloor, \quad M = \lceil T \rceil, \quad (9)$$

For non-integer  $T$ ,  $M = m + 1$  and is the index of the next omitted summand. At integer  $T$ ,  $M = m$  and refers instead to the last included summand. All “next summand” statements below are therefore restricted to non-integer  $T$  or interpreted through one-sided limits. Here  $T$  is a real number, while  $m = \lfloor T \rfloor$  is an integer. So now  $m$  and  $t$  depend on  $T$ , the reverse of Siegel’s (8). The single function  $I$  therefore determines both the imaginary part of the input and the integer summation cutoff; for this reason we call  $T$  “the index” even though it is real.

**Remark 3.1.** The index is similar to, but not exactly, the classical Riemann–Siegel cutoff  $\sqrt{t/2\pi}$ . Expanding the logarithm in (9) gives

$$\frac{I(T)}{2\pi} = T^2 + T + \frac{1}{6} + O\left(\frac{1}{T^2}\right) = \left(T + \frac{1}{2}\right)^2 - \frac{1}{12} + O\left(\frac{1}{T^2}\right),$$

and taking square roots,

$$\sqrt{\frac{t}{2\pi}} = T + \frac{1}{2} - \frac{1}{24(T + \frac{1}{2})} + O\left(\frac{1}{T^3}\right).$$

So  $T$  is approximately, but not exactly,  $\sqrt{t/2\pi} - \frac{1}{2}$ , and the approximation improves as  $T$  grows: the difference between  $T + \frac{1}{2}$  and  $\sqrt{t/2\pi}$  is  $O(1/T)$ , with explicit leading term  $\frac{1}{24}(T + \frac{1}{2})^{-1}$ . (At  $T = 6$ , for instance,  $\sqrt{t/2\pi} = 6.4936$  while  $T + \frac{1}{2} - \frac{1}{24 \cdot 6.5} = 6.4936$  as well.)

The abbreviations of (9) are in force for the rest of the paper: every object below depends on  $\sigma$  and  $T$  through them, and we do not carry the arguments. In this notation Siegel’s formula (7) becomes

$$\zeta = \Sigma_1 + \Sigma_2 + R, \quad (10)$$

where

$$\Sigma_1 = \sum_{n=1}^m n^{-s} \quad (\text{the main sum}), \quad (11)$$

$$\Sigma_2 = \chi(s) \sum_{n=1}^m n^{s-1} \quad (\text{the dual sum}), \quad (12)$$

where  $s = \sigma + iI(T)$  and  $m = \lfloor T \rfloor$ . Using the identity

$$\frac{(2\pi)^s e^{\frac{\pi i s}{2}}}{\Gamma(s)(e^{2\pi i s} - 1)} = \frac{\chi(s)}{e^{i\pi s} - 1}, \quad (13)$$

the remainder becomes

$$R = \frac{\chi(s)}{e^{i\pi s} - 1} \int_{C_2} g(x) dx, \quad (14)$$

where  $s = \sigma + i I(T)$ .

For any integer  $q \geq 0$ , let

$$R_q(s) := \zeta(s) - \sum_{n=1}^q n^{-s} - \chi(s) \sum_{n=1}^q n^{s-1}.$$

Siegel's standard choice is  $q = \lfloor \sqrt{t/(2\pi)} \rfloor$ , whereas here  $t = I(T)$  and  $q = m = \lfloor T \rfloor$ ; below we write  $R := R_m$ . Thus the map  $t = I(T)$  is exact, but the construction combines that exact reparameterization with a change from Siegel's standard cutoff.



## 4 Decomposing the remainder: $R = R_1 + R_2$

We now split Siegel's single remainder  $R$  into two exact pieces, named  $R_1$  and  $R_2$ , stated in the variables  $t$ ,  $s$ ,  $m$ , and  $M$  fixed in (9). Define

$$R_1 := d_1 M^{-it}, \quad (15)$$

$$R_2 := d_2 M^{it} \frac{\chi}{|\chi|}, \quad (16)$$

with real weights

$$d_1 := |R| \frac{\sin(\omega - \arg R + \arg \chi)}{\sin(2\omega + \arg \chi)}, \quad (17)$$

$$d_2 := |R| \frac{\sin(\omega + \arg R)}{\sin(2\omega + \arg \chi)}, \quad (18)$$

where  $R$  is Siegel's remainder (14),  $\chi = \chi(s)$ , and

$$\omega := t \log M. \quad (19)$$

For non-integer  $T$  the angle  $\omega = t \log(m+1)$  carries a geometric meaning: it is the total *clockwise* rotation accumulated by the summands of the first Dirichlet sum, so that  $-\omega$  is the absolute direction angle, measured from the positive real axis, of the next summand: the summand  $n^{-s} = n^{-\sigma} e^{-it \ln n}$  rotates by  $\theta_n = -t \ln \frac{n+1}{n}$  relative to its predecessor, and these relative rotations telescope,

$$\theta_1 + \theta_2 + \cdots + \theta_m = -t \ln \frac{2}{1} - t \ln \frac{3}{2} - \cdots - t \ln \frac{m+1}{m} = -t \ln(m+1) = -\omega, \quad (20)$$

so the  $(m+1)$ st summand points along  $e^{-i\omega}$ . (At integer  $T$ , where  $M = m$ , the direction  $e^{-i\omega}$  is instead that of the last *included* summand; the two directions are exactly antiparallel there, and the proof of Theorem 4.2 shows the ceiling is the orientation under which the decomposition holds at the integers as well.)

Writing  $\psi = \arg \chi(s)$ , the definitions (15)–(16) express  $R$  as a combination of two fixed *unit* directions,

$$R = d_1 e^{-i\omega} + d_2 e^{i(\omega+\psi)}, \quad (21)$$

namely the phases of the next summands of the two Dirichlet sums. The following proposition collects the basic facts about the two weights.

**Proposition 4.1** (Reality and positivity of the weights  $d_1$  and  $d_2$ ). *Let  $\psi = \arg \chi(s)$  and  $x = \{T\}$ , and let  $T$  be non-integer.*

- (i) *Each remainder lies along the next summand of its Dirichlet sum.*
- (ii) *For every  $\sigma$  and every  $T$  with  $\sin(2\omega + \psi) \neq 0$ , the weights  $d_1$  and  $d_2$  are real.*
- (iii) *On the critical line  $\sigma = \frac{1}{2}$  they are equal and positive.*
- (iv) *Fix  $\sigma \in (0, 1)$ ,  $\sigma \neq \frac{1}{2}$ . For all  $T \geq T_0(\sigma)$ , both weights are positive except on two windows of width  $O_\sigma(1/T)$ , centered  $O(1/T)$  from  $x = \frac{1}{4}$  and  $\frac{3}{4}$ . Inside each window one weight is negative, with  $d_1/d_2 \rightarrow -1$  at the pole between them, so the windows cannot be removed.*

The proof, which occupies Appendix A, reduces everything to a sign criterion on the two numerators of (17)–(18). Geometrically, positivity is the statement that each fractional summand points *forward* along its partial sum, a positive fraction of the next term.

**Theorem 4.2.** *Suppose  $T > 0$  and  $\sin(2\omega + \psi) \neq 0$ . Then*

$$R_1 + R_2 = R.$$

*Consequently,*

$$\zeta = \Sigma_1 + R_1 + \Sigma_2 + R_2.$$

*At a zero of  $\sin(2\omega + \psi)$ , the combined expression on a punctured neighborhood has limiting value  $R$ . At an off-line pole the individual terms  $R_1$  and  $R_2$  diverge and are not assigned finite values there.*

The qualifier in the last sentence costs nothing. On the critical line  $\sigma = \frac{1}{2}$  the vanishing of the denominator  $\sin(2\omega + \psi)$  is always cancelled by the numerators of (17)–(18), so  $d_1$  and  $d_2$  have no poles there at all (Appendix A). Off the line the two weights do have genuine poles at those points, but the divergences are equal and opposite and cancel in the sum, which stays finite and equal to  $R$  (Appendix A.5).

*Proof.* Write  $\phi = \arg R$ ,  $\psi = \arg \chi$ , and  $\omega = t \log M$ , so that  $R = |R|e^{i\phi}$  and  $\chi/|\chi| = e^{i\psi}$ . Since  $M$  is a positive real base and  $\omega = t \log M$  by (19), the substitution  $M^{\pm it} = e^{\pm i\omega}$  is valid for *every*  $T > 0$ , integer or not.

**Step 1 (substitute).** From (15)–(18),

$$R_1 = |R| \frac{\sin(\omega - \phi + \psi)}{\sin(2\omega + \psi)} e^{-i\omega}, \quad R_2 = |R| \frac{\sin(\omega + \phi)}{\sin(2\omega + \psi)} e^{i\omega} e^{i\psi}.$$

**Step 2 (add).**

$$R_1 + R_2 = \frac{|R|}{\sin(2\omega + \psi)} \left[ e^{-i\omega} \sin(\omega - \phi + \psi) + e^{i(\omega + \psi)} \sin(\omega + \phi) \right].$$

**Step 3 (expand via  $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$ ).**

$$\begin{aligned} e^{-i\omega} \sin(\omega - \phi + \psi) &= \frac{e^{i(\psi - \phi)} - e^{i(-2\omega + \phi - \psi)}}{2i}, \\ e^{i(\omega + \psi)} \sin(\omega + \phi) &= \frac{e^{i(2\omega + \psi + \phi)} - e^{i(\psi - \phi)}}{2i}. \end{aligned}$$

**Step 4 (add the expansions).** The  $e^{i(\psi - \phi)}$  terms cancel, leaving

$$\frac{1}{2i} \left[ e^{i(2\omega + \psi + \phi)} - e^{i(\phi - \psi - 2\omega)} \right].$$

**Step 5 (factor  $e^{i\phi}$ ).**

$$= \frac{e^{i\phi}}{2i} \left[ e^{i(2\omega + \psi)} - e^{-i(2\omega + \psi)} \right] = e^{i\phi} \sin(2\omega + \psi).$$

**Step 6 (substitute back).**

$$R_1 + R_2 = \frac{|R|}{\sin(2\omega + \psi)} \cdot e^{i\phi} \sin(2\omega + \psi) = |R|e^{i\phi} = R.$$

**Integer  $T$ .** No case split is needed: Steps 1–6 used only trigonometric algebra and the substitutions recorded at the start of the proof, all valid for every  $T > 0$ . What deserves spelling out is the bookkeeping that the ceiling in (19) handles. At integer  $T$  one has  $M = \lceil T \rceil = m$  while  $\lfloor T + 1 \rfloor = m + 1$ , and by the definition (9) of  $I$  the two candidate angles differ by

$$t \log \frac{m+1}{m} = t \log \left(1 + \frac{1}{T}\right) = \pi(2T + 1),$$

exactly, which for integer  $T$  is an *odd* multiple of  $\pi$ . Passing from  $\omega$  to  $\omega' = \omega + \pi(2T + 1)$  therefore flips the sign of both unit directions,  $e^{\mp i\omega'} = -e^{\mp i\omega}$ , and simultaneously flips the sign of both weights: each numerator in (17)–(18) shifts by an odd multiple of  $\pi$  and changes sign, while the common denominator shifts by the even multiple  $2\pi(2T + 1)$  and does not. The two flips cancel in the products, so the partial summands  $R_1$  and  $R_2$ , and with them the identity  $R_1 + R_2 = R$ , come out the same under either convention, *provided* the weights and the directions are computed from the same angle. The definitions (15)–(18) do exactly that: both use  $M$ , through (19). (Mixing the conventions, with weights from  $\lfloor T + 1 \rfloor$  against directions from  $M$ , would flip only one of the two signs and produce  $-R$  at the integers.)

**Exceptional points.** The algebra above divides by  $\sin(2\omega + \psi)$  and so applies wherever that denominator is nonzero. Its zeros are isolated in  $T$ , and  $R$  is continuous, so on a punctured neighborhood of each zero the identity  $R_1 + R_2 = R$  holds and forces the limit of  $R_1 + R_2$  to exist and equal  $R$  there, which is the limiting sense adopted in the statement. At an off-line pole this does not assign finite values to the two individual terms.  $\square$

A formalization of the algebra of Theorem 4.2 in Lean is given in Appendix B. It assumes no axioms of its own; the appendix records what its three hypotheses are and what is left outside its scope. A second Lean file, reproduced in Appendix C, formalizes the critical-line statements that descend from Proposition 4.1(iii): the reflection (70) and the equal-leg corollaries of §9.2. Since  $R = R_1 + R_2$ , we have in particular proved the exact formula

$$\zeta(s) = \Sigma_1 + R_1 + R_2 + \Sigma_2. \tag{22}$$

## 5 The remainders are “one more partial summand”

Theorem 4.2 is more than an algebraic identity. Unlike Siegel’s integral  $R$ , which bears no resemblance to the terms of the Dirichlet sums, each of  $R_1$  and  $R_2$  is *exactly the next term* of its partial sum, scaled by a real number (positive in the sense of Proposition 4.1). We call such a term a *fractional summand*.

Concretely, the last term actually included in  $\Sigma_1$  is the  $m$ -th term. For non-integer  $T$ , the very next term is the  $M$ -th, and  $R_1$  lies along its direction, scaled by a real signed coefficient; similarly for  $R_2$  along the next term of  $\Sigma_2$ . It is literally shortened only when the corresponding hatted coefficient lies in  $[0, 1]$ . Table 1 uses this non-integer convention.

|       | Last<br>summand   | Next<br>summand   | Fractional<br>amount                                                 | Partial<br>summand                       |
|-------|-------------------|-------------------|----------------------------------------------------------------------|------------------------------------------|
| $R_1$ | $m^{-s}$          | $M^{-s}$          | $\times (d_1 M^\sigma = \hat{d}_1)$                                  | $= d_1 M^{-it}$                          |
| $R_2$ | $\chi(s) m^{s-1}$ | $\chi(s) M^{s-1}$ | $\times \left(d_2 \frac{M^{1-\sigma}}{ \chi(s) } = \hat{d}_2\right)$ | $= d_2 M^{it} \frac{\chi(s)}{ \chi(s) }$ |

Table 1: Each remainder is the  $M$ -th (“next”) summand of its Dirichlet sum, scaled by a real fractional amount; here  $s = \sigma + it$  with  $t = I(T)$ ,  $m = \lfloor T \rfloor$ , and  $M = \lceil T \rceil$ . The shaded column is the last summand actually included in the sum, shown for reference; the remaining columns display the identity (next summand)  $\times$  (fractional amount) = (partial summand), the fractional amount being exactly the hatted weight  $\hat{d}_1$  or  $\hat{d}_2$  of (24), and the product simplifying to  $R_1$  and  $R_2$  respectively.

Reading (22) in this light, the zeta function takes the clean “one more summand” form

$$\zeta(s) = \sum_{n=1}^m n^{-s} + \hat{d}_1 M^{-s} + \chi(s) \sum_{n=1}^m n^{s-1} + \hat{d}_2 \chi(s) M^{s-1} \quad (23)$$

with  $\hat{d}_1, \hat{d}_2$  real numbers, positive on the critical line (and off it outside the pole windows of Proposition 4.1). In words: the two finite Dirichlet sums, each carried exactly one fractional term past its cutoff, together reproduce  $\zeta$  with no other remainder term.

The coefficients here are the *fractional amounts* of Table 1, not the weights  $d_1, d_2$  of (15)–(16). The hat is what distinguishes them, and the two differ by the length of the summand they scale:

$$\hat{d}_1 = M^\sigma d_1, \quad \hat{d}_2 = \frac{M^{1-\sigma}}{|\chi|} d_2. \quad (24)$$

The unhatted weights are signed coefficients satisfying  $|R_1| = |d_1|$  and  $|R_2| = |d_2|$ . The hatted quantities are signed normalized coefficients; they are literal fractions of their summands only where  $0 \leq \hat{d}_i \leq 1$ . At  $\sigma = \frac{1}{2}$ ,  $T = 6.18$  for instance,  $d_1 = 0.0875$  while  $\hat{d}_1 = \sqrt{7} d_1 = 0.2315$ .

## 6 Periodicity in $T$

Exact periodicity is rare in the study of  $\zeta$ . As a function of  $t$ , each summand  $n^{-it} = e^{-it \log n}$  has the fixed frequency  $\log n$ . What grows with  $t$  is the cutoff and hence the largest active frequency,  $\log m = \frac{1}{2} \log(t/(2\pi)) + o(1)$ . The closest classical notions are Bohr's almost periodicity in the half-plane  $\sigma > 1$  [22, Ch. XI] and the fixed coefficient functions  $C_k(p)$  of the Riemann–Siegel expansion, which depend only on the fractional part  $p = \{\sqrt{t/2\pi}\}$ . The index  $T$  exposes structure of the second kind: nothing here is exactly periodic. The unnormalized weights decay with  $T$ , while their normalized profiles converge to a common unit-cell limit. This section studies that asymptotic repetition and attempts to explain its origin.

### 6.1 $d_1$ and $d_2$ on the critical line

Figure 1 plots the weight  $d_1$  and the fractional amount  $\hat{d}_1 = \sqrt{m+1} d_1$  on the critical line for  $1 < T < 7$ ; on the line the two weights coincide,  $d_2 = d_1$  (Proposition 4.1(iii)), so the figure shows both at once. A pattern is visible:  $\hat{d}_1$  traces nearly the same waveform in every unit interval of  $T$ , while  $d_1$  repeats the same shape with slowly decaying amplitude and a small jump at each integer, where the summand it scales hands off from one term to the next.

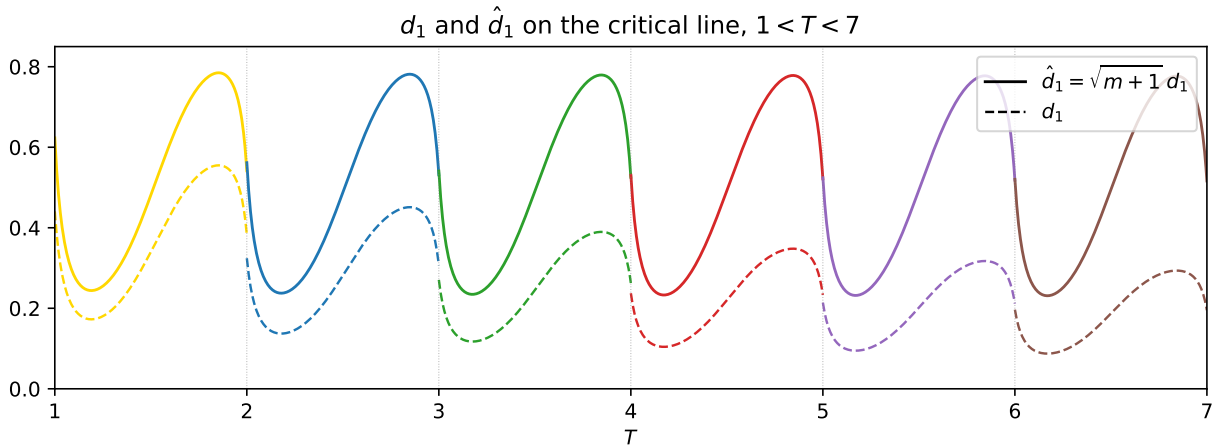


Figure 1:  $d_1$  (dashed) and  $\hat{d}_1 = \sqrt{m+1} d_1$  (solid) on the critical line for  $1 < T < 7$ , one color per integer interval of  $T$ ; this color scheme is held throughout the paper, a separate color for each interval  $[m, m+1]$ , to show the periodicity in  $T$ . Computed from the exact reduction  $d_1 = r/(2 \cos u)$  of (199). In each unit interval  $\hat{d}_1$  sweeps nearly the same waveform, while  $d_1$  decays like  $T^{-1/2}$  and jumps at the integers. Generated by `fig.d1_periodicity.py`.

### 6.2 A continuous and periodic normalization

The near-repetition has an exact limiting shape. Figure 2 overlays the six unit intervals of Figure 1 after two normalizations that put the waves on a common footing.

The first normalization rescales and centers: on  $[m, m+1]$  the centered coordinate

$$P(T) = T^{1/2} \left( d_1(T) - \Phi\left(-1, \frac{1}{2}, m+1\right) \right), \quad \Phi\left(-1, \frac{1}{2}, m+1\right) = \sum_{k \geq m+1} \frac{(-1)^{k-m-1}}{\sqrt{k}},$$

undoes the  $T^{-1/2}$  decay and measures  $d_1$  against the alternating sum of the remaining summand lengths ( $\Phi$  is the Lerch transcendent).

The second normalization pins both ends of the cycle (which fall exactly at integer  $T$ ) so they begin and end at zero: the endpoint values of  $P$  on  $[m, m+1]$  are  $+\varepsilon_m$  and  $-\varepsilon_{m+1}$ , where  $\varepsilon_n = n^{1/2}(\Phi(-1, \frac{1}{2}, n) - d_1(n^-))$ .<sup>2</sup> Subtracting the chord through them,

$$\mathcal{W}(T) = P(T) - (1-x)\varepsilon_m + x\varepsilon_{m+1}, \quad x = T - m,$$

pins each wave to zero at the two integer endpoints of its unit interval. Plotted against the fractional part  $x = \{T\}$ , the six waves nearly coincide, and as  $T \rightarrow \infty$  they converge to the tangent waveform

$$\mathcal{W}_\infty(x) = -\frac{1}{2} \tan(2\pi x) \tan(2\pi(x - \frac{1}{4})(x - \frac{3}{4})), \quad (25)$$

at rate  $O(1/T)$ ; the convergence is proved in Appendix A.6 (Corollary A.6).

Figure 3 records the same convergence for the fractional amount itself: at fixed fractional part  $x = \{T\}$  the normalized weight  $\hat{d}_1 = \sqrt{m+1} d_1$  tends to a closed-form profile  $d(x) = \frac{1}{2} + \mathcal{W}_\infty(x)$ , derived in the next subsection.

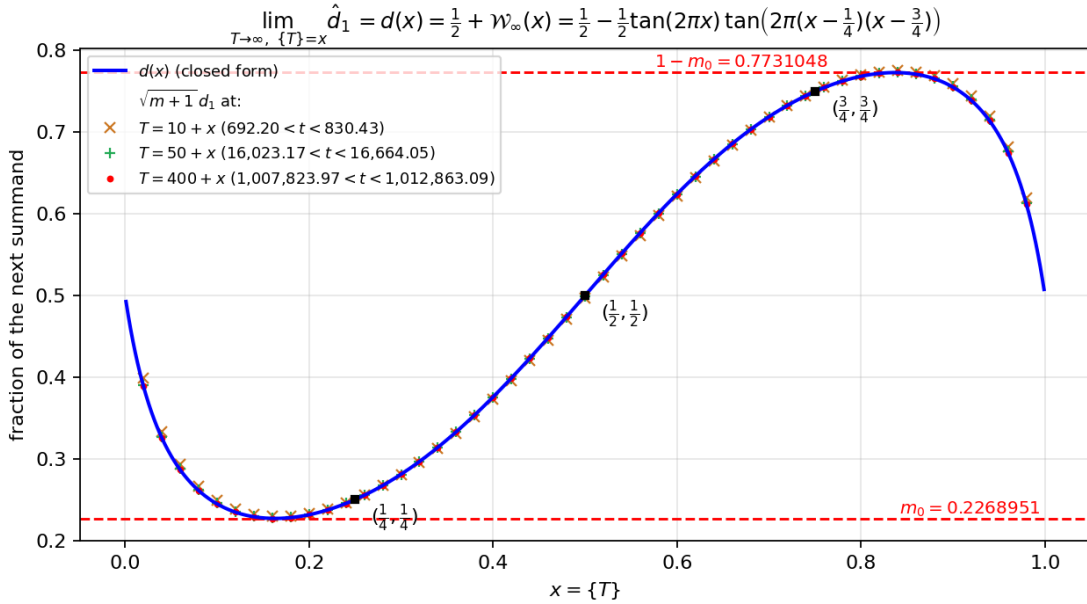


Figure 3: The closed-form profile  $d(x)$  (blue) against the exact  $\sqrt{m+1} d_1$  sampled at  $T = 10 + x$ ,  $50 + x$ ,  $400 + x$  on the critical line. Already at  $T = 10 + x$  the samples sit on the curve; the marked squares are the exact values  $(\frac{1}{4}, \frac{1}{4})$ ,  $(\frac{1}{2}, \frac{1}{2})$ ,  $(\frac{3}{4}, \frac{3}{4})$ , and the dashed red lines are the extremes  $m_0 = 0.2268951 \dots$  and  $1 - m_0 = 0.7731048 \dots$  of the profile. Generated by `fig.d_limit.py`.

### 6.3 Derivation of the tangent limit from Siegel's $F(\tau)$

The tangent waveform (25) is not an empirical fit: it can be derived from a function as old as Riemann's unpublished notes. Evaluating the elementary integral in equation 5 of Siegel's 1932

<sup>2</sup>The right endpoint is the definition of  $\varepsilon_{m+1}$ . The left endpoint uses the exact handoff relation  $d_1(m^+) + d_1(m^-) = m^{-1/2}$ , visible as the jump at each integer in Figure 1, together with  $\Phi(-1, \frac{1}{2}, m) + \Phi(-1, \frac{1}{2}, m+1) = m^{-1/2}$ . The handoff relation follows from the reduction  $d_1 = r/(2 \cos u)$  of (199): at  $T = m$  the angle  $u$  jumps by  $I(m) \log \frac{m+1}{m} = \pi(2m+1)$ , an odd multiple of  $\pi$ , so  $\cos u$  changes sign, while  $r$  drops the newly included summand's contribution  $2m^{-1/2} \cos u$ ; dividing gives  $d_1(m^+) = m^{-1/2} - d_1(m^-)$ .

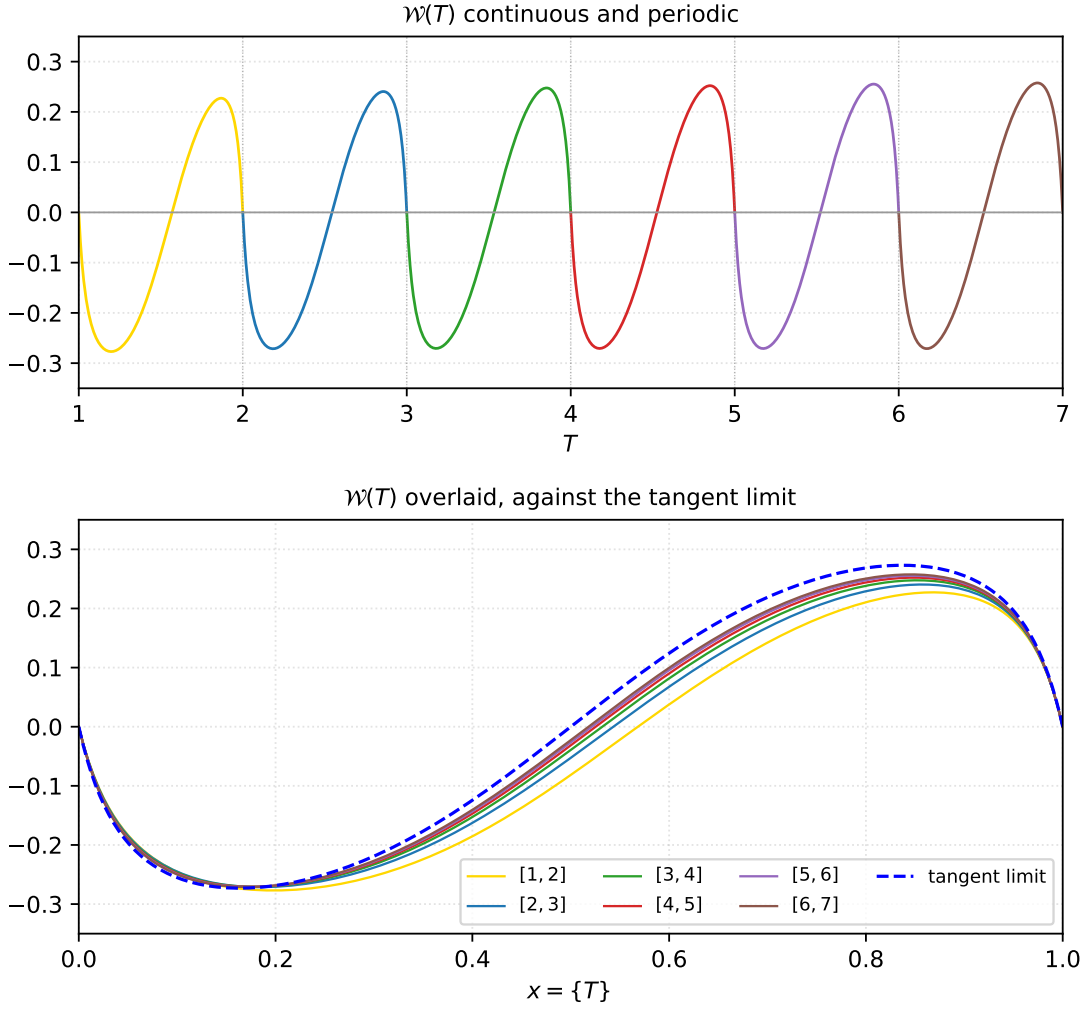


Figure 2: Top: the pinned waves  $\mathcal{W}(T)$  for  $1 < T < 7$ , one color per unit interval, the same range as Figure 1; the normalizations have removed the decay and the jumps, leaving a curve that is zero at every integer. Bottom: the six unit intervals overlaid, each pinned wave on  $[m, m + 1]$ ,  $m = 1, \dots, 6$ , plotted against the fractional part  $x = \{T\}$  in the color of the top panel, with the tangent limit  $\mathcal{W}_\infty$  dashed. The deviation from the limit falls from 0.087 on  $[1, 2]$  to 0.025 on  $[6, 7]$ . Generated by `fig-pinned_waves.py`.

paper [21], Siegel arrives at a closed form he calls the “desired result”,

$$F(\tau) = \frac{1}{1 - e^{-2\pi i \tau}} - \frac{e^{i\pi \tau^2}}{e^{i\pi \tau} - e^{-i\pi \tau}}. \quad (26)$$

For real  $\tau$  its complex conjugate has the compact form

$$\overline{F(\tau)} = \frac{e^{-i\pi \tau^2} - e^{-i\pi \tau}}{2i \sin(\pi \tau)} = \frac{e^{-i\pi \tau^2} - e^{-i\pi \tau}}{e^{i\pi \tau} - e^{-i\pi \tau}}, \quad (27)$$

and it is the conjugate we work with below. The apparent poles at integer  $\tau$  are removable:  $\tau^2 - \tau$  is even there, so the numerator of (27) vanishes with the denominator, and  $\overline{F(\tau)} \rightarrow \frac{1}{2} - n$  as  $\tau \rightarrow n$ . Figure 4 traces the curve.

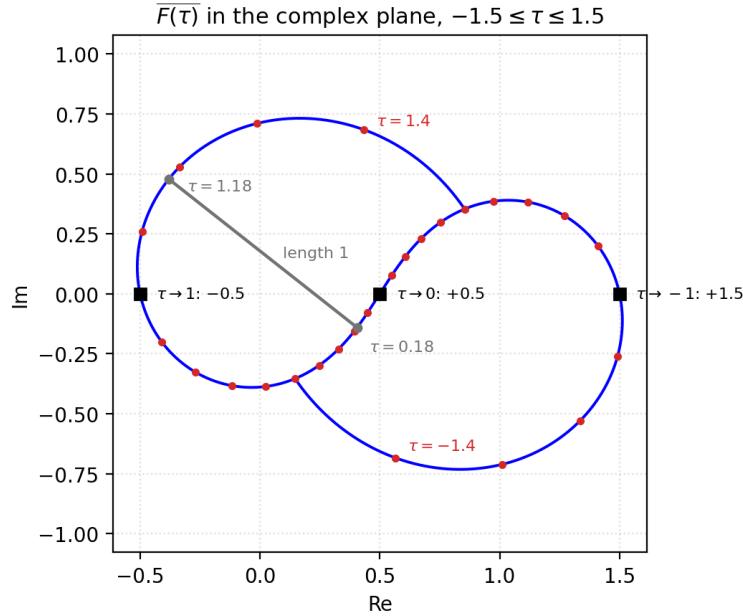


Figure 4: The trace of  $\overline{F(\tau)}$  of (27) in the complex plane for  $-1.5 \leq \tau \leq 1.5$ , with red dots every 0.1 in  $\tau$ . The apparent poles at the integers are removable (black squares), with  $\overline{F(\tau)} \rightarrow \frac{1}{2} - n$  as  $\tau \rightarrow n$ : the values  $-\frac{1}{2}, +\frac{1}{2}, +\frac{3}{2}$  at  $n = 1, 0, -1$ . Points of the curve one apart in  $\tau$  are exactly one unit apart in the plane:  $\overline{F(\tau+1)} - \overline{F(\tau)} = -e^{-i\pi(\tau^2+\tau)}$ , a unit vector; the gray chord joins  $\overline{F(0.18)}$  to  $\overline{F(1.18)}$  and has length exactly 1. Generated by `fig F.curve.py`.

**Remark 6.1** (regarding  $F$  as the source of Siegel’s  $f(s)$ ).  $F$  is more than an intermediate step in Siegel’s derivation: it is the source of his function  $f$ , the function that carries the first half of zeta,  $f(s) \approx \Sigma_1 + R_{1ak}$  (§8.2 records the details as (61)). In his §3 Siegel multiplies his equation 5, that is (26), by  $u^{-s}$  and integrates along the bisector of the first quadrant, and under that operation the two terms of  $F$  separate into the two halves of zeta. The pole term  $1/(1 - e^{-2\pi i u})$  yields  $-(2\pi)^{s-1} e^{i\pi(1-s)/2} \Gamma(1-s) \zeta(1-s)$ , while the Gaussian term yields  $f$  itself:

$$f(s) = (e^{\pi i s} - 1) \int_0^{e^{i\pi/4}\infty} \frac{u^{-s} e^{\pi i u^2}}{e^{\pi i u} - e^{-\pi i u}} du \quad (\sigma < 0), \quad (28)$$



the restriction  $\sigma < 0$  being needed only for convergence at the endpoint  $u = 0$ ;  $f$  is continued analytically from there, and the contour form (62) of §8.2 converges for every  $s$ . The same separation can be read off (27): since  $1/(1 - e^{2\pi i\tau}) = -e^{-\pi i\tau}/(2i \sin \pi\tau)$ , the numerator term  $e^{-\pi i\tau^2}$  of  $\overline{F}$  is the  $f(s)$  side and  $-e^{-\pi i\tau}$  is the  $\zeta(1-s)$  side, so the two lobes of Figure 4 carry the two halves in their numerator. So  $F$  is the common source of both  $\Sigma_1 + R_{1ak}$  and its reflected counterpart (the yin-yang connection is drawn in §6.4), and this common origin is one route into the convergence proved by other means in Theorem 6.5.

Everything we need is contained in the polar form of  $\overline{F}$ . Factoring the mean exponent  $e^{-i\pi(\tau^2+\tau)/2}$  out of the numerator turns the difference of exponentials into a sine:

$$\overline{F}(\tau) = -\frac{\sin(\frac{\pi}{2}(\tau^2 - \tau))}{\sin(\pi\tau)} e^{-\frac{i\pi}{2}(\tau^2+\tau)}. \quad (29)$$

Now rescale. Let  $a = \sqrt{t/2\pi}$ ,  $N = \lfloor a \rfloor$ , and  $\hat{p} = a - N$  be the classical Riemann–Siegel quantities (the same ones used throughout Appendix A), and substitute  $\tau = 2\hat{p} - \frac{1}{2}$  into (29). The three pieces collapse:  $\sin(\pi\tau) = -\cos(2\pi\hat{p})$ ,  $\frac{\pi}{2}(\tau^2 - \tau) = 2\pi(\hat{p}^2 - \hat{p} - \frac{1}{16}) + \frac{\pi}{2}$ , and  $\frac{\pi}{2}(\tau^2 + \tau) = 2\pi\hat{p}^2 - \frac{\pi}{8}$ , so that

$$\overline{F}(2\hat{p} - \frac{1}{2}) = C_0(\hat{p}) e^{-iA}, \quad A = 2\pi\hat{p}^2 - \frac{\pi}{8}, \quad C_0(\hat{p}) = \frac{\cos(2\pi(\hat{p}^2 - \hat{p} - \frac{1}{16}))}{\cos(2\pi\hat{p})}. \quad (30)$$

In words: evaluated at  $\tau = 2\hat{p} - \frac{1}{2}$ , twice the fractional part less one half,  $\overline{F}$  has amplitude exactly  $C_0(\hat{p})$ , the leading coefficient of the Riemann–Siegel correction, and phase  $-A$ ; both reappear, together with  $B = 2\pi\hat{p}$ , in Lemma A.1.

The limit now assembles from the critical-line machinery of Appendix A. On the line the exact reduction (199) reads  $d_1 = r/(2 \cos u)$ ; Gabcke’s form of the Riemann–Siegel expansion (201) gives  $r$  the leading term  $(-1)^{N-1} a^{-1/2} C_0(\hat{p})$ , the amplitude of (30); and the phase expansion (197) gives  $\cos u = (-1)^{m+1} \cos \beta + O(1/T)$ , where  $\beta = 2\pi(x - \frac{1}{4})(x - \frac{3}{4})$  and  $x = \{T\}$ . In each half of the unit interval the offset is  $\hat{p} = x \pm \frac{1}{2}$  (196), and the sign factors combine as in Steps 3 and 4 of the proof of Theorem A.2, so that

$$\hat{d}_1 = \sqrt{m+1} d_1 \longrightarrow \frac{C_0(x + \frac{1}{2})}{2 \cos \beta} \quad (x < \frac{1}{2}), \quad \hat{d}_1 \longrightarrow 1 - \frac{C_0(x - \frac{1}{2})}{2 \cos \beta} \quad (x > \frac{1}{2}),$$

the extra 1 in the second case being the newly included summand. Two lines of the product formula  $\cos(A \mp B) = \cos A \cos B \pm \sin A \sin B$  turn both branches into one and the same expression,

$$\hat{d}_1 \longrightarrow \frac{1}{2}(1 - \tan(2\pi x) \tan \beta) = \frac{1}{2} + \mathcal{W}_\infty(x) =: d(x),$$

the tangent waveform of (25) lifted by  $\frac{1}{2}$ .

Written in branch form, with  $g$  recording the  $x < \frac{1}{2}$  branch,

$$d(x) = \begin{cases} g(x), & x < \frac{1}{2}, \\ 1 - g(1-x), & x \geq \frac{1}{2}, \end{cases} \quad g(y) = \frac{-\cos(2\pi y^2 - \frac{5\pi}{8})}{2 \cos(2\pi y) \cos(2\pi y(1-y) - \frac{3\pi}{8})}. \quad (31)$$

In (31) the point  $y = \frac{1}{4}$  is a removable 0/0 singularity of  $g$  (the numerator vanishes together with  $\cos(2\pi y)$ ), filled in by continuity with  $g(\frac{1}{4}) = \frac{1}{4}$ ; this supplies the values of  $d$  at  $x = \frac{1}{4}$  and, through the second branch, at  $x = \frac{3}{4}$ . The  $d(x)$  limit profile passes through  $(\frac{1}{4}, \frac{1}{4})$ ,  $(\frac{1}{2}, \frac{1}{2})$ ,  $(\frac{3}{4}, \frac{3}{4})$  and stays inside the band  $[m_0, 1 - m_0]$  with  $m_0 = 0.2268951 \dots$  (at finite  $T$  the excursions are slightly larger

but still bounded between  $\frac{1}{5}$  and  $\frac{4}{5}$ ; Theorem A.7 in Appendix A.7); in particular the fractional amount never approaches either end of its summand.

Figure 3 plots the  $d(x)$  limit profile against the exact normalized weight  $\hat{d}_1 = \sqrt{m+1} d_1$ : the convergence is at rate  $O(1/T)$ , and the samples at  $T = 10 + x$  already sit on the curve. Theorem A.5 in Appendix A.6 makes the derivation of this subsection rigorous, controlling every error term including those at the transition points  $x = 0, \frac{1}{2}, 1$ .

#### 6.4 Separating $\overline{F}$ into two curves: yin and yang at infinity

Because the curve in Figure 4 resembles a *taijitu* (太极图), we call its two component curves yin and yang. The pair of curves turns the limit profile (31) into a geometric picture. Write  $x = \{T\}$  for the fractional part, so that  $x$  runs from 0 to 1 across each unit interval of  $T$ , and write

$$\Psi(x) := \frac{\cos(2\pi(x^2 - x - \frac{1}{16}))}{\cos(2\pi x)} \quad (32)$$

for the Riemann–Siegel correction function: it is the same function as  $C_0$  of (30) (we keep Gabcke’s symbol  $C_0$  for the analytic input and  $\Psi$  for the limit curve), and it appears in §15.3 of Titchmarsh’s book [22] and in §5.4.1 of Pugh’s thesis [20]. Define the two curves

$$\text{Yin}_\infty(x) := \Psi(x) \left( -\cos(2\pi(x^2 - \frac{1}{16})) + i \sin(2\pi(x^2 - \frac{1}{16})) \right) + 1 = 1 - \Psi(x) e^{-2\pi i(x^2 - \frac{1}{16})} \quad (33)$$

and

$$\text{Yang}_\infty(x) := -e^{4\pi i x} \text{Yin}_\infty(x) + e^{4\pi i x} = e^{4\pi i x} (1 - \text{Yin}_\infty(x)). \quad (34)$$

Both are built from  $\overline{F}$  alone: the exponent  $2\pi(x^2 - \frac{1}{16})$  is the phase  $A$  of (30), so  $\Psi(x) e^{-iA} = \overline{F}(2x - \frac{1}{2})$  and

$$\text{Yin}_\infty(x) = 1 - \overline{F}(2x - \frac{1}{2}), \quad \text{Yang}_\infty(x) = e^{4\pi i x} \overline{F}(2x - \frac{1}{2}). \quad (35)$$

As  $x$  runs over the unit interval the argument  $\tau = 2x - \frac{1}{2}$  runs from  $-\frac{1}{2}$  to  $\frac{3}{2}$ , from one lobe-meeting point of Figure 4 to the other, covering both lobes once. The yin curve is the point reflection of that arc through  $\frac{1}{2}$ ; the yang curve is the same arc spun by the unit factor  $e^{4\pi i x}$ , two full turns per unit interval. Figure 5 draws the pair.

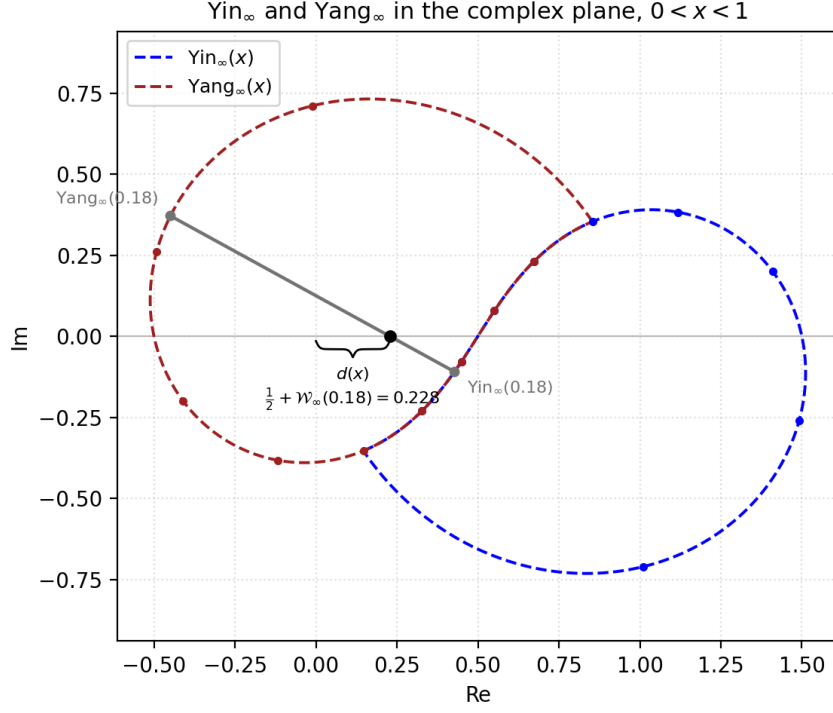


Figure 5: The curves  $\text{Yin}_\infty(x)$  (blue, dashed) and  $\text{Yang}_\infty(x)$  (dark red, dashed) for  $0 < x < 1$ , with dots every 0.1 in  $x$ ; both are the single curve  $\bar{F}$  of Figure 4 repackaged as in (35). The gray chord joins the two points at  $x = 0.18$ ; it crosses the real axis at  $d(0.18) = \frac{1}{2} + \mathcal{W}_\infty(0.18) \approx 0.228$  (black dot), as Proposition 6.2 asserts for every  $x$ . Generated by `fig_yinyang_infinity.py`.

The pair of curves is more than a repackaging: the chord between them crosses the real axis at exactly the limit profile  $d(x)$ .

**Proposition 6.2** (the chord recovers the limit profile). *For  $x \in (0, 1)$  with  $x \neq \frac{1}{4}, \frac{3}{4}$ , the line through  $\text{Yin}_\infty(x)$  and  $\text{Yang}_\infty(x)$  meets the real axis at the single point*

$$\frac{1}{2} + \mathcal{W}_\infty(x) = d(x),$$

*the limit profile of (31).*

*Proof.* Abbreviate  $A = 2\pi x^2 - \frac{\pi}{8}$ ,  $B = 2\pi x$ , and  $\Psi = \Psi(x)$ , so that  $\text{Yin}_\infty = 1 - \Psi e^{-iA}$  and  $\text{Yang}_\infty = \Psi e^{i(2B-A)}$ . The imaginary parts are  $\Psi \sin A$  and  $\Psi \sin(2B - A)$ , the real parts  $1 - \Psi \cos A$  and  $\Psi \cos(2B - A)$ . The line through points  $z_1, z_2$  with  $\text{Im } z_1 \neq \text{Im } z_2$  crosses the real axis at

$$x_0 = \frac{\text{Re } z_1 \text{Im } z_2 - \text{Re } z_2 \text{Im } z_1}{\text{Im } z_2 - \text{Im } z_1}.$$

Substituting, the denominator is  $\Psi(\sin(2B - A) - \sin A) = 2\Psi \cos B \sin(B - A)$ , and the numerator is  $\Psi(\sin(2B - A) - \Psi \sin 2B)$ . Since  $\Psi = \cos(A - B)/\cos B$  (the form of  $C_0$  used in Lemma A.1) and  $2 \sin B \cos(A - B) = \sin A + \sin(2B - A)$ , the numerator collapses to  $-\Psi \sin A$ , so

$$x_0 = \frac{-\sin A}{2 \cos B \sin(B - A)}.$$

Now let  $\beta = 2\pi(x - \frac{1}{4})(x - \frac{3}{4})$  as in §6.3. Then  $A - B = \beta - \frac{\pi}{2}$ , so  $\sin(B - A) = \cos \beta$ ; and  $B + \beta = A + \frac{\pi}{2}$ , so  $-\sin A = \cos(B + \beta)$ . Therefore

$$x_0 = \frac{\cos(B + \beta)}{2 \cos B \cos \beta} = \frac{1}{2}(1 - \tan(2\pi x) \tan \beta) = \frac{1}{2} + \mathcal{W}_\infty(x) = d(x).$$

Since  $|\beta| \leq \frac{3\pi}{8} < \frac{\pi}{2}$  on  $[0, 1]$ , the factor  $\cos \beta$  never vanishes; the excluded points  $x = \frac{1}{4}, \frac{3}{4}$  are exactly where  $\cos B = 0$ , and there  $\sin A = \sin(2B - A) = 0$ : both ends of the chord land on the real axis itself, and the crossing point is recovered instead by continuity,  $d(\frac{1}{4}) = \frac{1}{4}$  and  $d(\frac{3}{4}) = \frac{3}{4}$ .  $\square$

## 6.5 General yin and yang curves

What are the yin and yang curves when  $\sigma$  is not  $\frac{1}{2}$  and  $T$  is not infinite? Nothing in their construction requires either limit. Return to the exact decomposition  $\zeta = \Sigma_1 + \Sigma_2 + R$  of (10) and, as in §4, write  $t = I(T)$ ,  $s = \sigma + it$ , and  $M = \lceil T \rceil$ . The natural frame in which to watch the remainder is the one that makes the next summand of the main sum the unit segment: translate the end of the main sum,  $\Sigma_1$ , to the origin, and divide by the summand vector  $M^{-s}$ , that is,

$$z \mapsto (z - \Sigma_1) M^s. \quad (36)$$

In this coordinate frame, the next summand of the main sum runs from 0 to 1. Define two points in this frame,

$$\text{Yin}(s) := R M^s \quad (37)$$

and

$$\text{Yang}(s) := \text{Yin}(s) - \chi(s) M^{2s-1}. \quad (38)$$

The first point is where the local coordinate frame map (36) sends  $\Sigma_1 + R = \zeta - \Sigma_2$ , the end of the remainder. The second point is where the transformation to the local frame sends  $\Sigma_1 + R - \chi M^{s-1}$ , the same point after the dual sum takes one more step  $n = M$ : subtracting the next dual summand  $\chi M^{s-1}$  and applying (36) contributes the  $\chi M^{2s-1}$  term. As  $T$  sweeps with  $\sigma$  fixed, the two points trace the general yin and yang curves.<sup>3</sup>

The chord property of Proposition 6.2 is not special to the limit: it is the  $T \rightarrow \infty$  limit of an identity that holds exactly at every finite  $T$ , which we state next. Figure 6 draws the pair for  $1 \leq T \leq 2$  on the critical line over the limit curves of §6.4, with the chord at  $T = 1.18$ .

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<sup>3</sup>The construction has an exact mirror. Sending the next summand of the *dual* sum to the unit segment instead (translate  $\Sigma_1 + R$  to the origin and divide by the dual summand vector  $-\chi M^{s-1}$ , that is,  $z \mapsto (\Sigma_1 + R - z) M^{1-s}/\chi$ ), the two ends of the next *main* summand land at  $R M^{1-s}/\chi$  (the image of  $\Sigma_1$ ) and  $R M^{1-s}/\chi - M^{1-2s}/\chi$  (the image of  $\Sigma_1 + M^{-s}$ ), and the mirrored chord crosses the real axis at  $\hat{d}_2 = d_2 M^{1-\sigma}/|\chi|$ , the fractional amount of the dual sum's next summand.

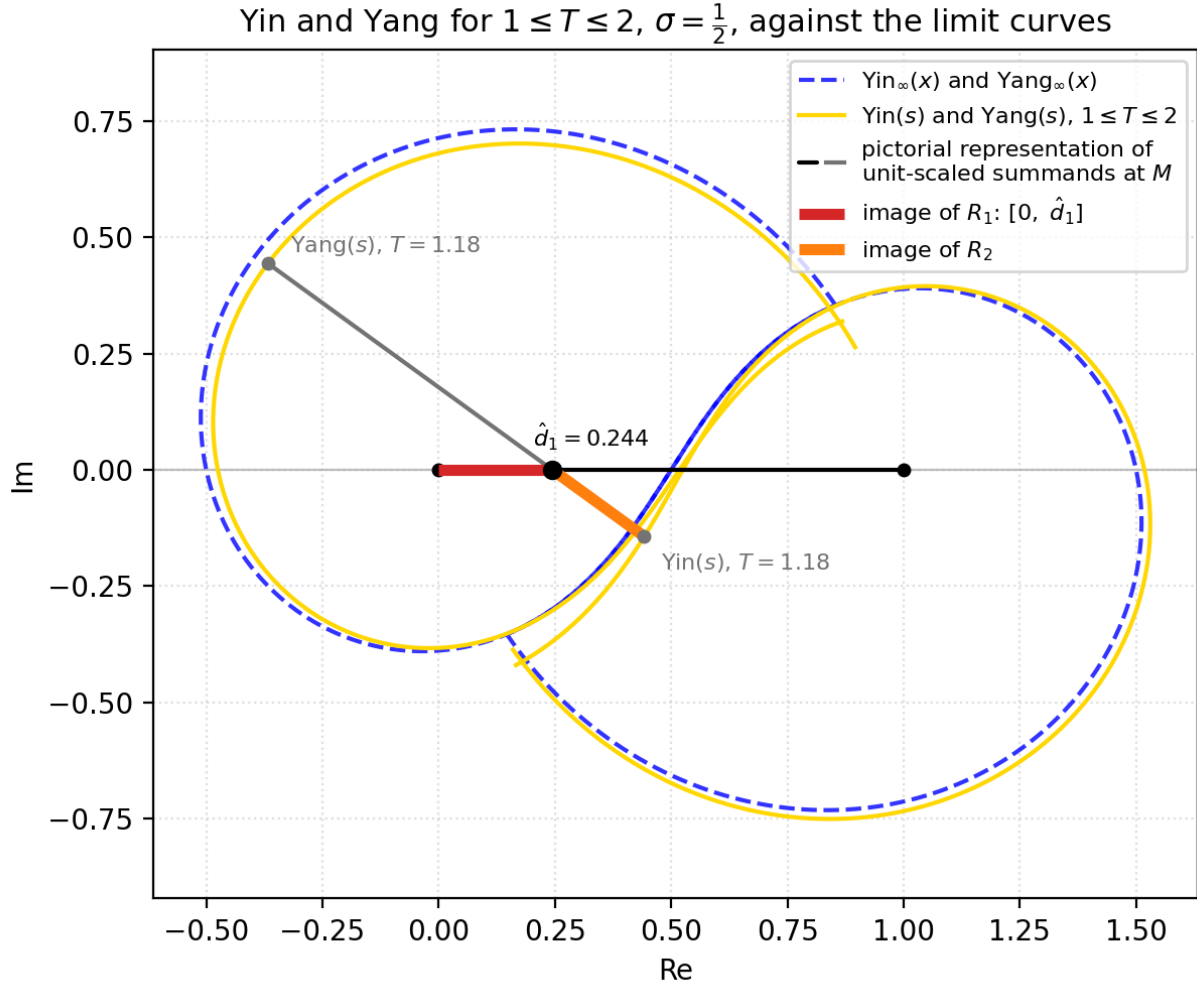


Figure 6: The general yin and yang curves for  $1 \leq T \leq 2$  at  $\sigma = \frac{1}{2}$  (solid, the color of the  $[1, 2]$  interval of Figure 2), drawn over the limit curves  $\text{Yin}_\infty$  and  $\text{Yang}_\infty$  of §6.4 (dashed, both in the blue of Figure 4): even on this very first interval the finite curves track their limits closely. The gray chord joins the two points at  $T = 1.18$  and crosses the real axis at the black dot  $\hat{d}_1 = 0.244$ . The red piece  $[0, \hat{d}_1]$  of the axis is the in-frame image of  $R_1$ , and the orange piece from the crossing to the yin point is the image of  $R_2$ . The black segment from 0 to 1 and the gray chord are together a pictorial representation of the unit-scaled summands at  $M$ : the black segment is the next summand of the main sum carried to  $[0, 1]$  by the local coordinate frame transformation, and the chord is the next summand of the dual sum in the same frame (on the critical line both have length exactly 1). Generated by `fig-yinyang-general.py`.

**Proposition 6.3** (the chord recovers the fractional amount of the next main summand). *Fix  $\sigma$  and non-integer  $T$  with  $\sin(2\omega + \psi) \neq 0$ , where  $\omega = t \log M$  and  $\psi = \arg \chi(s)$  as in §4. Then  $\text{Yin}(s)$  and  $\text{Yang}(s)$  both lie on the line through the real point  $\hat{d}_1 = d_1 M^\sigma$  with direction  $e^{i(2\omega + \psi)}$ :*

$$\text{Yin} = \hat{d}_1 + (d_2 M^\sigma) e^{i(2\omega + \psi)}, \quad \text{Yang} = \hat{d}_1 + (d_2 M^\sigma - |\chi| M^{2\sigma-1}) e^{i(2\omega + \psi)}, \quad (39)$$

*with real coefficients. In particular the line through the two points meets the real axis at the single*

point

$$\hat{d}_1 = d_1 M^\sigma = \frac{R_1}{M^{-s}},$$

the fractional amount of Table 1: the ratio is real because  $R_1$  lies along the next main summand  $M^{-s}$ , and it equals  $|R_1|/|M^{-s}|$  whenever  $d_1 > 0$ .

*Proof.* Since  $M$  is a positive real base,  $M^s = M^\sigma e^{i\omega}$ . Substituting the two-direction form (21) of the remainder,  $R = d_1 e^{-i\omega} + d_2 e^{i(\omega+\psi)}$ , into (37),

$$\text{Yin} = R M^\sigma e^{i\omega} = d_1 M^\sigma + d_2 M^\sigma e^{i(2\omega+\psi)}.$$

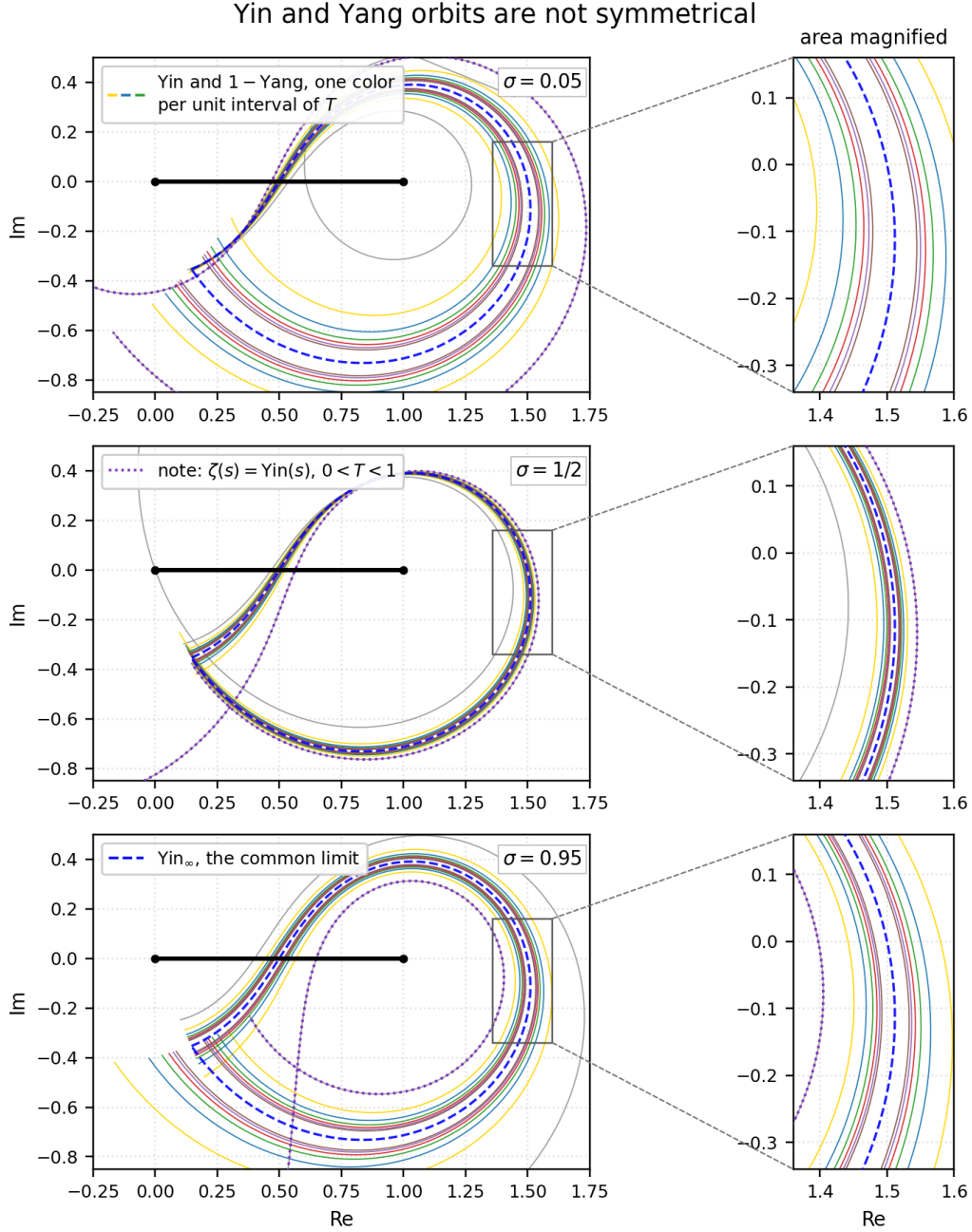
For the yang point,  $\chi M^{2s-1} = |\chi| M^{2\sigma-1} e^{i(2\omega+\psi)}$ , and subtracting it from Yin gives the second formula of (39). The coefficients  $d_1 M^\sigma$ ,  $d_2 M^\sigma$ , and  $d_2 M^\sigma - |\chi| M^{2\sigma-1}$  are real by Proposition 4.1(ii), and the direction  $e^{i(2\omega+\psi)}$  is not horizontal exactly when  $\sin(2\omega+\psi) \neq 0$ , so the line meets the real axis at the single point  $d_1 M^\sigma = \hat{d}_1$ .  $\square$

Whenever  $0 < d_2 < |\chi| M^{\sigma-1}$ , that is, whenever the dual fractional amount is strictly between 0 and 1, the two coefficients in (39) have opposite signs: the yin and yang points then lie on opposite sides of the real axis and the segment itself, not merely its extension, crosses at  $\hat{d}_1$ .

Nothing in the construction is tied to the critical line, and the curves converge to the limit curves of §6.4 at every  $\sigma$ . Figure 7 draws the yin curve against the *reflected* yang curve  $1 - \text{Yang}(s)$  over  $0 \leq T \leq 7$  at three values of  $\sigma$ . The point reflection  $z \mapsto 1 - z$  exchanges the two ends of the unit segment, and under it the yang family lands on top of the yin family; the two limit traces coincide exactly, since a direct computation from (33)–(34), using  $\Psi(1 - x) = \Psi(x)$ , gives

$$1 - \text{Yang}_\infty(x) = \text{Yin}_\infty(1 - x). \quad (40)$$

So a single dashed teardrop, the trace of  $\text{Yin}_\infty$ , is the limit of both families.



The convergence visible in Figure 7 is a theorem, and the machinery used in Appendix A already contains most of its proof: by (37) the yin point is  $R$  carried through the frame map (36), and Appendix A knows the size, the sign, and the phase of  $R$  to exactly the accuracy needed. The one genuinely new ingredient is a polar form of the limit curves.

**Lemma 6.4** (the two limit curves are really a single curve, described explicitly:  $\text{Yang}_\infty(x) = \text{Yin}_\infty(x - \frac{1}{2})$ ). Write  $A(x) = 2\pi(x^2 - \frac{1}{16})$  as in (30) and  $\theta(x) = 2\pi x(1 - x) - \frac{3\pi}{8}$ , the phase of (197). Then for all real  $x$ , with removable points filled in by continuity,

$$\text{Yin}_\infty(x) = 1 - \Psi(x) e^{-iA(x)} = \Psi(x + \frac{1}{2}) e^{i\theta(x)}, \quad (41)$$

and the yang limit is the yin limit half a unit behind in the argument:

$$\text{Yang}_\infty(x) = \text{Yin}_\infty(x - \frac{1}{2}) = \text{Yin}_\infty(x) - e^{2i\theta(x)}. \quad (42)$$

*Proof.* For (41), multiply through by  $\cos(2\pi x)$  and write  $\varphi = 2\pi x$ . On the left,  $\Psi(x) \cos(2\pi x) = \cos(A - \varphi)$ , since  $2\pi(x^2 - x - \frac{1}{16}) = A(x) - \varphi$ . On the right,  $\cos(2\pi(x + \frac{1}{2})) = -\cos(2\pi x)$  and  $(x + \frac{1}{2})^2 - (x + \frac{1}{2}) - \frac{1}{16} = x^2 - \frac{5}{16}$ , so  $\Psi(x + \frac{1}{2}) \cos(2\pi x) = -\cos(A(x) - \frac{\pi}{2}) = -\sin A$ ; also  $\theta(x) = \varphi - A - \frac{\pi}{2}$ , so  $e^{i\theta} = -ie^{i(\varphi - A)}$ . The claim therefore reads

$$\cos \varphi - \cos(A - \varphi) e^{-iA} = (-\sin A) (-i) e^{i(\varphi - A)} = i \sin A e^{i(\varphi - A)},$$

and expanding the left side settles it: the real part is  $\cos \varphi - \cos(A - \varphi) \cos A = \sin A \sin(A - \varphi)$  (product formula), the imaginary part is  $\sin A \cos(A - \varphi)$ , and  $\sin A (\sin(A - \varphi) + i \cos(A - \varphi)) = i \sin A e^{-i(A - \varphi)}$ , which is the right side.

For (42), the definition (34) and the first equality of (41) give  $\text{Yang}_\infty(x) = e^{2i\varphi} (1 - \text{Yin}_\infty(x)) = \Psi(x) e^{i(2\varphi - A)}$ . Applying (41) at  $x - \frac{1}{2}$  instead,  $\text{Yin}_\infty(x - \frac{1}{2}) = \Psi(x) e^{i\theta(x - \frac{1}{2})}$ , and  $\theta(x - \frac{1}{2}) = 2\pi(x - \frac{1}{2})(\frac{3}{2} - x) - \frac{3\pi}{8} \equiv 2\varphi - A \pmod{2\pi}$ , so the first equality holds. For the second,  $e^{2i\theta} = -e^{2i(\varphi - A)}$ , and

$$\begin{aligned} \text{Yin}_\infty(x) - \text{Yang}_\infty(x) &= 1 - \Psi(e^{-iA} + e^{i(2\varphi - A)}) = 1 - 2\Psi \cos \varphi e^{i(\varphi - A)} \\ &= 1 - 2 \cos(A - \varphi) e^{i(\varphi - A)} = -e^{2i(\varphi - A)} = e^{2i\theta(x)}, \end{aligned}$$

using  $\Psi \cos \varphi = \cos(A - \varphi)$  in the middle step.  $\square$

**Theorem 6.5** (the Yin and Yang curves converge to the limit curves,  $\text{Yin}_\infty$  and  $\text{Yang}_\infty$ ). Fix  $x \in (0, 1)$  and  $\sigma \in (0, 1)$ . Then

$$\lim_{\substack{T \rightarrow \infty \\ \{T\}=x}} \text{Yin}(s) = \text{Yin}_\infty(x), \quad \lim_{\substack{T \rightarrow \infty \\ \{T\}=x}} \text{Yang}(s) = \text{Yang}_\infty(x),$$

at rate  $O(1/T)$ , uniformly for  $x$  and  $\sigma$  in compact subsets of  $(0, 1)$ ; the limit does not depend on  $\sigma$ .

The proof needs only Gabcke's bound (201) on the critical line, and off it additionally Siegel's first-order asymptotics (44), the same input as in the proof of Proposition 4.1(iv).

*Proof.* Throughout,  $C_0$  and  $\Psi$  are the same function (§6.4).

*Step 1 (the yin point in asymptotic closed form).* By (37),  $\text{Yin} = R M^s = R(m+1)^\sigma e^{i\omega}$ , with  $M = m+1$  for non-integer  $T$ . On the critical line  $R = e^{-i\vartheta} r$  with  $r$  real (Step 1 of the proof of Proposition 4.1(iii)), so  $\text{Yin} = \sqrt{m+1} r e^{i\omega}$ , and inserting (200)–(201),

$$\text{Yin} = \left(\frac{m+1}{a}\right)^{1/2} (-1)^{N-1} (C_0(\hat{p}) + E) e^{i\omega} + [N = m+1] (1 + e^{2i\omega}), \quad (43)$$



where the extra summand of (200) became the closed pair  $2 \cos u e^{iu} = 1 + e^{2iu}$ . Off the line, the first term of Siegel's asymptotic expansion of the remainder at fixed  $\sigma$  [21], the same input used in the proof of Proposition 4.1(iv), gives

$$R(s) = (-1)^{N-1} \frac{C_0(\hat{p})}{2} \left( a^{-\sigma} e^{-i\tilde{\vartheta}(t)} + \chi(s) a^{\sigma-1} e^{+i\tilde{\vartheta}(t)} \right) + [N = m+1] \left( (m+1)^{-s} + \chi(m+1)^{s-1} \right) + O_\sigma(a^{-\sigma-1}), \quad (44)$$

with the elementary phase  $\tilde{\vartheta}(t) = \frac{t}{2} \log \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} = \vartheta(t) + O(1/t)$ . The error is additive because Siegel's expansion advances in powers of  $a^{-1}$ , one step of which is all we keep. Against the displayed term it is a relative  $O(1/a) = O(1/T)$ , the order of Gabcke's  $E$  in (201): the two pieces of the parenthesis align rather than cancel, since  $|\chi(s)| a^{\sigma-1} = a^{-\sigma} (1 + O(1/t))$  and their phases differ by  $\arg \chi + 2\tilde{\vartheta} = O_\sigma(1/t)$ , so the parenthesis has modulus  $2a^{-\sigma} (1 + o(1))$ , while  $C_0$  stays between  $\cos \frac{3\pi}{8} = 0.3826 \dots$  and  $\cos \frac{\pi}{8} = 0.9238 \dots$  on  $[0, 1]$ . The frame factor  $(m+1)^\sigma e^{i\omega}$  carries the two pieces of the parenthesis to  $\left(\frac{m+1}{a}\right)^\sigma e^{i(\omega-\tilde{\vartheta})}$  and  $|\chi| a^{\sigma-1} (m+1)^\sigma e^{i(\omega+\psi+\tilde{\vartheta})}$ , and since  $|\chi(\sigma+it)| = a^{1-2\sigma} (1 + O(1/t))$  (by Stirling's formula) and  $\psi = -2\vartheta + O_\sigma(1/t)$ , both tend to  $e^{iu}$ ; likewise the Iverson pair maps to  $1 + |\chi|(m+1)^{2\sigma-1} e^{i(2\omega+\psi)} \rightarrow 1 + e^{2iu}$ . In every case

$$Y_{\text{in}} = (-1)^{N-1} C_0(\hat{p}) e^{iu} (1 + O_\sigma(1/T)) + [N = m+1] (1 + e^{2iu}) (1 + O_\sigma(1/T)).$$

*Step 2 (the phase, kept as an angle).* The two case expansions of  $u$  displayed before (197), kept as angles rather than used only through their cosines, combine with (196) into the single statement

$$u \equiv \pi(m+1) + \theta(x) + O(1/T) \pmod{2\pi},$$

while  $\hat{p} \rightarrow x + \frac{1}{2}$  (with  $N = m$ ) for  $x < \frac{1}{2}$  and  $\hat{p} \rightarrow x - \frac{1}{2}$  (with  $N = m+1$ ) for  $x > \frac{1}{2}$ .

*Step 3 (case  $x < \frac{1}{2}$ ).* Here  $N = m$ , the Iverson term is absent, and the sign  $(-1)^{N-1} = (-1)^{m-1}$  cancels against  $e^{i\pi(m+1)} = (-1)^{m+1}$ . With  $\frac{m+1}{a} \rightarrow 1$  (by (195)) and  $C_0 = \Psi$ ,

$$Y_{\text{in}} \rightarrow \Psi\left(x + \frac{1}{2}\right) e^{i\theta(x)} = Y_{\text{in}\infty}(x)$$

by the polar form (41).

*Step 4 (case  $x > \frac{1}{2}$ ).* Here  $N = m+1$ , the sign product is  $-1$ , and the Iverson pair is present:

$$Y_{\text{in}} \rightarrow 1 - \Psi\left(x - \frac{1}{2}\right) e^{i\theta(x)} + e^{2i\theta(x)}.$$

Since  $\theta(x) = -A(x - \frac{1}{2})$  exactly, the first two terms are  $Y_{\text{in}\infty}(x - \frac{1}{2})$ , and the single-curve relation (42) finishes:  $Y_{\text{in}} \rightarrow Y_{\text{in}\infty}(x - \frac{1}{2}) + e^{2i\theta(x)} = Y_{\text{in}\infty}(x)$ .

*Step 5 (the yang point).* By (38),  $\text{Yang} - Y_{\text{in}} = -\chi M^{2s-1}$ , whose modulus  $|\chi|(m+1)^{2\sigma-1} = \left(\frac{m+1}{a}\right)^{2\sigma-1} (1 + O(1/t)) \rightarrow 1$  and phase  $2\omega + \psi = 2u + O_\sigma(1/t) \equiv 2\theta(x) + O(1/T)$ , so the subtracted vector tends to  $e^{2i\theta(x)}$  and  $\text{Yang} \rightarrow Y_{\text{in}\infty}(x) - e^{2i\theta(x)} = \text{Yang}_\infty(x)$ , again by (42).

Every ingredient contributes  $O(1/T)$ : the amplitude ratio  $\frac{m+1}{a} = 1 + O(1/T)$ , Gabcke's  $E = O(1/a)$ , the phase remainder of Step 2, and off the line the  $O_\sigma(1/t)$  terms of  $\psi$  and  $|\chi|$  together with the  $O_\sigma(a^{-\sigma-1})$  of (44), which against a leading term of size  $a^{-\sigma}$  is again  $O(1/T)$ . At the seam  $x = \frac{1}{2}$  the two case formulas agree:  $\Psi(1) = \Psi(0) = \cos \frac{\pi}{8}$  and  $1 + e^{i\pi/4} = 2 \cos \frac{\pi}{8} e^{i\pi/8}$ , so both give  $\cos \frac{\pi}{8} e^{i\pi/8}$ , which is  $Y_{\text{in}\infty}(\frac{1}{2})$ ; the narrow boundary intervals of (196) near  $x = 0, 1$  are excluded by keeping  $x$  in a compact subset of  $(0, 1)$ .  $\square$

Three attributes of the yin and yang curves stand out.

1. At any  $\sigma$ , the yin and yang curves approach  $\text{Yin}_\infty$  and  $\text{Yang}_\infty$  as  $T \rightarrow \infty$ : this is Theorem 6.5, at rate  $O(1/T)$ , with a limit that does not depend on  $\sigma$ . Figure 7 shows the successive orbits settling onto the limit curves as  $T \rightarrow \infty$  for each of  $\sigma = 0.05$ ,  $\frac{1}{2}$ , and  $0.95$ . The chord joining the pair is a *unit needle*: its length is  $|\chi|M^{2\sigma-1}$ , exactly 1 at every  $T$  on the critical line and  $1 + O(1/T)$  at any fixed  $\sigma$ . On the critical line, and at  $T \rightarrow \infty$ , the two curves travel at constant unit separation.
2. The curves are not symmetrical (when not at infinity), and this becomes obvious once the yang curve is reflected to  $1 - \text{Yang}(s)$  and compared with the yin curve: by (40) the two limit traces coincide exactly, but at finite  $T$  the two families only interleave, closing on one another as  $T$  grows and coinciding at no finite  $T$ .<sup>4</sup> The global asymmetry coexists with an instantaneous balance: on the critical line (39) with  $d_1 = d_2$  makes the crossing point equidistant from 0 and the yin point (both distances  $\hat{d}_1$ ) and from 1 and the yang point (both  $1 - \hat{d}_1$ ), so the pair rides two concentric circles centered at the moving crossing, with radii summing to 1.
3. Which family runs outside depends on  $\sigma$ . For  $\sigma$  close to 1 the yin curve runs inside the reflected yang curve; for  $\sigma$  close to 0 it runs outside; and on the critical line the two swap sides near  $\{T\} = \frac{1}{2}$ , one crossing outward as the other crosses inward: at matched positions on the teardrop the two offsets from the limit trade signs. The swap is forced by an exact duality: since  $\chi(s)\chi(1-s) = 1$  exchanges the main and dual sums,  $R(1-s) = \chi(1-s)R(s)$ , and with  $\lambda = \chi(1-s)M^{1-2s}$ ,

$$\text{Yin}(1-s) = \lambda \text{Yin}(s), \quad \text{Yang}(1-s) = \lambda(\text{Yin}(s) - 1), \quad \lambda(\text{Yin} - \text{Yang})(s) = 1,$$

so reflecting  $\sigma \mapsto 1 - \sigma$  (at conjugate height) interchanges yin and yang and swaps the needle with the unit segment.

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<sup>4</sup>The asymmetry is provable in the strongest sense: over one period the two curves have different arc lengths (at  $\sigma = \frac{1}{2}$  the yin loop of  $[6, 7]$  is longer by 0.029; at  $\sigma = 0.05$  by 0.32; at  $\sigma = 0.95$  the yang loop is longer by 0.26; in each case whichever family runs outside, in the sense of the next attribute, is the longer one), so no isometry of the plane (reflection, rotation, or point reflection) carries one curve exactly onto the other. What does hold exactly, at each instant, is a *moving* mirror: at any  $\sigma$  the two points are the ends of the in-frame next dual summand, hence point reflections of each other through its midpoint, and on the critical line  $\text{Yin} = 1 + e^{2iu} \overline{\text{Yang}}$  exactly, a glide reflection whose axis direction  $e^{iu}$  turns with  $T$  (both inputs are exact there: with  $u = \omega - \vartheta(t)$  as in Appendix A, Yin is a real multiple of  $e^{iu}$ , and  $\text{Yang} = \text{Yin} - e^{2iu}$  since  $|\chi M^{2s-1}| = 1$ ). The nearest fixed map, the time-reversed point reflection  $\text{Yang}(T) \approx 1 - \text{Yin}(2m+1-T)$  at the same  $\sigma$ , becomes exact only as  $T \rightarrow \infty$ .

## 7 Summands as links and joints

Equation (22) presents  $\zeta$  as the sum of the four terms  $\Sigma_1 + R_1 + \Sigma_2 + R_2$ , two of which are partial sums. Read geometrically, each summand of a partial sum is one *link* of a planar chain, and each remainder  $R_1, R_2$  is represented by one additional signed partial-summand link. When  $0 \leq \hat{d}_i \leq 1$ , the coefficient is the fraction of that link used; a negative coefficient reverses its direction, and a coefficient exceeding one extends it past the full summand. With  $t, s, m$ , and  $M$  as in (9) and  $\omega = t \log M$  as in (19), denote two complex numbers by

$$B_1 = \Sigma_1 + R_1, \quad B_2 = \Sigma_2 + R_2, \quad \zeta = B_1 + B_2. \quad (45)$$

The pair  $B_1, B_2$  will be used heavily in what follows:  $B_1$  is the bisector point of §9.1, the two legs of §9.2 run from the origin to  $B_1$  and from  $B_1$  to  $\zeta$ , and the angles read off those legs drive the zero-counting functions of §12.

For non-integer  $T$ , each remainder is a single extra signed link appended to its partial sum, pointing along the direction of the  $(m+1)$ st summand with real coefficient  $d_1$  or  $d_2$ , which makes each remainder the partial summand of §5:

$$R_1 = d_1 M^{-it} = d_1 e^{-i\omega}, \quad R_2 = d_2 M^{it} \frac{\chi}{|\chi|} = d_2 e^{i(\omega + \arg \chi)}, \quad (46)$$

with  $d_1, d_2$  as in (17)–(18); the phasor form restates (21).

### 7.1 Sum form

We now write zeta in different forms, first as a sum and then, in the next subsection, as a product, so the two forms can be compared side by side. Splitting  $\zeta$  into its partial sum plus this last fractional link,

$$B_1 = \sum_{n=1}^m n^{-s} + d_1 e^{-i\omega}, \quad (47)$$

$$B_2 = \chi(s) \sum_{n=1}^m n^{s-1} + d_2 e^{i(\omega + \arg \chi)}. \quad (48)$$

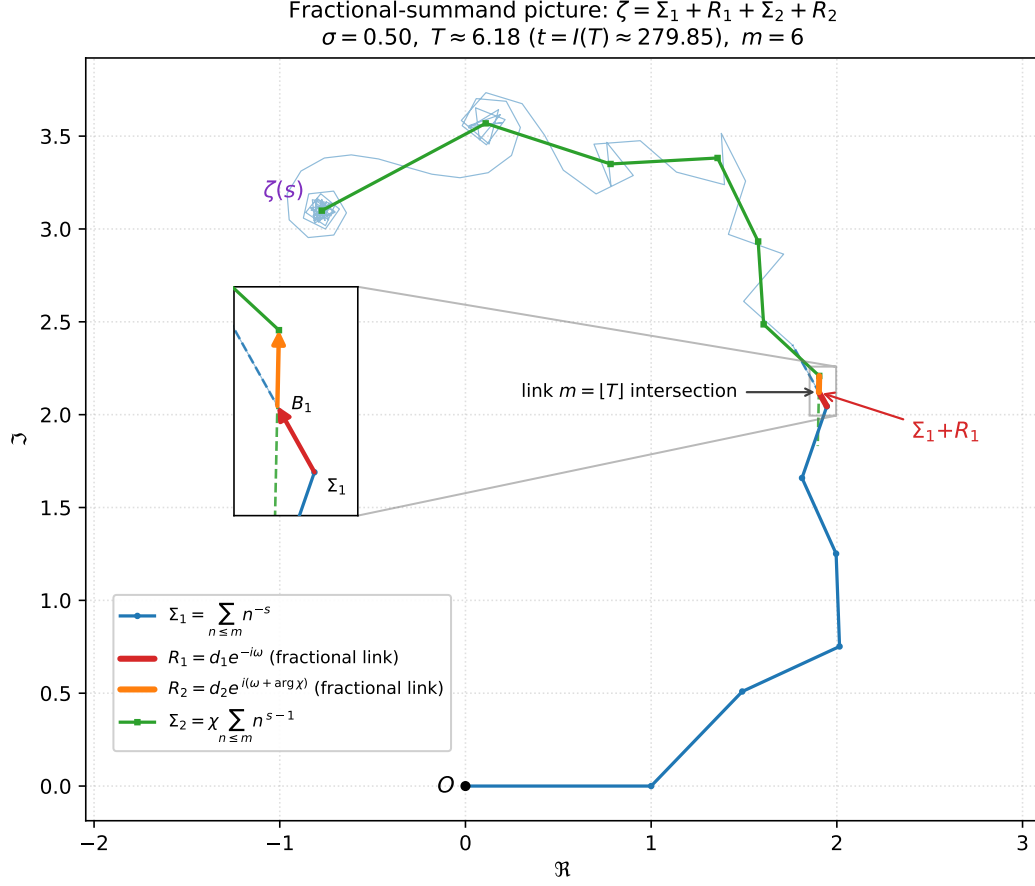


Figure 8: The fractional-summand picture: pictorial (spiral) representation of the four pieces of (23) in the complex plane for  $\sigma = 0.5$  and  $T \approx 6.18$  (equivalently  $t = I(T) \approx 279.85$ ,  $m = 6$ ). The blue path is the partial-sum spiral  $\Sigma_1 = \sum_{n \leq m} n^{-s}$ ; the green path, laid tip-to-tail from the point  $\Sigma_1 + R_1$ , is  $\Sigma_2 = \chi \sum_{n \leq m} n^{s-1}$ . The short red and orange segments  $R_1 = d_1 e^{-i\omega}$  and  $R_2 = d_2 e^{i(\omega + \arg \chi)}$  lie along link  $m$  of each spiral (the faint dashed vectors show the full next summand, of which each remainder is a “fractional” part), so that the four pieces add up to the point  $\zeta(s)$ . Underneath, the thin faint blue path continues the  $\Sigma_1$  chain past  $m$  out to link  $\lfloor I(T)/\pi \rfloor = 89$ , winding into its spiral near  $\zeta$ . On the critical line  $d_1 = d_2$ , so the two remainders have equal length. The fractional links are short here ( $d_1 = d_2 \approx 0.087$ ), so the inset zooms in on them. Generated by `fig1.spiral.summands.py`.

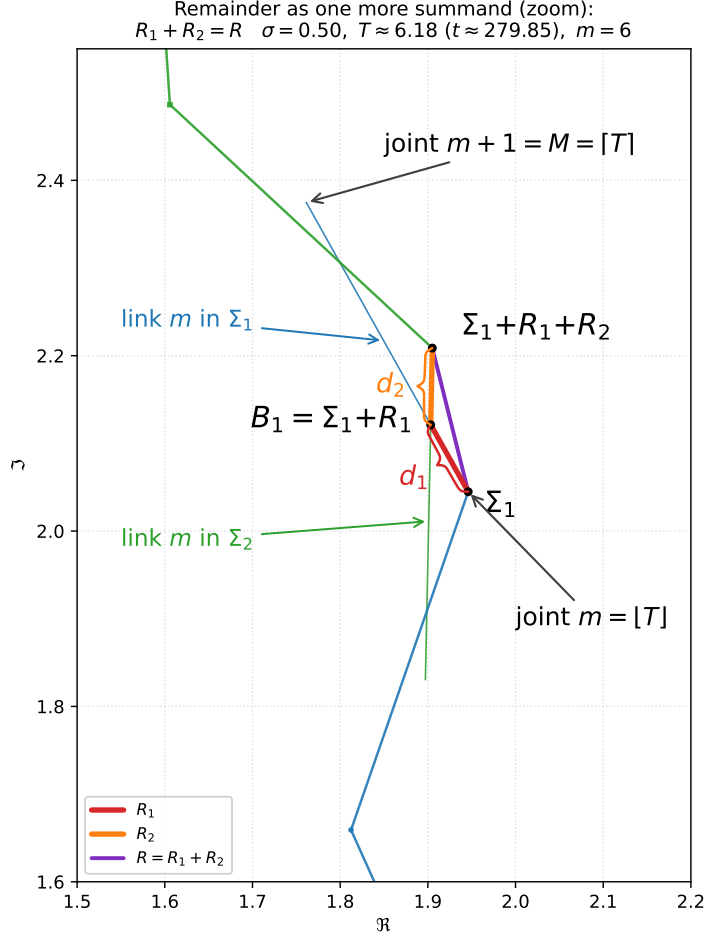


Figure 9: Zoom into the joint region of Figure 8 (imaginary part  $\approx 1.5$ – $2.6$ ), showing the theorem  $R_1 + R_2 = R$  as a small triangle. Starting from the joint  $\Sigma_1$  (end of the sum-1 partial sum), the red link  $R_1$  reaches  $\Sigma_1 + R_1$  and the orange link  $R_2$  reaches  $\Sigma_1 + R_1 + R_2$ ; the straight purple vector from  $\Sigma_1$  to that last joint is Siegel’s remainder  $R = R_1 + R_2$  (see the legend). The curly braces labeled  $d_1$  and  $d_2$  span the two fractional links, marking their lengths, i.e. how far each reaches from its own joint. The thin blue and green lines show the entire extent each fractional summand would have if it were not shortened to the lengths  $d_1$  and  $d_2$ : each extends its fractional link to the full  $(m + 1)$ st summand. The faint blue and green chains are the neighboring links of the two partial sums, arriving at and departing from the joint region. Generated by `fig2_remainder_zoom.py`.

## 7.2 Product form

This subsection is a tangent, and a reader in a hurry can skip it: nothing later rests on the matrix form, which is motivated by an analogy with multi-jointed planar robot manipulators, and a desire for a product form of the zeta function but over the integers, not over the primes.

The usual product form of the zeta function is the Euler product over the primes, one multiplication for each prime,

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}} \quad (\Re s > 1). \quad (49)$$

The goal of this subsection is to write each link as a homogeneous coordinate transformation matrix, and then to group the links into a single transformation matrix, so that  $\zeta$  takes the form of a matrix product with four components: the two sums  $\Sigma_1, \Sigma_2$  and the two partial-summand remainders  $R_1, R_2$ .

Each summand is a link in the plane. We number the links of each chain from zero: link  $n$  runs from joint  $n$  to joint  $n+1$  (joint  $n$  being the value of the partial sum after  $n$  summands), so link  $n$  represents the  $(n+1)$ st summand. The  $m$  summands of a partial sum are thus links  $0, \dots, m-1$ , and each remainder is the *fractional link*  $m$ . We encode one link by the  $3 \times 3$  matrix with link length  $l$  and rotation  $\theta$  from the previous link,<sup>5</sup>

$$\mathbf{L}(\theta, l) = \begin{pmatrix} \cos \theta & -\sin \theta & l \cos \theta \\ \sin \theta & \cos \theta & l \sin \theta \\ 0 & 0 & 1 \end{pmatrix}, \quad (50)$$

where the accumulated position is the top two entries of the last column. The link matrix rotates by  $\theta$  and then translates forward  $l$  along the new heading. Its inverse is again a link-shaped matrix, of a second type that translates *back*  $l$  along the current heading and then undoes the rotation:

$$\mathbf{L}(\theta, l)^{-1} = \begin{pmatrix} \cos \theta & \sin \theta & -l \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (51)$$

Forward links build chain 1; inverses are what let chain 2 be walked backwards.

*Chain 1.* Collect the  $m$  summand links of  $\Sigma_1$  into a single product and give the fractional link its own name:

$$P_{\Sigma_1} = \prod_{n=0}^{m-1} \mathbf{L}(\theta_n, (n+1)^{-\sigma}), \quad \mathbf{L}_{R_1} = \mathbf{L}(-t \ln \frac{m+1}{m}, d_1), \quad (52)$$

where link  $n$  carries the two parameters  $\theta_n$ , its rotation from the previous link, and  $l_n = (n+1)^{-\sigma}$ , its length: as in (20),  $\theta_0 = 0$  and  $\theta_n = -t \ln \frac{n+1}{n}$  for  $n \geq 1$ . Written out, with the first two factors and the last two shown, the chain that reaches  $B_1$  is

$$P_{\Sigma_1} \mathbf{L}_{R_1} = \underbrace{\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link } 0} \underbrace{\begin{pmatrix} \cos \theta_1 - \sin \theta_1 \frac{\cos \theta_1}{2\sigma} \\ \sin \theta_1 & \cos \theta_1 \frac{\sin \theta_1}{2\sigma} \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link } 1} \cdots \underbrace{\begin{pmatrix} \cos \theta_{m-1} - \sin \theta_{m-1} \frac{\cos \theta_{m-1}}{m^\sigma} \\ \sin \theta_{m-1} & \cos \theta_{m-1} \frac{\sin \theta_{m-1}}{m^\sigma} \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link } m-1} \underbrace{\begin{pmatrix} \cos \theta_m - \sin \theta_m d_1 \cos \theta_m \\ \sin \theta_m & \cos \theta_m d_1 \sin \theta_m \\ 0 & 0 & 1 \end{pmatrix}}_{\text{fractional link } m}, \quad (53)$$

the leading factor being the unit step  $1^{-s} = 1$  along the real axis, where  $\theta_0 = 0$  and  $l_0 = 1$  collapse (50) to a pure translation, and the trailing factor being  $\mathbf{L}_{R_1}$ , whose rotation  $\theta_m = -t \ln \frac{m+1}{m}$  continues the same pattern while its length is  $d_1$  rather than a power of  $m+1$ . The product carries the frame from the origin to the point  $B_1 = \Sigma_1 + R_1$ , whose coordinates are the top two entries of the last column, and it arrives with accumulated rotation  $-t \ln(m+1) = -\omega$  by the telescoping (20).

*Chain 2.* The second chain is traversed tip-first, exactly as Figure 8 draws the green chain: the fractional link  $R_2$  leaves  $B_1$ , and the summand links follow in descending order. Walking a

<sup>5</sup>This is the planar case of the Denavit–Hartenberg convention of robot kinematics [8], in which each link of a serial chain is encoded by the  $4 \times 4$  homogeneous transformation  $\text{Rot}_z(\theta_i) \text{Trans}_z(d_i) \text{Trans}_x(a_i) \text{Rot}_x(\alpha_i)$  built from four parameters: the joint angle  $\theta_i$ , the joint offset  $d_i$ , the link length  $a_i$  and the link twist  $\alpha_i$ . A planar chain has  $d_i = \alpha_i = 0$ , which leaves the two parameters  $(\theta_i, a_i)$  and the  $3 \times 3$  matrix (50); (57) is then the forward-kinematic product of the chain. One collision of names to watch: the Denavit–Hartenberg offset  $d_i$  has nothing to do with the weights  $d_1, d_2$  of (17)–(18).

link backwards is exactly its inverse (51); regrouping each rotation with the backward translation that follows it returns everything to the forward type (50), with negated parameters and each rotation paired with the preceding link's length. Two bookkeeping factors appear at the ends of the backward walk: a leading pure translation  $\mathbf{L}(0, -d_2)$ , and a trailing pure rotation  $\mathbf{L}(-\arg \chi, 0)$ , with the phase rotation associated with  $\chi$  peeled off on its own; the modulus  $|\chi|$  remains in the link lengths. The leading translation absorbs the zero-length joint  $\mathbf{L}(2\omega + \arg \chi + \pi, 0)$  that must sit between the two chains (a rotation followed by a translation is a single link), defining the backward fractional link

$$\mathbf{L}_{R_2} := \mathbf{L}(2\omega + \arg \chi + \pi, 0) \mathbf{L}(0, -d_2) = \mathbf{L}(2\omega + \arg \chi + \pi, -d_2); \quad (54)$$

the backward summand links collect into

$$P_{\Sigma_2} := \prod_{n=m}^1 \mathbf{L}(\theta_n, -|\chi| n^{\sigma-1}) \quad (n \text{ descending, } \theta_n = -t \ln \frac{n+1}{n}); \quad (55)$$

and the trailing rotation is simply dropped, since a post-multiplied zero-length joint never moves the position column, only the final orientation.

Written out as (53) was, with the first two factors and the last two shown and with  $\psi = \arg \chi$  as before, the second chain is

$$\begin{aligned} \mathbf{L}_{R_2} P_{\Sigma_2} = & \underbrace{\begin{pmatrix} \cos(2\omega + \psi + \pi) & -\sin(2\omega + \psi + \pi) & -d_2 \cos(2\omega + \psi + \pi) \\ \sin(2\omega + \psi + \pi) & \cos(2\omega + \psi + \pi) & -d_2 \sin(2\omega + \psi + \pi) \\ 0 & 0 & 1 \end{pmatrix}}_{\text{fractional link } m} \underbrace{\begin{pmatrix} \cos \theta_m & -\sin \theta_m & -\frac{|\chi| \cos \theta_m}{m^{1-\sigma}} \\ \sin \theta_m & \cos \theta_m & -\frac{|\chi| \sin \theta_m}{m^{1-\sigma}} \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link } m-1} \cdots \\ & \cdots \underbrace{\begin{pmatrix} \cos \theta_2 & -\sin \theta_2 & -\frac{|\chi| \cos \theta_2}{2^{1-\sigma}} \\ \sin \theta_2 & \cos \theta_2 & -\frac{|\chi| \sin \theta_2}{2^{1-\sigma}} \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link 1}} \underbrace{\begin{pmatrix} \cos \theta_1 & -\sin \theta_1 & -|\chi| \cos \theta_1 \\ \sin \theta_1 & \cos \theta_1 & -|\chi| \sin \theta_1 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link 0}}, \end{aligned} \quad (56)$$

the leading rotation being the merged joint of (54). The links run backwards, from the fractional link  $m$  down to link 0, each brace naming the link whose length that factor carries; the rotation it carries is the  $\theta_n$  of the link after it, which is the regrouping. Its last column is not  $B_2$  but  $e^{i\omega} B_2$ , chain 2 read in the frame chain 1 leaves behind, whose orientation is  $-\omega$ ; multiplying on the left by  $P_{\Sigma_1} \mathbf{L}_{R_1}$  turns that back onto world axes and adds  $B_1$ , which is (57). All three columns agree to 28 digits at  $\sigma = \frac{1}{2}$ ,  $T = 6.18$  (`check_chain2_matrix.py`).

**The full  $3 \times 3$  equation for  $\zeta$ .** With these four names the two chains combine into one serial product, in a single link type:

$$\mathbf{T}_\zeta = P_{\Sigma_1} \mathbf{L}_{R_1} \mathbf{L}_{R_2} P_{\Sigma_2} = \begin{pmatrix} \cos \alpha_\zeta & -\sin \alpha_\zeta & \Re \zeta \\ \sin \alpha_\zeta & \cos \alpha_\zeta & \Im \zeta \\ 0 & 0 & 1 \end{pmatrix}, \quad \alpha_\zeta = \pi + \arg \chi. \quad (57)$$

The last column is  $\zeta$  and the rotation block records  $\alpha_\zeta$ , discussed below. The factors line up one to one with the terms of (22), repeated here for comparison:

$$\zeta(s) = \Sigma_1 + R_1 + R_2 + \Sigma_2. \quad (22)$$

Figure 10 draws (57) with one arrow per factor.

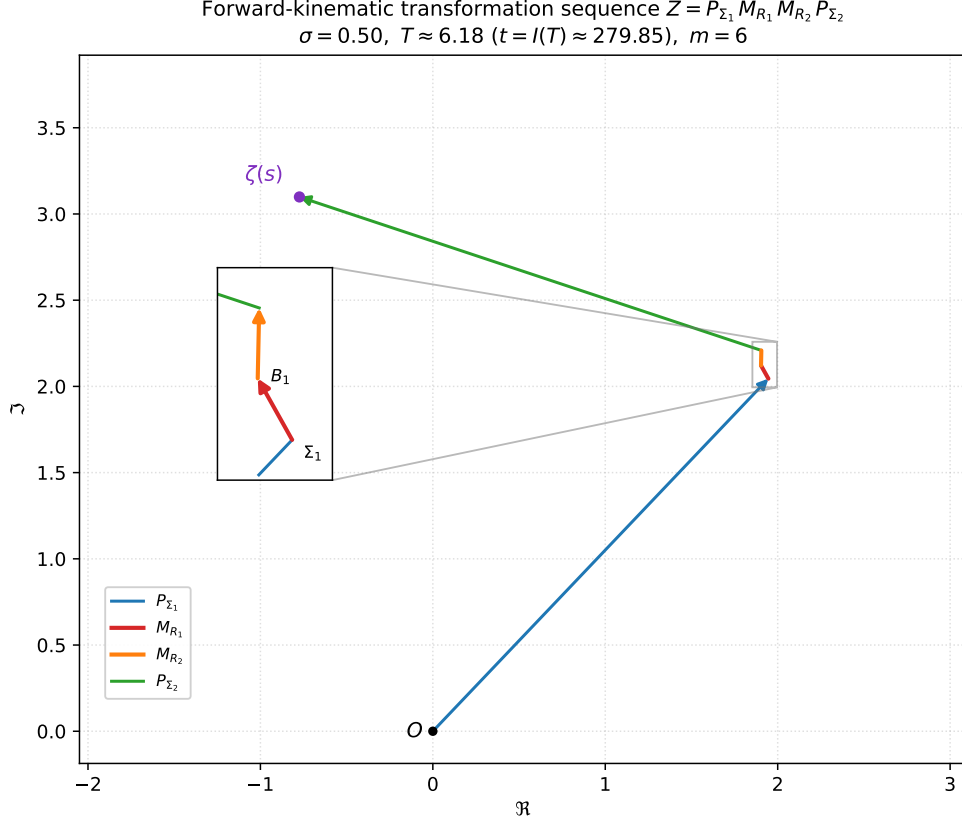


Figure 10: The serial chain of (57) at  $\sigma = \frac{1}{2}$ ,  $T = 6.18$  ( $t = I(T) \approx 279.85$ ,  $m = 6$ ), one arrow for the net displacement of each matrix factor:  $P_{\Sigma_1}$  (blue,  $O$  to  $\Sigma_1$ ),  $L_{R_1}$  (red, to  $B_1$ ),  $L_{R_2}$  (orange, to  $B_1 + R_2$ ); its rotation parameter carries the merged joint  $2\omega + \arg \chi + \pi$ , and  $P_{\Sigma_2}$  (green, to  $\zeta$ ). The four arrows are the four terms of  $\zeta = \Sigma_1 + R_1 + R_2 + \Sigma_2$ ; the two fractional links are short here ( $d_1 = d_2 \approx 0.087$ ), so the inset zooms in on them. Colors match Figure 8. Generated by `fig_zeta_chain.py`.

Three features of (57) deserve comment. First, the rotation  $2\omega + \arg \chi + \pi$  carried by  $L_{R_2}$  has three jobs at once:  $+\omega$  cancels chain 1's accumulated rotation  $-t \ln(m+1) = -\omega$ , a further  $+(\omega + \arg \chi)$  pre-rotates against the counter-rotation of the backward-walked chain 2, and the final  $+\pi$  rotates the backward translations to point forward again; together  $\omega + (\omega + \arg \chi) + \pi = 2\omega + \arg \chi + \pi$ . Second, the links of  $P_{\Sigma_2}$  carry exactly the rotation parameters  $\theta_n$  of chain 1: in this form the two chains use identical rotations and differ only in their lengths,  $(n+1)^{-\sigma}$  forward versus  $-|\chi| n^{\sigma-1}$  backward. Third, the net displacements of the four factors  $P_{\Sigma_1}$ ,  $L_{R_1}$ ,  $L_{R_2}$ ,  $P_{\Sigma_2}$  are exactly the four terms  $\Sigma_1$ ,  $R_1 = d_1 e^{-i\omega}$ ,  $R_2 = d_2 e^{i(\omega + \arg \chi)}$ ,  $\Sigma_2$  of (22), and the final orientation  $\alpha_\zeta = \pi + \arg \chi$  is the direction of  $-\chi/|\chi|$ : the trailing rotation  $L(-\arg \chi, 0)$  (the phase rotation associated with  $\chi$ , peeled off automatically when the base link of chain 2 is regrouped; the modulus  $|\chi|$  remains in the link lengths) was discarded, so  $\chi$ 's direction is left showing in the orientation block, and  $T_\zeta$  packages the pair  $(\zeta, \arg \chi)$  in one matrix. (Appending the discarded  $L(-\arg \chi, 0)$  would leave the position column untouched and make the orientation the constant  $\pi$ .)



## 8 Other remainders

The split  $R = R_1 + R_2$  of §4 is not the only way to dissect Siegel’s remainder in two. This section introduces two others, so that three splits are in play, all reassembling the same  $R$ : exactly in two of the three cases and to high accuracy in the third. We first name the three, then drill down on the newest of them, Kuznetsov’s; a closing subsection (§8.3) records a scaling observation. The remainder figures at different heights are nearly scaled copies, with the residual amplitude discrepancy tending toward zero as  $T$  grows.

### 8.1 The three splits and their names

To keep the remainders straight we name them with subscripts. Siegel’s remainder integral is written  $R$  when unsplit; each named split below recovers that same  $R$  (exactly, or approximately for the Kuznetsov form):

1. **Riemann–Siegel half-split.**

$$R = R_{rs} = R_{1rs} + R_{2rs}, \quad R_{1rs} = R_{2rs} = \tfrac{1}{2}R.$$

Here  $R_{1rs}$  is simply half the remainder integral.

2. **Partial-summand split** (§4–§5), the split of this paper.

$$R = R_{ps} = R_{1ps} + R_{2ps}.$$

Here  $R_{1ps}$  is a partial next summand: the next Dirichlet term after  $\Sigma_1$  stops, the  $M$ -th, shortened to length  $d_1$ . The unsubscripted  $R_1, R_2$  used everywhere else in this paper are exactly  $R_{1ps}, R_{2ps}$ ; we attach the  $ps$  subscript only where several splits stand side by side, as in this section and §9.

3. **Kuznetsov / Siegel  $f$  split** (§8.2).

$$R \approx R_{ak} = R_{1ak} + R_{2ak}.$$

Here  $R_{1ak}$  is, to high accuracy, Siegel’s  $f(s)$  minus the first partial sum,  $R_{1ak} \approx f(s) - \Sigma_1$ , with equality if this is taken as the definition (§8.2).

All three are measured from the same pair of joints, both at index  $m = \lfloor T \rfloor$ : each  $R_{1\bullet}$  leaves joint  $m$  of the  $\Sigma_1$  chain of §7, which is the partial sum  $\Sigma_1$  itself, and each  $R_{2\bullet}$  arrives at joint  $m$  of the  $\Sigma_2$  chain, which is the point  $\Sigma_1 + R$ . A split therefore chooses nothing but the point  $\Sigma_1 + R_{1\bullet}$  at which the two halves meet (for the  $ps$  split this meeting point is the  $B_1$  of §7); the anchoring joints are common to all three. Named for their splits, the three such points are

$$B_{1rs} = \Sigma_1 + \tfrac{1}{2}R, \quad B_{1ps} = \Sigma_1 + R_{1ps} = B_1, \quad B_{1ak} = \Sigma_1 + R_{1ak} \approx f(s). \quad (58)$$

### 8.2 Kuznetsov’s remainders

It is worth contrasting our remainders  $R_{1ps}, R_{2ps}$  with the remainder of Alexey Kuznetsov [13], whose paper appeared on the arXiv in March 2025 and in print in 2026. It gives a simple and accurate algorithm for computing  $\zeta$ , which we have found useful in our own numerical work. There are two differences. The first is one of *kind*: Kuznetsov’s remainder is an *approximation* to  $R$ , whereas  $R_{1ps} + R_{2ps} = R$  is an *exact* identity (Theorem 4.2), though, as we shall see below, if  $R_{1ak}$

is read as  $f(s) - \Sigma_1$  the *ak* split is exact too, the approximation moving into the numerical evaluation of  $f$ . The second is one of *direction*: neither of his two halves lies along the next summand of its Dirichlet sum, while each of ours is exactly that summand, shortened.

Where his pair comes from is worth a word, since Siegel has one remainder and Kuznetsov has two. Siegel's §2 opens at his equation 8, one Dirichlet sum plus one contour integral, and the residue theorem carries that into his equation 13, our (1): still one integral, now standing beside both sums. One integral, one remainder, and that single  $R$  is what we split in two. Kuznetsov opens instead at the symmetric formula of Siegel's §3, Siegel's equation 56, the one he says makes the functional equation evident. It carries two integrals, one in  $s$  and its reflection in  $1 - s$  weighted by  $\chi$ , so his two halves arrive with the starting point rather than from a later choice. Gabcke's rederivation [10] begins in the same place, and Kuznetsov reaches it through the quadrature forms of Galway [11] and Arias de Reyna [1] rather than from Siegel directly.

Writing Kuznetsov's approximate remainder as  $R_{ak}$  (the subscript "*ak*" for Alexey Kuznetsov, and to distinguish it from Siegel's exact, unsubscripted  $R$ ), his method gives

$$R \approx R_{ak} = -\frac{1}{2}(-1)^m (I_1 + \chi(s) I_2), \quad (59)$$

where  $I_1, I_2$  are the two integrals of Kuznetsov's construction, each evaluated as a short finite series in precomputed coefficients  $\omega_0, \omega_1[n], \lambda[n]$  (his notation; this  $\omega_0$  is unrelated to the direction angle  $\omega$  of (19)); in some of our calculations we use 8 such coefficients, which already gives a dozen or more digits of accuracy across the range of interest. Just as we split  $R$  into two pieces, the two terms of (59) define Kuznetsov's two half-remainders

$$R_{ak} = R_{1ak} + R_{2ak}, \quad R_{1ak} = -\frac{1}{2}(-1)^m I_1, \quad R_{2ak} = -\frac{1}{2}(-1)^m \chi I_2, \quad (60)$$

so that, in parallel with (22),

$$\zeta \approx \Sigma_1 + R_{1ak} + \Sigma_2 + R_{2ak}.$$

The split (60) mirrors the functional equation:  $R_{1ak}$  pairs with the first Dirichlet sum  $\Sigma_1$  and  $R_{2ak}$  carries the functional-equation factor  $\chi$  alongside  $\Sigma_2$ . Figure 11 compares the two routes from  $\Sigma_1$  to  $\Sigma_1 + R$ : the exact fractional links on one side and Kuznetsov's dashed pair on the other.

This pairing is exactly Siegel's: in his 1932 paper [21] Siegel writes  $\zeta$  as  $f(s)$  plus its reflected counterpart, and the first side is

$$f(s) \approx \Sigma_1 + R_{1ak}, \quad (61)$$

i.e. the first partial sum completed by an approximation to the first half of the remainder. The sign is  $\approx$  rather than  $=$  because  $R_{1ak}$  is the quantity  $-\frac{1}{2}(-1)^m I_1$  of (60), with  $I_1$  evaluated as a truncated series; read the other way round, as the *definition*  $R_{1ak} := f(s) - \Sigma_1$ , the relation is an identity and the approximation moves into the numerical evaluation of  $R_{1ak}$ .

The form of (61) is Siegel's own, and it is worth recording where in his paper to look for it. He defines  $f$  in his §3, equation 58, as the contour integral

$$f(s) = \int_{0 \swarrow 1} \frac{x^{-s} e^{\pi i x^2}}{e^{\pi i x} - e^{-\pi i x}} dx, \quad (62)$$

where his symbol  $0 \swarrow 1$  marks a straight path crossing the real axis between 0 and 1, running from upper right to lower left parallel to the bisector of the first and third quadrants; it is the reflection in the real axis of the path  $0 \searrow 1$  carrying the integral  $\Phi$  of his §1. The Gaussian factor  $e^{\pi i x^2}$

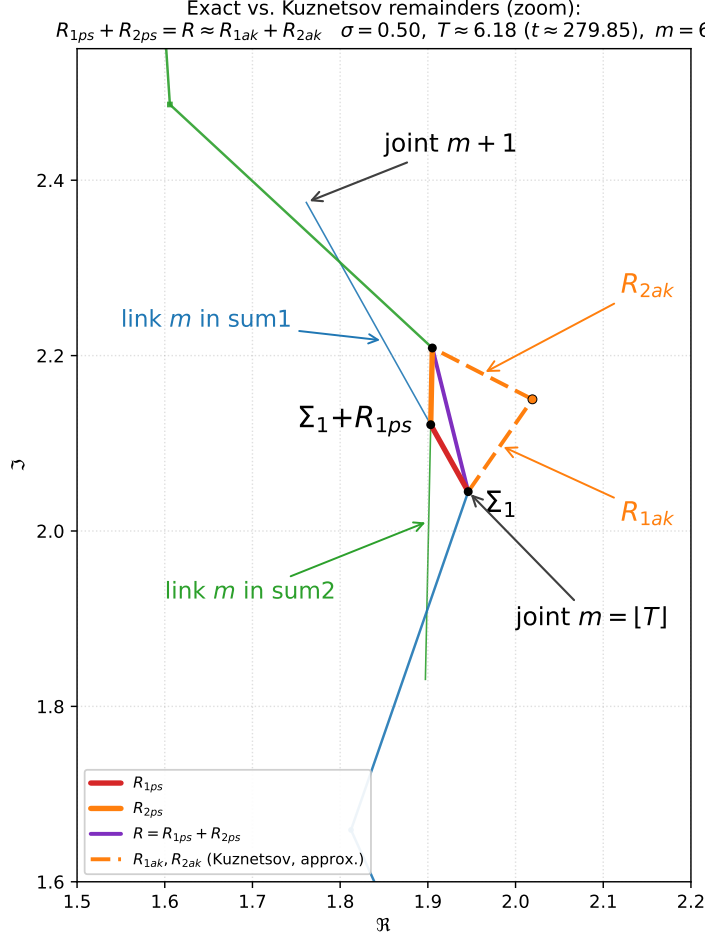


Figure 11: Exact versus Kuznetsov remainders, in the same zoom region as Figure 9. The exact fractional summands  $R_{1ps}$  (red) and  $R_{2ps}$  (orange) run from  $\Sigma_1$  to  $\Sigma_1 + R$  on one side of the resultant  $R$  (purple); Kuznetsov’s approximate half-remainders  $R_{1ak}, R_{2ak}$  are drawn as the orange dashed pair. Since  $R_{1ak} + R_{2ak} \approx R$  (here to within  $10^{-14}$ ), the dashed path lands on  $\Sigma_1 + R$ ; unlike the exact links, the  $ak$  links do not lie along a summand direction. Computed with 8 Kuznetsov coefficients and generated by `fig4_kuznetsov_zoom.py`.

decays in both directions along that line, so (62) converges. Siegel then develops  $f$  asymptotically in his §4, where his equation 84 reads

$$f(s) = \sum_{n=1}^{m_1} n^{-s} + O(|s|/2\pi e)^{-\sigma/2} \quad (\sigma \geq 0, t > 0), \quad (63)$$

with  $m_1 = [\Re \eta_1 - \Im \eta_1]$  and  $\eta_1 = \sqrt{s/2\pi i}$ , which to leading order is Siegel’s cutoff  $[\sqrt{t/2\pi}]$  of (8). Earlier in the same section he states it in words as well: developing  $f$  asymptotically yields “as a principal term a sum of  $[\sqrt{t/2\pi}]$  summands, that is  $\sum_{n=1}^m n^{-s}$ ”. So “first partial sum plus a remainder” is Siegel’s own description of  $f$ , and it shows the remainder is genuinely needed: at  $\sigma = \frac{1}{2}$  the bare partial sum misses  $f$  by  $O(t^{-1/4})$ , the gap that  $R_{1ak}$  closes. One caution on the cutoff: because  $\sqrt{t/2\pi} = T + \frac{1}{2} - \frac{1}{24T} + O(T^{-2})$  by Remark 3.1, while  $\Re \eta_1 - \Im \eta_1 = \sqrt{t/2\pi} + \frac{\sigma}{4\pi T} + O(T^{-2})$ , Siegel’s  $m_1$  is one more than our  $m = [T]$  whenever  $\{T\} > \frac{1}{2} + \frac{1}{24T} - \frac{\sigma}{4\pi T}$ ; at  $\sigma = \frac{1}{2}$  the two

corrections nearly cancel, the coefficient  $\frac{1}{24} - \frac{1}{8\pi}$  being about 0.0019. This does not disturb (61), since  $R_{1ak}$  carries the cutoff as a parameter and re-centers with it. Siegel’s choice aligns the cutoff with the saddle and yields the asymptotic remainder stated above. Equation numbers cited here are those of Siegel’s paper as reprinted in his collected works, the numbering carried over by the Barkan–Sklar translation [2].<sup>6</sup>

Neither Siegel nor Kuznetsov names the other side, but it is  $\Sigma_2 + R_{2ak}$ , and adding the two recovers  $\zeta$  to the same accuracy. The contrast with our result is that  $\Sigma_1 + R_{1ps}$  and  $\Sigma_2 + R_{2ps}$  are exact however they are read, and each remainder there is a single fractional summand (§5).

### 8.3 Scaling the remainders with the fractional part of $T$ unchanged

This subsection is a diversion into an observation we made about the three remainders just discussed. At a fixed fractional part  $\{T\}$ , the remainder figures at different heights are nearly scaled copies, with the residual amplitude discrepancy tending toward zero as  $T$  grows. This scaling is the unit- $T$  periodicity of the yin and yang curves (§6.5) expressed in the raw remainders: the frame map there multiplies by  $M^s$ , which removes exactly the  $M^{-\sigma}$  size factor and the  $\omega$  rotation observed here, so a figure that repeats with  $\{T\}$  in that frame must reappear here as a nearly scaled copy at each  $\{T\}$ ; and since all three splits assemble the same  $R$ , the copy carries all three. Nothing in the rest of the paper uses this observation.

Write  $T = [T] + \{T\}$  for the integer and fractional parts of the index. Empirically, the *shape* of the remainder figure depends almost entirely on the fractional part  $\{T\}$ . By shape we mean the relative angles among  $R$ ,  $R_{1ps}$ ,  $R_{2ps}$ ,  $R_{1ak}$ , and  $R_{2ak}$  once the configuration is rotated into the frame in which  $R$  lies along the positive real axis. The integer part  $[T]$  mainly retunes the overall size: the lengths scale like  $M^{-\sigma}$  times a slowly varying amplitude. Figure 12 makes this visible. Each panel shows only the remainders, drawn in the  $R$ -frame (rotated so  $R$  lies on the real axis and translated so the midpoint of  $R$  sits at the origin), with a common axis scale. The top row holds  $\{T\} = 0.18$  at  $T = 6.18$ ,  $50.18$ , and  $100.18$ ; the bottom row holds  $\{T\} = 0.72$  at  $T = 6.72$ ,  $50.72$ , and  $100.72$ . Across a row the figure shrinks but keeps its shape; across a column (same  $[T]$ , different  $\{T\}$ ) the shape changes.

---

<sup>6</sup>One incidental finding while checking (61) against the source: the translation’s equation 83 prints the sum as  $\sum n^{s-1}$ , while equation 84, derived from it, has  $\sum n^{-s}$ . The numerics say  $n^{-s}$  is correct, so equation 83 carries a typo in the Barkan–Sklar translation.

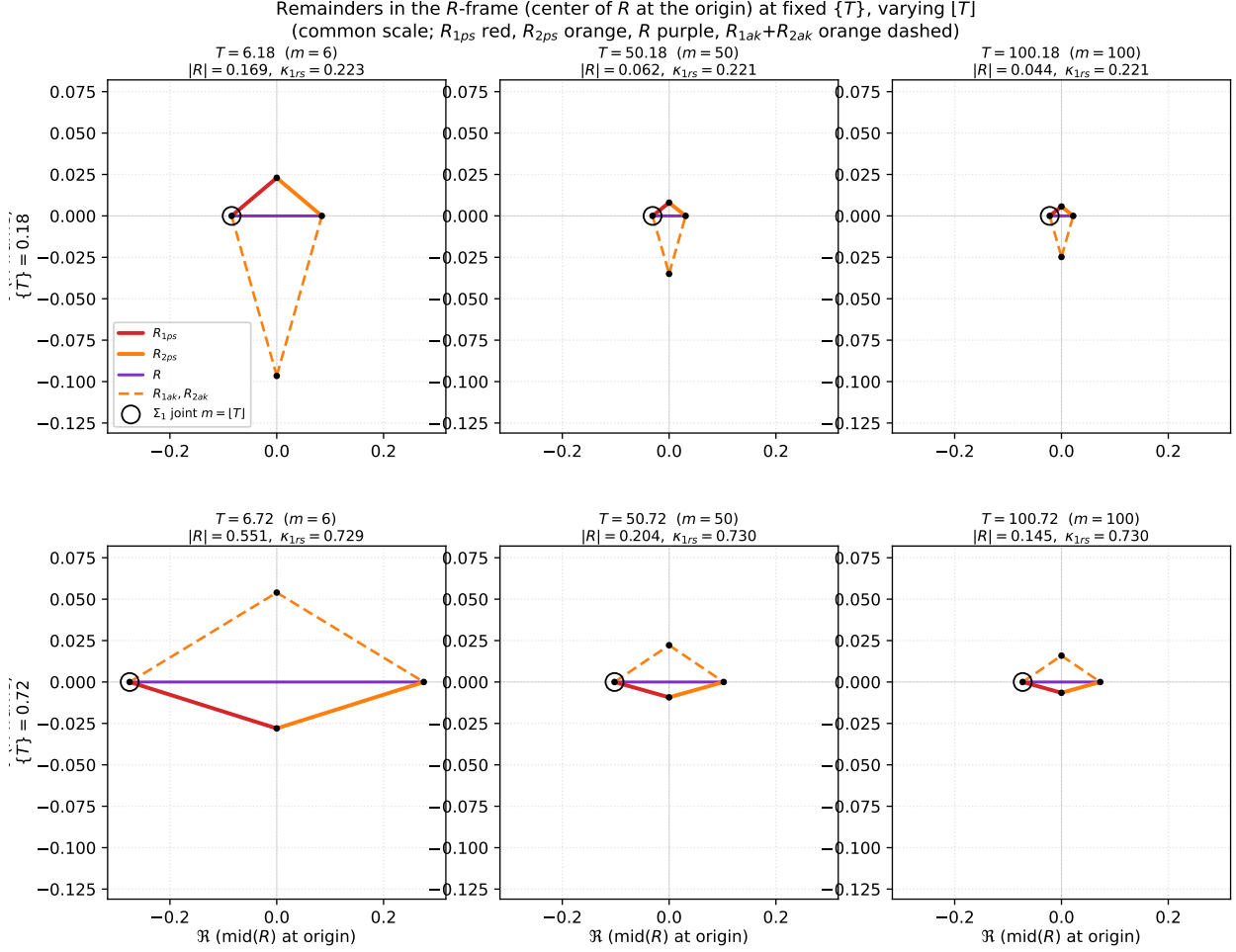


Figure 12: Remainders in the  $R$ -frame at fixed fractional part  $\{T\}$ , varying  $[T]$ . The frame is centered on the midpoint of  $R$  (so  $R$  runs from  $-|R|/2$  to  $+|R|/2$  on the real axis). Top row:  $\{T\} = 0.18$  at  $T = 6.18, 50.18, 100.18$ . Bottom row:  $\{T\} = 0.72$  at  $T = 6.72, 50.72, 100.72$ . Colors match Figure 11:  $R_{1ps}$  red,  $R_{2ps}$  orange,  $R$  purple,  $R_{1ak}$  and  $R_{2ak}$  orange dashed. The large open circle marks the left endpoint of  $R$ , which is the anchoring joint  $\Sigma_1$  at  $m = [T]$  that all three splits leave from. All six panels share the same axis scale. Generated by `fig_remainder_scale_panels.py`.

That figure holds  $\{T\}$  fixed and climbs the strip. The complementary view holds the chain fixed and lets  $T$  sweep a whole unit, which is Figure 13: the same  $R$ -frame in sixteen snapshots from  $T = 4$  to  $T = 5$  at  $m = 4$ . Two things show up there that a fixed  $\{T\}$  cannot show. First, as  $T$  sweeps, both apexes stay pinned to the perpendicular bisector of  $R$ : on the critical line  $|R_{1ps}| = |R_{2ps}|$  exactly (by Proposition 4.1(iii), since  $d_1 = d_2$  and the two directions are unit), likewise for the  $ak$  pair, and this holds at every instant of the motion, so the state of a split is at all times the single number giving the height of its apex above the chord. Second, the two splits move quite differently: the  $ak$  apex marches steadily from below  $R$  to well above it, while the  $ps$  apex swings down onto the chord, crosses to the far side, and comes back, touching the chord exactly at the two heights where  $\sin(2\omega + \arg \chi)$  vanishes (Appendix A.5). There the two unit directions of (21) have become parallel and no longer span the plane, and on the critical line the split collapses into the plain halving  $R_{1ps} = R_{2ps} = \frac{1}{2}R$ : the  $d_1$  pole of (17) is removable at  $\sigma = \frac{1}{2}$ , its numerator

vanishing with its denominator, which is why the sweep passes through those heights smoothly. Off the line it is a genuine pole: at  $\sigma = 0.6$  and  $\sigma = 0.75$ ,  $d_1$  runs through infinity and changes sign there, and that is what the windows of Appendix A.5 are.

The remainder splits in the  $R$ -frame,  $T = 4 \rightarrow 5$  at  $\sigma = 1/2$  (chain length  $m = 4$ , common scale)

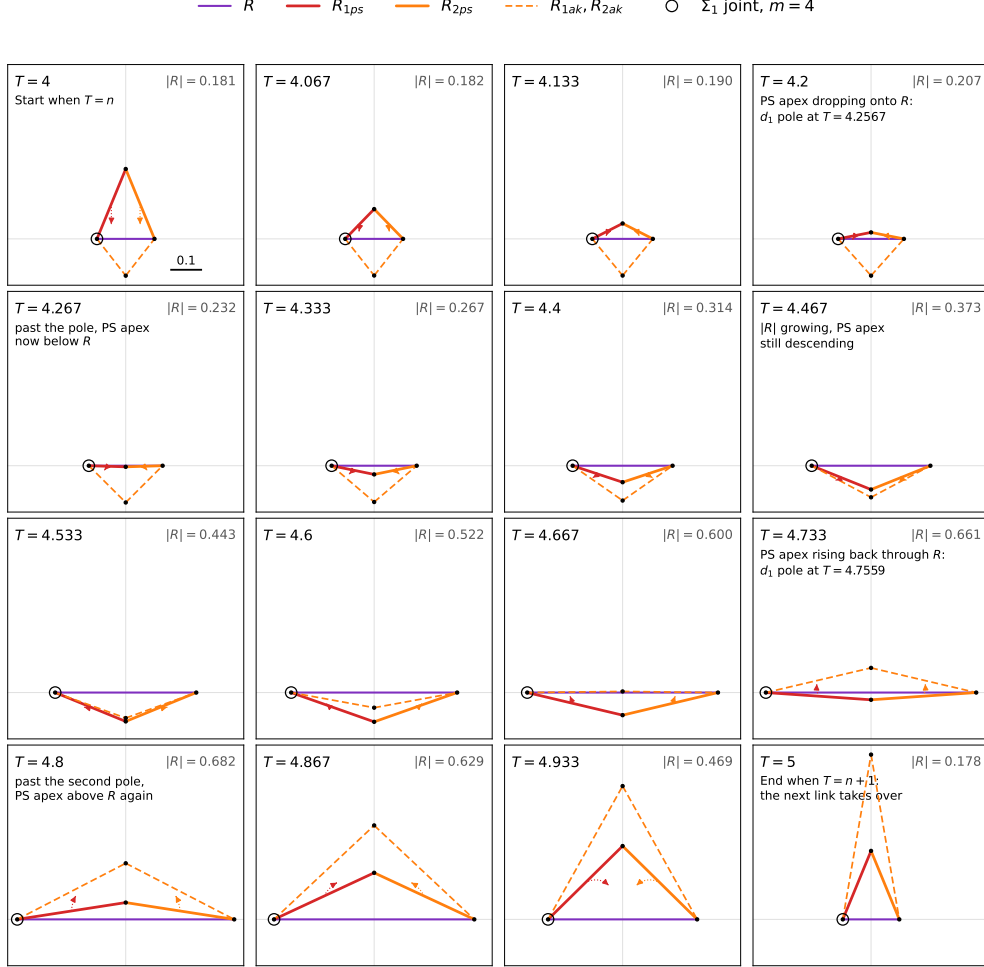


Figure 13: The remainder splits in the  $R$ -frame as  $T$  sweeps one unit,  $T = 4 \rightarrow 5$  at  $\sigma = \frac{1}{2}$ . Frame and colors as in Figure 12:  $R$  purple along the real axis, midpoint at the origin;  $R_{1ps}$  red and  $R_{2ps}$  orange from the  $\Sigma_1$  joint (open circle) to the far end of  $R$ ;  $R_{1ak}$ ,  $R_{2ak}$  orange dashed; all panels to one scale,  $|R|$  given in each. The chain length is held at  $m = 4$  throughout, so  $T = 5$  is the handoff. Dotted arrows trace the midpoints of  $R_{1ps}$  and  $R_{2ps}$  from each panel to the next. The  $ps$  apex crosses the chord between panels four and five and again between twelve and thirteen, the two heights of (203) where the split is the plain halving; the  $ak$  apex crosses once, between ten and eleven. Generated by `fig_remainder_frame_sweep.py`.

To make the scaling precise, factor out the leading  $M^{-\sigma}$  from each half-remainder length by writing

$$|R_{1\bullet}| = \kappa_{1\bullet} M^{-\sigma}$$

for the three first halves  $R_{1\bullet} \in \{R_{1rs}, R_{1ps}, R_{1ak}\}$  of §8.1. Each split is then on the same footing:  $R = R_{1rs} + R_{2rs} = R_{1ps} + R_{2ps} \approx R_{1ak} + R_{2ak}$ , and for the  $rs$  half-split one has  $R_{1rs} = R_{2rs} = \frac{1}{2}R$

so  $\kappa_{2rs} = \kappa_{1rs}$ . The three amplitudes admit the forms

$$\begin{aligned}\kappa_{1rs} &= \frac{1}{2} M^\sigma |R| \approx \frac{1}{4} M^\sigma |I_1 + \chi(s) I_2|, \\ \kappa_{1ps} &= |d_1| M^\sigma = 2\kappa_{1rs} \left| \frac{\sin(\omega - \arg R + \psi)}{\sin(2\omega + \psi)} \right|, \\ \kappa_{1ak} &= \frac{1}{2} M^\sigma |I_1|,\end{aligned}\tag{64}$$

with  $\psi = \arg \chi(s)$  and  $I_1, I_2$  the Kuznetsov integrals of §8.2; the “ $\approx$ ” in the first line holds when  $R$  is evaluated by Kuznetsov’s method (59), and the second equality in the  $\kappa_{1ps}$  line is exact with  $\kappa_{1rs}$  read as  $\frac{1}{2} M^\sigma |R|$ . Note that  $\kappa_{1ps} = |\hat{d}_1|$  is a nonnegative normalized length. The signed normalized coefficient remains  $\hat{d}_1 = d_1 M^\sigma$ , as in (24). Consequently, at fixed  $\sigma$  and fixed  $\{T\} = f$ , the passage from  $T_1 = N_1 + f$  to  $T_2 = N_2 + f$  (write  $M_i = N_i + 1$ ) multiplies every half-remainder length by

$$\frac{|R_{1\bullet}|_{T_2}}{|R_{1\bullet}|_{T_1}} = \left( \frac{M_1}{M_2} \right)^\sigma \frac{\kappa_{1\bullet}(T_2)}{\kappa_{1\bullet}(T_1)}.\tag{65}$$

When the fractional part is held fixed the ratio of  $\kappa$ ’s is nearly 1, so the leading factor  $(M_1/M_2)^\sigma$  already accounts for almost all of the size change. Figure 14 checks this in the  $R$ -frame: the left panel is  $T = 6.18$ ; the right panel is  $T = 50.18$  with all remainder vectors scaled by the single factor  $\lambda = |R|_{6.18}/|R|_{50.18}$ , the  $rs$  instance  $(M_2/M_1)^\sigma \kappa_{1rs}(T_1)/\kappa_{1rs}(T_2)$  of (65). Scaling by  $\lambda$  makes the two  $R$ ’s match by construction; the test is that the split halves nearly coincide too, and they do,  $|R_{1ps}|$  agreeing to about 0.3%.

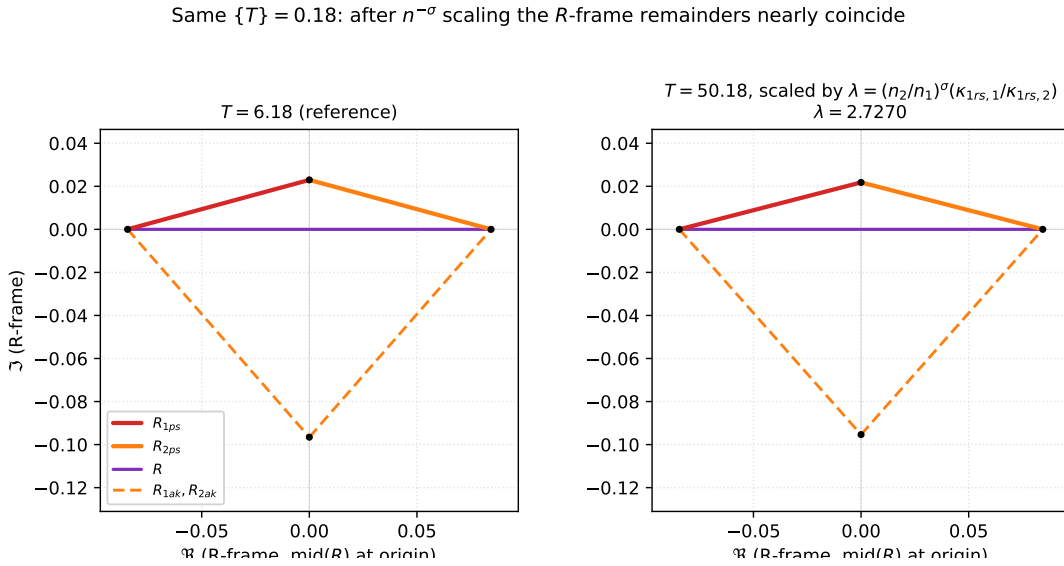


Figure 14: Same fractional part  $\{T\} = 0.18$ :  $R$ -frame remainders at  $T = 6.18$  (left) and at  $T = 50.18$  after scaling by  $\lambda = (M_2/M_1)^\sigma \kappa_{1rs}(T_1)/\kappa_{1rs}(T_2)$  (right). The scaled figure matches the reference to high accuracy ( $|R_{1ps}|$  agrees to about 0.3%). Generated by `fig_remainder_scale_panels.py`.

If one wants exactness rather than a close empirical match, divide by  $R$  itself. By (17) the modulus of  $R$  cancels identically, leaving

$$\frac{R_{1ps}}{R} = \frac{\sin(\omega - \arg R + \arg \chi)}{\sin(2\omega + \arg \chi)} e^{-i(\omega + \arg R)},\tag{66}$$

a function of the angles alone, exact at every  $T$  and every  $\sigma$  (off the line  $\arg R$  must itself be computed; on the line it is  $-\vartheta$  modulo  $\pi$ ): the size is then gone exactly, not almost. On the critical line the equal weights of Proposition 4.1(iii) force  $\operatorname{Re}(R_{1ps}/R) = \frac{1}{2}$ , so the shape is carried by the single number  $\operatorname{Im}(R_{1ps}/R)$ , and that number is a function of  $\{T\}$  up to  $O(1/T)$ , which is the content of Theorem A.5.

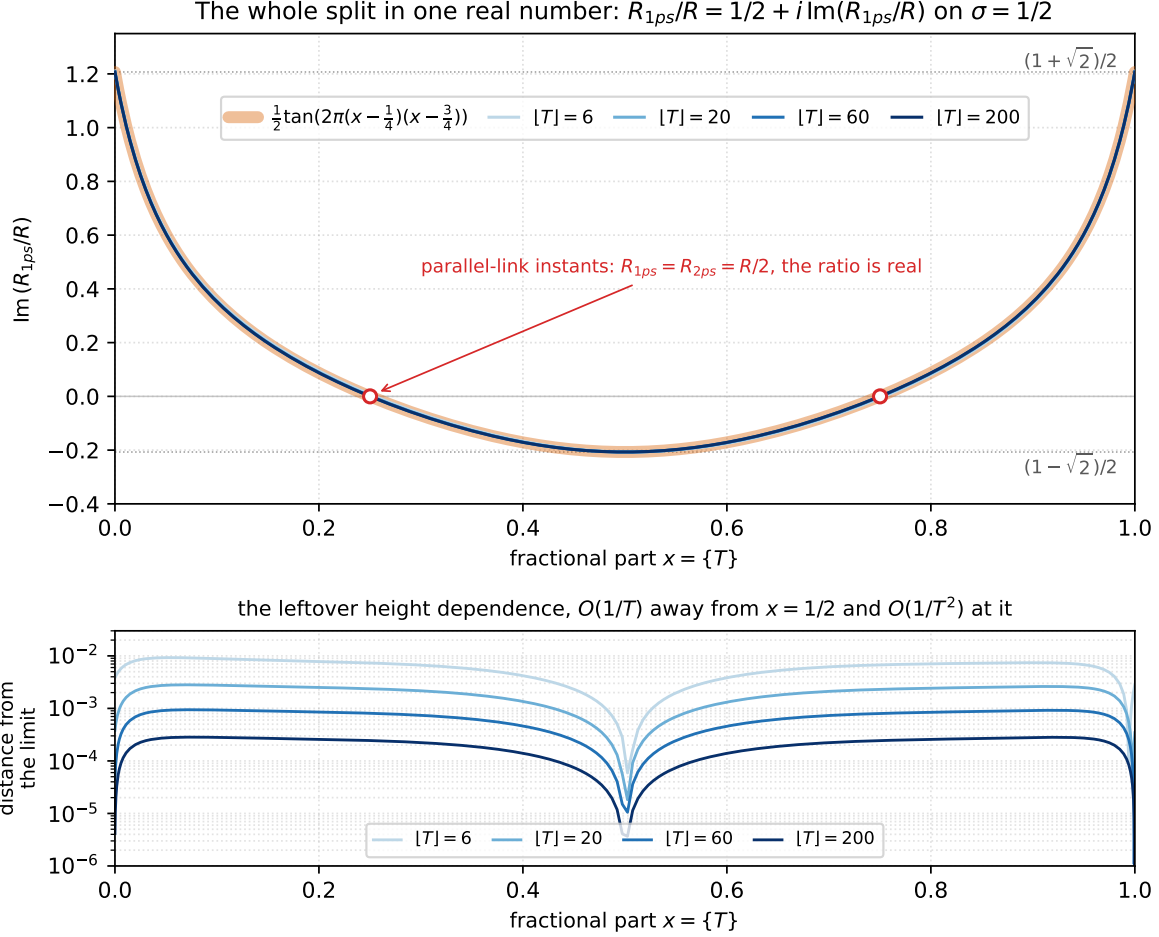


Figure 15: The split after division by  $R$ , on the critical line. Top:  $\operatorname{Im}(R_{1ps}/R)$  against  $x = \{T\}$  at  $|T| = 6, 20, 60, 200$ , each computed through  $\zeta$  and the Cramer solution (17), laid over the limit  $\frac{1}{2} \tan(2\pi(x - \frac{1}{4})(x - \frac{3}{4}))$  (thick orange). The limit falls from  $(1 + \sqrt{2})/2$  at the integers to  $(1 - \sqrt{2})/2$  at  $x = \frac{1}{2}$ , a swing of exactly  $\sqrt{2}$ , and vanishes at the two parallel-link instants (circles;  $x = 0.250156$  and  $0.750155$  at  $|T| = 200$ ), where  $R_{1ps} = R_{2ps} = \frac{1}{2}R$ . Bottom: distance from the limit,  $O(1/T)$  in general and  $O(1/T^2)$  at  $x = \frac{1}{2}$ . Generated by `fig_ratio_shape.py`.

Figure 15 dissects that single number. On the critical line  $\arg R = \frac{1}{2} \arg \chi$  and  $\chi = e^{-2i\vartheta}$ , with  $\vartheta$  the Riemann–Siegel theta function of Appendix A.2, so  $\omega - \arg R + \arg \chi = \omega - \vartheta$  and  $2\omega + \arg \chi = 2(\omega - \vartheta)$ , the  $\pi$  ambiguity in  $\arg R$  flipping the sine and the exponential of (66) together. That equation therefore collapses to

$$\frac{R_{1ps}}{R} = \frac{e^{-i(\omega - \vartheta)}}{2 \cos(\omega - \vartheta)} = \frac{1}{2} - \frac{i}{2} \tan(\omega - \vartheta), \quad (67)$$



which is the exact critical-line form  $d_1 = r/(2 \cos u)$  of (199) divided by  $R$ . Three things follow, and Figure 15 shows all three. First, the real part is  $\frac{1}{2}$  identically, which is Proposition 4.1(iii) once more; the plotted values were computed the long way, through  $\zeta$  and (17), and at all 964 samples their real parts agree with  $\frac{1}{2}$  to thirty digits. Second, everything that is left is the one angle  $\omega - \vartheta$  modulo  $\pi$ , and that angle tends to  $-\beta$ , where  $\beta = 2\pi(x - \frac{1}{4})(x - \frac{3}{4})$  and  $x = \{T\}$ ; this is the same reduction that turns  $2 \cos(\omega - \vartheta)$  into  $2 \cos \beta$  in the proof of Theorem A.5. So  $\text{Im}(R_{1ps}/R) \rightarrow \frac{1}{2} \tan \beta$ , which is the second of the two tangent factors in the limit waveform (25): that waveform is  $-\tan(2\pi x)$  times this very quantity. The four curves lie within  $9.3 \times 10^{-3}$ ,  $2.8 \times 10^{-3}$ ,  $9.4 \times 10^{-4}$  and  $2.8 \times 10^{-4}$  of the limit at  $\lfloor T \rfloor = 6, 20, 60, 200$ , shrinking in proportion to the index, which is the  $O(1/T)$  rate of the theorem seen directly. Third, the range is fixed by  $\beta$ , which runs from  $\frac{3\pi}{8}$  at the integers to  $-\frac{\pi}{8}$  at  $x = \frac{1}{2}$ : the curve therefore lies between  $(1 + \sqrt{2})/2$  and  $(1 - \sqrt{2})/2$ , swinging by exactly  $\sqrt{2}$ , and the tangent stays  $\frac{\pi}{8}$  clear of its pole at  $\frac{\pi}{2}$ , which is the boundedness of  $d_1$  on the line read as an angle. It crosses zero where  $\beta$  does, at the parallel-link instants, where the split is the plain halving as above.

## 9 The geometry behind the result (experimental mathematics)

The decomposition  $R = R_{1ps} + R_{2ps}$  was not first derived; it was first *seen*. Over roughly nine years we built and used an interactive visualization tool, Zest,<sup>7</sup> which draws the partial-sum chains of  $\zeta$  in the complex plane and lets us sweep  $\sigma$  and  $T$  continuously while measuring what appears on the screen. This section reconstructs the series of observations made with that tool. It led, in order, to a distinguished link and point of the spiral (the *bisector link* and *bisector point*), and from their motion to the mapping  $I(T)$  of §3. The final observation in that sequence, the *yin* and *yang* curves from which the weights  $d_1, d_2$  of §4 were read off, was developed in §6.4 and §6.5; §10 reconstructs how both the mapping and the curves were found.

Throughout we use the link and joint numbering of §7: joint  $k$  of a chain is the value of its partial sum after  $k$  summands, and link  $k$  runs from joint  $k$  to joint  $k + 1$ , so that link  $k$  represents the  $(k + 1)$ st summand and each remainder is the fractional link  $m$ . The *forward* link chain  $(\Sigma_1)$  is anchored at joint 0, the origin, and it represents the main sum. The *reverse* chain  $(\Sigma_2)$ , representing the dual sum, is anchored at  $\zeta$  and traversed backwards, so its joint  $m$  lands at  $\zeta - \Sigma_2 = \Sigma_1 + R$ . In the pictures the chains are drawn well past joint  $m = \lfloor T \rfloor$ : the sum, though ultimately divergent, first winds into an Euler-like spiral whose center lies near  $\zeta$ ; it seemingly converges before it diverges. Figure 16 shows the picture: the forward chain winds from the origin into a spiral centered near  $\zeta$ , while the reverse chain winds from  $\zeta$  into a mirror spiral centered near the origin, and the two link chains cross. If more links were drawn, the links would continue to spiral outward around  $\zeta$  and cover the entire figure, because the zeta sum diverges on the critical strip, unlike the Dirichlet  $L$ -functions of §10.1.1, whose non-principal characters have bounded partial sums and so those chains close in on their limit instead of covering the plane.<sup>8</sup> We stop drawing links at link number

$$L_N(T, S_n) = \left\lfloor \frac{I(T)}{\pi(2S_n + 1)} \right\rfloor, \quad (68)$$

where  $S_n$  is the spiral number, spiral number 0 being the last one (the biggest, at  $\zeta$ ). Note that  $\zeta$  itself is not typically *on* this last link; the last link is the link closest to  $\zeta$ . The same formula locates the other spirals and pseudo-spirals too: with  $S_n = 1$ , for example, it yields the link number of the link most in the center of the second-to-last spiral.

<sup>7</sup>Source code and standalone executables (Mac and PC) and a web app at <https://github.com/paul-stahura/zest>.

<sup>8</sup>At  $t = 176.13$  on the critical line, the partial sums of  $L(s, \chi_3)$  come within 0.021, 0.0021 and 0.00019 of the value by  $N = 10^3$ ,  $10^5$  and  $3 \times 10^6$ , a drift inward like  $N^{-1/2}$ , while zeta's are off by 0.18, 1.80 and 9.83, a drift outward at  $\sqrt{N}/t$ . For a principal character modulo  $q$ ,

$$L(s, \chi_0) = \zeta(s) \prod_{p|q} (1 - p^{-s}),$$

rather than  $\zeta(s)$  itself. The convergence behavior of higher-degree  $L$ -function Dirichlet series depends on the family and normalization, so no blanket critical-strip claim is made here.

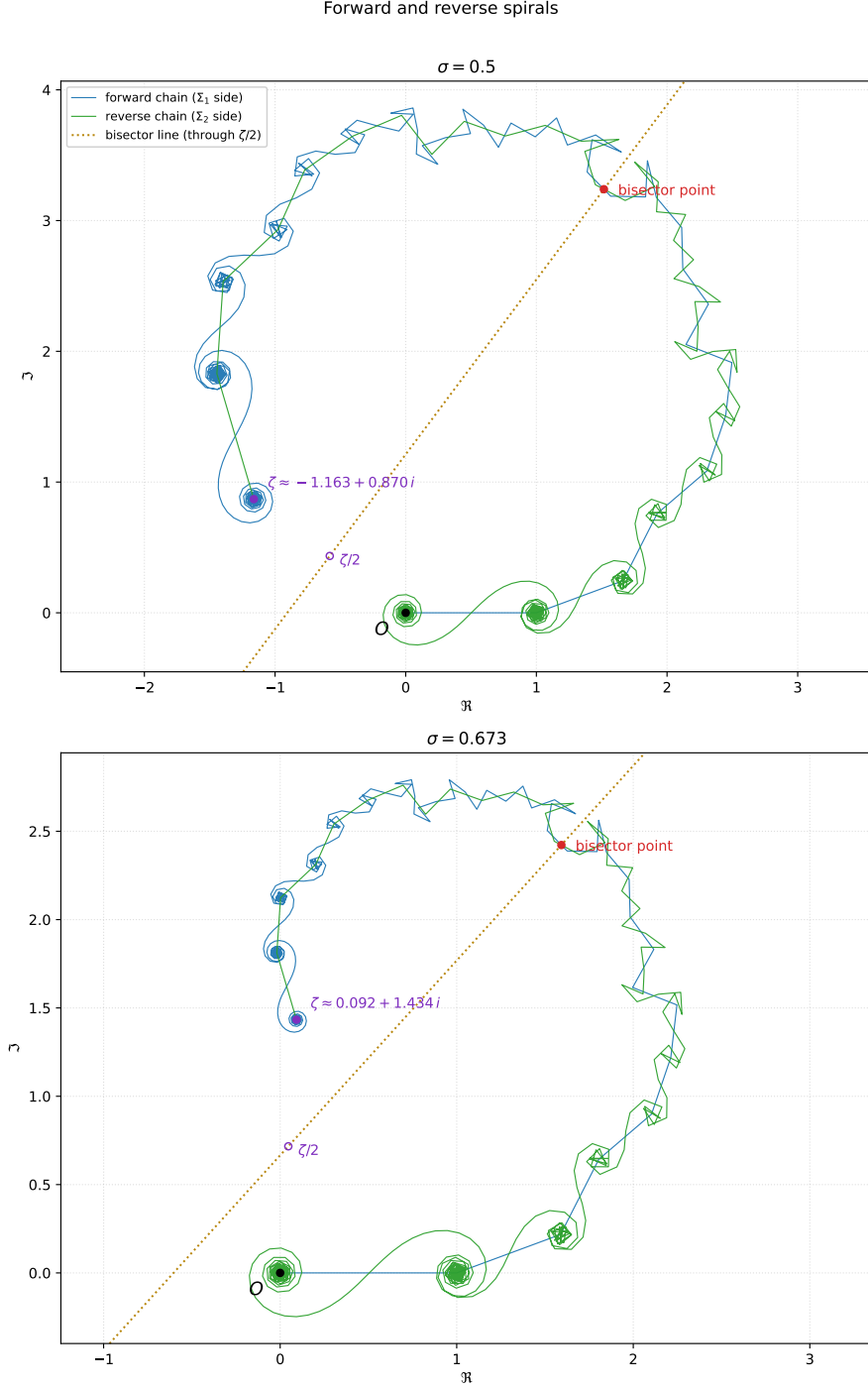


Figure 16: The forward spiral (blue, anchored at the origin  $O$ ) and the reverse spiral (green, anchored at  $\zeta$ ) for  $T = 14.3085$  (so  $t = I(T) \approx 1377.33$  and  $m = 14$ ), on the critical line  $\sigma = \frac{1}{2}$  (top,  $\zeta \approx -1.163 + 0.870i$ ) and off it at  $\sigma = 0.673$  (bottom,  $\zeta \approx 0.092 + 1.434i$ ). Each spiral is drawn out to link  $\lfloor I(T)/\pi \rfloor = 438$ , the link nearest its spiral center, for a total of 439 links per spiral (the count is independent of  $\sigma$ ). The red dot is the bisector point  $B_1 = \Sigma_1 + R_{1ps}$ , where the two chains cross at their bisector links (§9.1); the dotted line is the bisector line through the bisector point and  $\zeta/2$  (open circle). On the critical line (top) the whole configuration is symmetric about the bisector line; at  $\sigma = 0.673$  (bottom) the symmetry is lost. Generated by `fig_full_spirals.py`.

The last spiral, centered on  $\zeta$

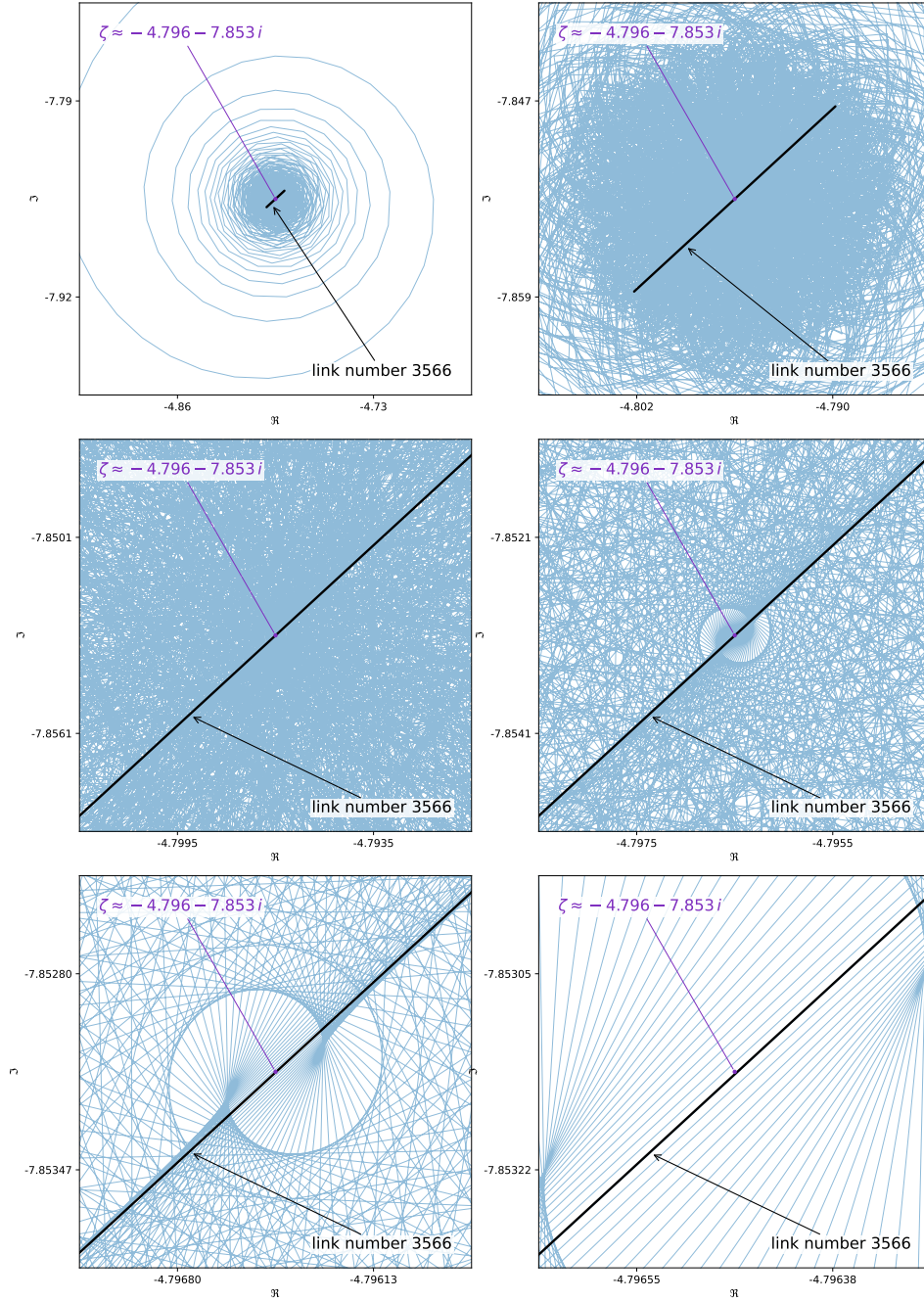


Figure 17: Zooming in on the last spiral, the spiral centered near  $\zeta$ , of the forward chain at  $T = 41.73$  (so  $t = I(T) \approx 11204.74$ ,  $\sigma = \frac{1}{2}$ ,  $\zeta \approx -4.796 - 7.853i$ ). The chain carries  $\lfloor I(T)/\pi \rfloor + 1 = 3567$  links; the last link is number 3566 by (68) with  $S_n = 0$ , and it is highlighted in black in every panel. Each panel is more zoomed in than the last (panels 1–6, left to right from the top). Near the spiral center each link turns by nearly  $\pi$ , so the links sweep back and forth across  $\zeta$  and the joints settle onto a tight ring of radius about half a link length: at coarse scale the spiral renders as a dense annulus (panels 1–3), and the near-diameter links envelope a lens-shaped clearing around  $\zeta$  (panels 4–5). Observe in panel 6 that  $\zeta$  is *not on* the last link: the last link is only the link closest to  $\zeta$ . Generated by `fig_last_spiral_zoom.py`.

## 9.1 The bisector link and the bisector point

Drawn at height  $T$ , each link chain exhibits spirals and pseudo-spirals, and (as others have also observed [9, 17, 12]) there is a one-to-one correspondence between the joints and the spiral centers, in a type of symmetry; Nickel puts it as the center-to-center distances and angles of the large linked spirals reproducing the initial steps exactly. The link at index  $m = \lfloor T \rfloor$  sits at the boundary between the two regimes, and we call it the *bisector link*. Each chain has one: the *forward bisector link* (link  $m$  of the  $\Sigma_1$  chain, running from  $\Sigma_1$  along the  $(m+1)$ st summand direction) and the *reverse bisector link* (link  $m$  of the  $\Sigma_2$  chain, running from  $\Sigma_1 + R$  along its  $(m+1)$ st summand direction).

The two spirals intersect at their bisector links, and we call the point where the forward and reverse bisector links cross the *bisector point*. One qualification: the crossing can momentarily fail. Twice per unit of  $T$ , near fractional parts  $\approx 0.25$  and  $\approx 0.75$ , the two bisector links become parallel and, off the critical line, do not intersect; these are the isolated instants at which  $\sin(2\omega + \arg \chi)$ , the denominator of (17)–(18), vanishes. Those are exactly the places where  $d_1$  and  $d_2$  have poles: the parallel-link condition and the pole condition are the same equation, which is why the precise locations were worked out in Appendix A.5. On the critical line the two links are then collinear rather than merely parallel, so the bisector point survives there. In the notation of the earlier sections the bisector point is exactly

$$B_1 = \Sigma_1 + R_{1ps} = \Sigma_1 + R - R_{2ps}; \quad (69)$$

it is visible as the joint between the red and orange fractional links in Figures 8 and 9.

## 9.2 Legs

When  $\sigma = \frac{1}{2}$  the spiral object has a type of symmetry about the line through the bisector point and  $\frac{\zeta}{2}$ , so the bisector point sits at the apex of an isosceles triangle whose base runs from the origin to  $\zeta$  and whose equal sides are the two legs  $B_1$  and  $\zeta - B_1$  (this is Corollary 9.1 below, proved from the reflection  $B_2 = \chi \overline{B_1}$  and verified in Lean in Appendix C; equivalently, the perpendicular projection of the bisector point onto the base  $\mathbb{R}e^{i\psi/2}$  carrying  $\zeta$ , where  $\psi = \arg \chi$ , is exactly  $\frac{\zeta}{2}$ , which is Corollary 9.2). Off the line we keep the same terminology even though the triangle's legs are then (usually) unequal.

The two legs: origin to bisector point, bisector point to  $\zeta$   
 $T = 14.3085$  ( $t = l(T) \approx 1377.33$ ),  $m = 14$

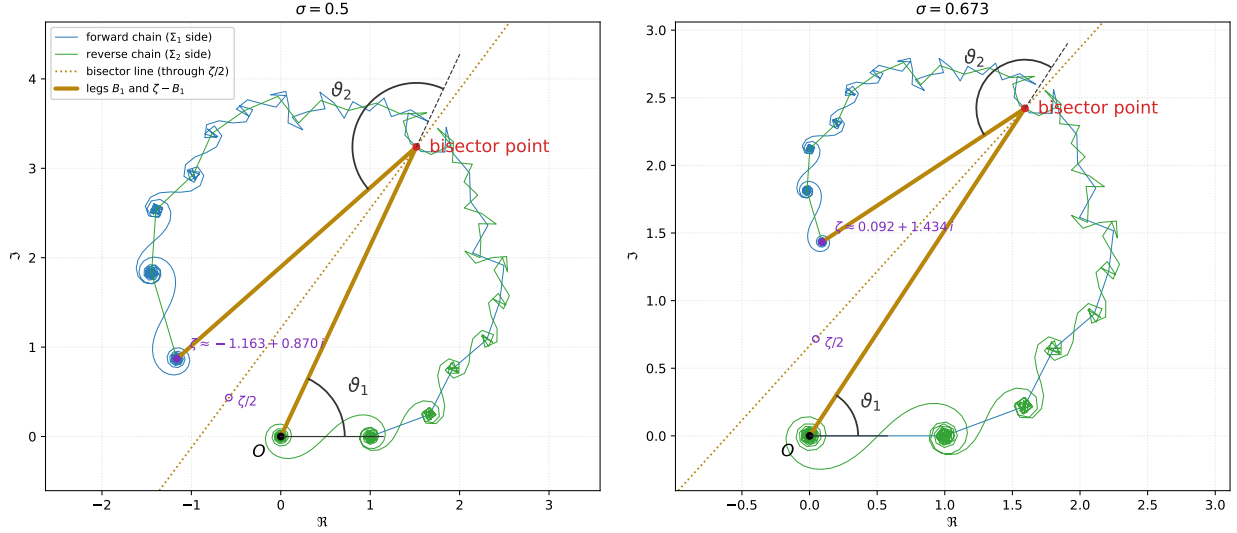


Figure 18: The two legs, drawn as thick dark-yellow segments over the spirals of Figure 16: one from the origin  $O$  to the bisector point ( $B_1$ ), the other from the bisector point to  $\zeta$  ( $\zeta - B_1$ ). On the critical line (left,  $\sigma = \frac{1}{2}$ ) the legs have equal length ( $|B_1| = |\zeta - B_1| \approx 3.577$ ): the bisector point is the apex of an isosceles triangle whose base runs from  $O$  to  $\zeta$ , and the dotted bisector line is its axis of symmetry. Off the critical line (right,  $\sigma = 0.673$ ) the legs are unequal ( $\approx 2.899$  and  $\approx 1.797$ ) at this  $T$ . Both panels also mark the two angles defined below:  $\vartheta_1$ , at  $O$  from the positive real axis to Leg 1, and  $\vartheta_2$ , at the bisector point from the extension of Leg 1 (dashed) to Leg 2. Generated by `fig-legs.py`.

On the critical line the two fractional summands are not independent: each is the other's complex conjugate, rotated by the functional-equation factor. Writing  $R_{1ps} = d_1 e^{-i\omega}$  and  $R_{2ps} = d_2 e^{i(\omega + \arg \chi)}$  as in §7 and using that  $|\chi| = 1$  when  $\sigma = \frac{1}{2}$ , the equality  $d_1 = d_2$  of Proposition 4.1(iii) is exactly equivalent to the compact relation

$$R_{2ps} = \chi \overline{R_{1ps}} \quad (\sigma = \tfrac{1}{2}). \quad (70)$$

Geometrically, (70) says the second fractional summand is the mirror image of the first (conjugation across the real axis) followed by the rotation  $\chi$ . Because  $\chi$  has unit modulus on the line this rotation is an isometry, so  $|R_{1ps}| = |R_{2ps}|$  immediately. This is the functional-equation reflection  $s \leftrightarrow 1 - s$  acting as a symmetry of the two remainders, and it is the source of the equal-leg phenomenon we now develop.

The two legs  $B_1 = \Sigma_1 + R_{1ps}$  and  $B_2 = \Sigma_2 + R_{2ps}$  of §7 meet at a “bisector point”: Leg 1 runs from the origin to  $\Sigma_1 + R_{1ps}$ , and Leg 2 from there to  $\zeta$ . Their lengths are

$$L_1 = |\Sigma_1 + R_{1ps}|, \quad (71)$$

$$L_2 = |\Sigma_2 + R_{2ps}|, \quad (72)$$

while  $\vartheta_1$  is the angle Leg 1 makes with the real axis and  $\vartheta_2$  is the angle from Leg 1 to Leg 2

(oriented so that a positive  $\vartheta_2$  rotates Leg 1 toward Leg 2):<sup>9</sup>

$$\vartheta_1 = \arg(\Sigma_1 + R_{1ps}), \quad (73)$$

$$\vartheta_2 = \arg\left(\frac{\Sigma_2 + R_{2ps}}{\Sigma_1 + R_{1ps}}\right). \quad (74)$$

With these,  $\zeta$  is simply the sum of the two legs,

$$\zeta = L_1 e^{i\vartheta_1} + L_2 e^{i(\vartheta_1 + \vartheta_2)}, \quad (75)$$

or equivalently, in matrix form,

$$\zeta = \begin{pmatrix} 1 & i \end{pmatrix} \begin{pmatrix} \cos \vartheta_1 & \cos(\vartheta_1 + \vartheta_2) \\ \sin \vartheta_1 & \sin(\vartheta_1 + \vartheta_2) \end{pmatrix} \begin{pmatrix} L_1 \\ L_2 \end{pmatrix}, \quad (76)$$

where the row  $\begin{pmatrix} 1 & i \end{pmatrix}$  turns the resulting real  $(x, y)$  components into the complex value  $x + iy$ .

**Corollary 9.1.** *When  $\sigma = \frac{1}{2}$  the two legs are always equal in length:  $L_1 = L_2$ .*

This is immediate from Proposition 4.1(iii): on the critical line every summand of  $\Sigma_2$  is  $\chi$  times the conjugate of the corresponding summand of  $\Sigma_1$ , and by (70), which is equivalent to  $d_1 = d_2$ , the same holds for the fractional summands. Hence  $\Sigma_2 + R_{2ps} = \chi \overline{\Sigma_1 + R_{1ps}}$ , and since  $|\chi| = 1$  on the line, the two legs have the same length. A Lean proof is recorded in Appendix C.

The same reflection locates the apex exactly, which is what the dotted bisector line of Figures 16 and 18 draws.

**Corollary 9.2.** *When  $\sigma = \frac{1}{2}$  the perpendicular projection of the bisector point  $B_1$  onto the line through the origin carrying  $\zeta$  is exactly  $\frac{\zeta}{2}$ : writing  $\psi = \arg \chi$ ,*

$$\Re(e^{-i\psi/2} B_1) e^{i\psi/2} = \frac{\zeta}{2}. \quad (77)$$

Indeed  $B_2 = \chi \overline{B_1}$  gives  $e^{-i\psi/2} \zeta = e^{-i\psi/2} B_1 + \overline{e^{-i\psi/2} B_1} = 2\Re(e^{-i\psi/2} B_1)$ , and  $e^{i\psi/2}$  is the direction of  $\zeta$  (this is the reality of the Riemann–Siegel  $Z$  function). So the isosceles triangle with base from  $O$  to  $\zeta$  has its apex over the midpoint of that base, and the equal-leg statement  $L_1 = L_2$  is the same fact read as a distance. This is also proved in Appendix C.

**Consequences for zeta zeros.** A zero of  $\zeta$  in the critical strip requires *both* of the following:

$$\frac{L_1}{L_2} = 1 \quad \text{and} \quad \vartheta_2 = \pi,$$

i.e. the two legs must be equal in length *and* Leg 2 must fold back onto Leg 1.<sup>10</sup> By Corollary 9.1 the length condition holds automatically on the critical line; there, a zero occurs exactly when

<sup>9</sup>The subscripted leg angles  $\vartheta_1, \vartheta_2$  are not to be confused with the Riemann–Siegel theta function  $\vartheta(t)$  of Appendix A.2.

<sup>10</sup>Tolkien’s subtitle for *The Hobbit, There and Back Again*, is the whole of the requirement. A hobbit leaves the Shire (the origin) for Mordor (the bisector point) and his journey must end where it began, and no choice of direction for the return trip from Mordor can bring the hobbit back to the Shire unless the way back is exactly as long as the way out. Equal legs is that necessity;  $\vartheta_2 = \pi$  is the further demand that the way back be the way he came.

$\vartheta_2 = \pi$ . Away from the critical line, the two legs are unequal except at exceptional points, so most of the strip is immediately ruled out for zeros. The leg-length ratio is symmetric across the line,

$$\frac{L_1(\sigma, T)}{L_2(\sigma, T)} = \frac{L_2(1 - \sigma, T)}{L_1(1 - \sigma, T)}. \quad (78)$$

Numerically we have never found a point with  $\sigma \neq \frac{1}{2}$  where the ratio is 1 and  $\vartheta_2 = \pi$  simultaneously; such a point would disprove the Riemann hypothesis.



Equal-leg points ( $L_1 = L_2$ ), zeta zeros (dots), and the critical line

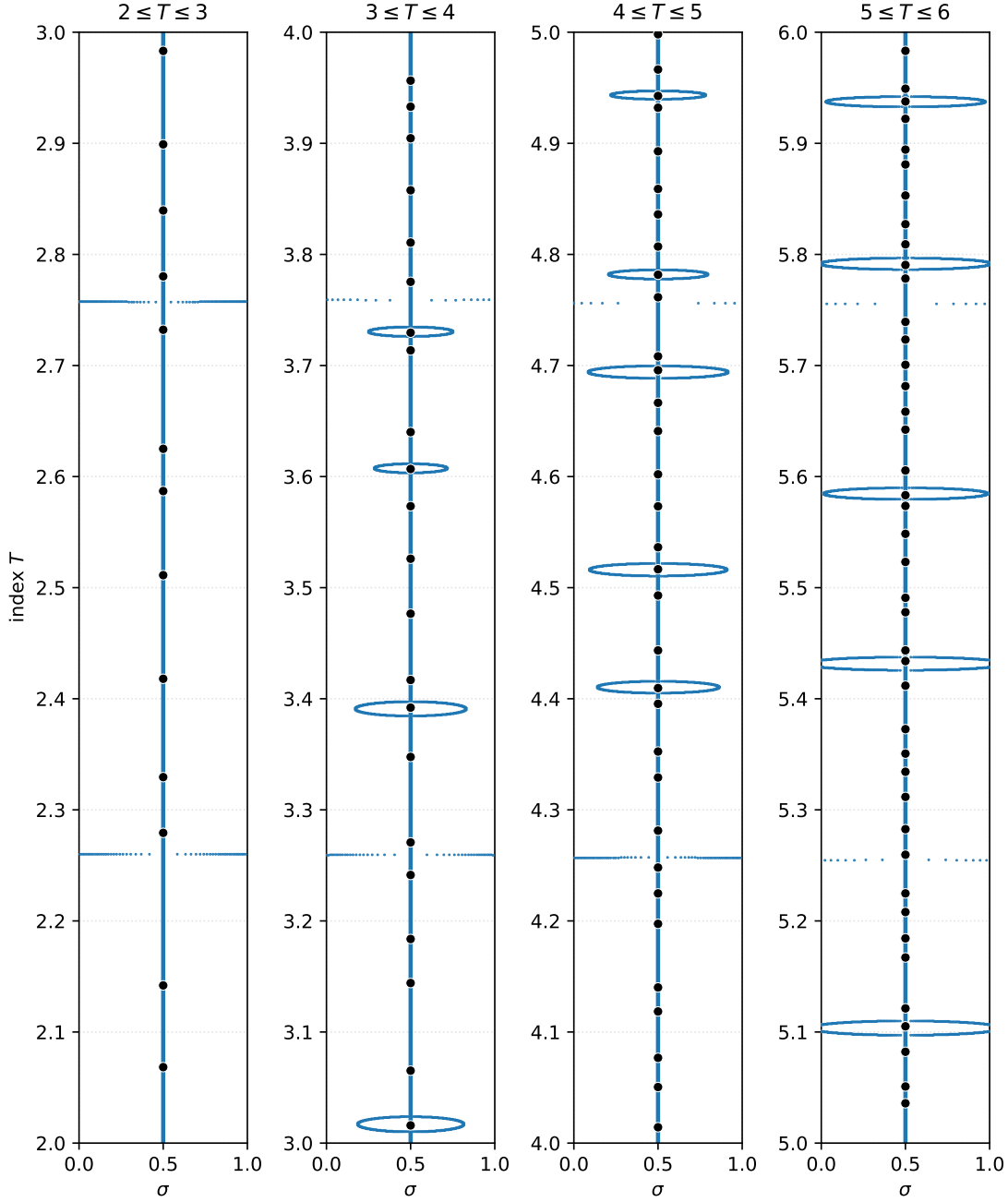


Figure 19: Four windows of the critical strip ( $0 \leq \sigma \leq 1$ , index  $T$  vertical):  $2 \leq T \leq 3$ ,  $3 \leq T \leq 4$ ,  $4 \leq T \leq 5$ , and  $5 \leq T \leq 6$ . The blue vertical line is the critical line  $\sigma = \frac{1}{2}$ , the black dots are the zeta zeros, and the small blue points are the equal-leg points: points  $(\sigma, T)$  where  $L_1 = L_2$ , i.e. where the bisector point  $B_1 = \Sigma_1 + R_{1ps}$  is equidistant from the origin and from  $\zeta$ . By Corollary 9.1 the entire critical line consists of equal-leg points; off the line the equal-leg locus forms isolated ovals and thin horizontal bands, and every zero sits on the critical line where the length condition is automatic. The horizontal bands, near fractional parts  $\approx 0.25$  and  $\approx 0.75$  of  $T$ , are the instants where the two bisector links become parallel and fail to intersect (§9.1): the bisector point has a pole there ( $d_1$  and  $d_2$  blow up), and as  $T$  passes through it the equal-leg point sweeps across the full width of the strip. The sweep gets faster as  $T$  grows, so the fixed-step sampling leaves fewer dots and the bands appear fainter. Generated by `fig.equal_legs_strips.py`.

Folded-leg points ( $\vartheta_2 = \pi$ ), zeta zeros (dots), and the critical line

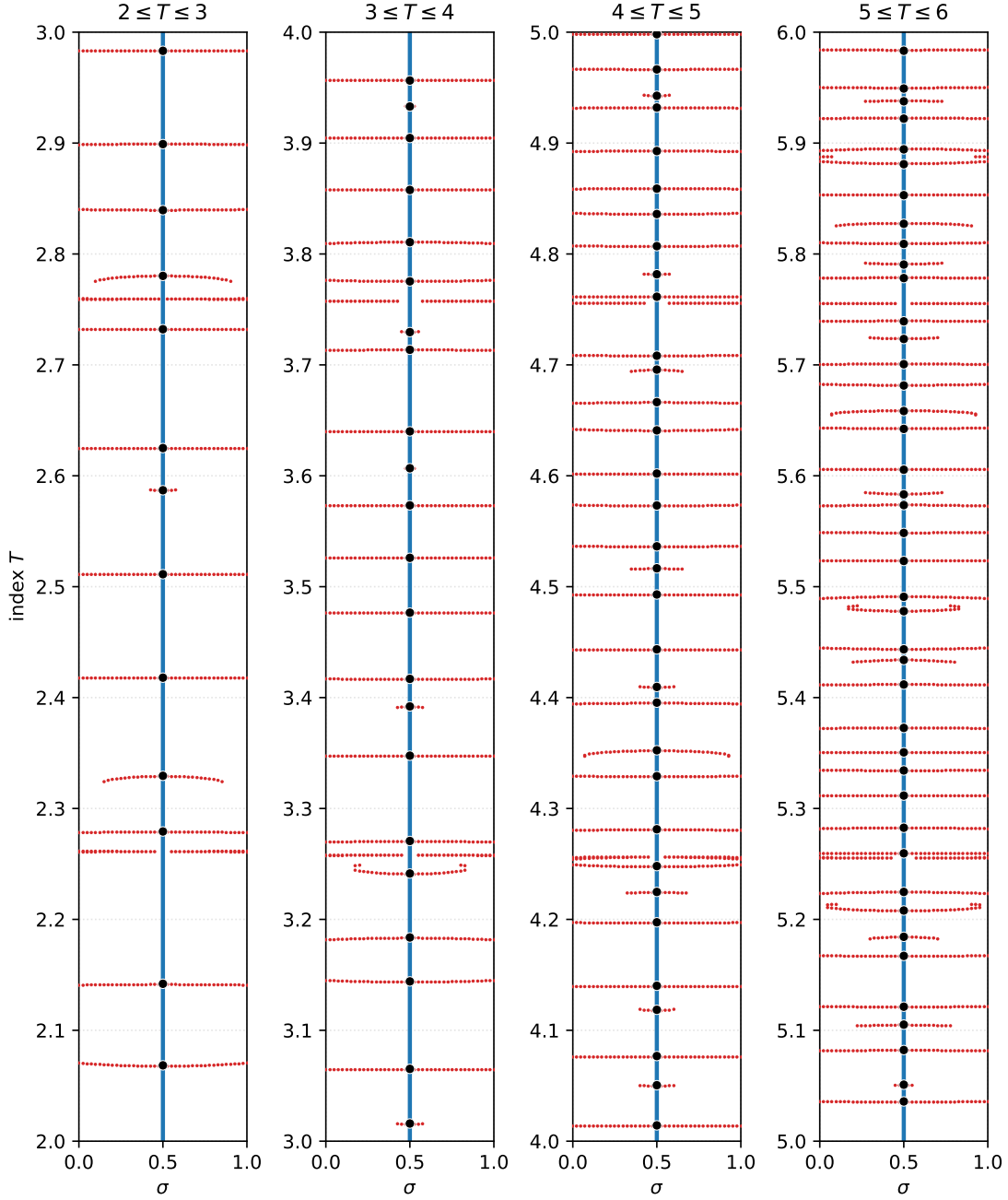


Figure 20: The same four windows of the critical strip as in Figure 19, now showing the *folded-leg* points (red): points  $(\sigma, T)$  where  $\vartheta_2 = \pi$ , i.e. where Leg 2 folds straight back onto Leg 1 and a zeta zero is therefore possible. The blue vertical line is the critical line and the black dots are the zeta zeros. The folded-leg locus forms nearly horizontal curves sweeping the strip; a zero requires *both*  $\vartheta_2 = \pi$  and  $L_1 = L_2$ , and since the length condition holds automatically on the critical line (Corollary 9.1), every crossing of a red curve with the critical line is a zeta zero. Generated by `fig_theta2_strips.py`.

Equal legs ( $L_1 = L_2$ , blue) and folded legs ( $\vartheta_2 = \pi$ , red): zeros where they cross

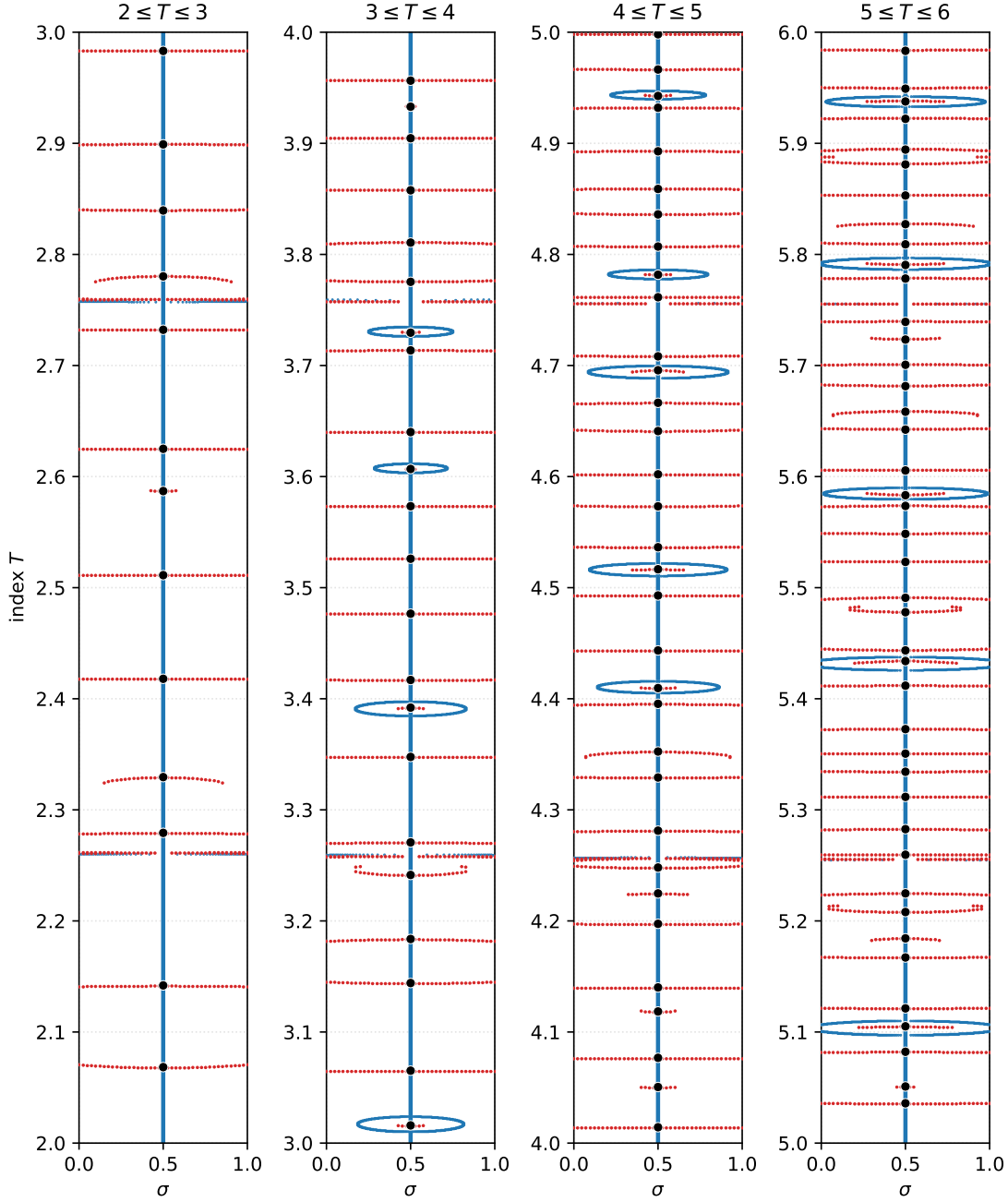


Figure 21: Figures 19 and 20 overlaid: the equal-leg locus  $L_1 = L_2$  (blue), the folded-leg locus  $\vartheta_2 = \pi$  (red), the critical line (blue, since it belongs to the equal-leg locus), and the zeta zeros (black dots). A zero requires both conditions at once, so zeros occur exactly where red meets blue. On the critical line the blue locus is the line itself (Corollary 9.1), and every red curve crossing it produces a zero; away from the line, the red curves and the blue ovals interleave without touching. A red/blue crossing off the critical line would be a counterexample to the Riemann hypothesis. Generated by `fig_combined_strips.py`.

Figure 21 makes the stakes plain: wherever a region of the critical strip contains no red and no blue dots, a zero of  $\zeta$  is impossible. On the critical line the two legs are always equal in length, so whenever  $\vartheta_2 = \pi$  there, a zero occurs. Off the line the legs are seldom equal in length, but sometimes they are (the blue ovals and bands). If one could prove that, off the critical line, in the rare times when the legs are equal, that  $\vartheta_2$  is simultaneously never  $\pi$ , one would have proven the Riemann hypothesis true.

### 9.3 PS, AK, and RS legs and angles

The equal-leg and folded-leg loci above were drawn with the exact bisector point  $B_1 = \Sigma_1 + R_{1ps}$ . The same two conditions  $L_1 = L_2$  and  $\vartheta_2 = \pi$  can be restated for any other choice of bisector point. Two natural alternatives come from Kuznetsov's split (60): take  $B_1 = \Sigma_1 + R_{1ak}$ , or take the midpoint remainder  $B_1 = \Sigma_1 + (R_{1ak} + R_{2ak})/2$ , which is the geometric analogue of Siegel's half-remainder  $\frac{1}{2}R$  once  $R_{1ak} + R_{2ak} \approx R$ .

Figure 22 magnifies a short window of the strip,  $4.65 \leq T \leq 4.80$ , and overlays the three choices side by side. On the left, equal-leg loci: PS in blue, AK in green, and  $R/2$  in purple. The three families form nested ovals about the critical line, but the nesting order is not fixed: already in this short window the lower oval has PS outermost and AK innermost, while the upper oval has AK outermost and  $R/2$  innermost, and across  $3 \leq T \leq 20$  all six orderings of PS, AK and  $R/2$  occur with no simple alternating pattern. On the right, folded-leg loci  $\vartheta_2 = \pi$ : PS in red, AK in orange, and  $R/2$  in purple (hollow). The zeros (black) still sit where a folded curve meets the critical line; the three remainder choices shift the off-line curves but agree on those critical-line crossings in this window.

Figure 23 is the same three-way overlay for a pair of neighboring equal-leg ovals near  $T \approx 9.44$ , drawn in the window  $9.415 \leq T \leq 9.465$ . The lower PS oval sits around the zero at  $T \approx 9.431$  and the upper around  $T \approx 9.451$ ; a third zero at  $T \approx 9.440$  lies in the gap between them (not inside either oval). The AK and  $R/2$  ovals nest with the PS locus as in Figure 22.

**Intersecting equal-leg ovals and the Riemann hypothesis.** The nested ovals in Figures 22 and 23 suggest a comparison that uses only exact remainders. Write

$$B_{1ps} = \Sigma_1 + R_{1ps}, \quad B_{1rs} = \Sigma_1 + R_{1rs}, \quad R_{1rs} := \frac{1}{2}R,$$

where  $R = R_{1ps} + R_{2ps} = \zeta - \Sigma_1 - \Sigma_2$  is Siegel's remainder, so  $R_{1rs}$  is the exact half-remainder ("rs" for Riemann–Siegel, parallel to the subscripts "ps" and "ak") and  $B_{1rs}$  is the midpoint of the remainder segment from  $\Sigma_1$  to  $\Sigma_1 + R$ . Let  $\mathcal{E}_{ps}$  and  $\mathcal{E}_{rs}$  be the corresponding equal-leg loci in the strip: the sets of  $(\sigma, T)$  at which  $|B_1| = |\zeta - B_1|$  for  $B_1 = B_{1ps}$  and  $B_1 = B_{1rs}$  respectively.

If  $\zeta(\sigma + iI(T)) = 0$ , then for either choice of bisector point one has  $L_2 = |\zeta - B_1| = |B_1| = L_1$ , so

$$\frac{L_1}{L_2} = 1$$

holds simultaneously for both  $B_{1ps}$  and  $B_{1rs}$ . Every zero therefore lies in  $\mathcal{E}_{ps} \cap \mathcal{E}_{rs}$ . On the critical line this is automatic: Corollary 9.1 puts the whole line in  $\mathcal{E}_{ps}$ , and the  $rs$  split is reflection-symmetric there as well:  $\Sigma_2 = \chi \overline{\Sigma_1}$  and  $R = \chi \overline{R}$  (the latter because  $e^{-i\psi/2}R$  is real on the line:  $R = e^{-i\vartheta}r$  with  $r$  real and  $\psi = -2\vartheta$ , Appendix A.2), so

$$\chi \overline{B_{1rs}} = \Sigma_2 + \frac{1}{2}R = \zeta - B_{1rs},$$

and since  $|\chi| = 1$  this gives  $|B_{1rs}| = |\zeta - B_{1rs}|$ : the whole line lies in  $\mathcal{E}_{rs}$  as well. (Note that  $B_{1rs} \neq \zeta/2$ ; only its perpendicular projection onto the base is  $\zeta/2$ , by Corollary 9.2.) The two loci therefore intersect everywhere along  $\sigma = \frac{1}{2}$ ; that common set is expected and does not constrain zeros beyond the folded-leg condition already discussed.

Off the critical line the situation is different. An off-line zero would be an off-line point of  $\mathcal{E}_{ps} \cap \mathcal{E}_{rs}$ : a point where the PS and  $rs$  equal-leg ovals meet. Consequently, an off-line intersection of  $\mathcal{E}_{ps}$  and  $\mathcal{E}_{rs}$  is *necessary* for an off-line zero (it is not sufficient: the folded-leg condition  $\vartheta_2 = \pi$ , or equivalently  $\zeta = 0$ , is still required). The contrapositive is the statement of interest:

PS, AK and  $R/2$  legs and angles,  $4.65 \leq T \leq 4.80$

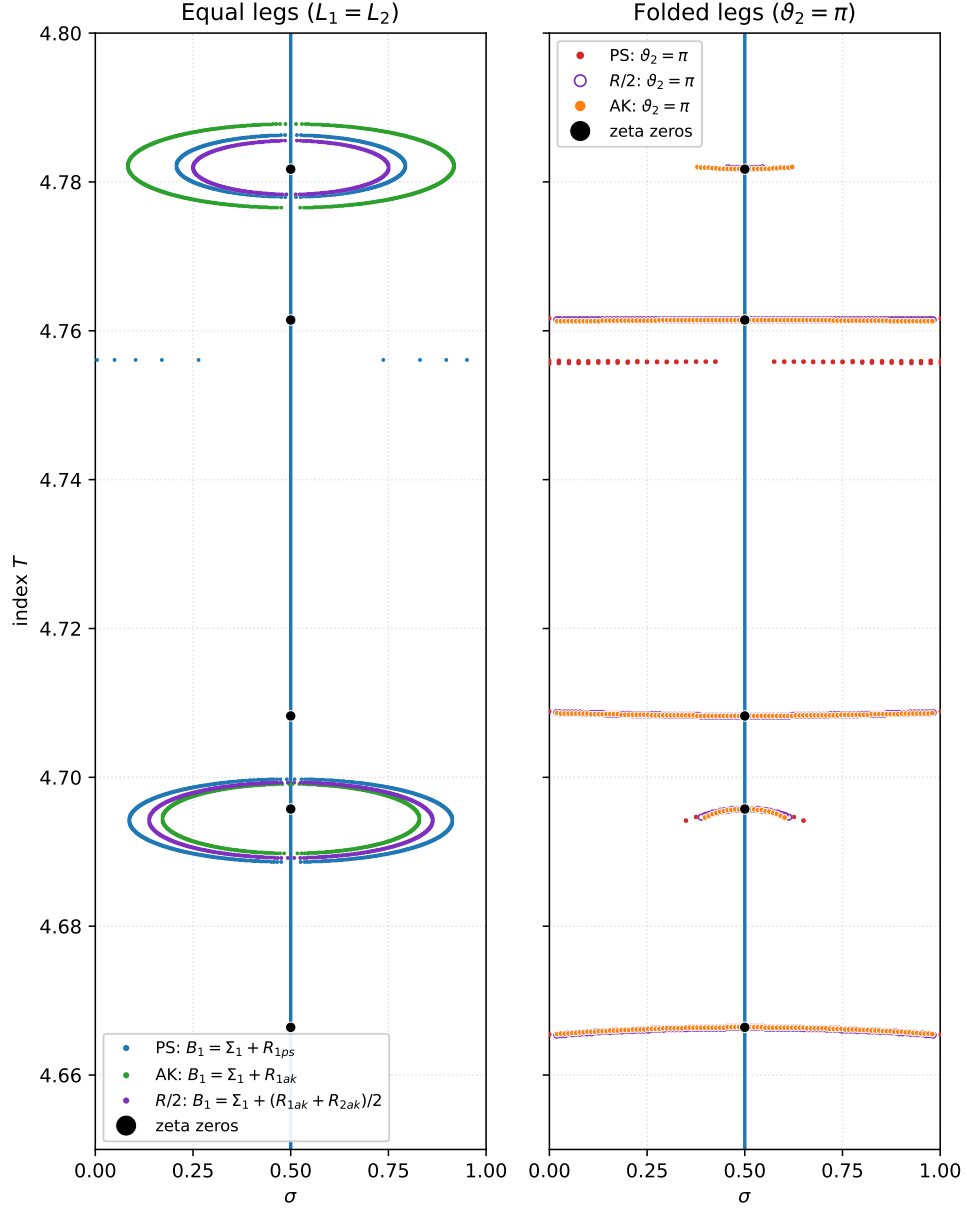


Figure 22: PS, AK and  $R/2$  legs and angles in the window  $4.65 \leq T \leq 4.80$ . Left: equal-leg loci  $L_1 = L_2$  for three choices of bisector point: PS  $B_1 = \Sigma_1 + R_{1ps}$  (blue), AK  $B_1 = \Sigma_1 + R_{1ak}$  (green), and  $R/2$   $B_1 = \Sigma_1 + (R_{1ak} + R_{2ak})/2$  (purple). Right: folded-leg loci  $\vartheta_2 = \pi$  for the same three choices (red, orange, and hollow purple). The blue vertical is the critical line; black dots are zeta zeros. Equal-leg and folded-leg loci computed from the definitions  $L_1 = L_2$  and  $\vartheta_2 = \arg((\zeta - B_1)/B_1)$  for each choice of  $B_1$ . Generated by `fig_ps_ak_r2_legs_angles.py`.

PS, AK and  $R/2$  legs and angles,  $9.415 \leq T \leq 9.465$

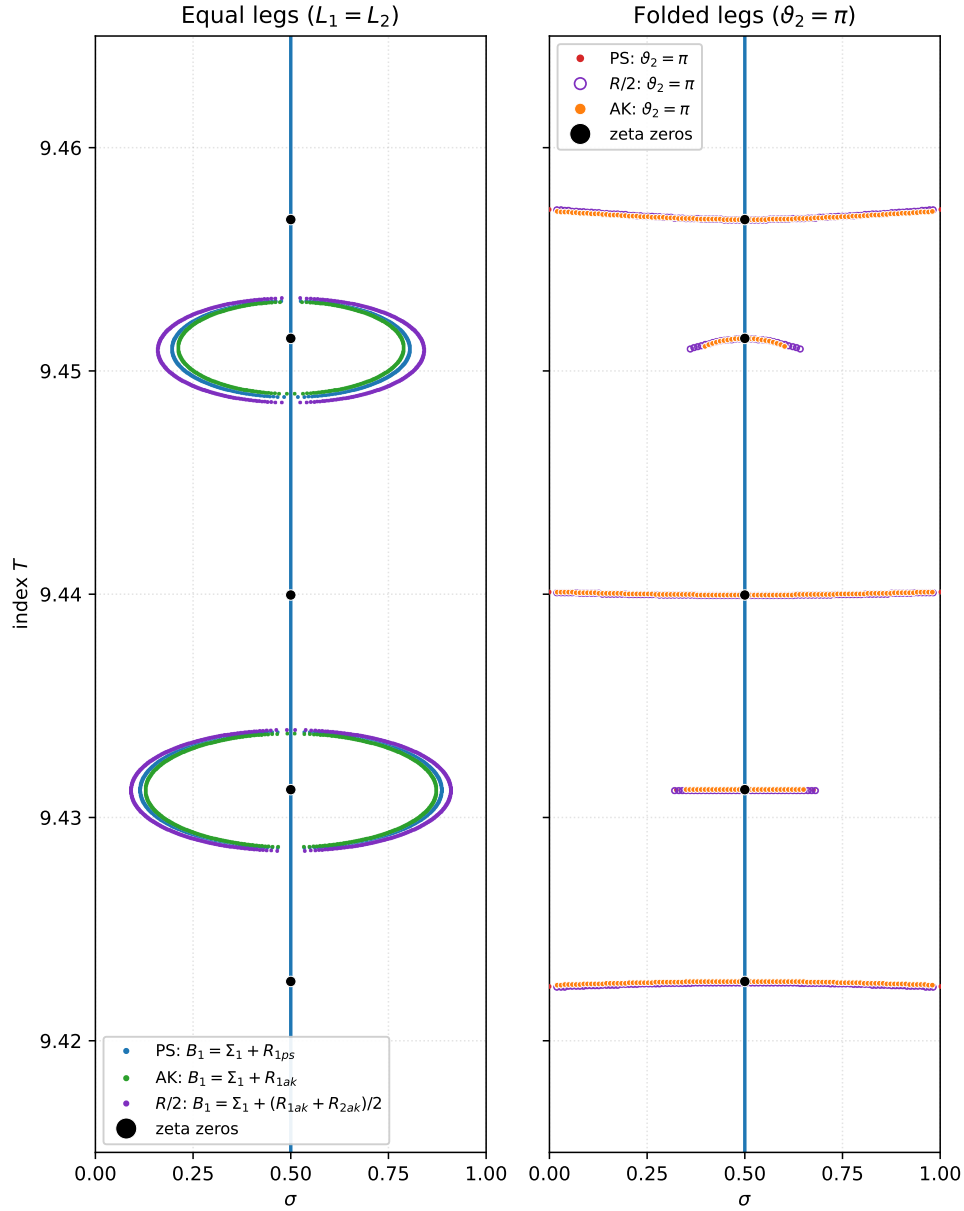


Figure 23: PS, AK and  $R/2$  legs and angles for two neighboring equal-leg ovals near  $T \approx 9.44$ , in the window  $9.415 \leq T \leq 9.465$ . Left: equal-leg loci  $L_1 = L_2$  for PS (blue), AK (green), and  $R/2$  (purple). Right: folded-leg loci  $\theta_2 = \pi$  for the same three choices (red, orange, and hollow purple). The blue vertical is the critical line; black dots are zeta zeros. The lower oval surrounds the zero at  $T \approx 9.431$  and the upper the zero at  $T \approx 9.451$ ; the zero at  $T \approx 9.440$  sits in the gap between them. Loci computed as in Figure 22. Generated by `fig-ps_ak_r2_legs_angles.py` (`--out fig-ps_ak_r2_oval_9441 --t-lo 9.415 --t-hi 9.465`).

If  $\mathcal{E}_{ps}$  and  $\mathcal{E}_{rs}$  have no point in common with  $\sigma \neq \frac{1}{2}$ , then  $\zeta$  has no zero off the critical line.

In particular, if the two loci are disjoint off the line throughout the strip, the Riemann hypothesis follows; if they are disjoint off the line for all  $T \leq T_{\max}$ , then there is no off-line zero with index at most  $T_{\max}$ . The figures above are consistent with a strict nesting (the ovals approach one another but do not cross off the line). A proof that this nesting persists for all  $T$ , with due care at the pole heights of  $d_1$  in Appendix A.5 where  $B_{1ps}$  ceases to be defined in the usual way, would therefore be a proof of the Riemann hypothesis.

In short: on the critical line the leg-length ratio is always one (Corollary 9.1, and likewise for  $B_{1rs}$ ), so the zeros there are determined by the legs folding back,  $\vartheta_2 = \pi$ , for either the  $ps$  or the  $rs$  choice of bisector point. Off the line the length condition is no longer free: a zero can occur only where  $L_1/L_2 = 1$  for the legs, and since that must hold for both choices it is a necessary condition that

$$\frac{L_1}{L_2} = 1 \quad (79)$$

simultaneously for the  $ps$  legs and for the  $rs$  legs.

**The mean leg-imbalance metric.** The equal-leg loci above are the zero sets of a continuous imbalance. For each of the three named remainder-splits of §8.1, write the two leg lengths

$$L_1 = |\Sigma_1 + R_1|, \quad L_2 = |\Sigma_2 + R_2|, \quad (80)$$

where  $(R_1, R_2)$  is  $(R_{1ps}, R_{2ps})$ ,  $(R_{1rs}, R_{2rs}) = (\frac{1}{2}R, \frac{1}{2}R)$ , or  $(R_{1ak}, R_{2ak})$  respectively. The normalized length imbalance of a split is

$$\delta = \frac{|L_1 - L_2|}{L_1 + L_2 + \varepsilon}, \quad \varepsilon = 10^{-18}, \quad (81)$$

so  $\delta = 0$  if and only if  $L_1 = L_2$  (the bisector condition for that choice of bisector point). Averaging the three splits gives the plotted color map

$$\bar{\delta} = \frac{\delta_{ps} + \delta_{rs} + \delta_{ak}}{3}. \quad (82)$$

On the critical line each split has  $L_1 = L_2$  (Corollary 9.1 and the parallel identities for  $rs$  and  $ak$ ), so  $\bar{\delta} \approx 0$  and the strip is dark purple along  $\sigma = \frac{1}{2}$ . Off the line  $\bar{\delta}$  rises, and the equal-leg ovals of Figures 22–23 appear as the dark cellular structures hugging the critical line in Figure 24.



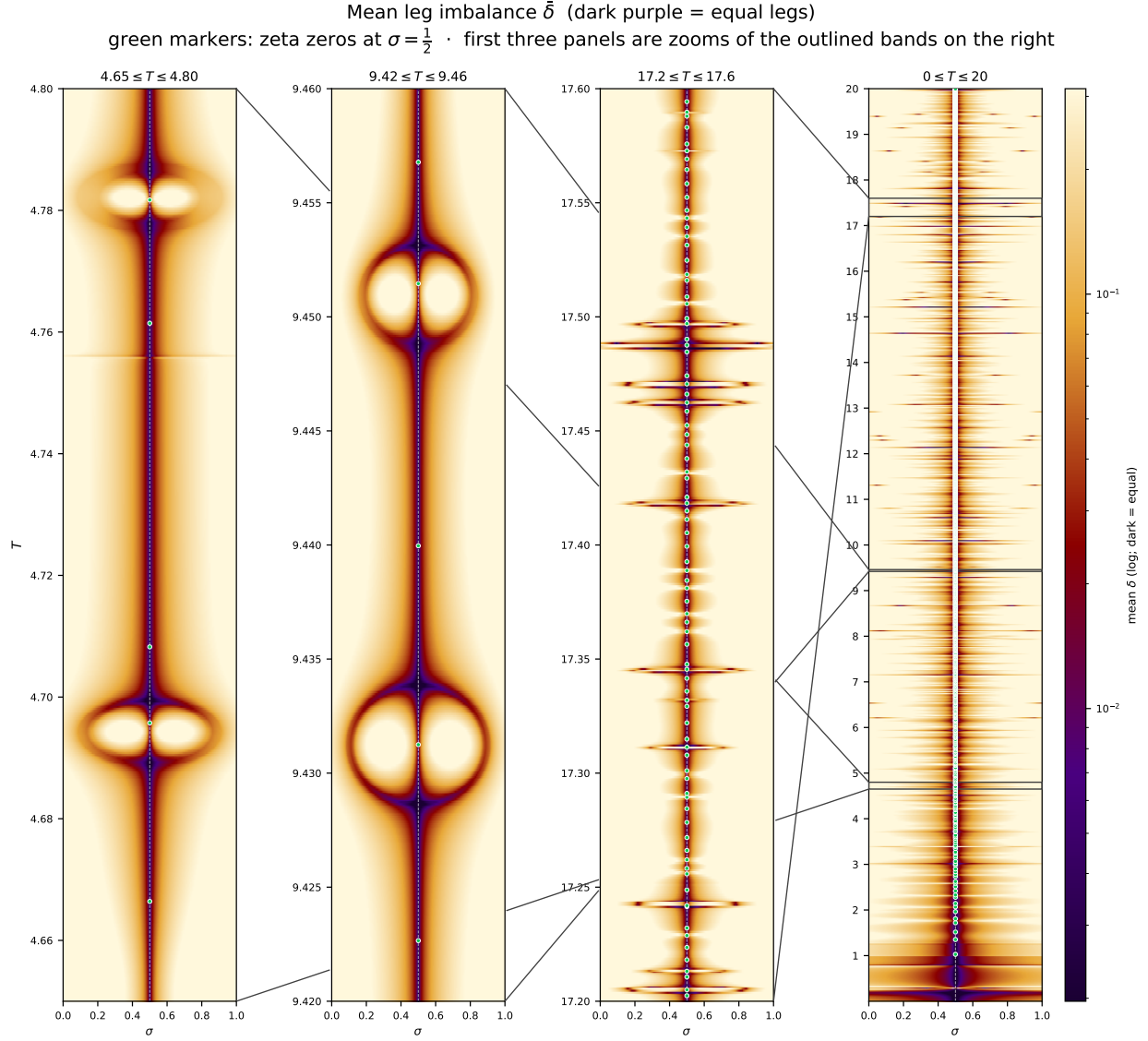


Figure 24: Mean leg imbalance  $\bar{\delta}$  of (82) (dark purple = equal leg lengths). Horizontal axis  $\sigma \in [0, 1]$ ; vertical axis the spiral index  $T$ . Color is  $\bar{\delta}$  on a log scale. Green markers are zeta zeros at  $\sigma = \frac{1}{2}$ . The rightmost panel is the full strip  $0 \leq T \leq 20$ ; the first three panels are zooms of the outlined bands ( $4.65 \leq T \leq 4.80$ ,  $9.42 \leq T \leq 9.46$ ,  $17.2 \leq T \leq 17.6$ ). The dark vertical band on  $\sigma = \frac{1}{2}$  is the simultaneous equal-leg condition for the  $ps$ ,  $rs$ , and  $ak$  splits. Generated by `plot_leg_equality_four_panel.py`.

**The mean fold-deviation metric.** We treat the folded-leg loci in a similar fashion to the equal-leg loci. For each of the three remainder splits write the two leg vectors  $v_1 = \Sigma_1 + R_1$  and  $v_2 = \Sigma_2 + R_2$ , and let  $\vartheta_2 = \arg(v_2/v_1) \in (-\pi, \pi]$  be the fold angle between them. The normalized fold deviation of a split is the scale-free quantity

$$\tau = \frac{\pi - |\vartheta_2|}{\pi}, \quad (83)$$

so  $\tau = 0$  if and only if  $\vartheta_2 = \pi$  (leg 2 folded straight back onto leg 1) and  $\tau = 1$  where the legs run parallel. Averaging the three splits gives the angle companion of  $\bar{\delta}$ ,

$$\bar{\tau} = \frac{\tau_{\text{ps}} + \tau_{\text{rs}} + \tau_{\text{ak}}}{3}, \quad (84)$$

plotted in Figure 25 over the same four windows as Figure 24. The two color maps organize themselves in opposite directions. The length condition is automatic on the critical line and rare off it, so  $\bar{\delta}$  concentrates its dark values in a vertical band. The fold condition instead is met along the nearly horizontal folded-leg curves that sweep the strip at the mean zero density, on and off the line alike (Figure 20), so  $\bar{\tau}$  concentrates its dark values in horizontal filaments crossing the full width of the strip. A zeta zero requires both: it is a point where a dark filament of Figure 25 crosses the dark vertical band of Figure 24, and every green marker sits on a filament at  $\sigma = \frac{1}{2}$ .

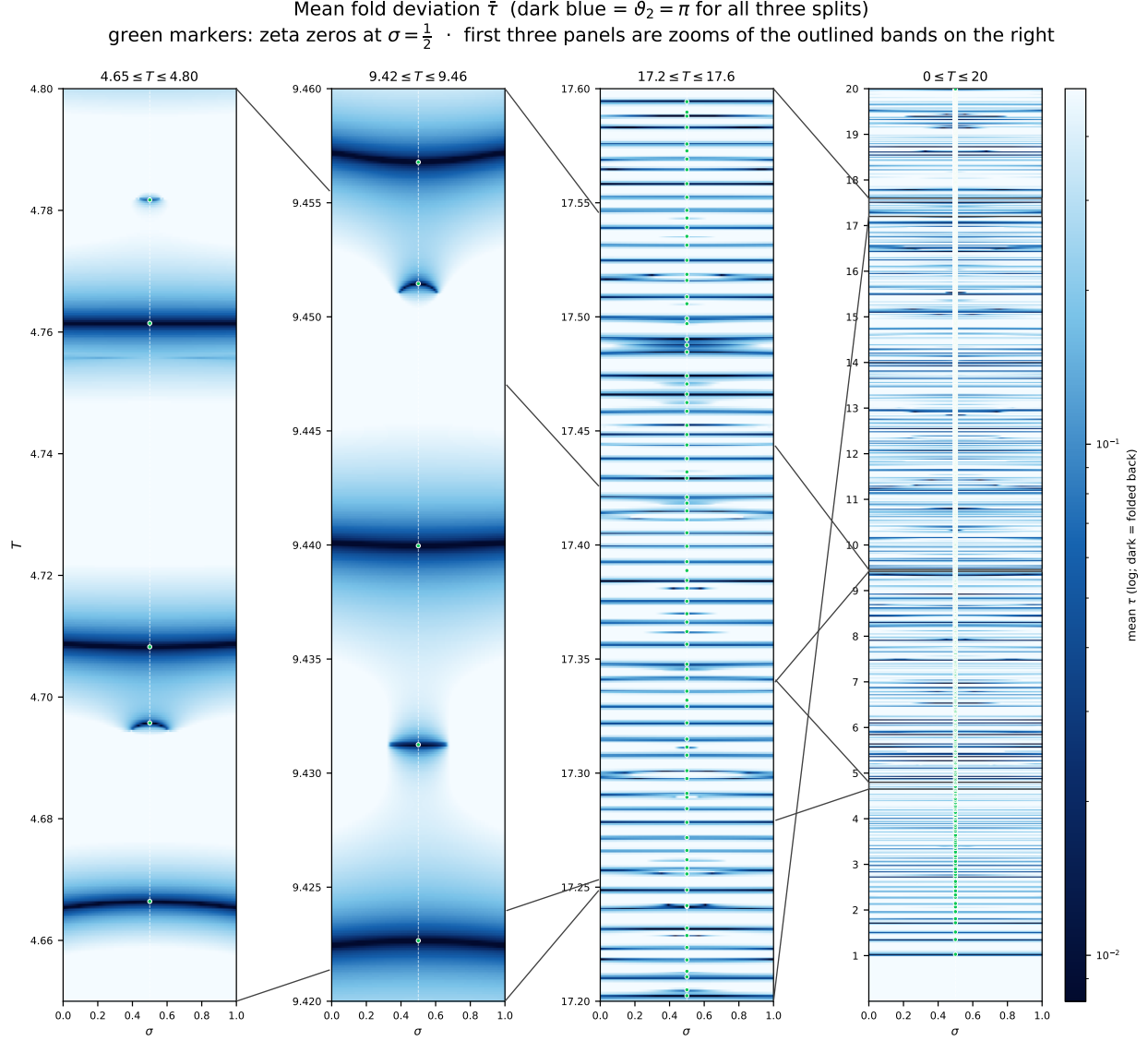


Figure 25: Mean fold deviation  $\bar{\tau}$  of (84) (dark blue =  $\vartheta_2 = \pi$  simultaneously for the *ps*, *rs*, and *ak* splits). Horizontal axis  $\sigma \in [0, 1]$ ; vertical axis the spiral index  $T$ ; color is  $\bar{\tau}$  on a log scale, in the four-panel layout of Figure 24: the rightmost panel is the full strip  $0 \leq T \leq 20$  and the first three panels are zooms of the outlined bands ( $4.65 \leq T \leq 4.80$ ,  $9.42 \leq T \leq 9.46$ ,  $17.2 \leq T \leq 17.6$ ). Green markers are zeta zeros at  $\sigma = \frac{1}{2}$ . Where the equal-leg color map of Figure 24 is dark along the critical line, the fold color map is dark along nearly horizontal filaments sweeping the strip at the mean zero density, and every green marker sits on one. Generated by `plot.theta2_pi_four_panel.py`.

Both color maps are exactly symmetric across the critical line: the leg ratio obeys  $L_1/L_2(\sigma, T) = L_2/L_1(1 - \sigma, T)$ , which leaves (81) unchanged, and the fold angle satisfies the corresponding reflection, so  $\bar{\delta}$  and  $\bar{\tau}$  take the same value at  $(\sigma, T)$  and  $(1 - \sigma, T)$ ; on the plotted grids the two halves agree to a relative  $10^{-10}$ . Half of each panel therefore carries the whole color map, and the two conditions can share one strip. Figure 26 splices them in the first two zoom windows: the left half

of each strip is the leg imbalance  $\bar{\delta}$  and the right half is the fold deviation  $\bar{\tau}$ , meeting at the seam  $\sigma = \frac{1}{2}$ . The two halves state the two conditions of a zero together: the length condition holds along the half-line itself and on the equal-leg ovals hugging it, the angle condition holds on the dark filaments arriving from both sides, and each filament reaches the seam exactly at a green zeta zero marker.

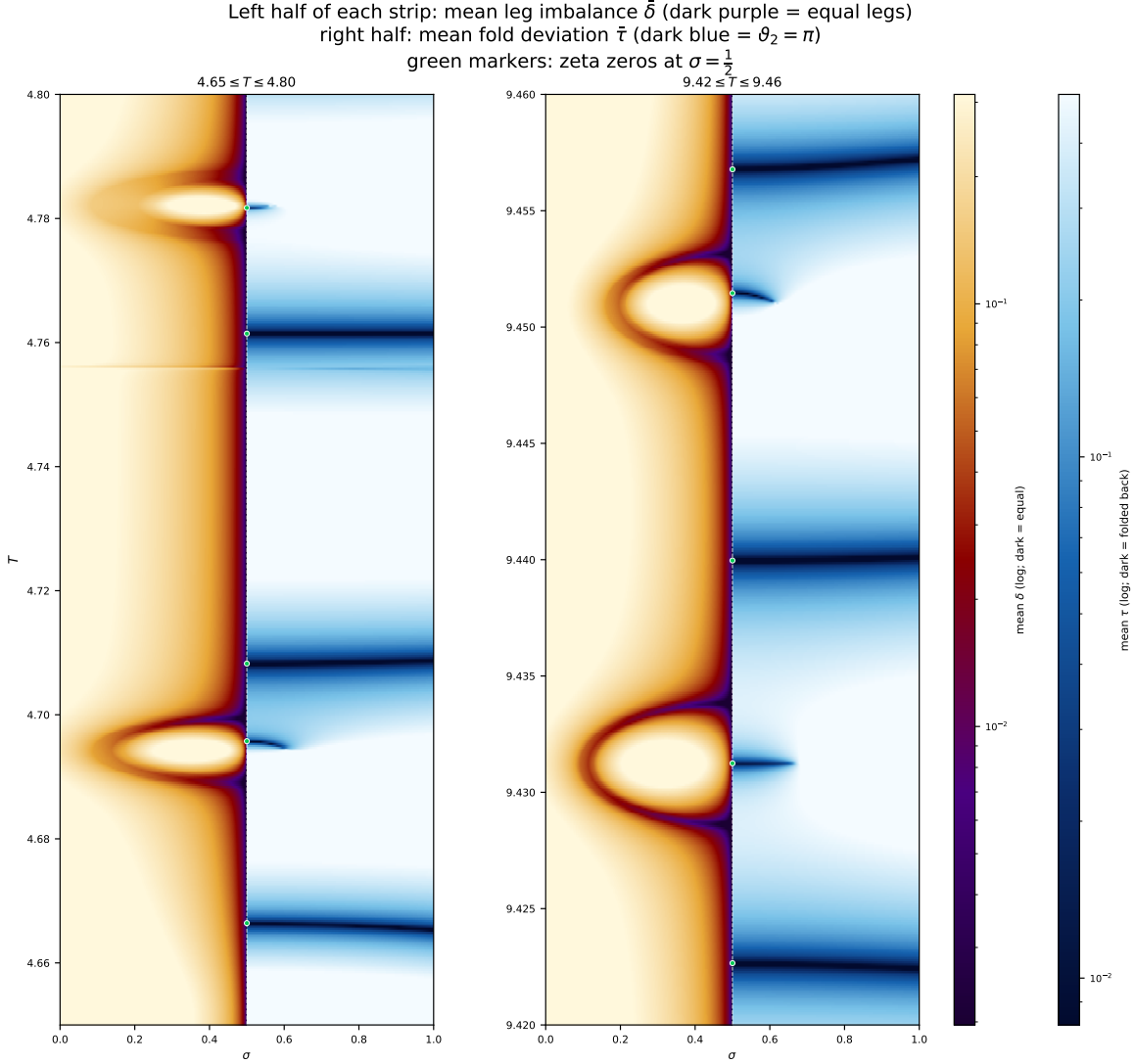


Figure 26: The two zoom windows of Figures 24 and 25 spliced at the critical line: in each strip the left half ( $\sigma < \frac{1}{2}$ ) shows the mean leg imbalance  $\bar{\delta}$  of (82) on the color scale of Figure 24 (dark purple = equal legs), and the right half ( $\sigma > \frac{1}{2}$ ) shows the mean fold deviation  $\bar{\tau}$  of (84) on the color scale of Figure 25 (dark blue = folded back). Both color maps are symmetric under  $\sigma \leftrightarrow 1 - \sigma$  (to a relative  $10^{-10}$  on these grids), so each half carries its whole color map and the splice loses nothing. Green markers are zeta zeros. The length condition is dark along the seam and on the ovals flanking it; the angle condition is dark on the horizontal filaments; a zero is a filament reaching the seam. Generated by `plot_leg_theta2_spliced.py`.

## 9.4 Toward a proof

The geometry assembled in this section suggests two routes to the Riemann Hypothesis, listed here in increasing order of how many remainder splits participate.

1. *Equal legs and fold-back together, for a single split.* A zero at  $(\sigma, T)$  forces, for any single choice of bisector point, both leg conditions of §9.2 at the same input:  $L_1 = L_2$  and  $\vartheta_2 = \pi$ . On the critical line the length condition is automatic (Corollary 9.1), so zeros there are cut out by the fold-back alone. Off the line the two conditions trace independent families of curves (the equal-leg ovals and the nearly horizontal folded-leg curves of Figure 21), and a proof that the two families never pass through the same off-line point, for even one split, would prove the Riemann hypothesis.
2. *Equal legs for two or more splits at once.* A zero makes the legs equal for *every* remainder split simultaneously: when  $\zeta = 0$ ,  $L_2 = |\zeta - B_1| = |B_1| = L_1$  for any  $B_1$ . An off-half-line zero would therefore be an off-half-line point of  $\mathcal{E}_{ps} \cap \mathcal{E}_{rs}$  (and of  $\mathcal{E}_{ak}$  as well). This route drops the angle condition entirely and asks only that two equal-leg ovals meet off the line; the figures of §9.3 show the ovals nesting without crossing. This route is developed in detail below.

**Route 2 in detail.** An off-half-line zero at  $(\sigma, T)$  would require both equal-leg identities at the same input:

$$|\Sigma_1 + R_{1ps}| = |\Sigma_2 + R_{2ps}|, \quad (85a)$$

$$|\Sigma_1 + \tfrac{1}{2}R| = |\Sigma_2 + \tfrac{1}{2}R|. \quad (85b)$$

Here (85a) is the *ps* condition and (85b) the *rs* condition with  $R_{1rs} = \tfrac{1}{2}R$ . Each is one real constraint on  $(\sigma, T)$ : an equality of two moduli is one real equation, so the pair is two real conditions on the two real coordinates of the strip, the right number of conditions to cut out isolated points (or nothing). On  $\sigma = \tfrac{1}{2}$  they are not independent, so they do not cut the half-line down to points. Off it, they are.

On  $\sigma = \tfrac{1}{2}$  both hold for every  $T$ , by Corollary 9.1. They are identities along the whole half-line and do not cut anything out there. Every point of  $\sigma = \tfrac{1}{2}$  is already a common root; that common set is expected and does not constrain zeros beyond the folded-leg condition, which this route does not use.

Off the half-line that automatic identity is gone. Then (85a) and (85b) are independent: they are not the same equation written twice, and satisfying one does not force the other. Each traces its own family of curves, the *ps* ovals  $\mathcal{E}_{ps}$  and the *rs* ovals  $\mathcal{E}_{rs}$ . A common root is a pair  $(\sigma, T)$  that solves both at once, hence a point of  $\mathcal{E}_{ps} \cap \mathcal{E}_{rs}$ . On the half-line that intersection is the whole line; off it, a meeting of the two oval families.

An off-half-line zero would be such a root: a  $(\sigma, T)$  that satisfies both (85a) and (85b) at once. At a zero the chain (the whole Dirichlet chain formed by the links of  $\Sigma_1$ , the remainder, and  $\Sigma_2$ ) closes,  $\zeta = 0$ , so for any  $B_1$

$$L_2 = |\zeta - B_1| = |-B_1| = |B_1| = L_1.$$

That does not depend on how  $B_1$  was chosen, so it holds for  $B_{1ps}$  and  $B_{1rs}$  at once. A zero therefore sits on both loci, on the half-line or off it; off the half-line that is exactly a common root of (85). The implication is one-way:

$$\text{off-half-line zero} \implies \text{off-half-line point of } \mathcal{E}_{ps} \cap \mathcal{E}_{rs}.$$

A meeting of the ovals is necessary for an off-half-line zero, not sufficient: the fold-back  $\vartheta_2 = \pi$  is still required. For proving there are no off-half-line zeros, necessity is what is needed.

A proof that (85) has no solution for  $\sigma \neq \frac{1}{2}$  (away from the poles of  $d_1$ , where the  $ps$  split is undefined) is a proof that those two loci never meet off the half-line: the off-half-line points of the intersection *are* the off-half-line common roots of (85). The contrapositive of the implication then gives the Riemann hypothesis. *If there is no off-half-line meeting, there cannot be an off-half-line zero, because such a zero would force a meeting.* In one line,

$$\begin{aligned} \text{no off-half-line solution of (85)} &\iff \text{no off-half-line point of } \mathcal{E}_{ps} \cap \mathcal{E}_{rs} \\ &\implies \text{no off-half-line zero} \iff \text{RH.} \end{aligned}$$

The figures of §9.3 already suggest the ovals nest and do not cross. Proving what those pictures represent, namely that (85) has no off-half-line solution, would be a proof of RH.

## 9.5 Equal legs density

If a zero were to exist off the critical line, it would have to lie on an equal-leg locus. The reason is the one used in §9.3: at a zero the chain closes,  $\zeta = 0$ , so the bisector point is equidistant from the two ends of the chain,

$$L_2 = |\zeta - B_1| = |-B_1| = |B_1| = L_1,$$

whichever split is used for  $B_1$ . Every zero, on the line or off it, is therefore an equal-leg point. On the line that says nothing, the whole line being equal-leg by Corollary 9.1; off the line it is a genuine restriction, because the off-line part of the locus is not the whole strip but the isolated ovals of Figure 19, symmetric about  $\sigma = \frac{1}{2}$ . An off-line zero has to sit on one of those ovals. Their number and size are therefore worth measuring: they are the only places in the strip where an off-line zero can be. This subsection reports a numerical census of them for  $0 \leq T \leq 401$  (which is  $t$  of  $1.0129 \times 10^6$ ).

**The imbalance and the scan.** The census counts sign changes of the *signed* imbalance

$$g(\sigma, T) = \frac{L_1 - L_2}{L_1 + L_2}, \tag{86}$$

which is (81) without the absolute value. The modulus is dropped on purpose:  $g$  changes sign across an oval, and it is the sign change that can be counted. Numerically  $g$  is odd under  $\sigma \mapsto 1 - \sigma$ , the reflection that exchanges the two legs, so it vanishes identically on the critical line (Corollary 9.1 again), and the ovals are symmetric about the line, as the figures show.

The scan fixes  $\sigma = \frac{1}{2} + \varepsilon$  and steps in  $T$ , recording sign changes of  $g$ . Each oval contributes two, a bottom and a top, and only bottoms are counted, so an oval straddling an integer  $T$  is counted once, to the interval holding its bottom. The pole bands are excluded: at the heights of Appendix A.5 the PS bisector point runs off to infinity and the equal-leg point sweeps the full width of the strip, which is not an oval. Those bands are a PS artifact, 599 of them against 1 for AK, whose remainder has no pole to escape through. Roots are refined by bisection to  $10^{-13}$ . The step is

$$\Delta T = 10^{-4}/m, \tag{87}$$

which holds the number of samples across an oval fixed, the median oval height falling like  $1/T$ :  $1.3 \times 10^{-3}$  in the interval at  $T = 25$  and  $5.2 \times 10^{-5}$  in the one at  $T = 400$ . Ovals thinner than the step are still caught, by minimizing  $g$  wherever it has a positive local minimum below a twentieth of the interval's median and testing whether the minimum dips below zero.

Four checks bound the error. First,  $\zeta$  evaluated as  $\Sigma_1 + \Sigma_2 + R_{ak}$  agrees with `mpmath` to  $2 \times 10^{-10}$  at  $T = 150$  and  $3 \times 10^{-8}$  at  $T = 1000$ , and  $g$  vanishes on the critical line to machine precision. Second, recounting the intervals at  $m = 250, 275, 299, 300, 350, 398, 399$  and  $400$  with  $\Delta T/2$  and  $\Delta T/4$  returns the same count every time, for both splits, and finds the same thinnest cross-section in each. Third, the window  $3 \leq T \leq 4$  reproduces Figure 19, the same four ovals at  $T = 3.0105, 3.3845, 3.6031$  and  $3.7257$ . Fourth, the  $\sigma$ -extent of each oval was measured by bisection, which is what decides whether a scan at a given  $\varepsilon$  can see it: at  $\varepsilon = 0.001$  every oval in the sample is reachable and at  $\varepsilon = 0.01$  some 99.95% are, so the count quoted below, taken at  $\varepsilon = 0.01$ , is not limited by the scan's reach.

**Are they ovals?** Drawn against the index they are hairlines. The component at  $T = 300$  in Figure 27 is 0.39 wide in  $\sigma$  and  $6.9 \times 10^{-5}$  tall in  $T$ , a ratio of 5600 to one, and every critical strip figure here stretches the vertical axis by some such factor to make the locus visible at all. That shape is an artifact of the coordinate. The index is a compressed ruler: at  $T = 300$ , one step in  $T$  is a step of  $I'(300) = 3789$  in  $t$ . Undo that and the component is  $6.9 \times 10^{-5} \times 3789 = 0.26$  tall against 0.39 wide, which is its shape in the  $s$ -plane, where  $\zeta$  is evaluated. It is an oval, and close to an ellipse: comparing its half-height at eight fractions of its half-width  $a$  against  $\sqrt{1 - (u/a)^2}$ , the one drawn departs by 1.8% rms. Over a sample of 300 of them at nine heights from  $T = 5$  to  $T = 400$ , drawn at random within each interval, the median departure is 0.5%, with 80% inside 2% and 94% inside 5%; the ratio of width to height has median 1.22 and ranges from 0.96 to 2.83, with no trend in height, the interval medians sitting between 1.13 and 1.27 throughout. In the sample of 300 closed one-zero components, the components are numerically smooth, convex, and nearly elliptical, with a median width-to-height ratio of 1.22. This description excludes the 23 two-zero components, which are hourglasses: two oval lobes joined at a neck, with one zero in each lobe, as in Figure 30. It also excludes the 72 components cut off by the edge of the strip instead of closing inside it, all of which lie below  $T = 100$ .

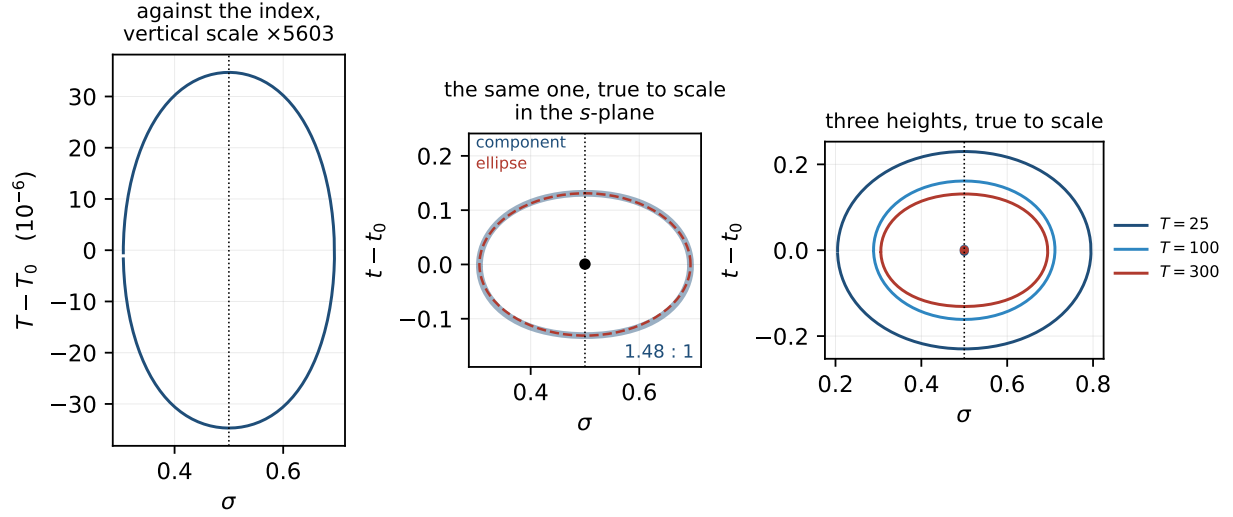


Figure 27: Why the components of the equal-leg locus are called ovals. Left: one of them at  $T = 300$  plotted against the index, as the strip figures plot them, the vertical axis stretched by 5600 to make it visible. Middle: the same component in the  $s$ -plane with the axes to a common scale, and the ellipse through its extremes dashed over it; the dot is the zero of  $\zeta$  it encloses. Right: components at  $T = 25, 100$  and  $300$  to a common scale, each drawn about its zero. They shrink with height and hold their shape. All three are the median-height component of their interval, taken among those whose zero sits nearest the middle, since the zero is central only on average. Generated by `fig_oval_shape.py` of the equal-leg density companion note in the Zest repository.

**The oval count.** Over  $0 \leq T \leq 401$ , scanning the line  $\sigma = \frac{1}{2} + \varepsilon$  with  $\varepsilon = 0.01$ , there are 328 511 ovals for the AK remainder split  $B_1 = \Sigma_1 + R_{1ak}$  and 328 191 for the PS split  $B_1 = \Sigma_1 + R_{1ps}$ , against 1 771 661 zeros of  $\zeta$  in the same range. The two splits converge on each other with height: the mean disagreement per unit interval grows from 1.5 ovals to 5.0, but falls from six percent of the count to 0.32%. The oval density of both splits tracks the density of the zeros. Writing the interval's zeros per oval as a function of height, and fitting over  $3 \leq T \leq 401$  weighted by the oval count,

$$\frac{\#\{\text{zeros in } (m, m+1)\}}{\#\{\text{ovals in } (m, m+1)\}} \approx 4.38 + \frac{11.14}{\ln(t/2\pi)} \quad (3 \leq T \leq 401). \quad (88)$$

This is a weighted empirical fit; above  $T = 100$ , its binned residuals are within  $\pm 0.01$ . The competing fit  $6.74 T^{-0.040}$  is indistinguishable on the measured data. If extrapolated, the displayed logarithmic model tends to 4.38, and with  $dN/dT = I'(T) \ln(t/2\pi)/2\pi \approx 4T \ln T$  would give an oval density near  $0.9 T \ln T$ . This extrapolation is not an established asymptotic law. Figure 28 plots both fits.



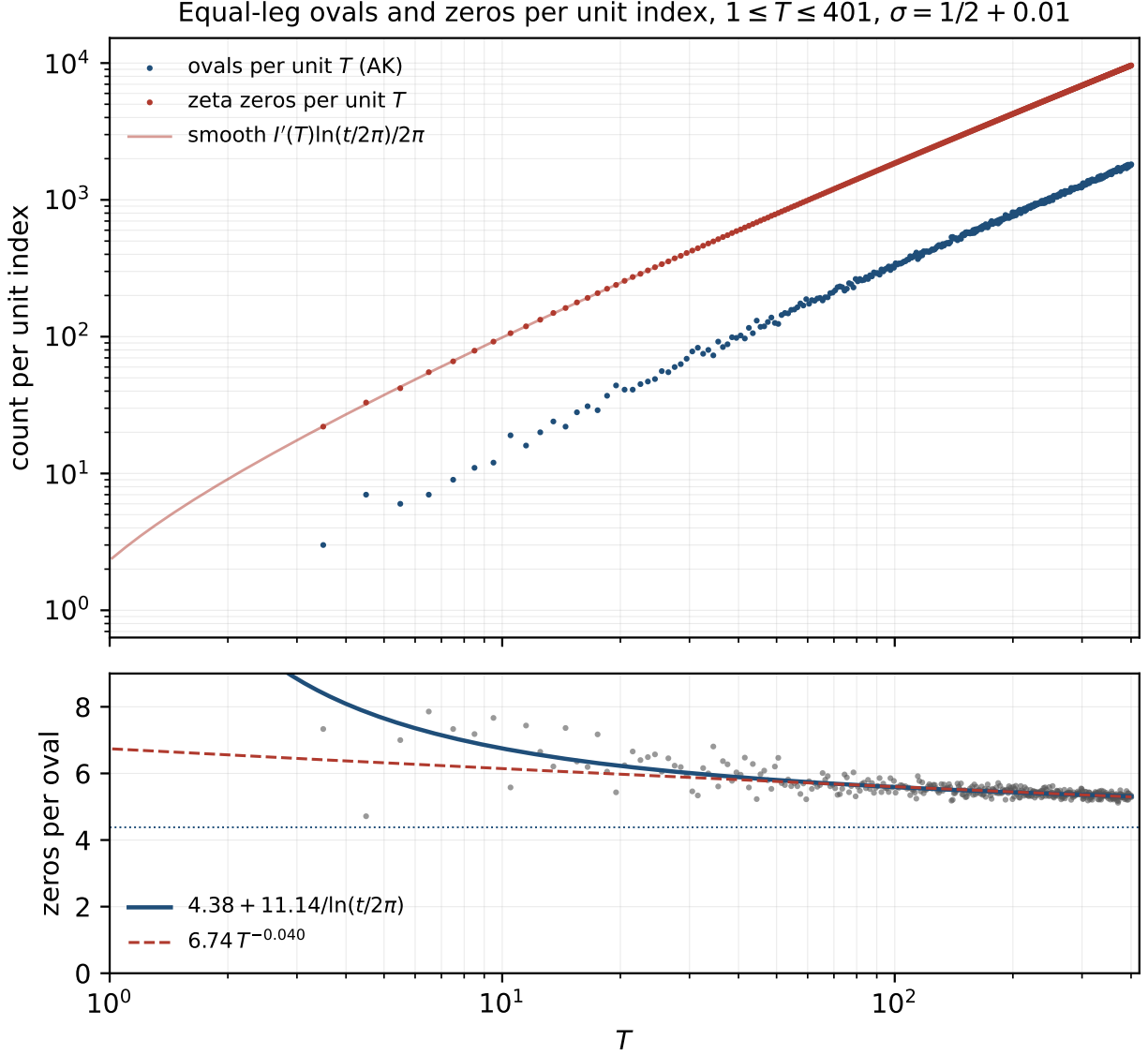


Figure 28: Above: equal-leg ovals and zeros of  $\zeta$  per unit interval of the index,  $1 \leq T \leq 401$ , at  $\sigma = \frac{1}{2} + 0.01$ , log-log. The two clouds are parallel. Below: the interval's zeros per oval, with (88) solid and its extrapolated limit 4.38 dotted, against the power law dashed. The two are indistinguishable over  $50 \leq T \leq 400$  and part company on either side. Generated by `census_analyze.py` of the equal-leg density companion note in the Zest repository.

**Where an off-line zero could be.** The observed oval extent identifies a narrow region for further searching, but the sampled cross-section does not provide a rigorous exclusion bound. In height: each oval is about 0.49 of a mean zero spacing tall, a ratio steady to within 0.01 above  $T = 100$ , and there are about a fifth as many ovals as zeros, so the ovals cover 8.8% of the index axis over  $350 \leq T < 401$ , up from 6.4% over  $3 \leq T < 25$ , a share growing like  $0.66 \ln T + 5.0$  per cent. On the sampled cross-section  $\sigma = \frac{1}{2} + 0.01$ , detected oval intervals therefore cover 8.8% of the  $T$ -axis over  $350 \leq T < 401$ . This is sampled cross-sectional coverage, not a bound on all possible off-line zero heights, because the components may extend farther in height nearer the critical line.

In  $\sigma$ : the ovals draw in toward the critical line as  $T$  grows. The median oval reaches  $\sigma = 1.01$  at  $T = 5$  but only  $\sigma = 0.67$  at  $T = 400$ ; the ninetieth percentile falls from  $\sigma = 1.11$  to  $\sigma = 0.77$ ; and of the 258 814 ovals above  $T = 200$ , just four reach  $\sigma = 0.95$ . Figure 29 shows both measurements.

The tail does not vanish, though, and the distinction matters. What retreats is the bulk of the ovals, not the widest of them: the widest oval in each interval still reaches  $\sigma \approx 0.9$  at every height measured (never below  $\sigma = 0.88$  for  $T \geq 50$ ), and the widest above  $T = 200$  gets to  $\sigma = 0.962$ . What thins is their number, from about eight ovals per unit interval reaching  $\sigma \geq 0.9$  over  $50 \leq T < 100$  to four over  $100 \leq T < 200$ , two over  $200 \leq T < 301$  and one over  $301 \leq T < 401$ , while the total count per interval grows five-fold. A zero well off the line at large  $T$  is therefore not excluded by any of this; it would have to lie on the one or two such ovals per unit interval rather than on any of the eighteen hundred, which is a sharper statement than the count alone gives.

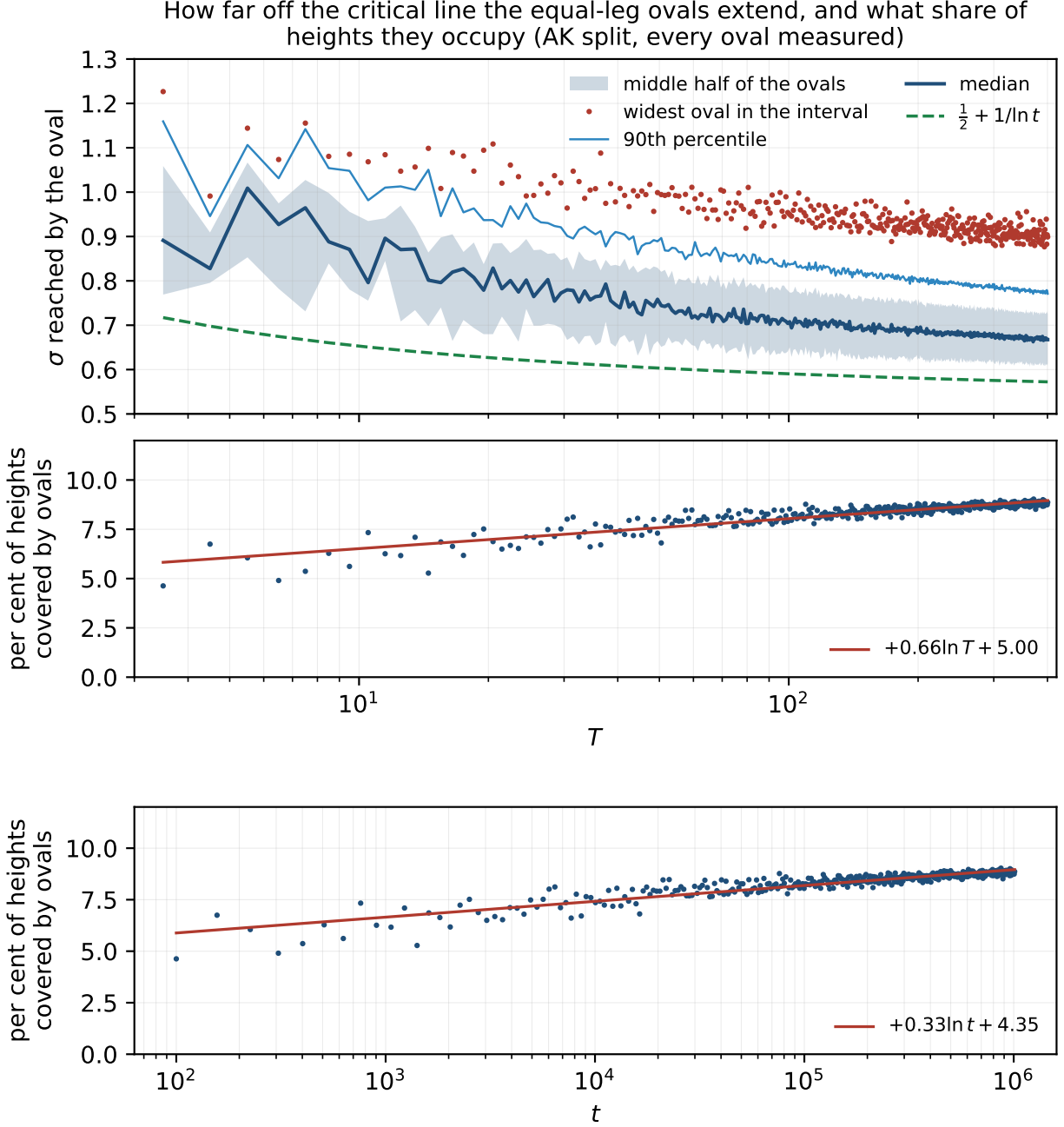


Figure 29: Above: how far off the critical line the ovals extend, from half-widths measured by bisection for every one of the 328 511 ovals in the census, one point per unit interval, from  $T = 3$  where the first ovals are. The band is the middle half of the ovals in that interval and the dots are the widest oval in it. The dashed green curve is  $\sigma = \frac{1}{2} + 1/\ln t$ , drawn on the index axis through  $t = I(T)$ . The bulk draws in toward the line as  $T$  grows while the widest ovals do not. Middle: the percentage of the index axis covered by ovals, which is the share of heights at which an off-line zero could sit, rising like  $0.66 \ln T + 5.0$ . Below: the same share read along  $t$ , where  $\zeta$  is evaluated. This is the middle panel with its horizontal axis relabeled in  $t$  instead of  $T$ , and its law reads  $0.33 \ln t + 4.35$ , the law above rewritten through  $\ln t = 2 \ln T + \ln 2\pi + O(1/T)$ . Measured directly in  $t$ , rather than relabeled, the share moves by under 0.01 of a percentage point above  $T = 25$ . Generated by `reach_census.py` and `fig_reach.py` of the equal-leg density companion note in the Zest repository.

**Usually, one zero per oval.** Almost every detected component brackets exactly one critical-line zero: 328 467 of the 328 511 AK components do so, while 23 contain two zeros and 21 fail the stated one-zero bracket test. Not every zero has a detected component, and that direction fails badly rather than by exception. The zero sits at the oval's center on average rather than individually. Its position within the oval has median 0.5000 and mean 0.5000 of the height, so there is no systematic offset, but the spread about that is 0.14. While 91% of zeros are in the middle half, one oval in eleven holds its zero outside it and one in fifty outside the range 0.16 to 0.84. The two-zero ovals are all unusually close pairs, separated by 0.25 to 0.65 of the local mean spacing, which is what an oval half a spacing tall would have to straddle to catch two. Those components are the hourglasses of the previous paragraph: two oval lobes and a neck, one zero in each lobe. Figure 30 draws the first of them, at  $T \approx 201.72$ , and another at  $T \approx 320.57$ , in the three-split style of Figure 22.

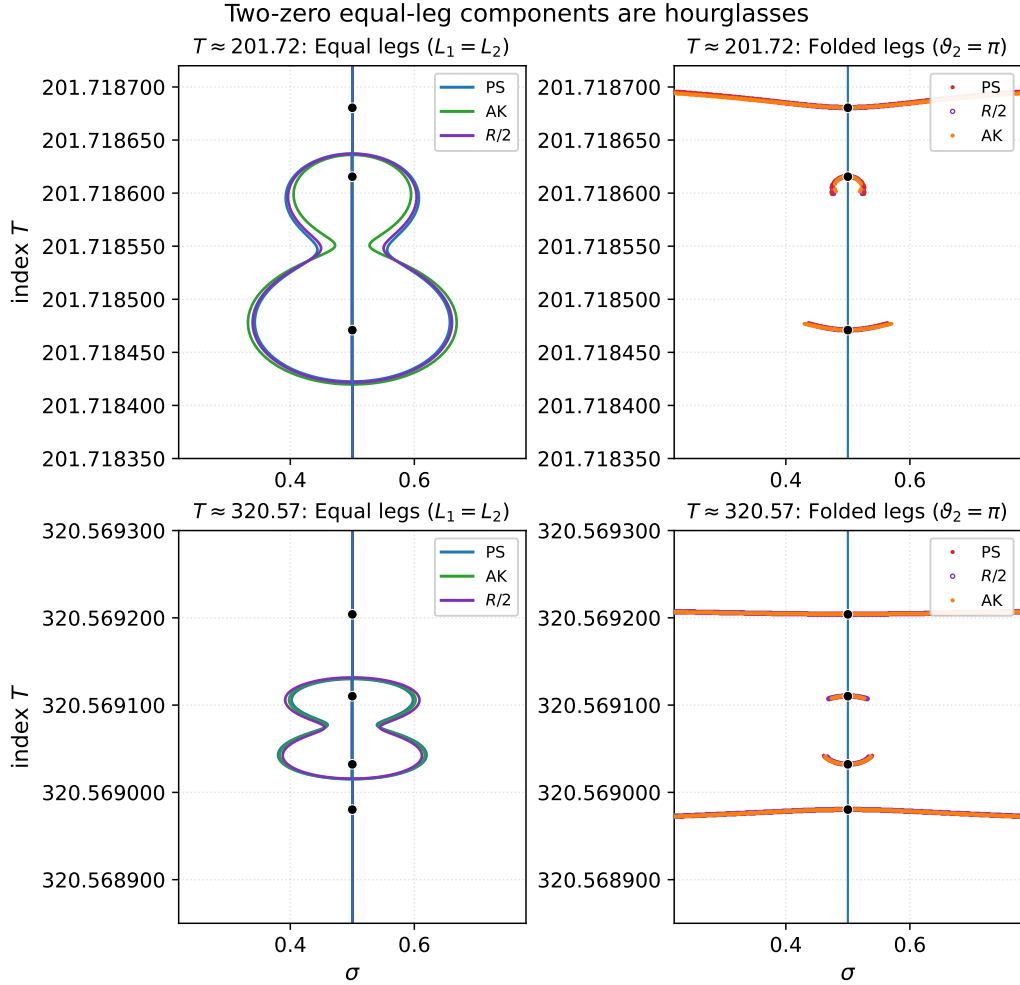


Figure 30: Two AK ovals that each enclose two zeros, drawn as in Figure 22. Top: the first such oval, at  $T \approx 201.72$ . Bottom: another at  $T \approx 320.57$ . Left: equal-leg loci  $L_1 = L_2$  for PS (blue), AK (green), and  $R/2$  (purple). Right: folded-leg loci  $\vartheta_2 = \pi$  for the same three choices (red, orange, and hollow purple). The blue vertical is the critical line; black dots are zeta zeros. Each equal-leg component is an hourglass, two oval lobes joined at a neck, with one zero in each lobe. The three splits nest and keep that shape. Generated by `fig_two_zero_hourglass.py`.

Of the twenty-one empties, eighteen are ovals that span an integer, where the chain gains a link, with their zero just across the integer- $T$  boundary; one is the degenerate interval below  $T = 1$ , where  $\Sigma_1$  is empty; and two, at  $T = 393$  and  $T = 398$ , are ovals whose  $\sigma$ -tip stops a hair past the scan line, at  $\sigma = 0.51004$  and  $\sigma = 0.51003$ , so that the cross-section the scan cuts is a sliver  $3 \times 10^{-7}$  tall at the very tip, lying just to one side of the zero's height. The first of those two misses below the zero and the second above it, by under  $10^{-7}$  either way. Both ovals do hold their zero: at the critical line each is some  $4 \times 10^{-6}$  tall with the zero inside, and the cross-sections hold it out to  $\sigma = 0.50997$ . The miss is in the bracket test at  $\varepsilon = 0.01$ , not in the oval. The exception rate falls from 0.104% over  $3 \leq T < 50$  to 0.007% over  $100 \leq T < 200$  and then edges back up to 0.014% over  $301 \leq T < 401$ , the rise being the close pairs, whose number grows with the zero count while staying under one oval in seven thousand.

The other direction is not close to one for one, and it is the more important half of the statement. There are 1 771 661 zeros in the range against 328 511 ovals, so approximately  $1 - 328\,511/1\,771\,661 = 81.46\%$ , or about four zeros in five, have no detected component around them. The relationship between detected components and zeros is an incidence relation, not a single-valued map: most components contain one zero, 23 contain two, and 21 fail the stated one-zero bracket test, while most zeros lie in no detected component. Which zeros acquire components is a question the census does not answer.

Three things follow. First, the off-line locus is tied to the zeros rather than being a structure independent of them. Almost every oval wraps one zero, centered on it in the mean, about half a mean spacing tall and a fifth wider than that; the ovals are a selection from the zeros rather than a phenomenon of their own, which is why their count obeys (88) instead of a law of its own.

Second, it makes the disjointness program of §9.3 and §9.4 local. What is needed for an off-line zero is that  $\mathcal{E}_{\text{ps}}$  and  $\mathcal{E}_{\text{rs}}$  meet off the line. Since an oval almost always encloses exactly one zero, the two ovals that have to be compared are the PS oval and the  $rs$  oval belonging to the *same* zero, and the global statement “the two loci have no common point off the line” becomes one nesting statement per bracketed zero, of which there are about  $0.9T \ln T$  per unit index. That is the right shape for an argument: a local inequality attached to each zero, to be proved uniformly, rather than a statement about two curves wandering the strip.

Third, the caution. None of this is evidence for or against the Riemann Hypothesis, and the one-zero-per-oval law in particular is not. The zero counted inside each oval is found *on* the critical line, by sign changes of Hardy's  $Z$ . The census evaluates an off-line numerical approximation to  $\zeta$  when computing the imbalance  $g$  on  $\sigma = 0.51$ , but it does not test the equation  $\zeta = 0$  along the detected components and does not search them for off-line zeros. A component already holding an on-line zero at its center is not forbidden from passing through one. No orphan component was detected among the components intersecting the scan line  $\sigma = 0.51$  over the stated range; this does not exclude components that fail to intersect that line. The census also points at the search that would test the hypothesis directly, since evaluating  $\zeta$  along the ovals is a one-dimensional problem (about  $0.9T \ln T$  closed curves per unit index), where searching the strip for off-line zeros is a two-dimensional one.

There is a converse worth stating. Almost every detected component brackets a zero, but a zero usually has no detected component: in this sample only about 18.54% of zeros acquire one, and what distinguishes those from the rest is open. The bracketing requires the transverse imbalance  $\partial g / \partial \sigma$  on the line to change sign on both sides of the zero, and nothing here explains why it does so for two zeros in nine and not for the others.

## 10 Geometric origins of the $I(T)$ and Yin and Yang functions

The two devices this paper leans on hardest, the index map  $t = I(T)$  of §3 and the yin and yang curves of §6.4–6.5, were both found in the bisector frame, by watching the link chains of §9 move. This section reconstructs the two discoveries.

### 10.1 Geometric origin of $I(T)$ functions

Change the point of view: hold the forward bisector link still. Translate joint  $m$  to the origin and rotate and scale so that the bisector link lies along the real axis from 0 to 1; everything else moves in this *bisector frame* as  $t$  increases. Watching the two links on either side of the bisector link, the link at  $m - 1$  behind it and the link at  $m + 1$  ahead of it, one sees a periodic picture: each neighboring link revolves around the stationary bisector link approximately *twice* (approaching exactly two revolutions as  $T$  grows), and then the whole pattern repeats. The instant the pattern repeats is precisely the handoff instant of §9.1: the neighboring link ***folds back*** onto the bisector link, the bisector point moves onto that next link and that next link becomes the new bisector link with its own two neighbors.

The neighboring links in the bisector frame,  $T = 4 \rightarrow 5$  (bisector link = link 4)

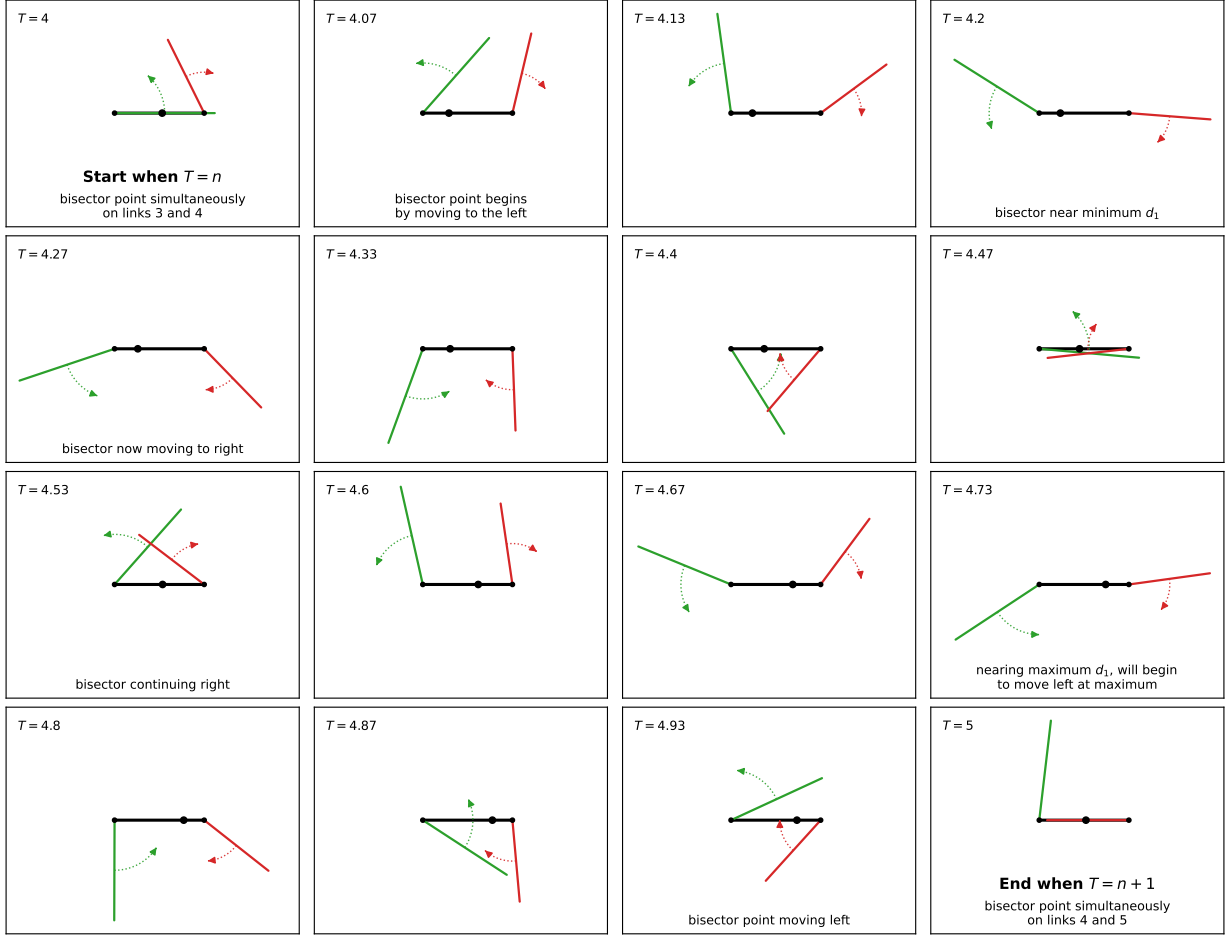


Figure 31: The two neighboring links in the bisector frame as  $T$  sweeps from 4 to 5 ( $\sigma = \frac{1}{2}$ ), sixteen snapshots read left to right from the top left ( $T = 4$ ) to the bottom right ( $T = 5$ ). In every panel the bisector link, link  $m = 4$  of the forward chain, is held fixed along the real axis from 0 to 1 (black); link 3 (green) hangs off joint 4 at the origin and link 5 (red) off joint 5 at the point 1, both scaled relative to the unit bisector link. The dotted curved arrow on each neighbor runs from the middle of the link to the middle of that link as the next panel depicts it, tracing the motion the link takes between panels. The black dot on the bisector link marks the bisector point, at the crossing fraction  $\hat{d}_1 = M^\sigma d_1$  along the unit link. As  $T$  advances, each neighbor revolves around the stationary bisector link approximately twice. At  $T = 4$  (Start) the green link is folded back onto the bisector link (unwrapped phase increment  $9\pi$ , hence geometric joint angle  $\pi \pmod{2\pi}$ ), so the bisector point lies on links 3 and 4 simultaneously: that is the handoff instant at which link 4 became the bisector link. At  $T = 5$  (End) it is the red link that folds back (unwrapped phase increment  $11\pi$ , hence geometric joint angle  $\pi \pmod{2\pi}$ ), the bisector point lies on links 4 and 5 simultaneously, and link 5 takes over as the new bisector link. Generated by `fig.bisector_frame.py`.

Besides the fact that the two links resemble the motion of the two different length hands of a clock but with one going clockwise and the other going counterclockwise, the *periodicity* begs for a clock that ticks once per bisector point handoff, and that is where  $I(T)$  came from: we sought a formula  $t = I(T)$  whose input  $T$  passes through an integer at exactly the moments the bisector

point moves from one link to the next. Two facts about the joint angle between the outgoing and incoming bisector links pin the formula down.

1. *From the zeta function.* Consecutive summands of  $\Sigma_1$  differ in phase by  $t \ln \frac{n+1}{n}$ , so the turn at the joint where link  $T-1$  meets link  $T$  (interpolating the index continuously) has unwrapped phase increment

$$J_{\text{unw}}(T) = t \log\left(\frac{1}{T} + 1\right). \quad (89)$$

2. *From the observed motion.* At a handoff the two links fold back onto one another, so the unwrapped phase increment there is an *odd multiple of  $\pi$*  (hence geometric joint angle  $\pi$ ); and the neighboring link revolves around the stationary bisector link *twice* per unit of the index, so the odd multiple advances by two with each unit of  $T$  (hence the factor 2). Interpolating between handoffs, define the unwrapped phase increment by

$$J_{\text{unw}}(T) = \pi(2T + 1). \quad (90)$$

The corresponding geometric joint angle is

$$J_{\text{geom}}(T) \equiv J_{\text{unw}}(T) \pmod{2\pi};$$

at every integer  $T$ ,  $J_{\text{geom}}(T) \equiv \pi \pmod{2\pi}$ .

Setting (89) equal to (90) and solving for  $t$  yields

$$\pi(2T + 1) = t \log\left(\frac{1}{T} + 1\right) \implies t = \frac{\pi(2T + 1)}{\log\left(\frac{1}{T} + 1\right)} = I(T), \quad (91)$$

which is the mapping adopted in §3: under  $t = I(T)$  the bisector point resides on link  $\lfloor T \rfloor$ , and  $T$  is an integer exactly when the bisector point moves from one link to the next.

**Remark 10.1.** Taylor-expanding the reciprocal logarithm in (91),  $1/\log(1 + \frac{1}{T}) = T + \frac{1}{2} - \frac{1}{12T} + \frac{1}{24T^2} - \dots$ , and collecting orders of  $T$  gives a family of invertible polynomial approximations to  $I(T)$ ; keeping successively more terms:

| terms | $I(T) \approx$                                   | solved for $T$                                                                    |
|-------|--------------------------------------------------|-----------------------------------------------------------------------------------|
| 2     | $2\pi(T^2 + T)$                                  | $T = -\frac{1}{2} + \sqrt{\frac{t}{2\pi} + \frac{1}{4}}$                          |
| 3     | $2\pi(T^2 + T + \frac{1}{6})$                    | $T = -\frac{1}{2} + \sqrt{\frac{t}{2\pi} + \frac{1}{12}}$                         |
| 4     | $2\pi(T^2 + T + \frac{1}{6} - \frac{1}{180T^2})$ | $T \approx -\frac{1}{2} + \sqrt{\frac{t}{2\pi} + \frac{1}{12} + \frac{\pi}{90t}}$ |

(The  $1/T$  term of the expansion cancels, so the fourth term is already  $O(T^{-2})$ .) The 2-term form misses  $I(T)$  by a constant  $\approx \pi/3$ ; the 3-term form is accurate to  $O(T^{-2})$  and the 4-term form to  $O(T^{-3})$ . At  $T = 20$  their errors in  $T$  after inversion are  $4 \times 10^{-3}$ ,  $3 \times 10^{-7}$  and  $4 \times 10^{-11}$  respectively. The first two rows invert exactly (quadratics); the 4-term row is inverted by substituting the 3-term inverse into the small  $T^{-2}$  correction. Figure 32 draws  $I(T)$  against the two-term form; the difference is invisible at scale, about one against values in the thousands.

The table also places Siegel's cutoff (8) on the index. The 3-term row is  $t/2\pi = (T + \frac{1}{2})^2 - \frac{1}{12} + O(T^{-2})$ , so

$$\sqrt{\frac{t}{2\pi}} = T + \frac{1}{2} - \frac{1}{24T} + O(T^{-2}), \quad (92)$$



which is right to  $5 \times 10^{-4}$  at  $T = 6.18$ ,  $5 \times 10^{-5}$  at  $T = 20$  and  $2 \times 10^{-6}$  at  $T = 100$ . Siegel's  $m$  therefore steps *near* the *half-integers* of  $T$ , a shade past them by  $1/24T$ , while ours steps *exactly* at the integers.

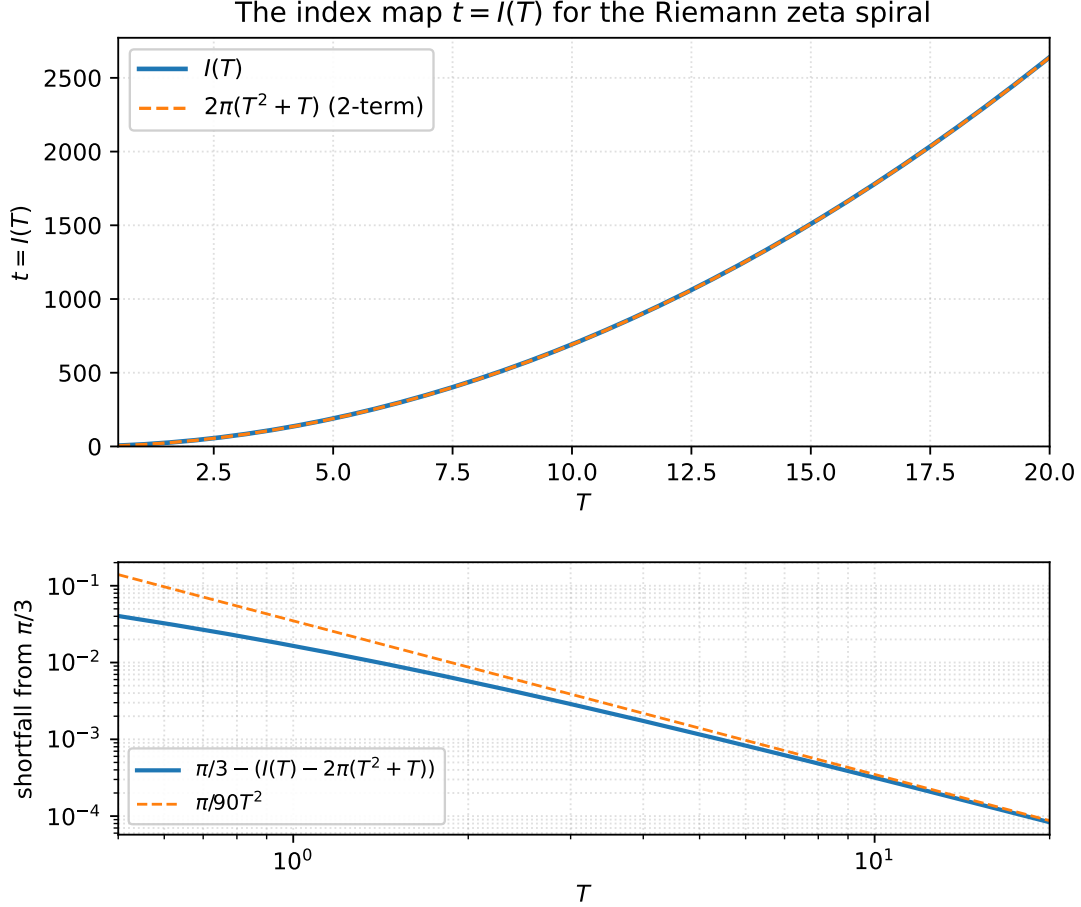


Figure 32: Above: the index map  $t = I(T)$  of (9) (solid), together with the two-term approximation  $2\pi(T^2 + T)$  from the Taylor table above (dashed). Below: the difference between the two, which the panel above cannot show, the curves differing by about one against values in the thousands. That difference is the constant  $\pi/3$  of the table, the  $\frac{1}{6}$  the third term restores, so what is plotted log-log is its distance from that constant, which is the fourth term: it follows  $\pi/90T^2$  (dashed) from  $T = 5$  on, and is  $8 \times 10^{-5}$  by  $T = 20$ . Generated by `fig.IT.py`.

We stress that the polynomial forms of the table serve only to compare with the familiar  $\sqrt{t/2\pi}$ : everywhere in this paper  $t = I(T)$  is used *exactly*. The defining property of the mapping, that the neighboring link folds back onto the bisector link *exactly* as  $T$  passes an integer and the bisector point transfers from one link to the next exactly at that instant, holds only for the exact  $I(T)$ . Under any approximation, including the one whose inverse is essentially Siegel's  $\sqrt{t/2\pi}$ , the fold instants drift off the integers by the corresponding error in the table: the clock still ticks once per handoff, but never precisely on the hour.

**Remark 10.2** (comparing the classical  $t$  rescaling so that zeros have unit mean spacing, and  $I(T)$ ). In random-matrix and zeta-statistics work, zero ordinates are unfolded using the smooth counting

function

$$\overline{N}(t) = \frac{t}{2\pi} \log \frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8}, \quad \tilde{\gamma}_n = \overline{N}(\gamma_n).$$

The ordinates  $\tilde{\gamma}_n$  have asymptotically unit mean spacing, which lets one compare zero statistics at different heights. The exact staircase instead satisfies  $N(\gamma_n) = n$  under the simple-zero convention and therefore removes the fluctuations by construction. Our  $I(T)$  is a change of variable in the same spirit, a reparameterization of the height  $t$ , but it runs in the opposite direction, compressing ever more zeros into each unit interval of the index  $T$ . For example, the interval  $[5, 6]$  carries 42 zeros,  $[10, 11]$  carries 106,  $[20, 21]$  carries 256, and  $[50, 51]$  carries 802. Rescaling, or what the random-matrix literature calls unfolding, the zeros flattens the count to unit spacing; the index does the reverse. Unfolding views the zeros one at a time, each in its own unit of the new variable; the index views them in groups of ever increasing size, one crowd of zeros per unit of  $T$ .

### 10.1.1 $I(T)$ functions for other $L$ -functions besides the zeta function

The same change-of-variable idea extends from  $\zeta$  to Dirichlet  $L$ -functions. This subsection is an aside recording another observation we made along the way; nothing in it is used elsewhere in this paper. Let  $p$  be an odd prime. On the critical line the partial sums of

$$L(s, \chi_p) = \sum_{n=1}^{\infty} \chi_p(n) n^{-s} \tag{93}$$

form a spiral analogous to zeta's, but now the character  $\chi_p(n) = \left(\frac{n}{p}\right)$  inserts periodic sign changes and, when  $p \mid n$ , zero-length *phantom* links. The series is conditionally, not absolutely, convergent on  $\Re s = \frac{1}{2}$ . Counting only the nonzero links as *links*, there are  $p - 1$  links and one phantom in every block of  $p$  summands, and the natural index  $T$  is the link index (phantoms receive no  $T$ ).

Watching these link chains as  $t$  varies, we noticed that the bisector point still moves from one link to the next, just as for zeta, but not always when links fold back on one another: sometimes the bisector point moves from link to link at a *joint*, as the bisector line sweeps through it. So  $L$ -functions have other event types that zeta does not. An *event* is the instant the bisector point moves from one link to the next. There are two ways this can happen: either the two links fold back onto one another (the turning angle at their common joint is an odd multiple of  $\pi$ ), so the bisector point slides across the joint from the old link onto the new one; or the bisector line sweeps through a joint, carrying the crossing from one link to its neighbor. For zeta only the first kind occurs: every handoff of §9.1 is a folded event, once per unit of  $T$ , and there are no bisector events. For  $L(s, \chi_p)$  both kinds occur, recurring with period  $\varphi(p) = p - 1$  in the integer index  $T$ :

- *folded* events, where the turning angle at the current joint is  $\pm\pi$  (the chain reverses), at  $T \equiv 0 \pmod{p-1}$ ;
- *bisector* events, where the joint lies on the perpendicular bisector of the segment from 0 to  $L(s, \chi_p)$ , at the remaining residues modulo  $p - 1$ .

For  $p = 3$  one has  $\varphi(3) = 2$ : even  $T$  are folds and odd  $T$  are bisectors. The map  $t = I_p(T)$  is defined so that integer  $T$  lands at the  $t$ -value of its event. Figure 33 draws the two link chains of  $L(s, \chi_3)$  at  $T = 6.125$ .

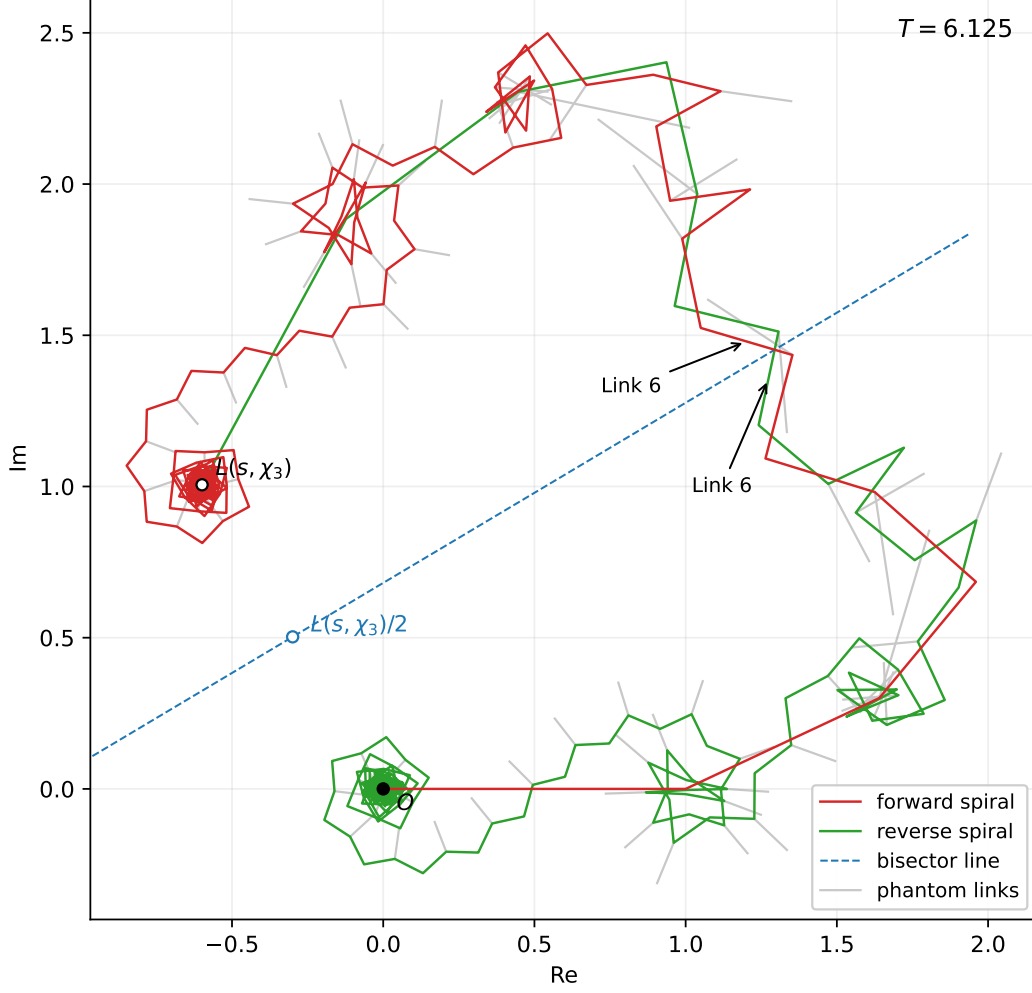


Figure 33: The two partial-sum spirals of  $L(s, \chi_3)$  at  $T = 6.125$  on the critical line ( $t = I_3(T) \approx 176.13$ ), drawn with  $\lfloor 2T(T+1)p \rfloor = 261$  summands. The forward chain (red) starts at the origin  $O$  and winds into its spiral around the analytic value  $L(s, \chi_3)$  (open dot); the reverse chain (green) is its mirror image across the bisector line (dashed blue), the perpendicular bisector of the segment from  $O$  to  $L(s, \chi_3)$  through the labeled midpoint  $L(s, \chi_3)/2$ , so the tail of the link chain sits on  $L(s, \chi_3)$  and its head lands at  $O$ . Every third summand has  $\chi_3(n) = 0$ : these zero-length phantom links are drawn light gray as the connectors the chain skips. The labeled link 6 of each chain is the current bisector link: the two chains cross on it, on the bisector line. Generated by `fig_L3_spirals.py`.

Because only a fraction  $(p-1)/p$  of the summands are links, the smooth *base* formula generalizing (9) is

$$I_p^{\text{base}}(T) = \frac{4\pi T}{(p-1) \log \frac{pT + (p-1)}{pT - (p-1)}}. \quad (94)$$

Asymptotically  $I_p(T) \sim \frac{2p}{(p-1)^2} \pi T^2$ : the leading coefficient is  $\frac{3}{2}\pi$  for  $p = 3$  and  $\frac{5}{8}\pi$  for  $p = 5$ , against  $2\pi$  for zeta's own  $I(T)$ . (Zeta is not the  $p = 1$  member of this family, since the factor  $2p/(p-1)^2$  diverges there; for zeta every summand is a link and there are no phantoms.) The

base does not run through the middle of the events: it touches  $I_p$  at the folds and lies below it in between, missing a one-signed bulge of period  $p - 1$  imposed by the character. For  $p = 3$  the contact is exact, since at even  $T$  the gate of (97) vanishes and  $c = 2$  turns (96) into the base; for  $p = 5$  the fitted correction returns to zero at  $T \equiv 0 \pmod{4}$  to within  $2 \times 10^{-3}$ . One restores the bulge by writing

$$I_p(T) = I_p^{\text{base}}(T) + C_p(T). \quad (95)$$

For  $p = 3$  the character is  $[1, -1, 0]$ . The base specializes to  $I_3^{\text{base}}(T) = 2\pi T / \log \frac{3T+2}{3T-2}$ , and the correction is built from the character-modulated map

$$c = \frac{3 + \cos(\pi T)}{2}, \quad I_3^{\text{geom}}(T) = \frac{c \pi T}{\log \frac{3T+c}{3T-c}}, \quad (96)$$

together with a small odd- $T$  offset  $\alpha(T) = 0.390914 + 0.01712 T^{-2}$  fitted to event scans:

$$I_3(T) = I_3^{\text{geom}}(T) + \frac{1}{2}(1 - \cos \pi T) \alpha(T). \quad (97)$$

(Equivalently  $I_3 = I_3^{\text{base}} + C_3$  with  $C_3 = I_3^{\text{geom}} - I_3^{\text{base}} + \frac{1}{2}(1 - \cos \pi T) \alpha$ ; the gate vanishes on even  $T$ , so  $C_3 = 0$  on folds.) On odd  $T = 1, \dots, 99$  the residual between scanned event times and (97) has RMSE about  $8 \times 10^{-6}$ .

For  $p = 5$  the same pattern gives  $I_5^{\text{base}}(T) = \pi T / \log \frac{5T+4}{5T-4}$  and a fitted period-4 correction

$$C_5(T) = 0.6206 - 0.6576 \cos \frac{\pi T}{2} + 0.0369 \cos \pi T + \frac{0.169}{T^{3.12}}. \quad (98)$$

Figure 34 compares  $I(T)$ ,  $I_3(T)$ , and  $I_5(T)$  on a common range of  $T$ .

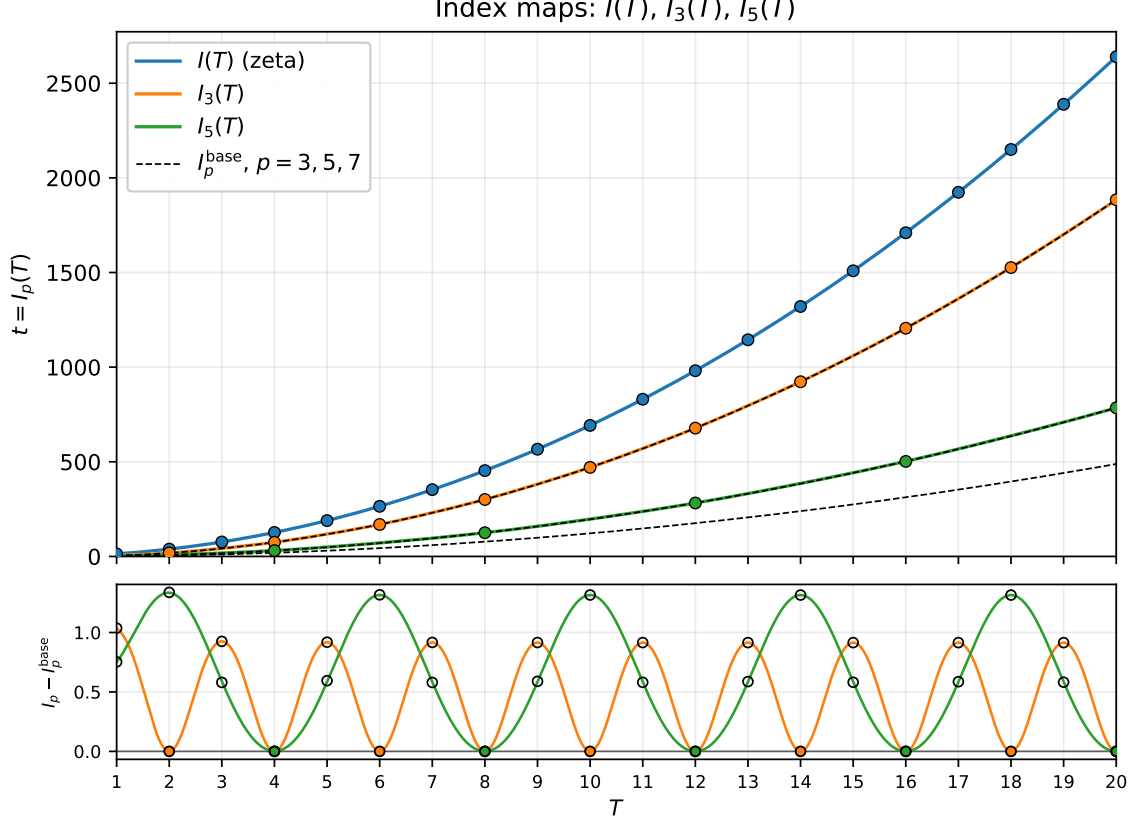


Figure 34: Top: index maps on  $T \in [1, 20]$ : zeta's  $I(T)$  (blue),  $I_3(T)$  for  $p = 3$  (orange), and  $I_5(T)$  for  $p = 5$  (green). Dots mark the fold events, at  $T \equiv 0 \pmod{p-1}$ : every integer on the zeta curve, every other integer on  $I_3$ , and every fourth integer on  $I_5$ . Leading growth is quadratic with coefficients  $2\pi$ ,  $\frac{3}{2}\pi$ , and  $\frac{5}{8}\pi$  respectively. Thin dashed black: the smooth base maps (94) for  $p = 3, 5, 7$ . The  $p = 3$  and  $p = 5$  bases are hidden under  $I_3$  and  $I_5$  at this scale; the  $p = 7$  base is the lowest curve, drawn for comparison, no  $I_7$  having been fitted here. Bottom: the character bulge on its own scale,  $I_p - I_p^{\text{base}}$ , which the top panel cannot resolve because it is of order 1 against values in the thousands. It is one-signed, returning to zero at the folds (dots) and rising to  $\approx 0.92$  between them for  $p = 3$  and  $\approx 1.33$  for  $p = 5$ , and its amplitude barely decays: the  $p = 3$  peaks are 0.9253, 0.9183, 0.9153 at  $T = 3, 5, 11$ . So the correction is a fixed additive offset, not a fixed relative one. Open black circles are the numerically scanned events at every integer  $T$ , measured against the same base: the folds sit on zero, the intermediate events ride the bulge, and all of them follow the modeled curves to within  $1.7 \times 10^{-5}$  for  $p = 3$  and  $1.0 \times 10^{-2}$  for  $p = 5$  on  $T \leq 20$ . Generated with the Dirichlet  $L$ -function companion note.

## 10.2 Geometric origin of the yin and yang curves

This subsection records the observation that led us to formulate the equations of the yin and yang curves; only later did we find their connection to equations in Siegel’s 1932 paper [21]. We attempt to explain what happens when the link chains are viewed in various coordinate frames, and how they move relative to one another as  $\{T\}$  is varied at various  $[T]$ ; this is hard to describe in words but easy to see with computer visualization assistance using the Zest program.

The story starts with watching the bisector point ride the chain. As we have said, as  $T$  increases at a steady rate, the bisector point stays on a single link, moving back and forth along it (at the fraction  $\hat{d}_1$  of its length) all while  $T$  is not an integer. Then at the instant it becomes an integer, the link the bisector point is on hands off the role of carrying the bisector point to the next link in the chain. When  $\sigma = \frac{1}{2}$ , there is no jump: the handoff happens exactly at the instant when the outgoing and incoming links fold back onto one another and momentarily occupy the same line in the plane (when  $\sigma = \frac{1}{2}$  the two bisector links are then coincident; for  $\sigma \neq \frac{1}{2}$  they are parallel). When  $\sigma = \frac{1}{2}$ , the path of the bisector point is therefore continuous both as a curve in the complex plane and as a position along the one-dimensional link structure. Off the line it is continuous only on regular intervals between parallel-link instants.

§10.1 watched the two links *neighboring* the bisector link in the bisector frame. Now watch the *reverse bisector link* in that same frame: holding the forward bisector link stationary while  $T$  increases, the reverse bisector link revolves around it, and each end of the revolving link follows a path reminiscent of a teardrop, the pair of paths together resembling a yin-yang (Figure 37). The paths of the two ends are exactly the yin and yang curves of §6.5: the bisector frame is the local coordinate frame (36), and the two ends of the revolving link are the points Yin( $s$ ) and Yang( $s$ ) of (37)–(38). That is where those definitions came from, and this subsection reconstructs the rest of the discovery: how the crossing of the revolving link with the real axis produced the weights  $d_1$  and  $d_2$  of §4. (The formulas for the curves, their asymmetry, and their  $T \rightarrow \infty$  limit were developed in §6.4–6.5 and are not repeated here.)

The motivating question is the one left open in §9.1: the two bisector links cross, but *where*? The answer matters because, by (69), the crossing (the bisector point) sits at distance  $d_1$  from the joint  $\Sigma_1$  along the forward bisector link and at distance  $d_2$  from the joint  $\Sigma_1 + R$  along the reverse bisector link. If we can *compute* where the crossing is, we obtain  $d_1$  and  $d_2$ , and with them a new formula for  $\zeta$  that uses the fractional summands in place of the usual remainders. Of course we could do this knowing  $\zeta$ , but the trick is to do it without knowing  $\zeta$  or either partial sum.

The frame did not begin as the abstraction of §6.5; it began as a way around a difficulty. In the world frame the reverse chain is anchored at  $\zeta$ , which is both moving and precisely the quantity we do not wish to assume known; worse, the reverse bisector link also moves relative to its own anchor. The resolution is that the reverse chain’s joint  $m$  sits at  $\Sigma_1 + R$ : relative to the forward joint  $\Sigma_1$  it is displaced by exactly  $R$ , and  $R$  has its own integral representation (14), independent of any evaluation of  $\zeta$ . Applying the frame map (36) to the two ends of the reverse bisector link gives exactly Yin( $s$ ) =  $R M^s$  and Yang( $s$ ) = Yin –  $\chi M^{2s-1}$  of (37)–(38). Because the frame map scales the bisector link to unit length, positions along it are *fractions*: the crossing we are about to compute is the fraction of the way along the link at which the bisector point sits, and multiplying that real fraction by the (unscaled) link vector recovers the fractional summand itself. The actual distance from the joint, the fraction times the true link length  $M^{-\sigma}$ , is the weight  $d_1$ .

**First, the modulus.** From its joint, how far along the forward bisector link do the two links intersect? This distance is  $|R_1|$ , because  $R_1$  is the fractional part of summand  $M$  of  $\Sigma_1$ , and that summand *is* the forward bisector link. In the frame the link lies along the real axis, so the

The bisector links at  $\sigma = 1/4$ ,  $T = 6.18$  ( $t = I(T) \approx 279.85$ ,  $m = 6$ )

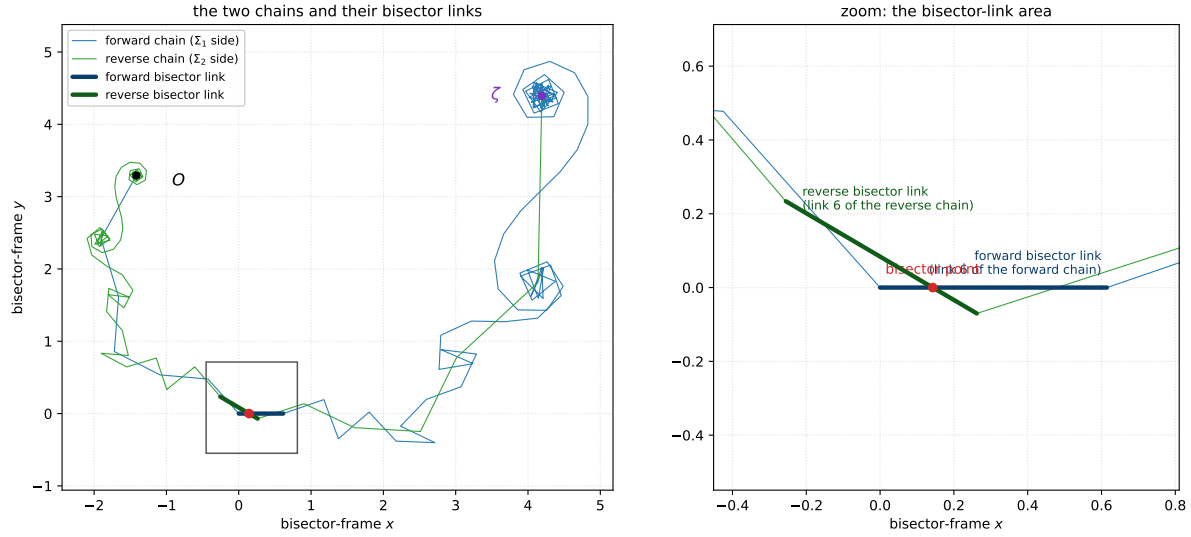


Figure 35: The two bisector links at  $\sigma = \frac{1}{4}$ ,  $T = 6.18$  ( $t = I(T) \approx 279.85$ ,  $m = 6$ ). Both panels are drawn in the coordinate frame of the *forward bisector link*: the whole picture is translated and rotated so that this link lies along the positive  $x$ -axis, with no rescaling. All lengths are true, so the link runs from 0 to  $7^{-1/4} \approx 0.615$ . *Left*: the forward chain (blue, starting at the world origin  $O$ ) and the reverse chain (green, ending at  $\zeta$ ), each drawn out to the link nearest its spiral center; link 6 of each chain (the forward and reverse bisector links) is overdrawn with a thick stroke, and the red dot where they cross is the bisector point. *Right*: zoom into the gray window: the two bisector links crossing at the bisector point, which sits on the  $x$ -axis at distance  $d_1$  from the origin. The bisector frame of this section additionally rescales this link to the unit interval  $[0, 1]$  and watches the thick reverse link revolve around it. Generated by `fig.yinyang.spirals.py`.

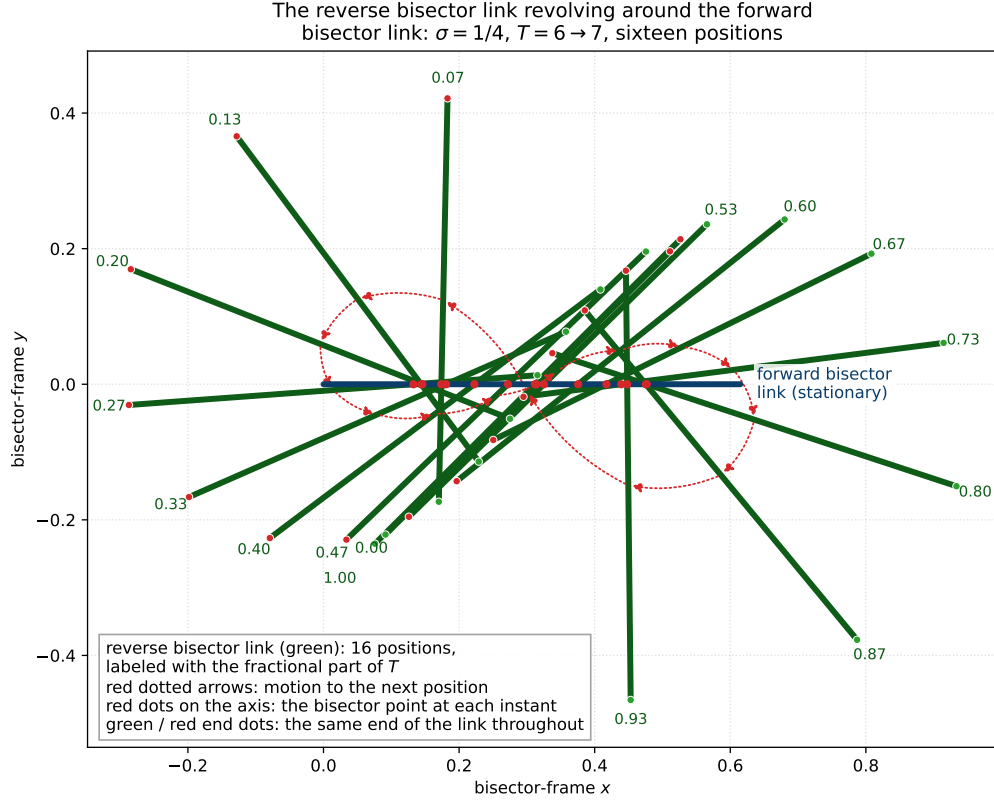


Figure 36: The two bisector links in the frame of the forward bisector link as  $T$  sweeps from 6 to 7 ( $\sigma = \frac{1}{4}$ , the frame of Figure 35, lengths unscaled). The forward bisector link is stationary (the single thick dark-blue segment on the  $x$ -axis) while the reverse bisector link (green, same thickness) is drawn at sixteen equally spaced values of  $T$ , each position labeled on its outer end with the fractional part of  $T$ ; over the sweep it revolves fully around the stationary link, and at  $T = 7$  (labeled 1.00) it returns nearly to its  $T = 6$  position; the heading turns through a half-revolution and then back, so the directed segment ends in the same orientation without having rotated all the way around. The red dotted arrows run from the middle of each position to the middle of the next, tracing the motion. The small green and red dots mark the two ends of the revolving link consistently: green is always the joint- $m$  end and red the joint- $(m+1)$  end; they trace the yin and yang curves of Figure 37, in the same colors. Each red dot on the  $x$ -axis is the bisector point at one instant: the crossing of that position with the  $x$ -axis, at  $(d_1, 0)$ . Near the parallel instants (fractional part of  $T$  near  $\frac{1}{4}$  and  $\frac{3}{4}$ ) the revolving link is nearly parallel to the stationary one and the crossing races along the axis (the poles of  $d_1$  of Appendix A.5). Generated by `fig_bisector_sweep.py`.



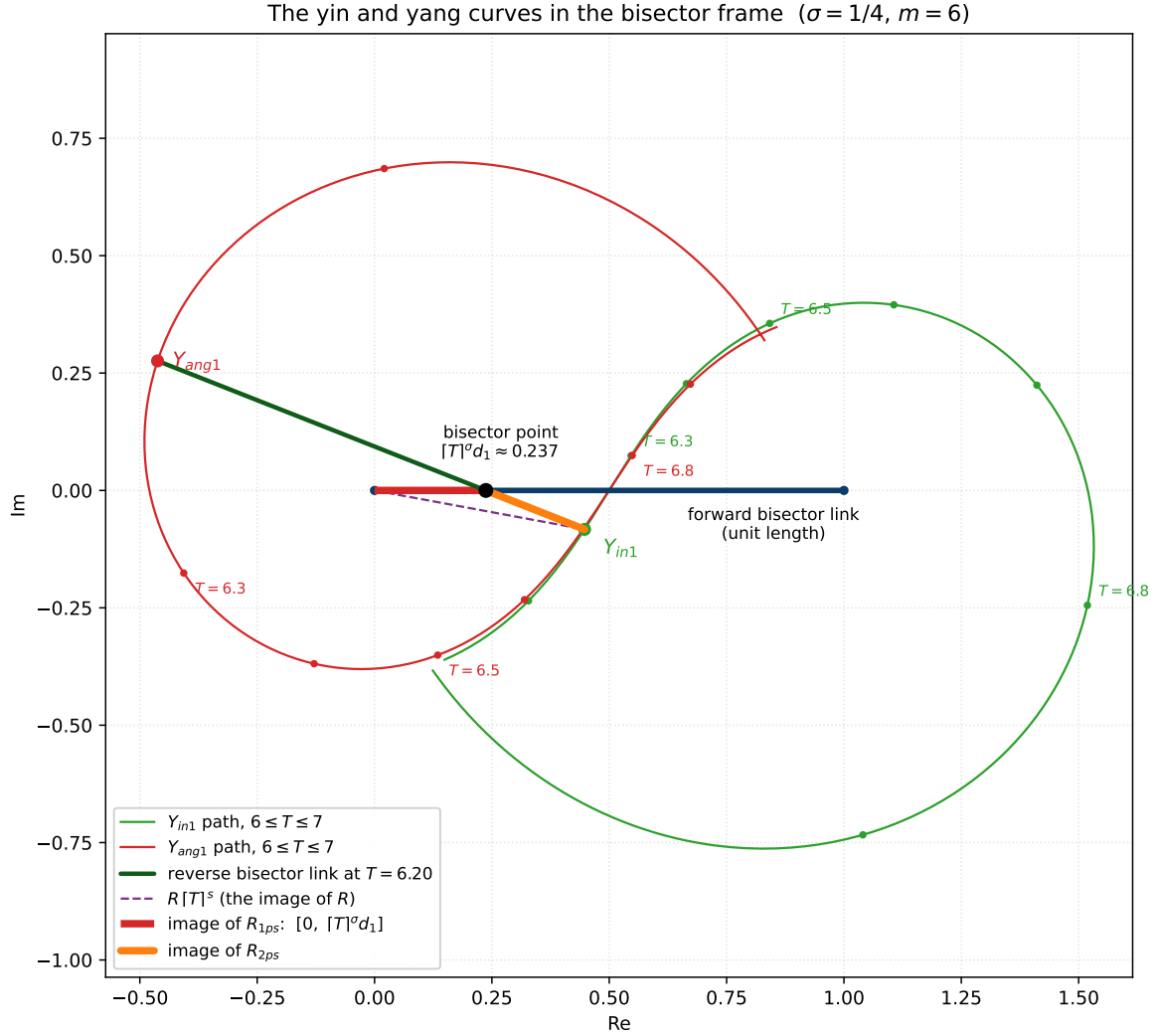


Figure 37: The yin and yang curves in the bisector frame,  $\sigma = \frac{1}{4}$ ,  $m = 6$ . The frame sends the forward bisector link to the unit interval  $[0, 1]$  (dark blue). As  $T$  sweeps  $6 \leq T \leq 7$  (Figure 36), the near end of the reverse bisector link traces the yin curve Yin (green, one dot every 0.1 in  $T$ ) and the far end traces the yang curve Yang (red); the two teardrop paths together resemble a yin-yang. The reverse bisector link itself is drawn at  $T = 6.20$  (dark green): it crosses the real axis at the bisector point (black dot), at the crossing fraction  $\hat{d}_1 = M^\sigma d_1 \approx 0.237$  of the way along the unit link. The dashed violet vector from 0 to Yin is the in-frame image of Siegel's remainder  $R$ ; the crossing splits it into the images of  $R_1$  (red, the piece of the unit link from 0 to the bisector point) and  $R_2$  (orange, from the bisector point to Yin). Generated by `fig.yinyang.py`.

intersection is the point where the segment from Yin to Yang crosses the axis. By the standard segment–axis intersection formula that crossing is

$$\text{Yin} = \frac{\Im \text{Yin}}{\Im \text{Yang} - \Im \text{Yin}} (\text{Yang} - \text{Yin}), \quad (99)$$

a real number. Substituting (37)–(38) and canceling  $\text{Yang} - \text{Yin} = -\chi M^{2s-1}$  from the ratio,

$$(99) = R M^s = \frac{\Im(R M^s)}{\Im(\chi M^{2s-1})} \chi M^{2s-1}. \quad (100)$$

Writing  $M^s = M^\sigma e^{i\omega}$ , expanding the imaginary parts in components, and finally writing  $\varphi = \arg R$  and  $\psi = \arg \chi$  and collapsing with the sine difference identity  $\sin(2\omega + \psi) \cos(\omega + \varphi) - \cos(2\omega + \psi) \sin(\omega + \varphi) = \sin(\omega - \varphi + \psi)$ , the whole expression reduces to

$$(99) = M^\sigma |R| \frac{\sin(\omega - \varphi + \psi)}{\sin(2\omega + \psi)} = M^\sigma d_1. \quad (101)$$

The reader will recognize the sine quotient: it is, symbol for symbol, the definition (17) of the weight  $d_1$ . *This is where that definition came from.* The crossing (101) is the crossing fraction (the “fractional amount” column of Table 1) along the unit link at which the reverse link crosses, and dividing out the frame’s scaling  $M^\sigma$  leaves the actual distance from the joint,  $d_1$ . (Proposition 6.3 is the same identity run in the other direction: there the definitions are taken as given and the chord is proved to cross at  $\hat{d}_1 = M^\sigma d_1$ .)

**Second, the direction.** The forward bisector link points along the next summand of  $\Sigma_1$ : extending the summation cutoff by one adds the single term  $M^{-s}$ , and that term is the link vector. Multiplying the fraction (101) by it undoes the frame map and yields the fractional summand itself:

$$R_1 = (M^\sigma d_1) \cdot M^{-s} = d_1 M^{-it}, \quad (102)$$

which is definition (15).

**The mirror frame gives  $d_2$  and  $R_2$ .** The same procedure in the mirror frame of §6.5 (the footnote there), where the *reverse* bisector link is sent to  $[0, 1]$  and the forward one revolves, locates the crossing at the fraction

$$\frac{M^{1-\sigma}}{|\chi|} |R| \frac{\sin(\omega + \varphi)}{\sin(2\omega + \psi)} = \frac{M^{1-\sigma}}{|\chi|} d_2 \quad (103)$$

of the way along the unit reverse link, again the “fractional amount” column of Table 1, and again the sine quotient is, symbol for symbol, the definition (18) of  $d_2$ : this is where  $it$  came from. (The two normalized fractions are nearly symmetric: since  $|\chi(\sigma + it)| = (t/2\pi)^{\frac{1}{2}-\sigma} (1 + O(1/t))$  and  $\sqrt{t/2\pi} \approx T + \frac{1}{2}$ , the prefactor  $M^{1-\sigma}/|\chi|$  is asymptotically  $M^\sigma$ , matching the forward side.) Multiplying by the reverse link vector  $\chi M^{s-1}$  undoes the frame map:

$$R_2 = \frac{M^{1-\sigma}}{|\chi|} d_2 \cdot \chi M^{s-1} = d_2 M^{it} \frac{\chi}{|\chi|}, \quad (104)$$

which is definition (16).

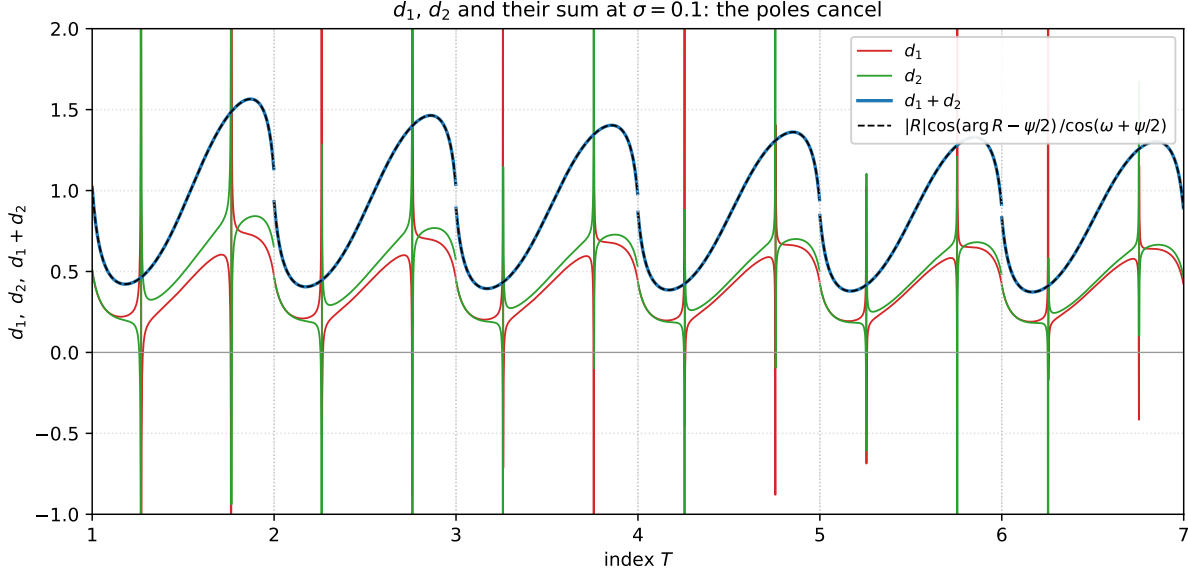


Figure 38:  $d_1$  (red),  $d_2$  (green) and their sum (blue) at  $\sigma = 0.1$ ,  $1 \leq T \leq 7$ . Twice per unit of  $T$  the individual weights blow up at the parallel-link poles of Appendix A.5, but the poles are equal and opposite and the sum passes smoothly through every one of them; the black dashed curve is the closed form (105), indistinguishable from the blue sum (they agree to machine precision on the plotted grid). Generated by `fig_d1_d2.sum.py`.

**The poles cancel in the sum.** The two weights share the denominator  $\sin(2\omega + \psi)$ , and Appendix A.5 locates the isolated heights at which it produces genuine poles. In the *sum*  $d_1 + d_2$  those poles cancel: adding the numerators with the sum-to-product identity  $\sin(\omega - \varphi + \psi) + \sin(\omega + \varphi) = 2 \sin(\omega + \frac{\psi}{2}) \cos(\varphi - \frac{\psi}{2})$  and writing  $\sin(2\omega + \psi) = 2 \sin(\omega + \frac{\psi}{2}) \cos(\omega + \frac{\psi}{2})$ , the offending factor divides out:

$$d_1 + d_2 = |R| \frac{\cos(\varphi - \frac{\psi}{2})}{\cos(\omega + \frac{\psi}{2})}. \quad (105)$$

The zeros of the remaining denominator  $\cos(\omega + \frac{\psi}{2})$  are matched by zeros of the numerator (at those instants  $R$  is parallel to the common link direction, forcing  $\varphi \equiv -\omega \pmod{\pi}$ ), so every singularity of (105) is removable: the sum is smooth across all the poles of its two terms, which is the closed form behind the qualitative cancellation remark following Table 4. Figure 38 shows this at  $\sigma = 0.1$ .

This is how the definitions of §4 were found: the sine expressions (17)–(18) are nothing other than the real-axis crossings of the yang segments, read off in the two bisector frames. Only afterwards was Theorem 4.2 proved, confirming that the two fractional summands so constructed add back to Siegel’s  $R$  exactly. Note the economy of the construction: the bisector point,  $d_1$ ,  $d_2$ , and both fractional summands are obtained from  $R$  and  $\chi$  alone. Numerically we computed  $R$  to high accuracy by Kuznetsov’s method (§8.2), which is how the pictures, and then the identity, were checked.

## 11 Conjecture: There is a Yin, Yang, $d_1$ , $d_2$ for every link

§10.2 watched one pair of links: the reverse bisector link revolving about the forward bisector link, its ends tracing the yin and yang curves, and the weights  $d_1, d_2$  as the real-axis crossings of that pair. *We conjecture that the same picture exists at every link of the chain.* On the critical line which reverse link crosses forward link  $k$  is a function of  $T$  alone. Off the line write  $C_\ell(k; \sigma, T)$ , since the segment intersection may depend on  $\sigma$ . Once that partner is named, the yin and yang formulas, the weights, and the remainder split are the same constructions with that pair in place of the bisector pair. The seven subsections record, in order:

1. first we need to know which main sum links cross which dual sum links (§11.1);
2. then we find the yin and yang curves for each crossing pair (§11.2);
3. then we use those yin-yang formulas to find the weights  $d_1$  and  $d_2$  for each link (§11.3);
4. then we use those weights to make the Dirichlet sums  $\Sigma_{1x}$  and  $\Sigma_{2x}$ , whose joints sit at the crossings (§11.4);
5. then we use the crossing points to rearrange the main sum into the two free-vector sums of the half-links (§11.5);
6. then we use the crossing pairing to generalize the partial-summand formula for  $\zeta$  to unequal cutoffs, whose two extreme cuts are example formulas built on the first link of either chain (§11.6);
7. and finally we list what in this section rests on numerics and is left unproved (§11.7).

### 11.1 Links crossing

§9.1 named one crossing of the two link chains: the forward bisector link meets the reverse bisector link, and where they meet is the bisector point  $B_1$ . All the other links of the forward chain are crossed by some link of the reverse chain as well, and which one it is depends on  $T$ . Watching the link at  $m - 1$  as  $T$ , for example, runs from 6 to 7, one reverse link crosses it, and that link is joined by its neighbor that subsequently also crosses over it, and by the end of the unit interval the crossing links have advanced by two.

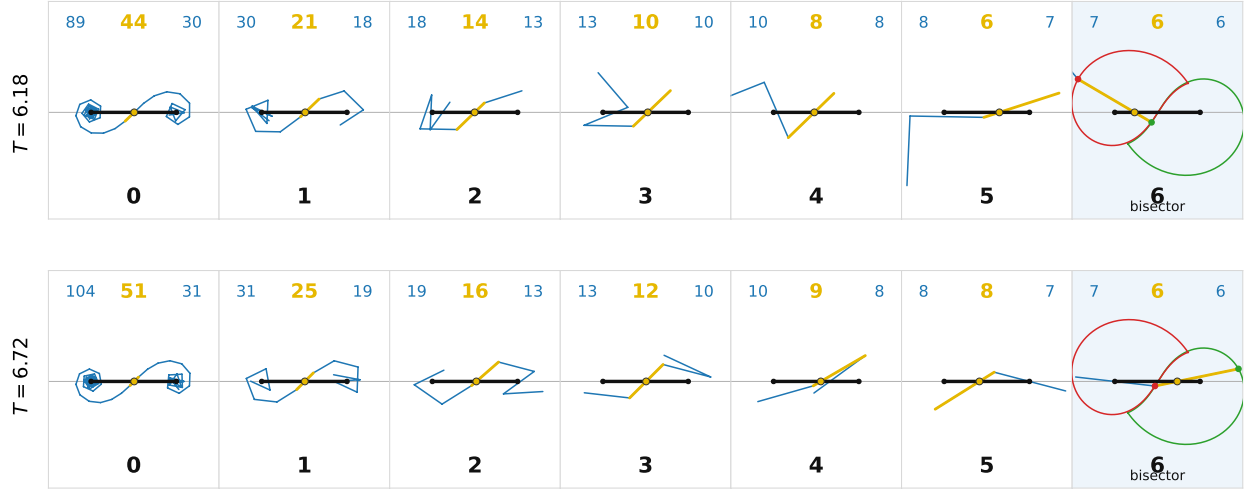


Figure 39: Local frames of the forward links at  $\sigma = \frac{1}{2}$ ,  $T = 6.18$  (top) and  $T = 6.72$  (bottom), the running example of Figure 8 and a later height in the same unit ( $t = I(T) \approx 279.85$  and  $327.01$ ,  $m = 6$ ). Each panel is the local coordinate frame for that forward link, sending one forward link to the unit interval along the  $x$ -axis; the number at the bottom is that link. The reflected reverse chain is drawn in the same frame: the blue links are the part of it that belongs to the particular forward link, numbered at the two ends with the outer (higher) link on the left, and the yellow link is the single reverse link that crosses the forward link, numbered in yellow, with a yellow dot at the crossing. The rightmost local frame is the bisector; the green and red curves are the yin and yang loci of the two ends of the reverse bisector link as  $T$  runs through  $(6, 7)$ . Generated by `fig_link_frames.py`.

Numbering summands from one, so that link  $k$  carries the summand  $n = k + 1$  (§7), forward link  $k$  and reverse link  $j$  cross when their two summands are reciprocal about the Riemann–Siegel cutoff  $a = \sqrt{t/2\pi}$  of Appendix A.2, as nearly as two integers can be. Under  $t = I(T)$  that cutoff is a function of the index alone, by (91), so the rule closes in  $T$ :

$$nn' = a^2 = \frac{t}{2\pi} = \frac{I(T)}{2\pi} = \frac{2T + 1}{2\log(1 + \frac{1}{T})}, \quad n = k + 1. \quad (106)$$

Nothing is approximated in  $a^2$ : it is a closed form in  $T$  rather than an asymptotic. But it is not an integer, so  $n' = a^2/n$  is the real partner, and the reverse link that actually crosses carries the nearest integer to it. The product of the two integers therefore misses  $a^2$ , by up to  $n/2$ ; the rounding is unbiased, its signed mean  $-9 \times 10^{-4}$  over the sample below, with a typical miss of a quarter of a unit. Write

$$j_0(k, T) = \left[ \frac{a^2}{k + 1} \right] - 1 = \left[ \frac{2T + 1}{2(k + 1)\log(1 + \frac{1}{T})} \right] - 1, \quad (107)$$

with  $[\cdot]$  the nearest integer. Nearest-integer alone misses at two families of half-integers. One is the offset (92) already recorded: Siegel’s cutoff  $a$  sits half a link past  $T$ , so it steps near the half-integers of  $T$  while the chains themselves step at the integers. The other is the joints of the hyperbola  $nn' = a^2$  itself. The *crossing link* is the rule that names the reverse link integer and

removes those misses,

$$C_\ell(k, T) = \begin{cases} m & \text{if } k = m, \\ j_0(k, T) & \text{if reverse link } j_0 \text{ crosses forward link } k, \\ j_0(k, T) - 1 & \text{otherwise,} \end{cases} \quad (108)$$

and the crossing summand is  $C_s(n, T) = C_\ell(n - 1, T) + 1$ . The first line is the self-pair:  $a^2/(m + 1)$  is near a half-integer when  $T$  is near an integer, so  $j_0$  names  $m \mp 1$  at the ends of the unit index,  $T$ , while the two bisector links cross at every height. The third line is the other family, the half-integers of the product  $a^2/n$  itself: when  $\{a^2/n\} \approx \frac{1}{2}$  the geometric crossing has already moved to the lower neighbor. Replacing  $a^2$  by  $T(T + 1)$  or  $(T + \frac{1}{2})^2$  does not remove either family; the product is already the exact  $I(T)$ , and it only moves the joints. The observed pairing suggests the unproved involutivity conjecture

$$C_s(C_s(n, T), T) = n.$$

At  $T = 6.18$ , where  $a^2 = 44.5389$ , (107) reads 44, 21, 14, 10, 8, 6, 5 for  $k = 0, \dots, 6$ , and all seven are real crossings (Figure 40). The sum rule  $k + j \approx 2T - 1$  that the links beside the bisector obey is this hyperbola's tangent there, and holds only there.

Two remarks on  $a$ , since it is easy to read it as a link number. It is not one: it is a real height, the self-dual summand  $n = n' = a$  the law pairs about, the one at which the forward chain hands over to the reflected one, and by (92) it sits half a link past the bisector link, at  $a^2 = (T + \frac{1}{2})^2 - \frac{1}{12} + O(T^{-2})$ . Nor is it a fixed number over a unit interval, as  $\lfloor T \rfloor$  is:  $a^2$  climbs from 42.17 to 56.17 as  $T$  runs from 6 to 7, and it is that climb that walks each link's partner along inside the interval. Replacing  $a^2$  by  $\lfloor T \rfloor^2 = 36$  or  $(\lfloor T \rfloor + 1)^2 = 49$  at  $T = 6.18$  would name reverse link 35 or 48 as crossing the unit step, and neither does.

**The reverse chain is the forward chain reflected in the bisector line.** Number the joints of both chains from zero as in §7.2,

$$J_n = \sum_{n' \leq n} n'^{-s}, \quad K_j = \zeta - \chi(s) \sum_{n \leq j} n^{s-1}, \quad (109)$$

so that  $J_0 = 0$ ,  $K_0 = \zeta$ , and  $K_m = \Sigma_1 + R$  is the near end of the reverse bisector link; link  $n$  of either chain runs from its joint  $n$  to its joint  $n + 1$ . On  $\sigma = \frac{1}{2}$  we have  $n^{s-1} = \overline{n^{-s}}$ , so  $K_j = \zeta - \chi \overline{J_j}$ , and with  $\chi = e^{-2i\vartheta}$  and  $Z = e^{i\vartheta} \zeta$  real (Hardy's  $Z$ , as in (200)),

$$e^{i\vartheta} K_j = Z - \overline{e^{i\vartheta} J_j}. \quad (110)$$

Write  $e^{i\vartheta} J_n = X_n + iY_n$ . Then (110) puts reverse joint  $j$  at  $(Z - X_j, Y_j)$ : in the  $\vartheta$ -rotated frame the reverse chain is the forward chain reflected in the vertical line  $X = Z/2$ , joint for joint and link for link. That line is the bisector line of §9.2: the rotation puts  $\zeta$  at  $Z$  on the real axis, so unrotated it is the perpendicular bisector of the base running from the origin to  $\zeta$ , drawn dotted in Figures 16 and 18 through  $\zeta/2$  and the bisector point  $B_1$  (Corollary 9.2). Doubling-back and the bisector line are two different things in the same neighborhood, and worth keeping apart: the place along the chain is the bisector link, where the chain doubles back; the bisector line is a line across the plane, the one the two chains reflect in. The bisector point is where the two meet, the point inside the bisector link at which the forward chain crosses the bisector line, as (120) works out below. This is the chain-level form of the reflection  $B_2 = \chi \overline{B_1}$  behind the equal legs of §9.2, and like it, it is a statement about the critical line only: off the line  $|\chi| \neq 1$  and the reverse chain is a scaled copy of the  $1 - \sigma$  chain instead of a reflection of this one.

**Crossings come in pairs.** Parameterize each chain continuously, the value  $\mu = k + p$  with  $0 \leq p < 1$  meaning the point  $p$  of the way along link  $k$ , and let  $X(\mu), Y(\mu)$  be the rotated coordinates of the forward chain there and

$$h(\mu) = X(\mu) - Z/2 \quad (111)$$

its signed distance to the bisector line. By (110) the reverse chain at  $\nu$  sits at  $(Z - X(\nu), Y(\nu))$ , so the forward chain at  $\mu$  meets the reverse chain at  $\nu$  exactly when

$$h(\mu) + h(\nu) = 0 \quad \text{and} \quad Y(\mu) = Y(\nu). \quad (112)$$

Both conditions are symmetric in  $\mu$  and  $\nu$ , which is why crossings come in pairs: reverse link  $j$  crosses forward link  $k$  if and only if reverse link  $k$  crosses forward link  $j$ , and any rule naming the partner must be symmetric too, as (106) is. The pair coincides only when  $\mu = \nu$ , and then  $h = 0$ ; so a self-paired crossing lies on the bisector line itself, there is exactly one (the bisector point  $B_1$ , inside the bisector link of each chain by Theorem A.7), and every other crossing sits off the line with its counterpart on the other side.

**Sweeps between spiral centers.** Along either chain the turn from one link to the next is  $t \ln(1 + 1/n) \approx t/n$ , decreasing as the chain is walked, so the joints where that turn is an odd multiple of  $\pi$  lie near the spiral centers (68) (the midpoint of the link they bound is closer still), and between two consecutive centers the chain makes one full turn. The middle of a turn is the link where the angle is the even multiple  $2\pi c$ ,

$$\frac{t}{n'} = 2\pi c \quad \Longleftrightarrow \quad n' = \frac{t}{2\pi c} = \frac{a^2}{c}, \quad (113)$$

and there successive links are parallel rather than antiparallel: their contributions add in phase, the chain stops circling and runs. Call that run the  $c$ -th *sweep*; it carries the chain from the spiral center  $S_n = c$  to the spiral center  $S_n = c - 1$ .

Heuristically, after inserting a smooth compact cutoff, Poisson summation separates the chain into oscillatory integrals. Stationary phase associates the  $c$ -th integral with the saddle  $x = a^2/c$  and gives the displayed vector only as a leading asymptotic:

$$\frac{1}{\sqrt{c}} \exp i\left(\vartheta + t - t \ln \frac{a^2}{c} + \frac{\pi}{4}\right) = \frac{1}{\sqrt{c}} e^{i(2\Theta - (\vartheta - t \ln c))}, \quad 2\Theta = 2\vartheta + t - t \ln a^2 + \frac{\pi}{4}. \quad (114)$$

No exact Poisson identity is asserted for the unwindowed critical-line series. Its modulus is that of the  $c$ -th summand,  $c^{-1/2}$ , and its phase is the  $c$ -th summand's phase  $\vartheta - t \ln c$  reflected in the fixed direction  $\Theta$ . Sweep by sweep, the tail of a chain reproduces the head of that chain reflected: this is the correspondence between joints and spiral centers recorded in §9.1, Nickel's center-to-center distances and angles reproducing the initial steps, now carrying an index; the  $c$ -th sweep answers to the  $c$ -th summand.

The reflection (110) undoes the reflection in  $\Theta$ . In the frame of the forward chain, therefore, the reverse chain's  $c$ -th sweep is the forward chain's own link  $c - 1$  traversed backwards: over the sampled heights the center-to-center step and that link agree to a median 1.4%, 2.7%, 3.1% and 5.2% of a link length for  $c = 1, 2, 3, 4$ . A copy of a link laid back along it must cross it, and the quarter turn in (114) is what tilts the copy so that it cuts across instead of lying flat: the two meet near the middle of the link, at some  $43^\circ$ . Which reverse link carries the  $c$ -th sweep is (113) with  $c = n$ , and that is (106). Panel (b) of Figure 40 is the case  $c = 1$ : the whole outermost turn of the reverse chain, links 30 to 89, laid back along the unit step and crossing it at link 44.

**The angle at a crossing.** How the two links lie relative to one another needs no estimate at all. It is forced, and what forces it is that the two summands multiply to a whole number. In the rotated frame forward link  $k$  points along  $e^{i(\vartheta - t \log n)}$  with  $n = k + 1$ , and by (110) reverse link  $j$  points along  $-e^{-i(\vartheta - t \log n')}$  with  $n' = j + 1$ , so the angle from the first to the second is

$$\theta(n, n') = \pi - 2\vartheta(t) + t \log(nn') \pmod{2\pi}, \quad (115)$$

exactly, whether or not the two links meet.

**Proposition 11.1.** *Fix  $C > 0$ . On  $\sigma = \frac{1}{2}$ , for positive integers  $n, n'$  satisfying  $|nn' - a^2| \leq Ca$ , the displayed expansion holds with error  $O_C(1/a)$ :*

$$\theta(n, n') \equiv \frac{5\pi}{4} - \frac{\pi(nn' - a^2)^2}{a^2} + O_C\left(\frac{1}{a}\right) \pmod{2\pi}. \quad (116)$$

*In particular, if  $|nn' - a^2| \leq a/2$  the two links are never parallel: they meet at an angle between  $\frac{3}{4}\pi$  and  $\pi$ , that is between  $135^\circ$  and  $180^\circ$ , at most a quarter turn away from antiparallel.*

The proof depends on  $N = nn'$  being an integer. Write  $N = a^2(1 + \epsilon)$ . Since  $t = 2\pi a^2$ ,

$$t\epsilon = \frac{t(N - a^2)}{a^2} = 2\pi N - t \equiv -t \pmod{2\pi}, \quad (117)$$

so in  $t \log N = t \log a^2 + t\epsilon - \frac{1}{2}t\epsilon^2 + \frac{1}{3}t\epsilon^3 - \dots$  the first-order term is  $-t$  modulo  $2\pi$ , whatever  $N$  is. The Stirling expansion of the Riemann–Siegel theta function  $\vartheta(t)$  (Appendix A.2) is  $2\vartheta(t) = t \log a^2 - t - \frac{\pi}{4} + \frac{1}{24t} + O(t^{-3})$ , so in (115) the whole of  $t \log a^2 - t$  cancels against it and the  $\frac{\pi}{4}$  survives:

$$\theta \equiv \pi + \frac{\pi}{4} - \frac{t\epsilon^2}{2} + O(t|\epsilon|^3) + O(t^{-1}).$$

Here  $\frac{1}{2}t\epsilon^2 = \pi(N - a^2)^2/a^2$ . Under the proposition's hypothesis,  $|\epsilon| \leq C/a$ , so  $t|\epsilon|^3 = O_C(1/a)$ , which proves (116). When  $|N - a^2| \leq a/2$  we have  $|\epsilon| \leq 1/2a$ , so the cubic term is at most  $\pi/12a$  and the quadratic one at most  $\pi/4$ . Read as an unsigned angle,  $\frac{5}{4}\pi$  is  $\frac{3}{4}\pi$ , and the correction can only carry it towards  $\pi$ .

The naming (107) stays inside that hypothesis: it makes  $|nn' - a^2| = n|n' - a^2/n| \leq n/2$ , and  $n \leq m < a$  for every forward link below the bisector, so the  $135^\circ$  is a floor and not an average: no link-crossing that the rule names can be sharper than that. Only the self-pair, being the bisector links, handled by the first line of (108), where the rule overrides the rounding, can fall outside it. The quarter turn is the  $-\frac{\pi}{8}$  of  $\vartheta$ , which is the Gaussian factor of (114) over again: the  $43^\circ$  by which a crossing misses antiparallel, the tilt that keeps the sweep from lying flat along the link it reproduces, is the stationary-phase quarter turn, read off exactly rather than measured. What the proposition fixes is the direction of the partner and not its position; that the two segments actually meet is verified numerically below but is not proved here.

**Near the bisector, a sum rule.** Near the self-dual link  $n = n' = a$  the hyperbola (106) is indistinguishable from its tangent. Writing  $n = a + x$ ,

$$n' = \frac{a^2}{n} = a - x + \frac{x^2}{a} - \dots, \quad \text{so} \quad k + j = n + n' - 2 = 2a - 2 + \frac{x^2}{a} - \dots, \quad (118)$$

and since  $2a = 2T + 1 - \frac{1}{12T} + O(T^{-2})$  by (92), the sum is  $2T - 1$  up to the quadratic term and a drift of order  $1/12T$ :  $k + j$  is one of the two integers straddling  $2T - 1$ , that is

$$j = \lfloor 2T \rfloor - 1 - k \quad \text{or} \quad j = \lfloor 2T \rfloor - k, \quad (119)$$



a rule needing nothing computed at all. The same rule comes out of the reflection (110), which is worth doing because it identifies the fixed point. Near the bisector the forward chain sweeps across the bisector line and  $h$  passes through a single zero there, close to straight over the few links either side; writing  $r^*$  for that zero and linearizing turns the first condition of (112) into  $\mu + \nu = 2r^*$ , the two arcs straddling the crossing symmetrically, so  $k + j$  straddles  $S = 2r^* - 1$ . And the crossing fraction is not a new quantity. By (200),  $h(m) = X_m - Z/2 = -r/2$  with  $r = e^{i\vartheta}R$  the real remainder of Appendix A.2, and link  $m$  of the forward chain, being  $(m+1)^{-s}$  rotated by  $e^{i\vartheta}$ , has horizontal extent  $X_{m+1} - X_m = \cos(\omega - \vartheta)/\sqrt{m+1}$ . So the zero falls inside link  $m$ , at the crossing fraction

$$r^* = m + \lambda, \quad \lambda = \frac{Z/2 - X_m}{X_{m+1} - X_m} = \sqrt{m+1} \frac{r}{2\cos(\omega - \vartheta)} = M^{1/2}d_1 = \hat{d}_1, \quad (120)$$

by the exact reduction (199) and the normalization (24). The place where the forward chain crosses the bisector line is therefore the bisector point  $B_1$  itself, sitting at the crossing fraction  $\hat{d}_1$  along the bisector link exactly as Figure 31 marks it. So  $S = 2r^* - 1 = 2m + 2\hat{d}_1 - 1$ , against  $2m + 2\{T\} - 1$  from the tangent. These are not the same number: the limit of  $\hat{d}_1$  is the waveform  $d(x)$  of §6, not  $x$  itself, and the two part by nearly  $\frac{1}{2}$  at the ends of the unit interval. But the straddling pair depends only on which side of  $\frac{1}{2}$  each falls, and on that the two agree. The branch form (31) gives  $d(x) + d(1-x) = 1$ , and on  $(0, \frac{1}{2})$  the curve stays below  $\frac{1}{2}$ : there  $d = \frac{1}{2} + \mathcal{W}_\infty$  with  $\mathcal{W}_\infty = -\frac{1}{2}\tan(2\pi x)\tan\beta$  (25), and the two tangents carry the same sign on each of  $(0, \frac{1}{4})$  and  $(\frac{1}{4}, \frac{1}{2})$ , both positive on the first and both negative on the second. So  $d$  crosses  $\frac{1}{2}$  exactly where  $x$  does, and the  $O(1/T)$  of §6 can disturb this only within  $O(1/T)$  of  $\{T\} = 0, \frac{1}{2}, 1$ , which the sample below does not see it do. With  $\frac{1}{5} < \hat{d}_1 < \frac{4}{5}$  from Theorem A.7 holding  $S$  within  $\frac{3}{5}$  of  $2m$ , the straddling integers are therefore  $\lfloor 2T \rfloor - 1$  and  $\lfloor 2T \rfloor$  whichever side of  $\frac{1}{2}$  the fractional part is on, and (119) follows either way: the tangent to the hyperbola and the linearization of  $h$  are the same line.

How far the tangent can be trusted is (118) again: it costs half a link once  $x^2/a = \frac{1}{2}$ , so the sum rule is good only within

$$|n - a| \lesssim \sqrt{a/2} \approx \sqrt{T/2} \quad (121)$$

of the bisector (three links either side at  $T = 17$ , fourteen at  $T = 400$ , and just under two at the  $T = 6.18$  of the figure), which is why  $k = 4, 5, 6$  obey it there and  $k = 3$  already does not. Inside that range the two indices trade off one for one: each step back along the forward chain moves the partner one step forward along the reverse chain, and over one unit of  $T$  the sum moves by two, so each partner advances by two, in two single steps, one when  $\{T\}$  passes  $\frac{1}{2}$  and one when  $T$  passes an integer, the two heights at which  $\lfloor 2T \rfloor$  jumps. Outside the range the product takes over and the partners run away as  $a^2/n$ , reaching the far end of the chain at the unit step.

The bisector link is the one place where the pairing is exact rather than asymptotic, and that is the first line of (108). It is the self-dual link,  $n = m + 1$  and  $x = \frac{1}{2} - \{T\}$ , so (107) rounds to  $m$  only over the middle half of the unit interval and to  $m \mp 1$  at its ends; but the two bisector links still cross, and the self-paired crossing  $B_1$  of (112) is there at every height. The bisector links always cross, as §9.1 has it, and the links  $j_0$  names near the ends of the interval are second crossings of the same forward link, not replacements for it.

Links crossing at  $\sigma = 1/2$ ,  $T = 6.18$  ( $t = I(T) \approx 279.85$ ,  $m = 6$ ,  $a = \sqrt{t/2\pi} = 6.674$ ,  $a^2 = 44.54$ )

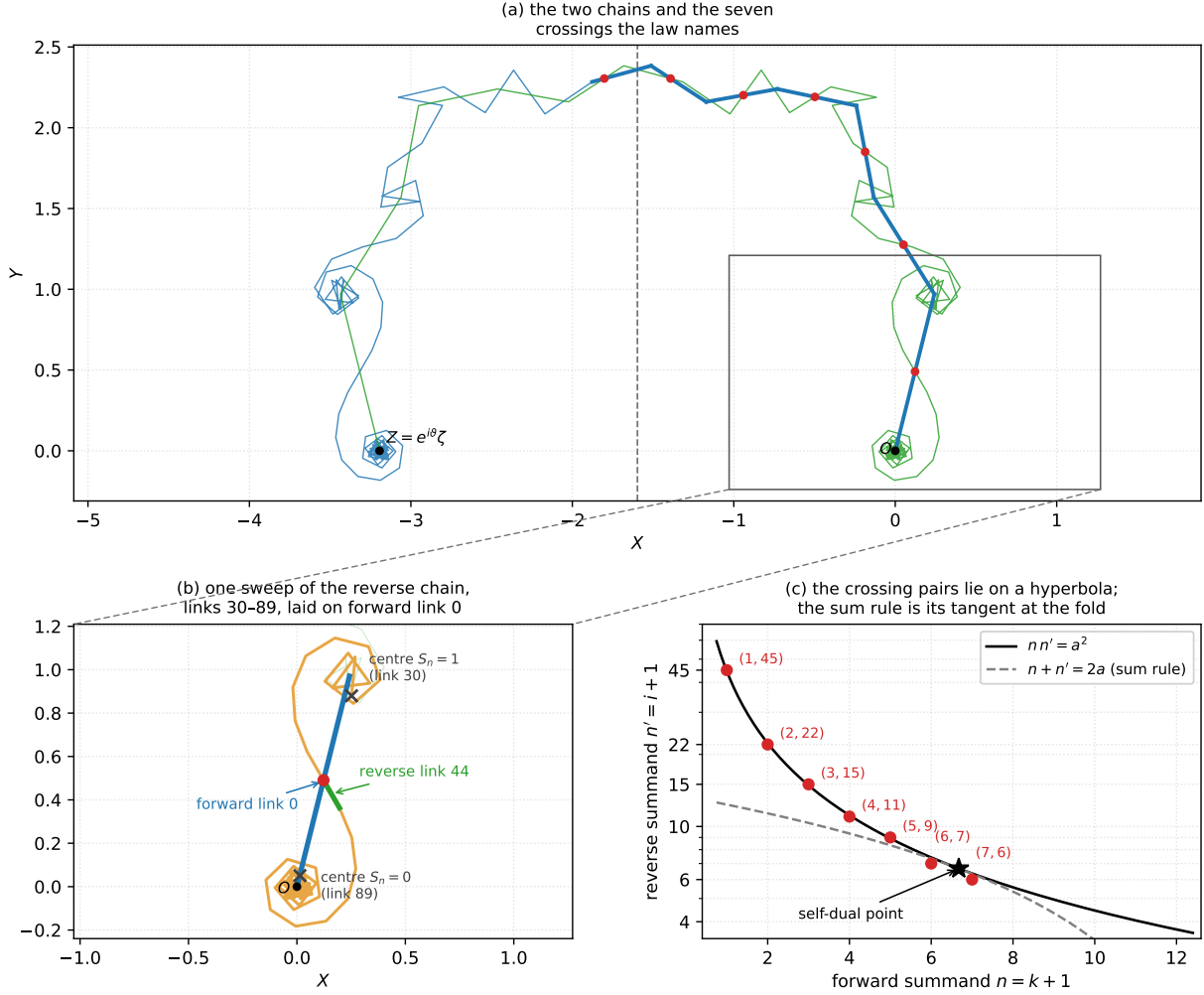


Figure 40: Links crossing at  $\sigma = \frac{1}{2}$ ,  $T = 6.18$ , the running example of Figure 8 ( $t = I(T) \approx 279.85$ ,  $m = 6$ ,  $a = \sqrt{t/2\pi} = 6.674$ ,  $a^2 = 44.54$ ; the first turn of each chain holds 90 links). (a) Both chains in the  $\vartheta$ -rotated frame, forward  $e^{i\vartheta}J_n$  in blue from  $O$  and reverse  $e^{i\vartheta}K_j$  in green from  $Z = e^{i\vartheta}\zeta$ , which by (110) is the blue chain reflected in the dashed bisector line  $X = Z/2$ . Forward links 0 through 6 are drawn heavy and the red dots are the seven crossings (107) names for them,  $j = 44, 21, 14, 10, 8, 6, 5$ ; they run the length of the chain, not just the neighborhood of the bisector. (b) The boxed region enlarged, showing why. The gold arc is one turn of the reverse chain, links 30 to 89, running from the spiral center  $S_n = 1$  to the spiral center  $S_n = 0$  (crosses, at the links (68) where the turn angle is an odd multiple of  $\pi$ ). By (114) that step reproduces forward link 0 reversed, so it lies back along the unit step and crosses it near the middle (red dot); the link carrying the sweep is 44, the middle of the turn, where the turn angle is  $2\pi$  and  $n' = a^2/1$ . (c) The pairs  $(n, n')$  of all seven crossings, on the hyperbola  $nn' = a^2$ . The self-dual point  $n = n' = a$  (star) is the summand the law pairs about, half a unit of the index past  $T$  by (92). These are summand axes, and the star is plotted here at  $(a, a)$  with  $a = 6.674$ , while the bisector link is link  $m = 6$ , which carries summand  $n = 7$ : the star therefore sits between summands 6 and 7, a third of a summand short of the bisector link's own value rather than past it. Near it the hyperbola is indistinguishable from its tangent  $n + n' = 2a$  (dashed), which is the sum rule (119), and away from it the two part company fast: the tangent gives the unit step a partner near  $n' = 11$  where the truth is 45. Generated by `fig_links_crossing.py`.

**How well it holds.** Sampling  $m \in \{8, 17, 40, 123, 400\}$  at 199 fractional parts each and testing every forward link  $0 \leq k \leq m$ , 118,007 cases in all, the link named by (108) crosses forward link  $k$  in every one of them. The named integer (107) alone does so in 117,870, 99.9%; the other 137 are the two half-integer families of that naming, 13 of them the self-pair (first line of (108)) and 124 the lower neighbor at  $\{a^2/n\} \approx \frac{1}{2}$  (third line), never the upper neighbor and never off by more than one link. The crossing sits at the middle of the forward link, median position 0.500 along it with median deviation 0.009 four or more links out from the bisector, widening to 0.148 at the bisector link, where the crossing is  $B_1$  and the position is  $\hat{d}_1$  instead. The two links meet at a median  $137^\circ$ , tenth to ninetieth percentile  $135^\circ$  to  $151^\circ$ , and never below  $135^\circ$  in any of them: that floor is Proposition 11.1 and not a feature of the sample, and the exact form (115) holds across the crossings to  $3.8 \times 10^{-9}$ , the accumulated double precision of the chains. The  $43^\circ$  of the median is the quarter turn of (114) that keeps the sweep from lying flat along the link it reproduces. The center-to-center steps match those links to a median 1.4%, 2.7%, 3.1% and 5.2% of a link length at  $c = 1, 2, 3, 4$ , and the self-pair  $(m, m)$  crossed at all 995 sampled heights while (107) named  $m$  at exactly half of them, the middle half of each unit interval, as the previous paragraph says. The sum rule (119), by contrast, names a real crossing in only 10.6% of the cases overall, but in 99.8% of those inside its range (121) and 5.1% of those outside it, which is what that range is worth. The identity  $\lambda = \hat{d}_1$  of (120) holds to  $6 \times 10^{-15}$ . Computed in `check_links_crossing.py`.

## 11.2 General yin and yang form for any link

§10.2 watched one link cross another: the reverse bisector link revolving about the forward bisector link, its two ends tracing the yin and yang curves (37)–(38). Now that (108) says which reverse link crosses forward link  $k$ , every link of the chain has such a pair, and they are all one formula. Like the bisector pair it carries no  $\zeta$ : the remainder  $R$ , the factor  $\chi$ , and  $\sigma$  and  $T$  are all it needs.

Send forward link  $k$  to  $[0, 1]$  exactly as (36) sends the bisector link: translate its joint  $J_k$  to the origin and divide by the link vector  $(k+1)^{-s}$ , writing  $s = \sigma + iI(T)$  throughout. Reverse joint  $j$  of (109) lands at

$$Y_k(j) = (K_j - J_k)(k+1)^s = (k+1)^s \left( \sum_{n=k+1}^m n^{-s} + R + \chi \sum_{n=j+1}^m n^{s-1} \right), \quad (122)$$

the second sum read as  $-\sum_{n=m+1}^j$  once  $j > m$ . What removes  $\zeta$  is the same fact the bisector frame turned on:  $K_m = \Sigma_1 + R$ , so measuring the reverse chain from its own joint  $m$  replaces the anchor  $\zeta$  by the remainder. The yin and yang points of link  $k$  are then the two ends of the link that crosses it, which are consecutive reverse joints:

$$Y_{in}(k, T) = Y_k(j), \quad Y_{ang}(k, T) = Y_k(j+1) = Y_{in}(k, T) - \chi (k+1)^s (j+1)^{s-1}, \quad j = C_\ell(k, T), \quad (123)$$

with  $j = m$  at the bisector by the first line of (108).

**Remark 11.2.** Equation (122) is the general frame formula, and on the bisector it *is* the yin formula. It holds there because the crossing rule sets  $j = m$  when  $k = m$ . Plug  $k = m$  and  $j = m$  into (122):

$$Y_m(m) = (m+1)^s \left( \sum_{n=m+1}^m n^{-s} + R + \chi \sum_{n=m+1}^m n^{s-1} \right).$$

The first sum runs from  $n = m + 1$  to  $n = m$  and is empty. The second sum runs over the same empty range (the joint is  $m$ , not past it, so the  $j > m$  sign-flip is not used). What remains is

$$Y_m(m) = (m + 1)^s (R) = R (m + 1)^s = \text{Yin}(s),$$

which is (37). Geometrically the same thing:  $J_m = \Sigma_1$  and  $K_m = \Sigma_1 + R$ , so  $(K_m - J_m)(m + 1)^s = R (m + 1)^s$ , and the frame map is (36) with  $M = m + 1$ . The yang end is the next reverse joint  $j = m + 1$ . Now  $j > m$ , so the second sum is  $-\sum_{n=m+1}^{m+1} = -(m + 1)^{s-1}$  and the first sum is still empty:

$$Y_m(m + 1) = (m + 1)^s (R - \chi (m + 1)^{s-1}) = R (m + 1)^s - \chi (m + 1)^{2s-1},$$

which is (38), since  $2s - 1 = -1 + 2(\sigma + iI(T))$ . So (122) at the bisector is not an analogue of yin; it *is* yin. The closed-form loops of §6.5 and §10.2 are this formula with the two empty sums already canceled.

Take  $k = m - 1$ . Its frame factor  $m^s$  multiplies the summand  $m^{-s}$ , so that product is 1 (forward joint  $m$  is the far end of that link), and

$$Y_{m-1}(j) = 1 + m^s \left( R - \chi \sum_{n=m+1}^j n^{s-1} \right). \quad (124)$$

Here the crossing changes hands, which is the one thing the bisector's does not do. By (106),  $a^2/m = m + 2\{T\} + 1 + O(1/m)$ , so the crossing summand  $n' = j + 1$  runs  $m + 1 \rightarrow m + 2 \rightarrow m + 3$  across the unit and the curves come in three arcs:

$$n' = m + 1 : \quad Y_{in} = 1 + m^s R, \quad Y_{ang} = 1 + m^s (R - \chi (m+1)^{s-1}), \quad (125)$$

$$n' = m + 2 : \quad Y_{in} = 1 + m^s (R - \chi (m+1)^{s-1}), \quad Y_{ang} = Y_{in} - \chi m^s (m+2)^{s-1}, \quad (126)$$

$$n' = m + 3 : \quad Y_{in} = 1 + m^s (R - \chi ((m+1)^{s-1} + (m+2)^{s-1})), \quad Y_{ang} = Y_{in} - \chi m^s (m+3)^{s-1}, \quad (127)$$

the three running over  $\{T\}$  in  $[0, 0.214]$ ,  $[0.215, 0.647]$  and  $[0.648, 1)$  at  $m = 6$  (Figure 41).

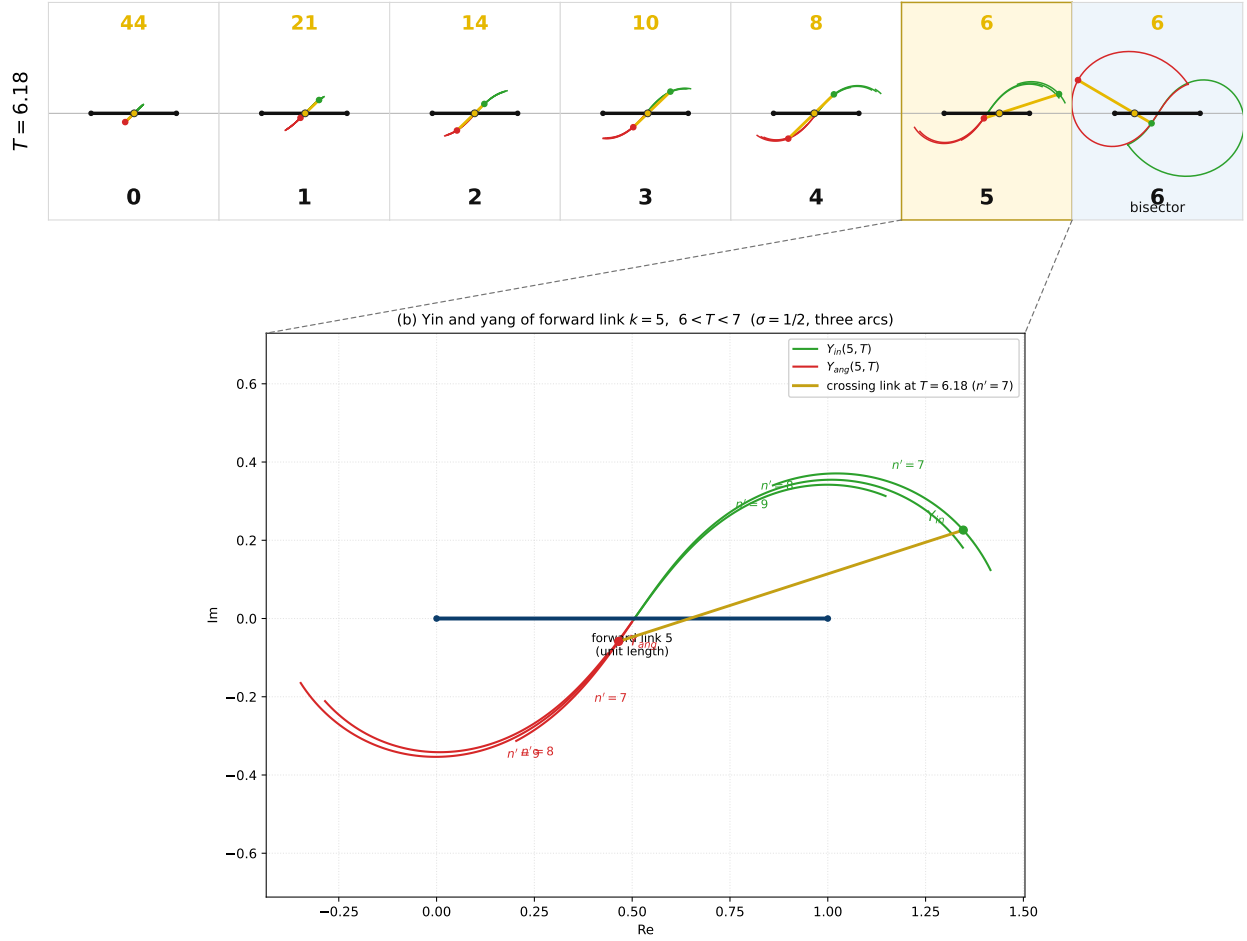


Figure 41: (a) Local frames of the forward links at  $\sigma = \frac{1}{2}$ ,  $T = 6.18$ , as in the top row of Figure 39, now with the yin and yang loci of every link as  $T$  runs through  $(6, 7)$ : green  $Y_{in}(k, T)$ , red  $Y_{ang}(k, T)$ . (b) The yin and yang curves of forward link  $k = 5$  over  $6 < T < 7$ , from (124)–(127). The frame sends that link to  $[0, 1]$  (dark blue). Each end traces three arcs, one per crossing summand  $n' = 7, 8, 9$ , kept apart where one reverse link gives way to the next: green is  $Y_{in}(5, T)$ , red is  $Y_{ang}(5, T)$ . The dark-yellow segment is the crossing link at the running example  $T = 6.18$ , where  $C_\ell(5, 6.18) = 6$  so  $n' = 7$ . Dashed lines carry the local coordinate frame for link 5 of (a) down to the magnification (b). Each arc's yin is the previous arc's yang, the same joint passing from one reverse link to the next. Generated by `fig_yinyang_link5.py`.

The same local frames with each non-bisector arc continued past the unit are Figure 42. The continuation holds the reverse link of that piece and steps half a unit of  $T$  past one end (the end farther from the local-frame origin for yin, the other end for yang). The bisector is left alone: its two curves already close.

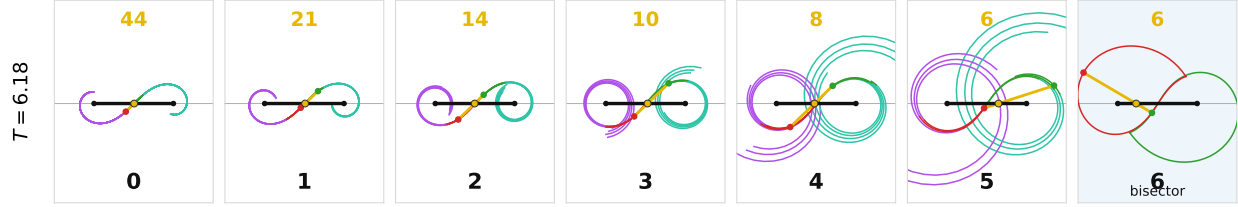


Figure 42: Local frames of the forward links at  $\sigma = \frac{1}{2}$ ,  $T = 6.18$ , as in panel (a) of Figure 41, with each non-bisector yin and yang arc continued as a solid curve (teal past the green end farther from the origin, purple past the other red end). The continuation holds the reverse link of that piece and runs half a unit of  $T$ . The bisector local frame is not extended. Generated by `fig_yinyang_extended.py`.

Each arc's yin is the preceding arc's yang, the same point at the same  $T$ : both are  $Y_k(j+1)$ , once as the far end of the outgoing crossing link and once as the near end of the incoming one. That is the crossing of §11.1 seen from the ends rather than the middle: one reverse link is dragged off the forward link and the next is dragged across it. The joint passing across the forward link belongs to both links at that instant, so the crossing itself goes through smoothly while the yin curve steps onto where the yang curve was and the yang steps one joint further out.

The count of reverse links dragged across forward link  $k$  in one unit of  $T$  is about  $2a/(k+1)$ :  $n' = a^2/(k+1)$  advances by  $2a/(k+1) = (2T+1)/(k+1) + O(T^{-1})$  over a unit of the index, and that is how many arcs the link gets. On the unit step,  $k+1=1$ , so about  $2a$  reverse links cross it; at  $T=6.18$ ,  $a \approx 6.67$ , so roughly a dozen. That is why the local coordinate frame for link 0 in Figures 41–42 is a thicket of short arcs, so close together they are not visible separately in the figure, while the links beside the bisector only get two, and the bisector none. Its partner is the other bisector link at every height, so its yin and yang alone close into single loops, and they alone have the limit teardrop of §6.4.

The two ends fix where the crossing reverse link sits along the forward link. In each separate local coordinate frame the link is  $[0, 1]$  on the real axis, so the crossing is where the chord from yin to yang crosses that axis, the computation (99) makes at the bisector. Writing  $v_k = \chi(k+1)^s(j+1)^{s-1}$  for the crossing link's own vector in the frame, which by (123) is the offset  $Y_{in} - Y_{ang}$ , that crossing fraction is

$$\lambda_k(T) = \operatorname{Re} Y_{in} - \operatorname{Im} Y_{in} \frac{\operatorname{Re} v_k}{\operatorname{Im} v_k} = \frac{\operatorname{Im}(\overline{Y_{in}} v_k)}{\operatorname{Im} v_k} = |Y_{in}| \frac{\sin(\beta - \alpha)}{\sin \beta}, \quad \alpha = \arg Y_{in}, \quad \beta = \arg v_k, \quad (128)$$

Let  $u_k$  be the corresponding parameter on the reverse segment. The finite segments intersect exactly when  $0 \leq \lambda_k \leq 1$  and  $0 \leq u_k \leq 1$ . The distance in the plane from joint  $k$  is  $\lambda_k(k+1)^{-\sigma}$ . At  $k=m$  this is (99) itself,  $\lambda_m = M^\sigma d_1$ , the bisector point's own crossing fraction, which on the critical line is  $\hat{d}_1$ . So the rightmost panel of Figure 44 is the quantity the rest of the paper is built on, and the panels to its left are that quantity's counterparts link by link.

Checked against the local frame construction in the Zest web app on the critical line at  $T = 6.18, 6.72, 12.4, 17.4, 40.05$  and  $434.37$ : (122) matches the frame to  $1.3 \times 10^{-15}$ , (128) matches the app's own reading to  $6.9 \times 10^{-14}$ , and  $\lambda_m$  matches  $\hat{d}_1$  to  $1.0 \times 10^{-12}$ , in `probe-yinyang-below-bisector.mjs`.

### 11.3 $d_1$ and $d_2$ for any link

§4 split Siegel's remainder along the two bisector links, and §10.2 found the weights  $d_1, d_2$  as the real-axis crossings of the yin and yang pair in those two frames. The same two crossings exist for every link: (123) is the pair, (128) is the crossing fraction, and the weights are that fraction read once along the forward link and once along the reverse one.

Write  $n = k + 1$ ,  $n' = j + 1 = C_s(n, T)$ , and

$$W_k = \sum_{n=k+1}^m n^{-s} + R - \chi \sum_{n=m+1}^j n^{s-1} \quad (129)$$

for the vector in the plane from forward joint  $k$  to reverse joint  $j$ , so that  $Y_{in}(k, T) = (k + 1)^s W_k$  by (122). At the bisector both sums are empty and  $W_m = R$ . Let

$$\omega_n = I(T) \log n, \quad \omega_{n'} = I(T) \log n', \quad \varphi_k = \arg W_k, \quad \psi = \arg \chi, \quad (130)$$

so  $-\omega_n$  is the absolute direction of summand  $n$  of  $\Sigma_1$ , the same telescoping that made  $\omega = \omega_{m+1}$  in (20), and  $\omega_{n'} + \psi$  is the absolute direction of summand  $n'$  of  $\Sigma_2$ .

**Measured along the forward link.** In the frame of link  $k$  the crossing fraction (128) is how far along that link it is from joint  $k$  to the crossing point. The frame scaled the link by  $(k + 1)^\sigma$ , so the distance in the plane is

$$d_1(k, T) = (k + 1)^{-\sigma} \lambda_k(T) = |W_k| \frac{\sin(\omega_{n'} - \varphi_k + \psi)}{\sin(\omega_n + \omega_{n'} + \psi)}. \quad (131)$$

The sine form is (128) with  $Y_{in} = (k + 1)^s W_k$  and  $v_k = \chi (k + 1)^s (j + 1)^{s-1}$ , the same reduction that took (99) to (17):  $\alpha = \omega_n + \varphi_k$ ,  $\beta = \omega_n + \omega_{n'} + \psi$ , and  $|Y_{in}| = (k + 1)^\sigma |W_k|$ . At  $k = m$  one has  $W_m = R$ ,  $n = n' = \lceil T \rceil$ ,  $\omega_n = \omega_{n'} = \omega$ , and (131) is (17).

**Measured along the reverse link.** The same crossing, read along the reverse link from joint  $j$ , is the parameter  $u_k = \Im Y_{in} / \Im v_k$  of the chord in that frame. The reverse link has length  $|\chi| (j + 1)^{\sigma-1}$ , so

$$d_2(k, T) = |\chi| (j + 1)^{\sigma-1} u_k = |W_k| \frac{\sin(\omega_n + \varphi_k)}{\sin(\omega_n + \omega_{n'} + \psi)}. \quad (132)$$

$d_1(k, T)$  and  $d_2(k, T)$  share a denominator, as  $d_1$  and  $d_2$  did. At  $k = m$  this is (18). The fractional amounts, the hat of (24), are the two parameters themselves:

$$\hat{d}_1(k, T) = \lambda_k(T) = (k + 1)^\sigma d_1(k, T), \quad \hat{d}_2(k, T) = u_k = \frac{d_2(k, T)}{|\chi| (j + 1)^{\sigma-1}}. \quad (133)$$

**The direction, and the two-direction form.** Forward link  $k$  points along  $e^{-i\omega_n}$  and reverse link  $j$  along  $e^{i(\omega_{n'} + \psi)}$ . The two fractional summands are those directions, shortened by the two weights,

$$R_1(k, T) = d_1(k, T) e^{-i\omega_n}, \quad R_2(k, T) = d_2(k, T) e^{i(\omega_{n'} + \psi)}, \quad (134)$$

and they add back to the vector they split:

$$W_k = d_1(k, T) e^{-i\omega_n} + d_2(k, T) e^{i(\omega_{n'} + \psi)}. \quad (135)$$

This is the two-direction form (21) with  $R$  replaced by  $W_k$  and the two bisector directions replaced by the two links that actually cross. The sine expressions (131)–(132) are again Cramer’s rule for that  $2 \times 2$  real system, which is why they are real, and why they reduce to (17)–(18) at the bisector. On the critical line the bisector pair still satisfies  $d_1(m, T) = d_2(m, T)$ , but a general pair need not: the two links have different lengths, and the crossing sits at  $\hat{d}_1 \approx \frac{1}{2}$  four or more links out from the bisector (§11.1), so  $d_1(k)$  and  $d_2(k)$  scale as  $n^{-\sigma}$  and  $|\chi|n'^{\sigma-1}$ . Figure 43 is the same decomposition, one local frame per forward link, in the local frames of the top row of Figure 41: the last local frame is the bisector,  $W_6 = R$ , which is Figure 9 seen in that frame.

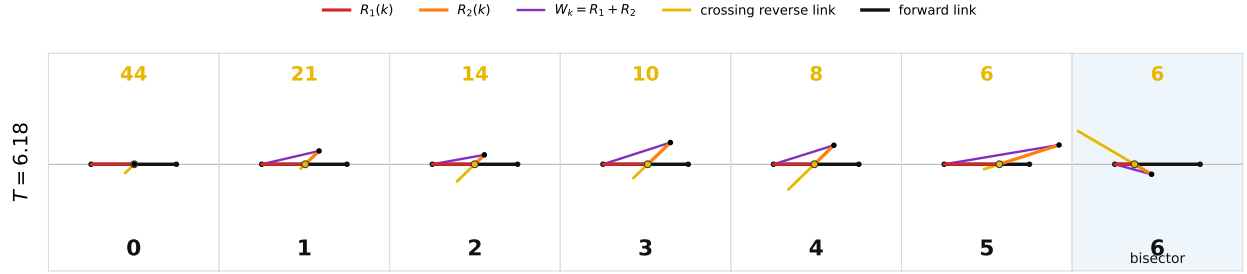


Figure 43: The two-direction form (135) at  $\sigma = \frac{1}{2}$ ,  $T = 6.18$ , in the local frame of each forward link  $k = 0, \dots, 6$ , the same local frames as the top row of Figure 41. Each local frame sends that link to  $[0, 1]$  on the real axis (black); the gold number is the reverse link that crosses it. Red is  $R_1(k)$  from the left joint to the crossing along the unit link, orange is  $R_2(k)$  from the crossing to reverse joint  $K_j$ , and purple is the resultant  $W_k$ . The last local frame is the bisector,  $W_6 = R$ . Generated by `fig_Wk_triangle.py`.

At the running example  $\sigma = \frac{1}{2}$ ,  $T = 6.18$ , the seven pairs are

| $k$ | $j$ | $d_1(k)$ | $d_2(k)$ | $\hat{d}_1(k)$ |
|-----|-----|----------|----------|----------------|
| 0   | 44  | 0.506    | 0.00185  | 0.506          |
| 1   | 21  | 0.360    | 0.159    | 0.510          |
| 2   | 14  | 0.286    | 0.0914   | 0.496          |
| 3   | 10  | 0.263    | 0.184    | 0.525          |
| 4   | 8   | 0.214    | 0.143    | 0.478          |
| 5   | 6   | 0.264    | 0.300    | 0.647          |
| 6   | 6   | 0.0875   | 0.0875   | 0.231          |

The last row is Table 1 at this  $T$ :  $d_1 = d_2 = 0.0875$  and  $\hat{d}_1 = \sqrt{7} d_1 = 0.231$ . The first six sit near the middle of their forward link, as §11.1 measured, and  $d_2(0)$  is the tiny reverse link 44. Figure 44 draws the same seven numbers against the whole unit of  $T$ .



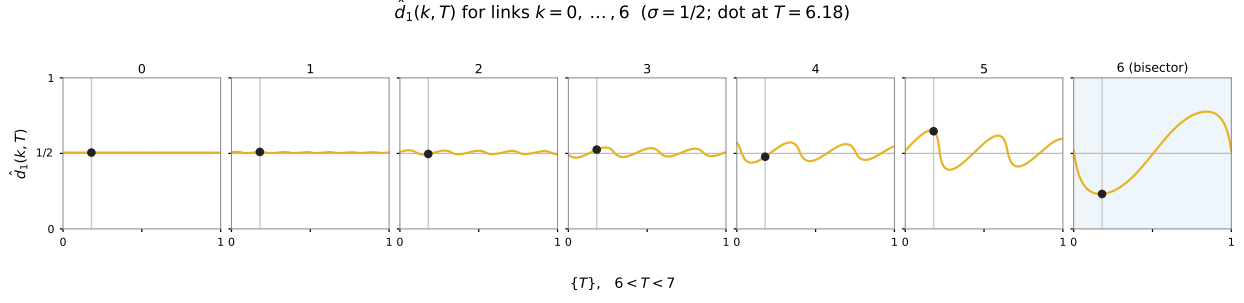


Figure 44:  $\hat{d}_1(k, T) = (k + 1)^\sigma d_1(k, T)$  for forward links  $k = 0, \dots, 6$  over  $6 < T < 7$ ,  $\sigma = \frac{1}{2}$ , one panel per link,  $x$  the fractional part of  $T$  and  $y$  the crossing fraction along that link from the left joint. The light horizontal is  $\frac{1}{2}$ ; the light vertical and the dot are the running example  $T = 6.18$ , the seven values of the table. Away from the bisector the track sits on  $\frac{1}{2}$  and barely moves; at  $k = 6$  it is the normalized  $d_1$  of Figure 1. Generated by `fig.d1.any.link.py`.

Checked at  $\sigma = \frac{1}{2}, \frac{1}{4}, \frac{7}{10}$  and  $T = 6.18, 6.72, 12.4, 17.4$ , for  $k = 0, 1, m-2, m-1, m$ : (131)–(132) at  $k = m$  match (17)–(18) to 0, (133) matches (128) to  $1.5 \times 10^{-27}$ , and (135) holds to  $4.4 \times 10^{-28}$ , in `check.d1.any.link.py`.

## 11.4 The crossings: $\Sigma_{1x}$ and $\Sigma_{2x}$ formula

§11.1 named the reverse partner of each forward link, and §11.3 read the crossing fraction  $\hat{d}_1(k, T)$  along that link. Those two data locate a point on every forward link  $0, \dots, m$ . Connecting the points in order is the same Dirichlet sum as  $\Sigma_1 + R_1$ , re-jointed: the joints sit at the crossings instead of at the integers, and the walk ends at the bisector point  $B_1$  rather than at  $\Sigma_1$ . Call that walk  $\Sigma_{1x}$ .<sup>11</sup>

The crossing fraction of link  $k$ , from joint  $k$  to the crossing point, is  $\hat{d}_1(k, T)$  of (133);  $T$  is fixed through what follows, so we write it  $\hat{d}_1(k)$ . Link  $k$  carries the summand  $n = k + 1$ , so the crossing on it is

$$P_k = J_k + \hat{d}_1(k) (k + 1)^{-s} = \sum_{n' \leq k} n'^{-s} + \hat{d}_1(k) (k + 1)^{-s}, \quad (136)$$

with  $J_k$  as in (109). The last crossing is the bisector point itself,  $P_m = B_1$ . The walk  $0 = P_{-1} \rightarrow P_0 \rightarrow \dots \rightarrow P_m$  therefore telescopes to  $B_1$  by construction. Its first step is  $P_0 = \hat{d}_1(0)$ , since  $J_0 = 0$  and  $1^{-s} = 1$ . Taking that step separately and substituting (136) in the rest, with the empty sum  $\sum_{n' \leq 0} = 0$ , removes the intermediate points and keeps  $0^{-s}$  out of the display:

$$\Sigma_{1x} = \hat{d}_1(0) + \sum_{k=1}^m \left( \sum_{n' \leq k} n'^{-s} + \hat{d}_1(k) (k + 1)^{-s} - \sum_{n' \leq k-1} n'^{-s} - \hat{d}_1(k-1) k^{-s} \right) = \sum_{n' \leq m} n'^{-s} + \hat{d}_1(m) (m + 1)^{-s}. \quad (137)$$

Each full summand  $k^{-s}$  is divided across two consecutive steps,  $\hat{d}_1(k-1) k^{-s}$  in one and  $(1 - \hat{d}_1(k-1)) k^{-s}$  in the next, and those two pieces add to one. What remains is the integer partial sum plus the last fractional stub, which is the ordinary first-half formula (47) once the stub is named.

That stub is  $\hat{d}_1(m) (m + 1)^{-s}$ . The bisector link is link  $m = \lfloor T \rfloor$ . It runs from  $J_m = \Sigma_1$  to  $J_{m+1} = \Sigma_1 + (m + 1)^{-s}$ , so the vector along it is the next Dirichlet term,  $n = M = m + 1$ . The factor  $(m + 1)^{-s}$  is that whole unused summand, the faint dashed continuation past  $\Sigma_1$  in Figure 8.

<sup>11</sup>The subscript  $x$  is for crossing, not a dummy index:  $\Sigma_{1x}$  is the crossing-joint form of  $\Sigma_1$ , and  $\Sigma_{2x}$  of  $\Sigma_2$ .

The number  $\hat{d}_1(m)$  is the crossing fraction: how far along the link the two bisector links meet. That meeting is  $B_1$ , and (120) already computed it: the crossing fraction on the bisector link is the remainder weight itself,  $\hat{d}_1(m) = \hat{d}_1$ . Therefore

$$\hat{d}_1(m) (m+1)^{-s} = \hat{d}_1 (m+1)^{-s} = R_1, \quad (138)$$

the “one more partial summand” of §5: not a new term, the next summand scaled by a real signed coefficient. The unhatted weight  $d_1$  multiplies a unit vector in the direction of  $(m+1)^{-s}$ , so  $|R_1| = |d_1|$ . The hatted quantity  $\hat{d}_1$  is a signed normalized coefficient, and is literally a fraction of the link when  $0 \leq \hat{d}_1 \leq 1$ ; it multiplies the full summand, whose length is  $M^{-\sigma}$ , and  $\hat{d}_1 \cdot M^{-\sigma} = d_1$  is the same remainder (24). At the running example  $\sigma = \frac{1}{2}$ ,  $T = 6.18$ , one has  $m = 6$ , next term  $7^{-s}$  of length  $1/\sqrt{7} \approx 0.378$ ,  $d_1 \approx 0.0875$ , and  $\hat{d}_1 = \sqrt{7} d_1 \approx 0.231$ : the walk uses about 23% of link 6, and that stub is  $R_1$ . The first six  $\hat{d}_1(k)$  in the table of §11.3 sit near  $\frac{1}{2}$ , as the saddle of §11.1 said they would; only  $\hat{d}_1(m)$  is the named remainder weight. So (137) is

$$\Sigma_{1x} = \sum_{n' \leq m} n'^{-s} + \hat{d}_1(m) (m+1)^{-s} = \Sigma_1 + R_1 = B_1, \quad (139)$$

and not  $\Sigma_1$  itself.

The same construction on the other Dirichlet sum gives  $\Sigma_{2x}$ . Write  $L_k = \chi(s) \sum_{n' \leq k} n'^{s-1}$  for the joints of  $\Sigma_2$  from the origin, and  $\hat{d}_2(k)$  for the crossing fraction of reverse link  $k$  from  $L_k$  to its crossing point, the  $u_k$  of (133). Then

$$Q_k = L_k + \hat{d}_2(k) \chi(s) (k+1)^{s-1}, \quad (140)$$

and the same telescoping, with first step  $Q_0 = \hat{d}_2(0) \chi(s)$  and the empty sum  $L_0 = 0$ , is

$$\begin{aligned} \Sigma_{2x} &= \hat{d}_2(0) \chi(s) + \sum_{k=1}^m \left( L_k + \hat{d}_2(k) \chi(s) (k+1)^{s-1} - L_{k-1} - \hat{d}_2(k-1) \chi(s) k^{s-1} \right) \\ &= \chi(s) \sum_{n' \leq m} n'^{s-1} + \hat{d}_2(m) \chi(s) (m+1)^{s-1}. \end{aligned} \quad (141)$$

The last stub is  $\hat{d}_2(m) = \hat{d}_2$ , so

$$\Sigma_{2x} = \Sigma_2 + R_2 = B_2. \quad (142)$$

Nothing here used  $\sigma = \frac{1}{2}$ . The split  $R = R_1 + R_2$  of §4 holds at every  $\sigma$ , the weights  $\hat{d}_1, \hat{d}_2$  of (24) are defined at every  $\sigma$ , and so

$$\Sigma_{1x} + \Sigma_{2x} = B_1 + B_2 = \zeta \quad (143)$$

off the critical line as well as on it. On  $\sigma = \frac{1}{2}$  the two walks are the two legs of §9.2; off it they are just the two summands of Theorem 4.2, re-jointed. Reflecting  $\Sigma_{1x}$  through  $\zeta/2$  sends the origin to  $\zeta$  and fixes  $B_1$ , which is the other way of seeing the second half on the critical line, not the walk of  $\Sigma_{2x}$  from the origin. The reverse partners of the forward links  $0, \dots, m$  are not consecutive, so the points  $P_k$  are not the joints of a walk along the reverse chain. This re-jointing is not Kapitons’ iterated-midpoint polygon [12]: that construction is the binomial transform of the partial sums and does not aim at  $B_1$  or  $B_2$ .

## 11.5 First-part and second-part sums

Here we use the places where each link is crossed to divvy up the main sum into two separate sums: one adds up the first parts of all the summands, the other all the second parts. The remainder is handled automatically, because it is itself the first part of a summand. This subsection is nothing but an exploration of a rearrangement of the terms of the main sum.

Each forward link  $k = 0, \dots, m-1$  is cut at its crossing  $P_k$  of (136) into two pieces: a first part  $\hat{d}_1(k) (k+1)^{-s}$  from the joint to the crossing, and a second part  $(1 - \hat{d}_1(k)) (k+1)^{-s}$  from the crossing to the next joint. The bisector link contributes only its first part, the stub  $\hat{d}_1(m) (m+1)^{-s} = R_1$  from  $\Sigma_1$  to  $B_1$ . Summing those pieces as free vectors,

$$V_1 = \sum_{k=0}^m \hat{d}_1(k) (k+1)^{-s}, \quad V_2 = \sum_{k=0}^{m-1} (1 - \hat{d}_1(k)) (k+1)^{-s}, \quad (144)$$

gives  $V_1 + V_2 = B_1$ : the full links  $0, \dots, m-1$  plus the bisector stub are  $\Sigma_1 + R_1$ . Figure 45 draws  $V_1$  from the origin and  $V_2$  from the tip of  $V_1$ .

First-part and second-part sums of  $\Sigma_1$  at  $\sigma = 1/2$ ,  $T = 6.18$  ( $V_1 + V_2 = B_1$  to  $4.4e-31$ )

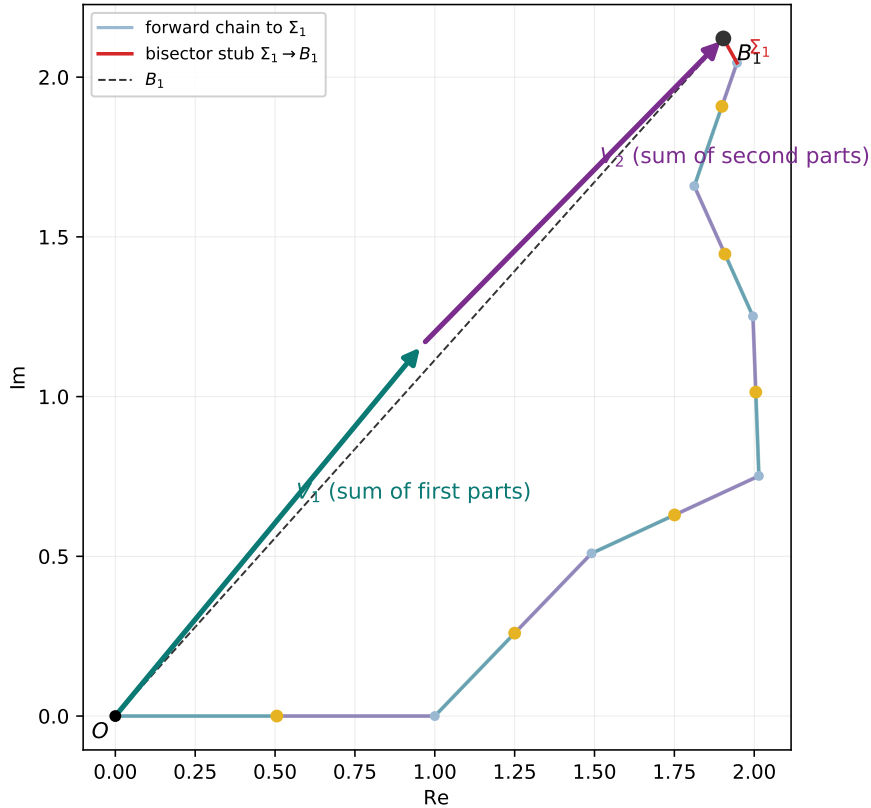


Figure 45: Each forward link cut at its yellow crossing: teal the first part, purple the second, red the bisector stub  $\Sigma_1 \rightarrow B_1$ . The teal arrow is  $V_1$ , the sum of the first parts including that stub; the purple arrow is  $V_2$ , the sum of the second parts, drawn from the tip of  $V_1$ . They meet at  $B_1$ . At  $\sigma = \frac{1}{2}$ ,  $T = 6.18$ . Generated by `fig_half_link_sums.py`.

## 11.6 General partial summand Riemann zeta function

The partial-summand Riemann zeta formula (23), our main result, presented near the start of this paper, has the same summation upper bound in both the main sum and the dual sum. But with the knowledge of which links cross, we can make those two upper bounds different. This subsection presents that new, more general form of the Riemann zeta function.

For any nonparallel pair of summand directions, Cramer's rule gives unique real coefficients and the displayed arbitrary-cut identity. The crossing conjecture is needed only to show that both coefficients lie in  $[0, 1]$ . For  $j = C_\ell(k, T)$ , the corresponding coefficient relation, written with the joints of (109), is

$$J_k + \hat{d}_1(k, T) (k+1)^{-s} = K_j - \hat{d}_2(k, T) \chi(s) (j+1)^{s-1}, \quad (145)$$

and moving the reverse sum inside  $K_j$  across gives

$$\boxed{\zeta(s) = \sum_{n=1}^k n^{-s} + \hat{d}_1(k, T) (k+1)^{-s} + \chi(s) \left[ \sum_{n=1}^j n^{s-1} + \hat{d}_2(k, T) (j+1)^{s-1} \right]}, \quad (146)$$

$$j = C_\ell(k, T).$$

This is (23) with the two upper summation bounds unequal, though not independent:  $j$  is the crossing partner of  $k$  named by the rule (108). At  $k = m$  the pairing is the self-pair,  $j = m$ , both cutoffs are  $\lfloor T \rfloor$ , and (146) is the boxed formula (23), hats and all. At any other  $k$  the two cutoffs differ: the forward sum stops at  $k$  while the reverse one runs to  $j \approx a^2/(k+1)$ , the two being reciprocal about  $a$  by (106). Reaching that far means carrying the reverse chain past its own bisector link, the continuation already written into (129).

For comparison, equation (4.13.1) of Titchmarsh's book [22, §4.13], the approximate functional equation, states: if  $h$  is a positive constant and

$$0 < \sigma < 1, \quad 2\pi xy = t, \quad x > h > 0, \quad y > h > 0,$$

then

$$\zeta(s) = \sum_{n \leq x} \frac{1}{n^s} + \chi(s) \sum_{n \leq y} \frac{1}{n^{1-s}} + O(x^{-\sigma} \log |t|) + O(|t|^{\frac{1}{2}-\sigma} y^{\sigma-1}). \quad (147)$$

The two sums are the same, and his constraint  $2\pi xy = t$ , that is  $xy = a^2$ , ties the two cutoffs exactly as the crossing product (106) does. Where he pays two error terms for the freedom to slide the cut, (146) pays with the two fractional terms, exactly.

The extreme cases are the ends of the pairing. Take  $j = 0$  in (146): the first reverse link is  $\chi(s) 1^{s-1} = \chi(s)$ , so the reverse sum is empty and it is the second half that collapses to a single term,

$$\boxed{\zeta(s) = \sum_{n=1}^k n^{-s} + \hat{d}_1(k, T) (k+1)^{-s} + \hat{d}_2(k, T) \chi(s)}, \quad k = \left\lceil \frac{I(T)}{2\pi} \right\rceil - 1 \text{ or one less,} \quad (148)$$

the forward cutoff being the partner of  $n' = 1$ , about  $(T + \frac{1}{2})^2$  links out. Which of the two it is the crossing test decides, exactly as in the third line of (108): the nearest integer names the paired link at most heights and its lower neighbor at the rest, and off the critical line the choice can turn on  $\sigma$  as well. Here  $\Sigma_1$  is walked whole, out to link  $k$  and a fraction of the next, and the walk lands on the

segment hanging off  $\zeta$  in the direction of  $-\chi$ . On the critical line that segment has length one, so the forward chain carried that far passes within one unit of  $\zeta$ , and what is left over is  $\hat{d}_2(k, T) \chi(s)$ , a real multiple of  $\chi$ . At  $\sigma = \frac{1}{2}$ ,  $T = 6.18$  the cutoff is  $k = 44$  and the two fractions are 0.012402 and 0.505537, reproducing  $\zeta$  to  $9.9 \times 10^{-32}$ .

For comparison, equation (4.11.1) of Titchmarsh's book [22, §4.11], the simplest approximation of  $\zeta(s)$  in the critical strip by a partial sum of its Dirichlet series, states:

$$\zeta(s) = \sum_{n \leq x} \frac{1}{n^s} - \frac{x^{1-s}}{1-s} + O(x^{-\sigma}), \quad (149)$$

uniformly for  $\sigma \geq \sigma_0 > 0$ ,  $|t| < 2\pi x/C$ , when  $C$  is a given constant greater than 1. Both close one long partial sum with a small correction: this is the smooth tail  $-x^{1-s}/(1-s)$  plus an error term, while (148) closes exactly with the two fractional terms. And the cutoffs match in scale: his condition puts  $x$  at order  $t/2\pi$  at least, which is where  $k = \lceil I(T)/2\pi \rceil - 1$  sits.

The cut at the other end of the pairing is the mirror of this one,  $k = 0$ . The first forward link runs from  $J_0 = 0$  to  $J_1 = 1^{-s} = 1$ : it is the unit segment of the real axis. So the forward sum is empty, the link is the number 1, and the entire first half of (146) collapses to a single real number,

$$\boxed{\zeta(s) = \hat{d}_1(0, T) + \chi(s) \left[ \sum_{n=1}^j n^{s-1} + \hat{d}_2(0, T) (j+1)^{s-1} \right]}, \quad j = C_\ell(0, T). \quad (150)$$

The reverse cutoff is the partner of the first summand,  $n' = a^2/1$ , so  $j$  is  $\lceil I(T)/2\pi \rceil - 1$  or one less, by (108) again. On the critical line that is the same integer as the forward cutoff in (148), the chain mirror (110) carrying the one cut to the other; off the line the two part, as the measurements at the end of this subsection show. And  $\hat{d}_1(0, T) = d_1(0, T)$  here: link 0 has length one, so the fraction of the link and the distance along it are the same number, and this is the only link on which the hat of (24) does nothing.

Read as a walk, (150) says this. Start at  $\zeta$  and come in along the reverse chain,  $j$  whole links and then a fraction  $\hat{d}_2(0, T)$  of the next. The walk lands on the real axis, between 0 and 1, and  $\hat{d}_1(0, T)$  is where. Equivalently,  $\zeta$  minus that whole reverse tail is a real number in the unit interval. It is not just any point of that interval either: it is essentially the midpoint, the  $k = 0$  panel of Figure 44 sitting on  $\frac{1}{2}$ . The crossing fraction has the closed form of (131) read at  $k = 0$ , where  $\omega_n = I(T) \log 1 = 0$  and  $W_0 = K_j$  is the reverse joint itself:

$$\hat{d}_1(0, T) = \operatorname{Re} K_j - \operatorname{Im} K_j \cot(\omega_{n'} + \psi), \quad n' = j + 1, \quad (151)$$

the crossing of the real axis by the line that carries reverse link  $j$ .

None of this is a shortcut to a value. The reverse sum now has about  $(T + \frac{1}{2})^2$  terms where the balanced cut of (23) has  $2\lfloor T \rfloor$  in all, and the two fractions are read off  $\zeta$  just as  $d_1$  and  $d_2$  were: (150) is one complex equation in the two real unknowns  $\hat{d}_1(0, T)$  and  $\hat{d}_2(0, T)$ , so it fixes them from  $\zeta$  and not the other way about. What it fixes is a position. Nor is the crossing the only one: the continued reverse chain crosses the unit segment many times, 87 times before link 400 at the running example, since both chains wind. The pairing law picks one of them out, the link paired with the first summand, and (151) says where that one lands.

At  $\sigma = \frac{1}{2}$ ,  $T = 6.18$ , one has  $a^2 = 44.5389$ , so  $j = 44$ , the tiny reverse link of the table in §11.3; then  $\hat{d}_1(0) = 0.505537$  and  $\hat{d}_2(0) = 0.012402$ , and the right side of (150) reproduces  $\zeta$  to  $4.6 \times 10^{-41}$  at forty digits. The same cut at  $k = 0, \dots, 6$  and that  $T$  gives seven formulas for the one number, with reverse cutoffs 44, 21, 14, 10, 8, 6, 6: the list of §11.1 with the self-pair in

last place. Checked on 500 samples of the critical line spread over  $2.05 \leq T \leq 60$ , and on 100 more at  $\sigma = 0.25, 0.35, 0.7, 0.9$ : the crossing lay inside both links at every sample, and the two sides of (150) agreed to  $3.2 \times 10^{-30}$  at thirty digits. The first crossing stayed near the middle of the unit segment throughout,  $\hat{d}_1(0, T)$  in  $[0.5006, 0.5233]$  on the line and in  $[0.4858, 0.5467]$  off it, and on the line it approaches  $\frac{1}{2}$  from above as  $T$  grows: the product  $T(\hat{d}_1(0, T) - \frac{1}{2})$  reads 0.0354, 0.0365, 0.0370, 0.0373 at  $T = 10.5, 20.5, 40.5, 80.5$ . The nearest-integer  $j_0$  was the paired link in 481 of the 500 line samples; at the other 19 it was  $j_0 - 1$ , the third line of (108), and at each of those the crossing sat at  $\hat{d}_2 \geq 0.98$ , at the far end of the reverse link and about to hand over. In `check_first_link.py`.

The two cuts are mirrors on the critical line: the fractions 0.012402 and 0.505537 of (148) are the pair of (150) exchanged, because the chain mirror (110) carries forward link  $n$  to reverse link  $n$  and so carries the one crossing to the other. Off the critical line the two formulas are independent, the mirror being what fails there: at  $\sigma = \frac{1}{4}$  and the same height the partner of reverse link 0 is forward link 43, with fractions 0.9721 and 0.5141, while the partner of forward link 0 stays at reverse link 44. The two cuts, which name one integer on the line, name two off it, and which of them drops turns over at  $\sigma = \frac{1}{2}$ : at  $\sigma = \frac{3}{4}$  and  $\sigma = \frac{9}{10}$  it is the reverse cutoff that reads 43 and the forward one 44.

## 11.7 Left unproved

Section 11 mixes identities that are proved, identities that are written down and then checked, and readings that are only measurements. Two statements about the crossings we would like to prove and have not.

1. Every forward link is crossed by a reverse link, and the partner is the summand  $n' = a^2/n$  of (106) (§11.1). The product law is motivated by a stationary-phase sweep and then checked on 118,007 samples. Proposition 11.1 settles the direction: the named partner is never parallel to the link and never more than a quarter turn from antiparallel, because  $nn'$  is an integer. What is missing is the position. A theorem would have to show that the two polylines meet, not only that the named integer usually works, and that needs the sweep (114) with an effective error term rather than as a stationary-phase heuristic.
2. The pairing is an involution:  $C_s(C_s(n, T), T) = n$ . Claimed as the symmetry of (106); if the previous item is proved, this is immediate.

## 12 $\vartheta_1, \vartheta_2$ and the zero-counting function

We direct our attention back to the legs of §9.2 and recall the two leg angles  $\vartheta_1$  and  $\vartheta_2$ . The angle  $\vartheta_1$  is the argument of the bisector point,  $B_{1ps}$ : it is the angle that Leg 1, running from the origin to that point, makes with the real axis. The angle  $\vartheta_2$  is the fold angle at the bisector point: the angle from Leg 1 to Leg 2,  $\vartheta_2 = \arg(B_2/B_1)$ , recording how far Leg 2 is folded back against Leg 1. Now isolate  $\vartheta_2$  by laying Leg 1 along the  $x$ -axis, so that our frame of reference is attached to Leg 1: in that frame we would observe  $\vartheta_2$  swing Leg 2 around while everything else stays stationary. Of course  $\zeta$  sits at the far end of Leg 2, and both legs have the same length when  $\sigma = \frac{1}{2}$ , so every time  $\vartheta_2$  passes  $\pi$ , Leg 2 folds straight back along Leg 1, its far end lands on the origin, and a zeta zero occurs. Therefore counting the turns of  $\vartheta_2$  would give us a perfect zeta-zero counting formula.

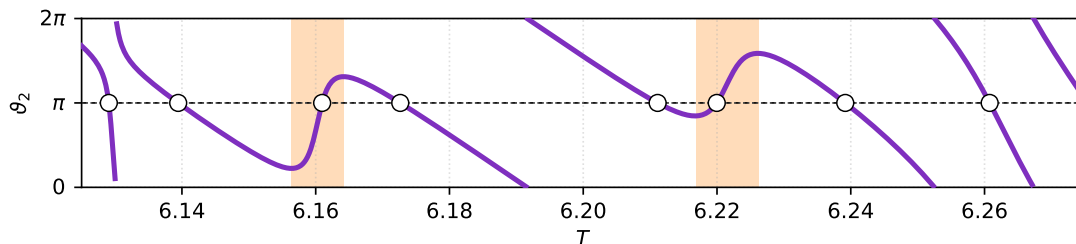


Figure 46: The fold angle  $\vartheta_2$  of the  $ps$  split on the critical line over  $6.125 \leq T \leq 6.275$ , reduced to  $[0, 2\pi)$ . Open circles mark the zeta zeros, where  $\vartheta_2$  passes through  $\pi$  (dashed); the shaded stretches are where  $\vartheta_2$  runs retrograde, in the shading of Figure 55, which returns to this window in detail. Generated by `fig_theta2_retro_strip.py`.

The problem is that  $\vartheta_2$  sometimes reverses its motion, and during those times the count would be off by one. We call the times when  $\vartheta_2$  reverses direction *retrograde*. We chose that name deliberately. When Claudius Ptolemy observed a planet's orbit from the earth and saw it run backwards in the sky (retrograde), he kept the earth at the center of the world frame, and so had to keep the complex epicycles to describe the orbits. We do not want to do that. We want instead to change our frame of reference to the sun and make our observation from there, where it is easier to understand that the orbits are not complicated: they are just ellipses. That is how we fix the  $\vartheta_2$  retrograde problem. We pick a different bisector point, which we call  $B_1^*$  below, with a different  $\vartheta_2$  that never goes retrograde, and from *that*  $\vartheta_2$  we build a new counting formula, which we call  $N_*$ . Unlike  $N$ ,  $N_*$  is continuous and always increasing (the increase is conditional on RH; Remark 12.4 makes this precise). And also unlike  $N$ , it counts only the zeros on  $\sigma = \frac{1}{2}$ , while  $N$  counts all the zeros: if there are any off the line,  $N$  will count them and  $N_*$  will not. We think these properties of  $N_*$  will prove useful. These properties, and the zeta sum's geometry from various frames of reference, deserve further study.

### 12.1 Review: the Riemann–von Mangoldt formula and $N_{ps}$

The well-known Riemann–von Mangoldt formula for the number of zeros of  $\zeta$  with imaginary part in  $(0, t]$ , in its exact form, reads

$$N(t) = \frac{\vartheta(t)}{\pi} + 1 + S(t), \quad S(t) = \frac{1}{\pi} \arg \zeta\left(\frac{1}{2} + it\right), \quad (152)$$

with  $\vartheta$  the Riemann–Siegel theta function of Appendix A.2. The argument here is the one fixed by continuous variation along the two straight segments from  $2$  to  $2 + it$  and then to  $\frac{1}{2} + it$ , starting from  $\arg \zeta(2) = 0$ . The branch has to be named: along any one path  $\arg \zeta$  is determined only up to a multiple of  $2\pi$ , and  $S$  only up to an even integer, so without the convention (152) would fix  $N$  only modulo 2. This is an identity ([22, Theorem 9.3]), obtained by applying the argument principle to  $\xi$  around a rectangle:  $\vartheta(t)/\pi + 1$  is the contribution of the  $\Gamma$  and  $\pi^{-s/2}$  factors, and  $S(t)$  is the contribution of  $\zeta$  itself along the critical line. Stated in terms of  $\vartheta$  the formula is exact: (152) carries no error term at all; approximation enters only when  $\vartheta$  is replaced by its expansion, and inexactness only when  $S$  is dropped. A word on usage before the counting begins: here and in the rest of the paper, an *ordinate* always means the ordinate of a zeta zero on the critical line, the imaginary part  $\gamma$  of a zero of  $\zeta(\frac{1}{2} + it)$ , equivalently a real zero of  $Z$ , with  $\gamma_1 < \gamma_2 < \dots$  numbering them up the line; under the Riemann hypothesis these are all the nontrivial zeros, and unconditionally they are the ones  $N_*$  counts. On the index the ordinate  $\gamma_n$  sits at  $T = I^{-1}(\gamma_n)$ . Figure 47 shows the two pieces over one unit of the index.

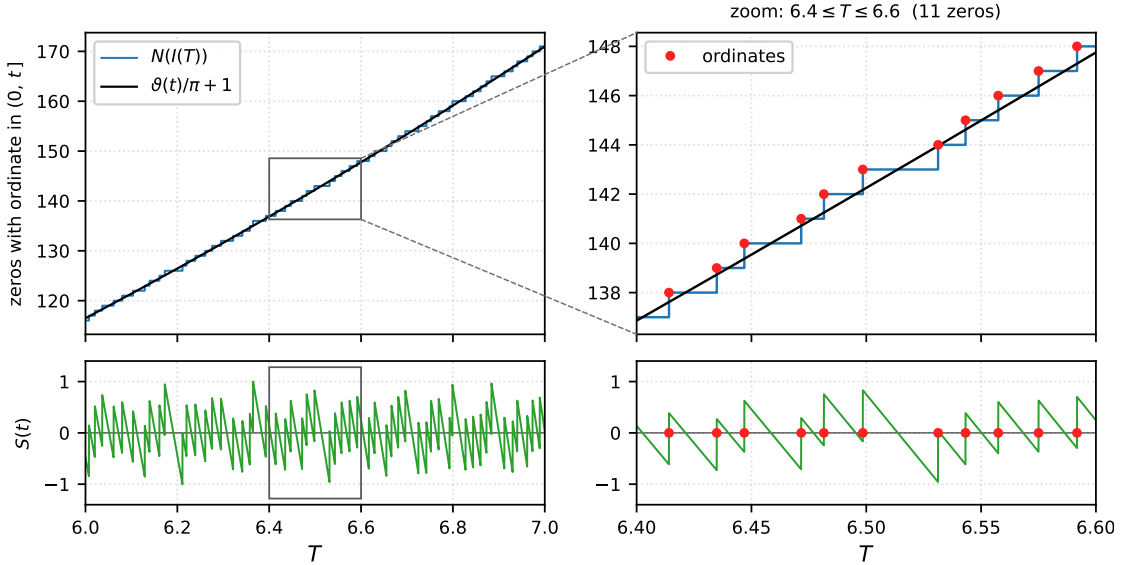


Figure 47: *Left*: the staircase  $N(I(T))$  (blue) over  $6 \leq T \leq 7$ , which is  $264.9 \leq t \leq 352.9$  and carries the 55 ordinates  $\gamma_{117}$  through  $\gamma_{171}$ , against the smooth part  $\vartheta(t)/\pi + 1$  of (152) (black). The vertical gap between them is  $S$ , which here stays within  $[-1.002, +1.001]$ . *Right*: the boxed window  $6.4 \leq T \leq 6.6$  magnified, with the eleven ordinates it holds,  $\gamma_{138}$  through  $\gamma_{148}$ , marked in red at the top of each riser and placed at  $T = I^{-1}(\gamma)$ . The risers are unevenly spaced while the smooth part climbs steadily. Below each panel is the gap itself,  $S$  (green), which at this magnification shows plainly as a sawtooth: a unit jump at every ordinate and a steady slide between them. Generated by `fig.staircase.67.py`.

**All of the discontinuity is in  $S$ .**  $N$  is a step function and  $\vartheta(t)/\pi + 1$  is analytic, so  $S$  carries every jump: at a height holding zeros it climbs by their total multiplicity, which is one at a simple ordinate and two at a symmetric off-line pair  $\rho, 1 - \bar{\rho}$ , the two sharing an imaginary part. In between it falls smoothly, at the rate

$$S'(t) = -\frac{\vartheta'(t)}{\pi} = -\frac{1}{2\pi} \log \frac{t}{2\pi} + O(t^{-2}), \quad (153)$$



which is minus the mean zero density. Across an ordinate  $S$  jumps up by one (at a simple zero); between ordinates it slides back down along that line. The sawtooth is why  $S$  has mean zero without being small: each unit jump is repaid by a decline whose steepness is the density itself. At an ordinate the definition of  $S$  fails, since  $\zeta$  vanishes there and its argument is undefined; one sets  $S(\gamma) = \frac{1}{2}[S(\gamma^-) + S(\gamma^+)]$ . Then (152) holds at every  $t > 0$ , reading  $N$  the same way at an ordinate, as  $\frac{1}{2}[N(\gamma^-) + N(\gamma^+)]$ , half a unit below the count of  $(0, \gamma]$ ; at every other height  $N$  is that count outright.

**Three counting curves against the staircase.** Expanding  $\vartheta$  by Stirling and dropping  $S$  turns (152) into the Riemann–von Mangoldt formula in its approximate form,

$$N_{\text{rvm}}(t) = \frac{t}{2\pi} \log \frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8} + \frac{1}{48\pi t} + \frac{7}{5760\pi t^3} \quad (154)$$

(the  $\frac{7}{8}$  collects the  $-\frac{\pi}{8}$  of the expansion, divided by  $\pi$ , with the standalone 1 of (152)). Nothing of consequence is discarded in truncating there: at  $t = 1000$  even the first correction retained,  $1/(48\pi t)$ , is only  $6.6 \times 10^{-6}$ , while  $S(1000) = +0.384$ . The whole of the error that matters is  $S$ . Set also

$$N_{\text{ps}}(T) = \frac{1}{\pi} \vartheta(I(T)) + \frac{1}{\pi} \vartheta_1\left(\frac{1}{2}, T\right) + \frac{3}{2}, \quad (155)$$

which replaces  $S$  by the angle  $\vartheta_1$  that leg 1 makes with the real axis (§9.2), the leg running from the origin to the bisector point  $B_1 = \Sigma_1 + R_{1ps}$ . These two behave quite differently against the true count  $N(I(T))$ , and a third curve  $N_*$ , built below in (162) from a different splitting of  $\zeta$ , repairs the one defect of  $N_{\text{ps}}$ .

$N_{\text{rvm}}$  is analytic and increasing, so it can only approximate the staircase in the mean, never landing on an integer at an ordinate, and its error is exactly the sawtooth  $S$ .  $N_{\text{ps}}$  instead meets an integer *exactly* at every ordinate, and only there.

**Proposition 12.1.** *On  $\sigma = \frac{1}{2}$ , at every  $T$  with  $B_1(T) \neq 0$ , so that  $\vartheta_1 = \arg B_1$  and hence  $N_{\text{ps}}$  is defined, the value  $N_{\text{ps}}(T)$  is an integer if and only if  $t = I(T)$  is an ordinate.*

The reason is already Corollary 9.2: on the critical line

$$B_1 e^{i\vartheta(t)} = \frac{1}{2}Z(t) + ih, \quad h \in \mathbb{R}. \quad (156)$$

The real part is  $|B_1| \cos(\vartheta + \vartheta_1) = \frac{1}{2}Z$ . That cosine is zero exactly when  $\vartheta + \vartheta_1 \equiv \frac{\pi}{2} \pmod{\pi}$ , which is exactly when  $N_{\text{ps}} = (\vartheta + \vartheta_1)/\pi + \frac{3}{2}$  is an integer. So  $N_{\text{ps}}$  is an integer if and only if  $Z = 0$ . (The cosine step needs  $B_1 \neq 0$ : since  $|B_1|^2 = \frac{1}{4}Z^2 + h^2$ , that would require  $Z$  and the offset  $h$  of (156) to vanish together, which happens nowhere in the ranges we computed.) Multiplying by  $e^{i\vartheta}$  is the  $\vartheta$ -frame of the glossary: it is the rotation  $e^{-i\psi/2}$  of Corollary 9.2 written with  $\vartheta = -\frac{\psi}{2}$ , since  $\chi = e^{-2i\vartheta}$  on the line, and it puts the base  $O \rightarrow \zeta$  on the real axis so that a split point reads as  $\frac{1}{2}Z + ih$ . This does not say  $N_{\text{ps}}$  equals the staircase  $N$ : retrograde ordinates still land on the wrong integer, as the next paragraph explains. It only says the value is an integer there, and nowhere else.

**Where  $N_{\text{ps}}$  misses, and why.** Two things keep (155) from being a counting function. First,  $\vartheta_1$  is an angle, so (155) is determined only up to an even integer, and each passage of  $\vartheta_1$  through  $\pm\pi$  breaks the curve; there are nine such wraps over  $3 < T < 5$ , and  $\frac{3}{2}$  is the constant that puts the

curve on the staircase. Second, and this is the real defect, the curve sometimes meets the wrong integer. Differentiating (156) at an ordinate, where  $Z$  vanishes, gives

$$\left. \frac{d}{dt} \arg(B_1 e^{i\vartheta}) \right|_{Z=0} = -\frac{Z'(t)}{2h}, \quad (157)$$

so the crossing carries the angle forward or backward with the sign of  $-hZ'$ . Consecutive simple zeros of  $Z$  alternate in the sign of  $Z'$ , so the crossings all turn the same way exactly when  $h$  alternates in sign too. Where  $h$  nearly vanishes it keeps its sign across two ordinates instead; the angle turns back on itself, the count fails to advance, and the second of the pair reads one too high. Over the first 400 ordinates  $|h|$  has median 1.081 at the 350 that  $N_{\text{ps}}$  places correctly and 0.319 at the 50 it misses, and those 50 are exactly the crossings that run backwards, which we call *retrograde*. They are also exactly the wraps: the nine wraps of  $\vartheta_1$  over  $3 < T < 5$  counted above sit one apiece on the nine retrograde ordinates of that range. Figure 50 shows the cost over a longer stretch: five retrograde ordinates in  $5.9 < T < 6.7$ , and after each one  $N_{\text{ps}}$  runs a unit above the staircase until the following ordinate restores it. The two ways of describing the failure, a broken branch and a backward crossing, are one event. Over  $3 < T < 5$ , which holds 55 zeros, both curves stay within one of the staircase all the same: the largest deviation is 0.946 for  $N_{\text{vm}}$  and 0.999 for  $N_{\text{ps}}$ .

Reading  $\vartheta_2$  in place of  $\vartheta_1$  cannot help. On the critical line  $B_2 = \chi \overline{B_1}$  and  $\arg \chi = -2\vartheta$ , so

$$\vartheta_2 = -2(\vartheta + \vartheta_1) \pmod{2\pi}, \quad (158)$$

whichever remainder defines the split. Every sawtooth built from  $\vartheta_2$  is therefore the same function as the one built from  $\vartheta_1$  and retrogrades in the same places: among the 62 ordinates of  $3.0 \leq T \leq 5.2$  the splits at  $R_{1\text{ps}}$ , at  $R/2$  and at  $R_{1ak}$  run backwards at 10, 9 and 12 of them. The retrograde belongs to the split, not to the branch.

## 12.2 Improving $N_{\text{ps}}$ : the velocity split and a curve that lands on every ordinate

The cure is to split  $\zeta$  so that the offset  $h$  of (156) alternates by construction. With  $\vartheta'$  as in (153), exactly  $\frac{1}{2} \operatorname{Re} \frac{\Gamma'}{\Gamma} \left( \frac{1}{4} + \frac{it}{2} \right) - \frac{1}{2} \log \pi$ , put

$$B_1^* = \zeta(s) + \frac{\zeta'(s)}{2\vartheta'(t)}, \quad B_2^* = \zeta(s) - B_1^* = -\frac{\zeta'(s)}{2\vartheta'(t)} = \frac{i}{2\vartheta'(t)} \frac{d\zeta}{dt}, \quad (159)$$

so leg 2 is the velocity of the chain's endpoint along the line, quarter-turned and scaled by  $1/2\vartheta'$ . The companion  $B_1^*$  already appears in the literature as the function Levinson calls  $G$  at his (1.9), where the weight is written in the gamma-factor form  $f'(s) + f'(1-s)$ , which on the line is this same real number  $2\vartheta'(t)$  [15]; §14.2 sets Levinson's use of  $G$  beside ours.

On  $\sigma = \frac{1}{2}$  this split is again reflection-symmetric,  $B_2^* = \chi \overline{B_1^*}$ , so its legs are exactly equal at every height,

$$\left| \zeta + \frac{\zeta'}{2\vartheta'} \right| = \frac{|\zeta'|}{2|\vartheta'|}, \quad (160)$$

which is the identity  $\operatorname{Re}(\zeta \overline{\zeta'}) = -\vartheta' |\zeta|^2$ , itself no more than the statement that  $Z = e^{i\vartheta} \zeta$  is real. Differentiating that statement gives  $e^{i\vartheta} \zeta' = -iZ' - \vartheta' Z$ , and hence (156) holds for the new split with

$$e^{i\vartheta} B_1^* = \frac{Z}{2} - \frac{iZ'}{2\vartheta'}, \quad h^* = -\frac{Z'}{2\vartheta'}. \quad (161)$$

Now  $h^*$  alternates because  $Z'$  does, and (157) returns  $-Z'/2h^* = \vartheta'$ : every crossing turns forward, and at exactly the mean density. Figure 48 draws the frame of (161) over one unit of the index: the path of  $e^{i\vartheta} B_1^*$  loops around the origin once per ordinate, crossing the imaginary axis exactly at the ordinates, and it is that winding we now turn into a counting curve. Figure 49 repeats the drawing on the hardest ground the count meets this low in the strip: the window  $32.87 \leq T \leq 32.91$  holding the Lehmer pair  $\gamma_{6709} = 7005.063$ ,  $\gamma_{6710} = 7005.101$ , whose two crossings arrive only  $9 \times 10^{-5}$  index units apart. The path still winds once per ordinate; between the two crossings of the pair it passes within  $|B_1^*| = 2.0 \times 10^{-3}$  of the origin, exactly half the famous shallow extremum  $Z = 0.00397$  of Lehmer's phenomenon, since  $h^*$  vanishes with  $Z'$  right where  $Z$  turns.

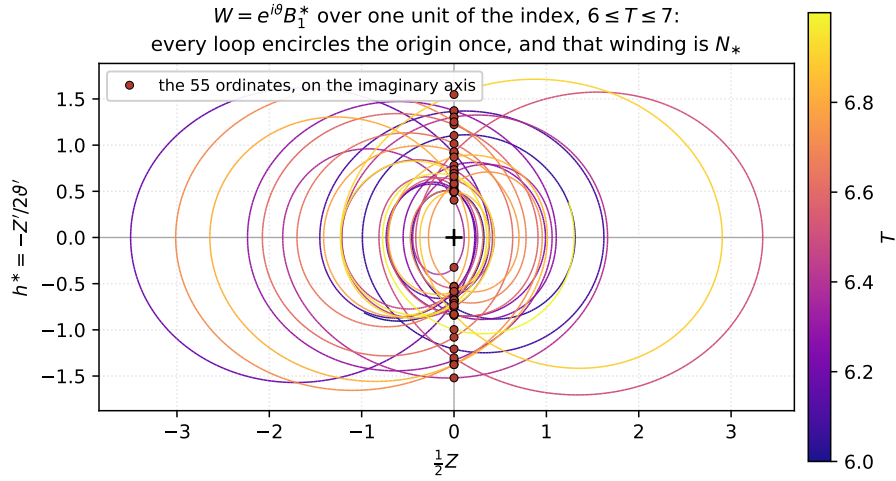


Figure 48: The velocity-split companion in the rotated frame of (161),  $W = e^{i\vartheta} B_1^* = \frac{1}{2}Z - iZ'/(2\vartheta')$ , over one unit of the index,  $6 \leq T \leq 7$  ( $264.9 \leq t \leq 352.9$ ), the color running with  $T$ . The horizontal coordinate is  $\frac{1}{2}Z$  and the vertical one is the offset  $h^* = -Z'/(2\vartheta')$ . Every loop encircles the origin (black cross) exactly once, and the path crosses the imaginary axis exactly at the 55 ordinates of the unit (red dots): that winding is the counting curve  $N_*$  of (163). Generated by `fig_b1_star_winding.py`.

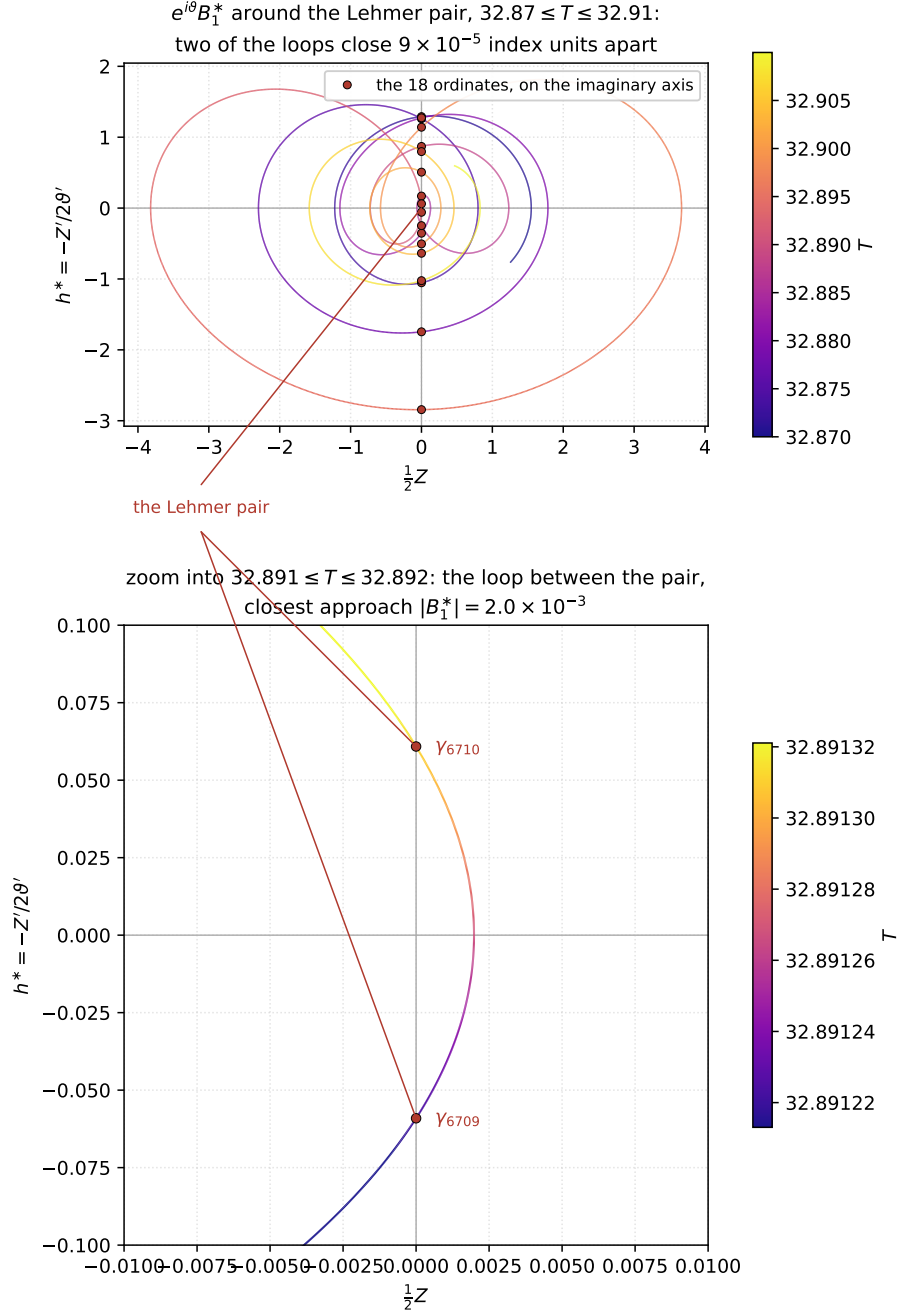


Figure 49: The same frame as Figure 48 over  $32.87 \leq T \leq 32.91$  ( $6996.2 \leq t \leq 7012.9$ ), the window around the Lehmer pair  $\gamma_{6709}, \gamma_{6710}$ . *Top*: the 18 ordinates of the window sit on the imaginary axis as before, but the two crossings of the pair (arrow) land almost on top of one another,  $9 \times 10^{-5}$  index units apart, and the loop that separates them is nearly invisible at this scale. *Bottom*: zoom into  $32.891 \leq T \leq 32.892$ , the thousandth of a unit holding the pair, on a horizontal scale stretched ten to one, with the color scale narrowed to the  $9 \times 10^{-5}$  index units just bracketing the pair. The two crossings are the red dots at  $h^* = \pm 0.06$ , and between them the path bows past the origin at  $|B_1^*| = 2.0 \times 10^{-3}$ , half the shallow extremum  $Z = 0.00397$  that makes this pair famous. The winding still advances by exactly one per ordinate, so  $N_*$  counts the pair correctly. Generated by `fig_b1_star_winding_lehmer.py`.

We set

$$N_*(T) = \frac{1}{\pi} \vartheta(I(T)) + \frac{1}{\pi} \vartheta_1^*(T) + \frac{3}{2}, \quad \vartheta_1^* = \arg B_1^*. \quad (162)$$

That is the Riemann–Siegel phase plus the argument of the companion  $B_1^* = \zeta + \zeta'/(2\vartheta')$ , or equivalently the Prüfer angle of  $(Z, -Z'/\vartheta')$ <sup>12</sup>:

$$\boxed{N_*(T) = \frac{1}{\pi} \arg\left(Z - \frac{iZ'}{\vartheta'}\right) + \frac{3}{2}.} \quad (163)$$

The argument is the continuous lift, not the principal value: by (161) it is  $\vartheta + \vartheta_1^*$  of (162), whose size is carried by  $\vartheta$ , and it is normalized so that  $N_* = 1$  at  $\gamma_1$ , which puts the lift at  $-\frac{\pi}{2}$  there. A principal argument would confine the whole right side to  $(\frac{1}{2}, \frac{5}{2}]$ . The Prüfer angle is Sturm–Liouville theory’s device for counting the zeros of a real oscillator  $y(t)$ : plot the point  $(y, y'/\omega)$  in the plane, with  $\omega$  the oscillator’s local frequency, and track its polar angle on a continuous branch. The point crosses the vertical axis exactly where  $y$  vanishes, so the angle passes a multiple of  $\pi$  at each zero and its winding is the count; dividing  $y'$  by  $\omega$  only rounds the orbit, so the angle advances steadily instead of in bursts. In (163) the oscillator is  $Z$  and the frequency is  $\vartheta'$ , the mean zero density. The resulting curve is integer-valued at the ordinates and nowhere else, since  $\operatorname{Re}(e^{i\vartheta} B_1^*) = \frac{1}{2}Z$  vanishes only there, and that advances by exactly one from each ordinate to the next. It lands on every one of the first 400 ordinates,  $N_*(I^{-1}(\gamma_n)) = n$  to  $2.8 \times 10^{-13}$ , against 50 misses for  $N_{\text{ps}}$ ; over  $0.6 \leq T \leq 8$  it is strictly increasing on a grid of 3000 points, no wrap occurs ( $|\vartheta_1^*|$  stays below  $0.60\pi$ ), and it rises by 237.64 across the 237 zeros of that range. Height costs it nothing and no branch correction is needed there either: at  $\gamma_{10000} = 9877.78$  and at  $\gamma_{50000} = 40433.69$  the same formula with the same constant  $\frac{3}{2}$  still returns exactly 10000 and 50000. Where  $N_{\text{rvm}}$  threads the staircase without touching it, and  $N_{\text{ps}}$  locates zeros without counting them,  $N_*$  does both. Like any count read off  $Z$  it sees the zeros on the line, which in the range plotted are all of them.

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<sup>12</sup>What we have not found in the literature is the argument of this companion used this way, as an interpolant of the count. The nearest classical objects are the exact count  $N = \vartheta/\pi + 1 + S$  of (152) and the Prüfer phase of an oscillator [19].

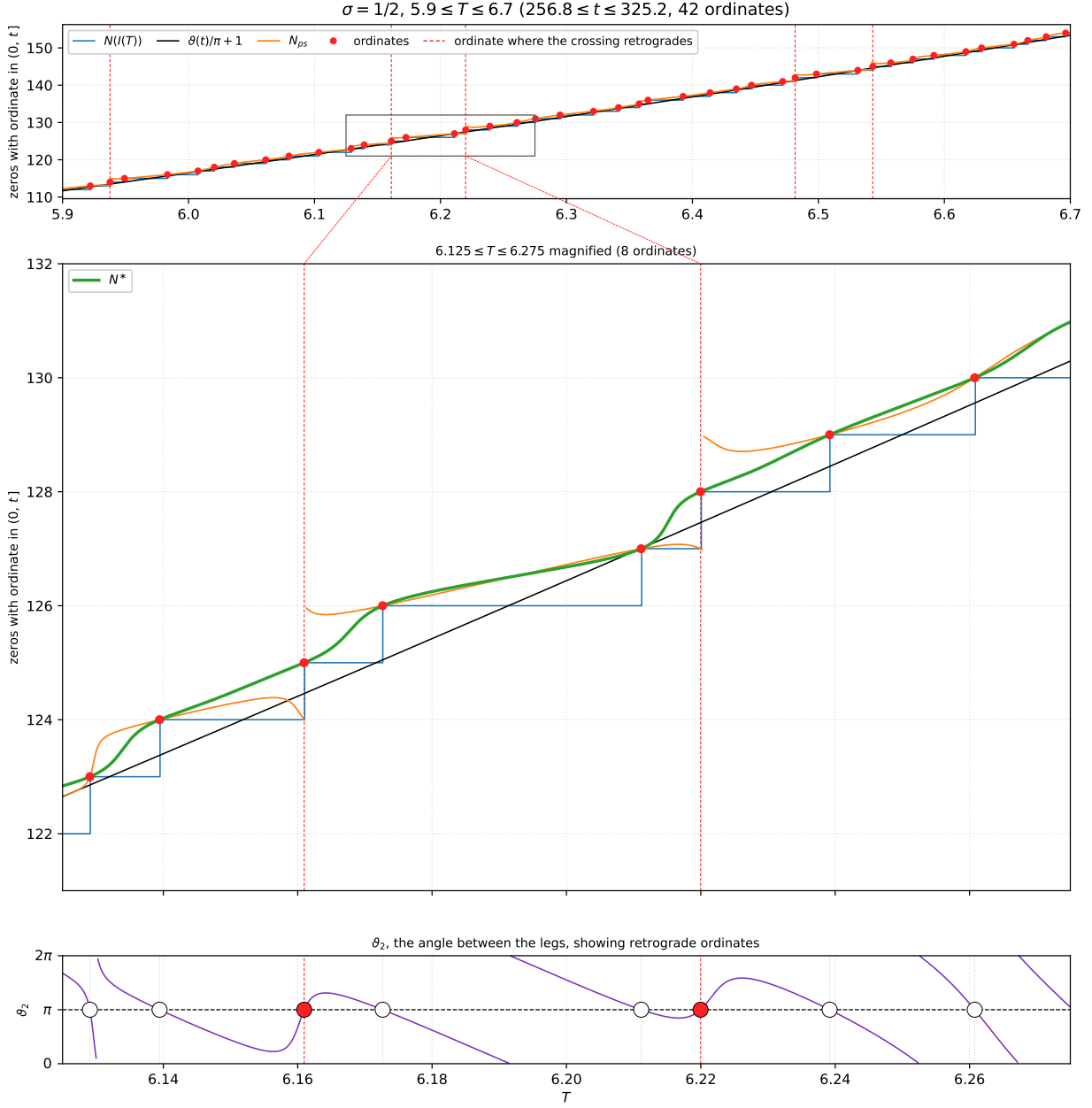


Figure 50: Staircase, in blue, is the exact Riemann–von Mangoldt formula (152), and the approximate Riemann–von Mangoldt formula (154) is the black curve, that formula with  $S$  dropped. *Top:*  $5.9 \leq T \leq 6.7$ , which is  $256.8 \leq t \leq 325.2$  and holds the 42 ordinates  $\gamma_{113}$  through  $\gamma_{154}$ , five of them retrograde, with  $N_{ps}$  of (155) in orange, broken at its wraps. *Middle:* the boxed window  $6.125 \leq T \leq 6.275$  magnified, holding 8 ordinates of which  $\gamma_{125}$  and  $\gamma_{128}$  are retrograde. The velocity split  $N^*$  of (162) is drawn in green. *Bottom:* the fold angle of the legs  $\vartheta_2$  of §9.2 on the same  $T$  range, reduced to  $[0, 2\pi)$ . Every passage through  $\pi$  (dashed) is at an ordinate; the two filled dots are the two retrograde ordinates of the panel above, where  $\vartheta_2$  rises through  $\pi$ , turns around, and comes back down through it. Generated by `fig_nps_staircase.py`.

**Remark 12.2** (the two offsets  $h_{R/2}$  and  $h^*$ ). The offsets  $h_{R/2}$  and  $h^*$  of the  $R/2$  split and the

velocity split are both drawn in Figure 52, and they differ only by a weighting of the links. The velocity offset has the closed form of (161),

$$h^*(t) = -\frac{Z'(t)}{2\vartheta'(t)}, \quad (164)$$

the slope of  $Z$  in units of  $2\vartheta'$ :  $h^*$  vanishes exactly at the critical points of  $Z$  and is largest where  $Z$  is steepest. For the  $R/2$  split  $e^{i\vartheta}R$  is real, so  $h$  is the sine companion of the main sum in the  $\vartheta$ -frame,  $h_{R/2} = \text{Im}(e^{i\vartheta}\Sigma_1) = \sum_{n \leq m} n^{-1/2} \sin(\vartheta - t \log n)$ , while differentiating the real part of that rotated main sum and dividing by  $-\vartheta'$  gives

$$h^* = \sum_{n \leq m} n^{-1/2} \left(1 - \frac{\log n}{\vartheta'}\right) \sin(\vartheta - t \log n) + \dots,$$

the tail being the remainder's derivative; the displayed sum by itself reproduces  $h^*$  to  $2.3 \times 10^{-2}$  over the first 400 ordinates. The weight is the phase rate  $\vartheta' - \log n$  of link  $n$  measured in units of  $\vartheta'$ , so it runs from 1 on the first link down to nearly 0 on the last, where  $\log m \approx \vartheta'$ . The counting angle is carried by the same links; each one contributes in proportion to how much of the rotation rate it retains.

**Remark 12.3** (the velocity split off the critical line). The weight  $2\vartheta'(t)$  in (159) is exactly  $-(\chi'/\chi)(\frac{1}{2} + it)$ , since  $\chi = e^{-2i\vartheta}$  on the line. Replacing it by that logarithmic derivative extends the split to the whole strip:

$$B_1^*(s) = \zeta(s) - \frac{\zeta'(s)}{(\chi'/\chi)(s)}, \quad B_2^* = \zeta - B_1^*, \quad \frac{\chi'}{\chi}(s) = \log 2\pi + \frac{\pi}{2} \cot \frac{\pi s}{2} - \psi(1-s) \quad (165)$$

( $\psi$  the digamma function, not the  $\arg \chi$  of §4). This is again Levinson's  $G$  and not an extension of it: his weight  $f'(s) + f'(1-s)$  equals  $-(\chi'/\chi)(s)$  at every  $s$  [15], the critical line being the place where it happens to be real. Applying the functional equation twice gives  $\chi(s)\chi(1-s) = 1$ , whose logarithmic derivative forces  $(\chi'/\chi)(s) = (\chi'/\chi)(1-s)$ ; differentiating  $\zeta(s) = \chi(s)\zeta(1-s)$  then yields the exact reflection

$$B_2^*(s) = \chi(s) B_1^*(1-s)$$

at every  $s$  in the strip, pairing the split at  $(\sigma, T)$  with its mirror image at  $(1-\sigma, T)$ . On the critical line the mirror point is the point itself, the weight is the real number  $2\vartheta'$ , and the identity collapses to  $B_2^* = \chi \overline{B_1^*}$  with its equal legs (160). Off the line the weight is complex (imaginary part  $\approx \mp 7 \times 10^{-4}$  at  $t \approx 280$ ), the offset  $h^*$  ceases to be a real scalar, and, as for every split, the equal-leg property survives only on the line. Verified numerically to twenty-four digits at  $\sigma = 0.3$ .

The  $\chi$ -weight also separates the two chains exactly. Write  $\zeta = F + \chi \tilde{G}$  with  $F = \Sigma_1 + R_1$  and  $\tilde{G} = (\Sigma_2 + R_2)/\chi$ , for any split whose two halves are holomorphic on the link interval, the  $R/2$  and velocity splits among them: the second half always carries the factor  $\chi$ , so peeling it off is natural. Differentiating, multiplying by  $\chi/\chi'$ , and subtracting cancels the  $\chi \tilde{G}$  term entirely; on the critical line, where  $\chi'/\chi = -2\vartheta'$ , the result is

$$B_1^* = \Sigma_1 + R_1 + \frac{(\Sigma_1 + R_1)' + \chi \tilde{G}'}{2\vartheta'}, \quad B_2^* = \Sigma_2 + R_2 - \frac{(\Sigma_1 + R_1)' + \chi \tilde{G}'}{2\vartheta'}. \quad (166)$$

The velocity split therefore hands the whole forward chain  $\Sigma_1 + R_1$  to leg 1 and the whole reflected chain  $\Sigma_2 + R_2$  to leg 2, exactly, with no mixing at the level of the sums; the two legs trade only the derivative corrections, damped by  $2\vartheta'$ . The reflected half influences  $B_1^*$  only through its rate

of change  $\tilde{G}'$ , never through its value. This is why  $B_1^*$  crowds against the other split points in Figure 52, and the taper weights  $1 - \log n/\vartheta'$  of Remark 12.2 are this cancellation read term by term. Verified to twenty-four digits at  $T = 6.18$ .

Incidentally, with the off-line definition (165) in hand, the zeros of  $B_1^*$  in and around the strip can be pinned down exactly by the argument principle: the winding number of  $B_1^*$  around the boundary of a box counts the zeros inside, and bisecting any box that contains zeros localizes them to arbitrary precision. The winding counts zeros and nothing else because  $B_1^*$  is analytic on these boxes: its only poles are at the zeros of  $\chi'/\chi$ , and that modulus stays above 3.7 throughout, while the pole of  $\zeta$  at  $s = 1$  lies at  $t = 0$ , far below. Figure 51 shows the result. The box  $-1 \leq \sigma \leq 1$ ,  $6 \leq T \leq 7$  contains exactly 40 zeros of  $B_1^*$ , every one strictly to the left of the critical line (the rightmost at  $\sigma \approx 0.416$ ), and each root checks directly,  $|B_1^*| < 10^{-10}$ . Two features are worth recording. First, no zero lies on or to the right of  $\sigma = \frac{1}{2}$ , consistent with the reflection  $B_2^*(s) = \chi(s)B_1^*(1-s)$  placing the mirror zeros to the right. Second, the window contains 55 ordinates but only 40 zeros, and only 22 of those sit inside the strip  $\sigma \in [0, 1]$ : the other 18 have drifted left of  $\sigma = 0$ , down to  $\sigma \approx -0.71$ . Running the same winding count over the box  $-20 \leq \sigma \leq -1$  turns up just 5 more, all between  $\sigma \approx -1.5$  and  $\sigma = -1$  (the count confirmed by an independent densely sampled winding integral around the full boundary), for 45 zeros right of  $\sigma = -20$ . That is ten short of the 55 ordinates of the window. We have no theorem saying the two totals should agree, so whether ten more lie further left, or the tally simply differs from the ordinate count, we have not determined.



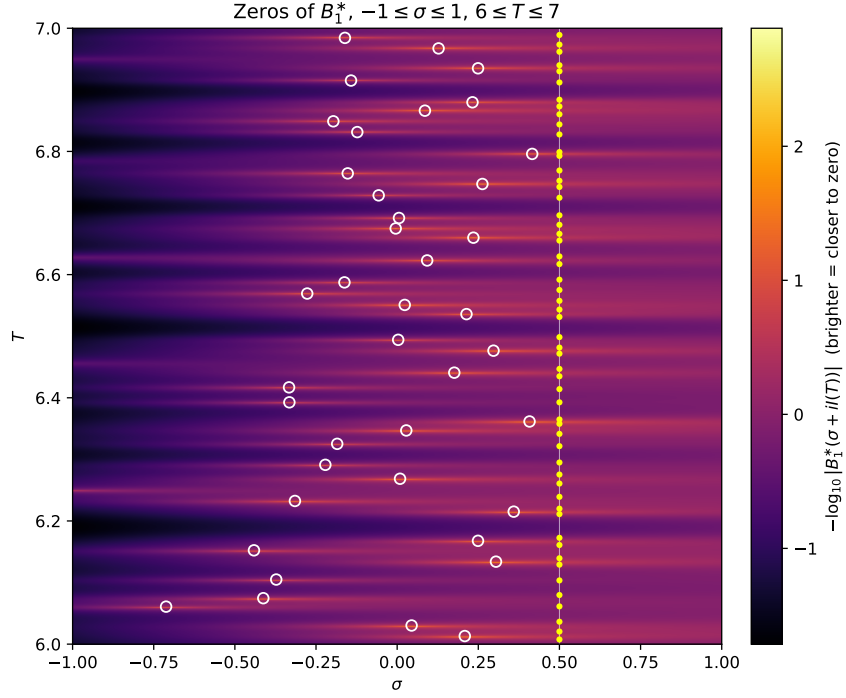


Figure 51: The 40 zeros of  $B_1^*$  in the box  $-1 \leq \sigma \leq 1$ ,  $6 \leq T \leq 7$  (white open circles), located by the argument principle: winding-number counts over recursively bisected boxes, each root then polished by Muller's method to  $|B_1^*| < 10^{-10}$ . The background heatmap, on a grid of 599 steps in  $\sigma \in [-1, 1]$  and 5000 steps in  $T$  with  $t = I(T)$  and the off-line definition (165), colors each input point by how close the output is to zero,  $-\log_{10} |B_1^*(\sigma + iI(T))|$ , brighter meaning closer; the thin white vertical line is the critical line, the yellow dots on it are the 55 zeta zeros of the window, and the dark horizontal bands are the ordinates' neighborhoods, where  $|B_1^*|$  stays comparatively large across the whole region. All 40 zeros lie strictly left of the critical line. Generated by `find_b1_star_zeros.py`.

The splits and the offset  $h$ , measured from  $\zeta/2$  along the line perpendicular to the base:  $\sigma = 0.50$ ,  $T = 6.18$  ( $t = 279.85$ ,  $m = 6$ )

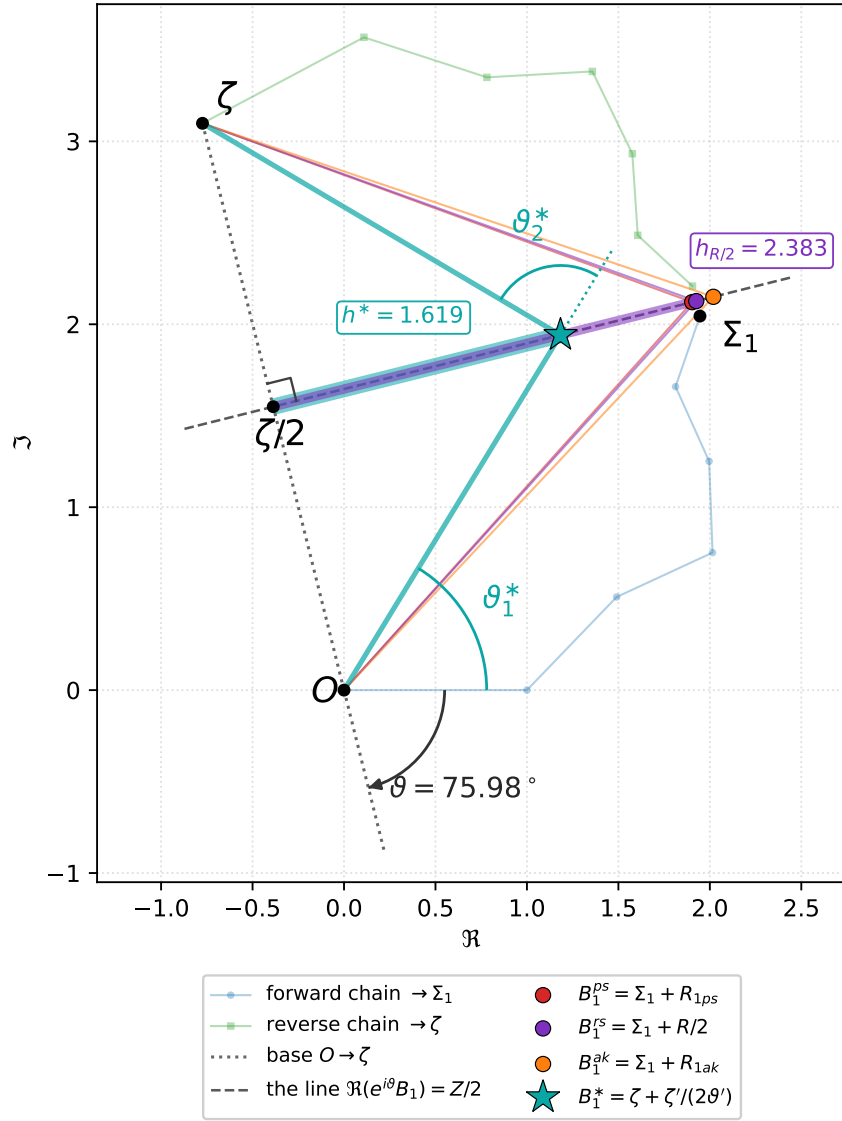


Figure 52: The offset  $h$  of (156), drawn at  $T = 6.18$ . Because  $\Re(e^{i\vartheta} B_1) = Z/2$  for every split, all of the split points lie on one line when  $\sigma = \frac{1}{2}$ , the dashed line through  $\zeta/2$  perpendicular to the base  $O \rightarrow \zeta$  (the right angle is marked). What distinguishes one split from another is  $h$ , the signed distance from  $\zeta/2$  along that line, drawn as a thick stretch of the line for the  $R/2$  split (purple,  $h_{R/2} = 2.383$ ) and for the velocity split (teal,  $h^* = 1.619$ ); the two legs of each split are drawn in its own color, the velocity split's legs thickest, and the two chains fainter still. The arc at  $O$  is  $\vartheta$ , swept clockwise off the positive real axis onto the base line, since  $\arg \zeta \equiv -\vartheta \pmod{\pi}$ : with  $\vartheta = 390.884$  reducing to  $75.98^\circ$  modulo  $\pi$ , the base line leaves the axis at  $-75.98^\circ$ , drawn dotted below  $O$ , and  $\zeta$  itself lies on the far side of  $O$  at  $104.02^\circ$  because  $Z = -3.194$  is negative at this height. The two teal arcs mark the angles of the velocity split:  $\vartheta_1^* = \arg B_1^*$  at  $O$ , and the fold angle  $\vartheta_2^* = \arg(B_2^*/B_1^*)$  at  $B_1^*$ , between the extension of leg 1 (dotted) and leg 2. Generated by `fig_splits_h.py`.

**Remark 12.4** ( $N_*$  is monotonically increasing if RH is true). Integer at the ordinates and nowhere else, and up by one from each to the next: what (162) leaves open is whether the curve ever runs backwards in between. It does not, and the reason is a known consequence of the Riemann hypothesis. With  $W = e^{i\vartheta} B_1^* = \frac{1}{2}Z - \frac{iZ'}{2\vartheta'}$  from (161) and  $\pi N_* = \arg W + \frac{3\pi}{2}$ , differentiating and measuring against the local mean density  $\vartheta'$  gives the slope in closed form,

$$\rho_* = \frac{1}{\vartheta'} \frac{d}{dt} \arg W = \frac{\pi}{\vartheta' I'(T)} \frac{dN_*}{dT} = \frac{Z'^2 - ZZ'' + \frac{\vartheta''}{\vartheta'} ZZ'}{\vartheta'^2 Z^2 + Z'^2}, \quad (167)$$

which returns 1 at  $Z = 0$  and  $-Z''/\vartheta'^2 Z$  at  $Z' = 0$ , as it must. Since  $\vartheta'$  and  $I'$  are positive,  $\rho_*$  and  $dN_*/dT$  carry the same sign; the denominator of (167) is positive and  $\vartheta''/\vartheta' = O(1/t \log t)$ , so up to that negligible term the curve increases exactly where the Laguerre expression  $Z'^2 - ZZ''$  is positive.

That expression belongs to  $\Xi$ . Writing  $\Xi(t) = \xi(\frac{1}{2} + it) = -g(t)Z(t)$  with  $g(t) = \frac{1}{2}(\frac{1}{4} + t^2)\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{it}{2})| > 0$ , algebra gives  $\Xi'^2 - \Xi\Xi'' = g^2[(Z'^2 - ZZ'') - (\log g)''Z^2]$ . If the hypothesis holds then every zero of  $\Xi$  is real, its Hadamard product gives  $-(\log \Xi)'' = \sum_k (t - \gamma_k)^{-2}$ , and therefore

$$Z'^2 - ZZ'' = Z^2 \left[ \sum_k \frac{1}{(t - \gamma_k)^2} + (\log g)'' \right], \quad (168)$$

the sum running over the zeros of  $\Xi$  of both signs. The bracket is positive with room to spare: it is of the order of the squared zero density, carried mostly by the two ordinates flanking  $t$ , while  $(\log g)'' = O(t^{-2})$  takes away almost nothing. At  $t = 100.5$  it reads 2.2003, of which the 600 zeros with  $|\gamma| \leq 542$  supply 2.1972 and  $(\log g)''$  removes  $1.7 \times 10^{-4}$ .

Monotonicity is what makes the curve a counting function outright, rather than a curve that merely meets the integers in the right places.  $N_*$  is continuous, it equals  $n$  exactly at  $\gamma_n$ , and it is an integer nowhere else; if it also increases then between consecutive ordinates it is trapped strictly between  $n$  and  $n + 1$ , and taking the floor recovers (152) on the nose,

$$\lfloor N_*(T) \rfloor = N(I(T)), \quad (169)$$

with no error term and no sawtooth. We find no exception at  $5721$  grid points across  $14 \leq t \leq 300$  nor at  $301$  across  $9877 \leq t \leq 9880$ . Three things are wanted for it, and each is worth naming. The count is read off  $Z$ , so what it counts is zeros on the critical line, and it is the hypothesis again that makes those all of them. The zeros must be simple: at a double zero  $Z$  and  $Z'$  vanish together,  $W = 0$ , the argument is undefined, and the curve would climb by one where the count climbs by two; RH does not supply simplicity. And  $\vartheta_1^*$  must be read on a continuous branch. A fourth is bookkeeping rather than mathematics:  $N$  on the right of (169) is the plain count of  $(0, t]$ , so that  $N(\gamma_n) = n$ , and not the midpoint value that §12.1 gives it at an ordinate when reading (152). Subtracting (154) from (162) gives  $\vartheta_1^* = \pi(N_* - N_{\text{rvm}} - \frac{1}{2})$ , so the principal branch serves precisely while the exact count and the smooth one stay within one of each other; since  $S$  is unbounded that must eventually fail, and past it the wrap count has to be carried explicitly, which is the bookkeeping  $N_{\text{ps}}$  already needs at every height. There is room to spare at these heights: over  $14 \leq t \leq 300$  the angle  $|\vartheta_1^*|$  reaches only  $0.506\pi$ .

Read the other way, (167) can only fail where  $ZZ'' > Z'^2$ , which at a critical point of  $Z$  means  $Z$  and  $Z''$  of one sign: a positive local minimum or a negative local maximum, a pair of zeros that comes to the axis and fails to reach it. The Laguerre inequality  $\Xi'^2 \geq \Xi\Xi''$ , which Laguerre proved for every real entire function with only real zeros, is precisely what forbids that. Being a consequence of the hypothesis and not a theorem about  $\zeta$ , it leaves the monotonicity conditional;

and being only the first of a family of inequalities of every order, which Csordas and Varga [7] show to be equivalent to RH when taken all together, it is weaker than the hypothesis. A failure of monotonicity would therefore disprove RH, while a proof of monotonicity would not prove it. Nothing is close to the boundary: on grids of spacing 0.1 in windows from  $t = 10$  up to  $\gamma_{10000}$  the slope never turns negative and never dips below 0.375. The near misses do not threaten it either; they push  $\rho_*$  the opposite way from what one might guess. At the shallow hump inside Lehmer's pair [14], where  $Z = 3.97 \times 10^{-3}$ , one finds  $\rho_* = 458$ , because  $-Z''/\vartheta'^2 Z$  grows without bound as a hump flattens onto the axis. What such a pair costs is margin, not sign:  $Z'^2 - ZZ''$  reads 0.089 at that hump against 1.53 and 1.38 just outside the pair. Verified by `check_nstar_mono.py`.

**Remark 12.5** (regarding the slope of  $N_*$  and the size of  $|Z|$ ). By (163),  $\pi N_* - \frac{3\pi}{2} = \arg(Z - iZ'/\vartheta')$  is the polar angle of the plane point  $(Z, -Z'/\vartheta')$ , the Prüfer angle [19] with  $\vartheta'$  in the role of the frequency, so the slope of the counting curve is fixed by  $Z$ ,  $Z'$  and  $Z''$  at that height alone; against the local mean density it is the  $\rho_*$  of (167). Two values of it are exact. At a simple ordinate, setting  $Z = 0$  in (167) leaves  $Z'^2/Z'^2$ , so  $\rho_* = 1$  (also via (157) with  $h^* = -Z'/2\vartheta'$ ), and the whole of the variation lives strictly between zeros. At an extremum of  $Z$ , where  $Z' = 0$  leaves  $W = \frac{1}{2}Z$  real,

$$\rho_* = -\frac{Z''}{\vartheta'^2 Z} \quad (Z' = 0), \quad (170)$$

so flatness at a hump is exactly a small curvature-to-height ratio, in units of  $\vartheta'^2$ : a broad tall hump gives  $\rho_*$  well below 1 and a sharp small one gives  $\rho_*$  well above it. The flatness of  $N_*$  therefore ranks the size of  $|Z|$  between the zeros: over three blocks of 100 gaps near  $\gamma_{100}$ ,  $\gamma_{1000}$  and  $\gamma_{10000}$ ,  $\rho_*$  at the hump ranks the height of that hump at  $-0.94$  to  $-0.99$ , and even the single number  $\rho'_*(\gamma_n)$ , read at the zero ahead of the hump, still ranks it at  $-0.66$  to  $-0.80$ . Figure 53 plots the slope against  $|Z|$  in two windows and then the two rankings. What the slope really measures is the local phase rate of  $Z$ , not its amplitude:  $Z = A \cos \varphi$  with  $\varphi' = \vartheta'$  gives  $\rho_* = 1$  whatever  $A$  may be, and  $\rho_* < 1$  says the phase turns more slowly than  $\vartheta'$ , which is a wide gap; it is the sparseness of the zeros that lets  $|Z|$  grow ( $\rho_*$  at the hump and the gap width are collinear to  $-0.99$  in rank). Verified in `check_nstar_slope.py`.

| slope at...           |                                               | in $t$ units                                                     | normalized by<br>the mean<br>density: the $\rho_*$<br>of (167) | in $T$ units                          |
|-----------------------|-----------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------|
| an ordinate           | $\left. \frac{dN_*}{dt} \right _{t=\gamma_n}$ | $\frac{\vartheta'(t)}{\pi} = \frac{1}{2\pi} \log \frac{t}{2\pi}$ | $\rho_* = 1$                                                   | $\frac{\vartheta'(I(T)) I'(T)}{\pi}$  |
| an extremum<br>of $Z$ | $\left. \frac{dN_*}{dt} \right _{Z'=0}$       | $-\frac{Z''}{\pi \vartheta' Z}$                                  | $\rho_* = -\frac{Z''}{\vartheta'^2 Z}$                         | $-\frac{Z'' I'(T)}{\pi \vartheta' Z}$ |

Table 2: The exact values of the slope of  $N_*$ . At every simple ordinate the slope is exactly the mean zero density. At a hump of  $Z$  the slope of  $N_*$  is the curvature-to-height ratio of that hump, (170), so one can use the flatness of  $N_*$  to rank the heights of the humps of  $|Z|$ .

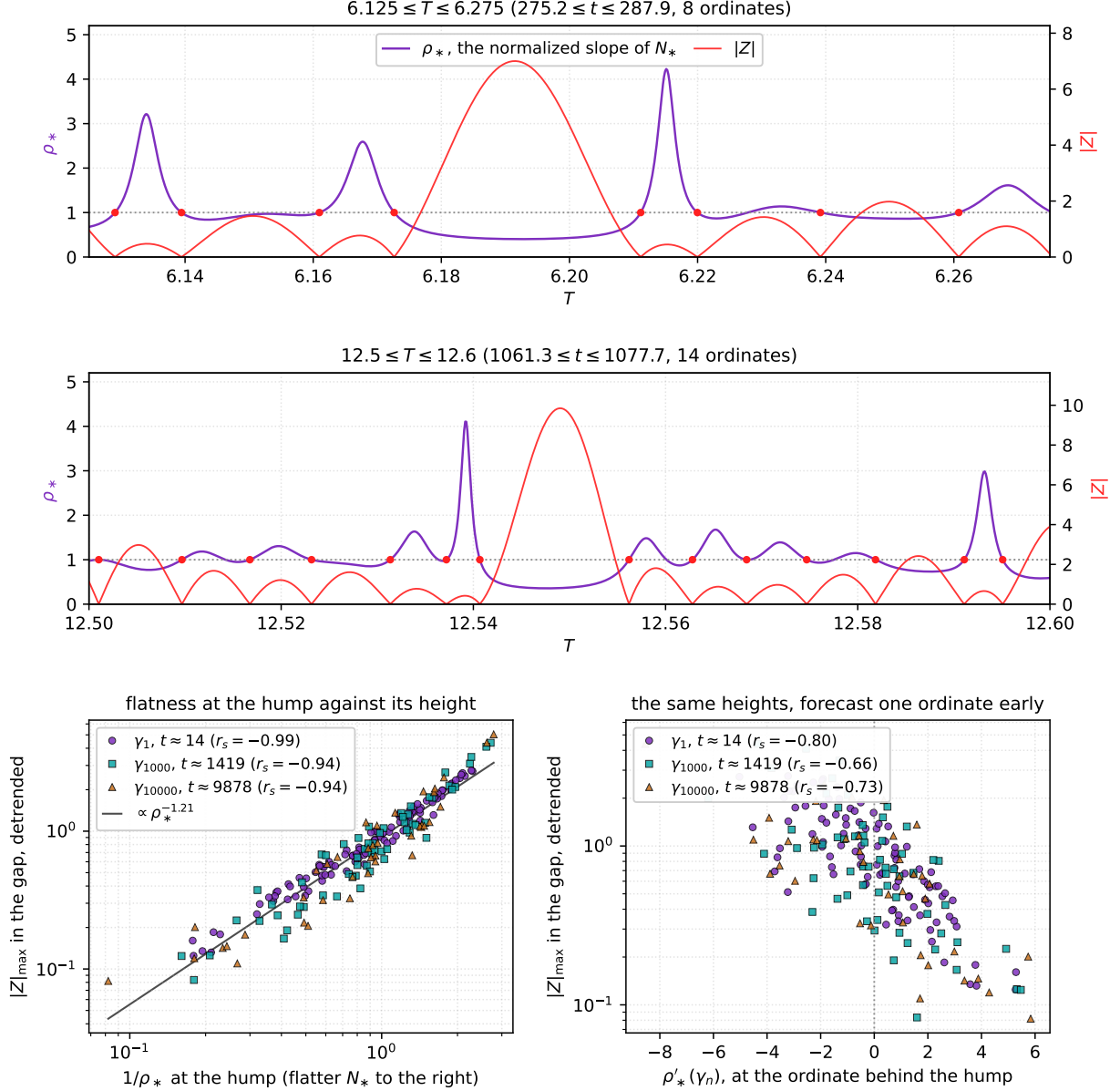


Figure 53: The normalized slope  $\rho_*$  of (167) against the size of  $|Z|$  between ordinates, everything at  $\sigma = \frac{1}{2}$ . *Top*:  $\rho_*$  in purple on the magnified window of Figure 50, with  $|Z|$  in red on the right-hand scale. The red dots mark the ordinates, where  $\rho_* = 1$  exactly; between them  $\rho_*$  dips, and it dips deepest where  $|Z|$  rises highest. *Second*: the same reading twice as high in the index, over  $12.5 \leq T \leq 12.6$ . The hump that reaches  $|Z| = 9.85$  near  $T = 12.549$  sits on the flattest stretch of the window, where  $\rho_*$  bottoms at 0.356, while each ordinary hump carries only a shallow arch of  $\rho_*$  just above 1 and the two narrowest gaps throw up spikes of 4.11 and 2.99. *Bottom left*: one point per gap for three blocks of ordinates spanning two decades of  $t$ , the height of the hump against  $1/\rho_*$  at that hump, both on log scales, with the heights divided by a 31-gap moving average so the slow growth of  $|Z|$  with height does not enter. The line is the fit over all three blocks; the legend quotes the rank correlation against  $\rho_*$  for each block separately. *Bottom right*: the same hump heights against  $\rho'_*(\gamma_n)$ , read at the ordinate behind the hump rather than at the hump itself, scaled by the mean spacing  $\pi/\vartheta'$ . Generated by `fig_nstar_slope.py`.

### 12.3 Connecting the count, $N_*$ , the angle $\vartheta_2$ , and the equal-leg loci

The retrograde ordinates of §12.1 were found by watching the count; the equal-leg ovals of §9 (Figure 19) were found by scanning the strip. They are the same events. Figure 55 puts the two side by side. Its top panel repeats the magnified window of Figure 50. The bottom panel draws, on the same window, the fold angles  $\vartheta_2$  of the  $ps$  split and  $\vartheta_2^*$  of the velocity split together with the equal-leg locus  $L_1 = L_2$  of the  $ps$  split in the surrounding strip: the vertical scale doubles as  $2\pi\sigma$ , with  $\sigma$  on the right-hand axis, so the critical line, which belongs to the locus at every height, is the horizontal blue line lying exactly on the dashed line  $\vartheta_2 = \pi$ , and the off-line components of the locus are the blue ovals straddling it. Shaded are the stretches of  $T$  where  $\vartheta_2$  runs *retrograde*: where the unwrapped angle increases against its prevailing descent.

The correspondence is one to one, three ways. Over  $5.9 \leq T \leq 6.7$  the angle  $\vartheta_2$  runs retrograde on exactly five stretches, each 0.008–0.010 of  $T$  wide. Each stretch contains exactly one ordinate, and those five are precisely the five retrograde ordinates  $\gamma_{114}, \gamma_{125}, \gamma_{128}, \gamma_{142}, \gamma_{145}$  at which  $N_{ps}$  miscounts in Figure 50. And the strip holds exactly five equal-leg ovals over the same window (the two falling in the zoom window are drawn in Figure 55; the other three lie outside it), one per stretch, coinciding with it end to end to about  $10^{-3}$  of  $T$ . The reading is not local to this window: twelve windows spread from  $T = 1$  to  $T = 50$ , holding 344 ordinates between them, give 45 retrograde stretches, 45 ovals and 45 retrograde ordinates, one for one, the stretches narrowing with the ordinate spacing from 0.013 of  $T$  at  $T \approx 3$  to 0.0007 at  $T \approx 50$  (`check_retro_ovals.py`). The only other off-line feature in the window, the thin band at  $T \approx 6.255$ , is not an oval but the pole band of Figure 19: at  $\{T\} \approx \frac{1}{4}$  the weight  $d_1$  passes through its off-line pole (§9.1),  $B_1$  sweeps out and back, and the legs pass through equality on the way out and on the way back at every  $\sigma$  at once, in a band far narrower than any oval. That last property tells the two apart at sight: a band still crosses at  $\sigma = 0.98$ , while an oval has closed by  $\sigma \approx 0.8$ .

The three readings are one event. A retrograde ordinate is one where the offset  $h$  is small (§12.1: median  $|h|$  of 0.319 at the missed ordinates against 1.081 at the rest), which puts the bisector point near  $\zeta/2$ ; there the two legs are nearly equal *and* nearly folded, so  $\vartheta_2$  sits near  $\pi$  and moves slowly, and the slow passage is what lets the angle turn back on itself. The same near-degeneracy is what an oval is. On the critical line the leg difference  $\delta = L_1 - L_2$  vanishes identically, so an off-line zero of  $\delta$  can open up only where the balance survives a small excursion in  $\sigma$ , and that margin exists exactly while the configuration lingers: the oval opens when the stall begins and closes when prograde motion resumes. During ordinary prograde motion the legs rotate past each other too quickly for the balance to hold anywhere off the line, which is why the ovals are confined to the shaded stretches. Read against §12.2, the correspondence also says why the velocity split has no such windows: its offset  $h^* = -Z'/2\vartheta'$  alternates by construction, no crossing is retrograde, and  $N_*$  inherits none of the stalls that mark  $N_{ps}$ . What is special about the velocity split is therefore not the definition of its fold angle  $\vartheta_2^* = \arg(B_2^*/B_1^*)$ , which is exactly as for the other splits, but its behavior: its legs are exactly equal at every height on the line, and  $\vartheta_2^*$  passes  $\pi$  prograde at every ordinate, with no retrogrades. That is precisely the property that makes  $\vartheta_1^*$ , and hence  $N_*$ , count correctly. Figure 54 draws the two fold angles over the same window.

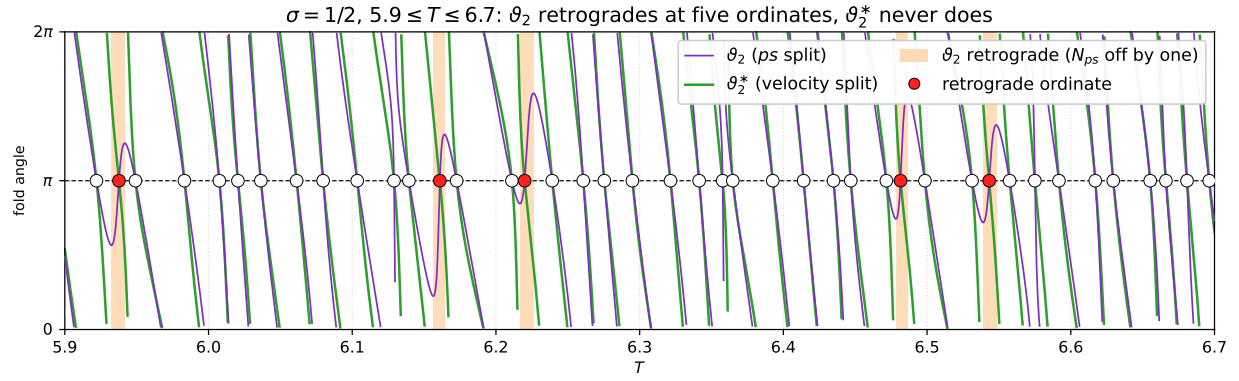


Figure 54: The two fold angles compared over  $5.9 \leq T \leq 6.7$  at  $\sigma = \frac{1}{2}$ :  $\vartheta_2$  of the *ps* split (purple) and  $\vartheta_2^*$  of the velocity split (green), both reduced to  $[0, 2\pi)$  and broken at their wraps. Dots mark the ordinates, where both angles pass through the dashed line at  $\pi$ , filled red at the retrograde ones. On the shaded stretches  $\vartheta_2$  runs retrograde: it stalls at  $\pi$ , turns back, and recrosses, and on that stretch  $N_{\text{ps}}$  is off by one. The green curve has no retrograde stretch anywhere: at every ordinate  $\vartheta_2^*$  passes  $\pi$  prograde, and  $N_*$  is never off the count. Generated by `fig_theta2_star_compare.py`.

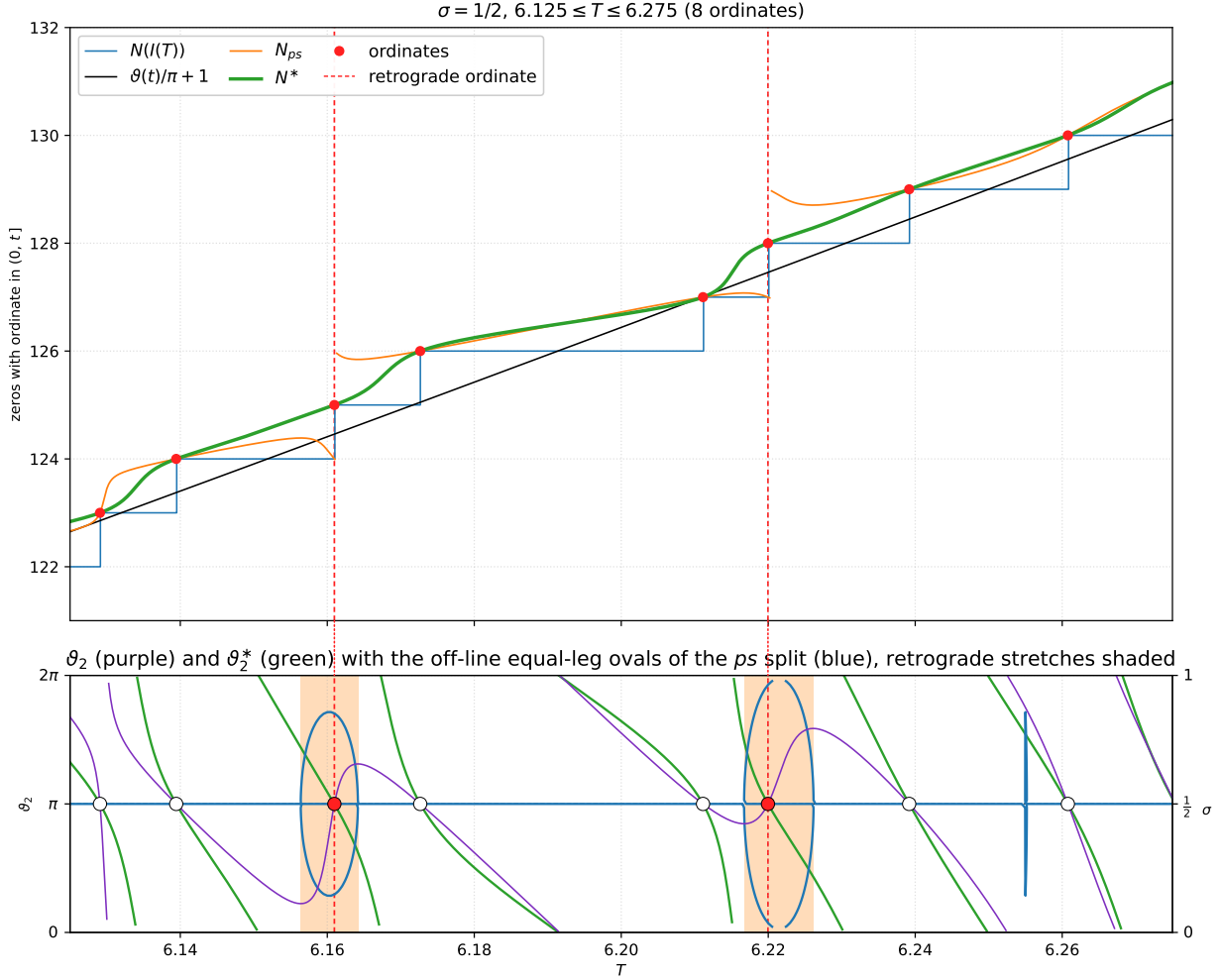


Figure 55: The count, the fold angle, and the equal-leg locus of the  $ps$  split read together, everything at  $\sigma = \frac{1}{2}$  except the locus. *Top:* the magnified window  $6.125 \leq T \leq 6.275$  of Figure 50: the staircase (blue), the smooth part (black),  $N_{ps}$  (orange),  $N_*$  (green), and the retrograde ordinates dashed in red. *Bottom:*  $\vartheta_2$  (purple) and  $\vartheta_2^*$  (green) on the same window, reduced to  $[0, 2\pi)$ , with the equal-leg locus  $L_1 = L_2$  of the  $ps$  split drawn in the surrounding strip (blue) on the shared vertical scale  $2\pi\sigma$ ,  $\sigma$  on the right-hand axis. The horizontal blue line on the dashed  $\vartheta_2 = \pi$  line is the critical line, which belongs to the locus at every height; the two blue ovals are the off-line components in this window. Dots mark the ordinates, where both angles pass through  $\pi$ , filled red at the retrograde ones; the shaded stretches are where  $\vartheta_2$  runs retrograde, and there  $\vartheta_2^*$  passes straight through. Each oval fills a shaded stretch and each stretch holds one retrograde ordinate. The thin blue band at  $T \approx 6.255$  is the pole band of Figure 19 at  $\{T\} \approx \frac{1}{4}$ , not an oval. Generated by `fig_count_theta2_ovals.py`.

## 12.4 Positivity of the slope under the Riemann hypothesis

Remark 12.4 recorded that the monotonicity of  $N_*$  is a known consequence of the Riemann hypothesis, through the Laguerre-type inequality (168). Here we give the proof in full, in a form that shows exactly where the hypothesis enters and how much room the inequality has.



**Theorem 12.6** (RH implies  $N_*$  is eventually strictly increasing). *Assume the Riemann hypothesis. Then there is a  $t_0$  such that  $\rho_*(t) > 0$  for every  $t \geq t_0$  that is not an ordinate; together with  $\rho_* = 1$  at the (simple) ordinates (Remark 12.5),  $N_*$  is strictly increasing there. Grid computations (Remarks 12.4 and 12.5) support positivity down through the first zero.*

*Proof. Step 1: reduction to a monotonicity statement (unconditional).* Between consecutive zeros  $Z \neq 0$ , so we may factor  $W = \frac{1}{2}Z(1 - iu)$  with

$$u(t) = \frac{Z'(t)}{\vartheta'(t)Z(t)}.$$

On such an interval  $\arg Z$  is constant (0 or  $\pi$ ), hence  $\arg W = \text{const} - \arctan u$  and

$$\rho_* = \frac{1}{\vartheta'} \frac{d}{dt} \arg W = \frac{-u'}{\vartheta'(1 + u^2)}.$$

Since  $\vartheta' > 0$ , proving  $\rho_* > 0$  is exactly proving that  $u$  is strictly decreasing on every gap ( $u$  falls from  $+\infty$  just after a zero to  $-\infty$  just before the next).

*Step 2: the Hadamard product (first use of RH).* Let  $\Xi(t) = \xi(\frac{1}{2} + it)$ , where  $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ . Then  $\Xi = -gZ$  with  $g(t) = \frac{1}{2}(t^2 + \frac{1}{4})\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{it}{2})|$  the smooth positive factor of Remark 12.4. From the Hadamard product  $\xi'/\xi(s) = \sum_{\rho} (s-\rho)^{-1}$  (zeros paired symmetrically), the Riemann hypothesis puts every zero at  $\rho = \frac{1}{2} + i\gamma_k$ , so on  $s = \frac{1}{2} + it$  the differences  $s-\rho = i(t-\gamma_k)$  are purely imaginary and

$$\frac{\Xi'}{\Xi}(t) = \sum_k \frac{1}{t - \gamma_k}, \quad (\log \Xi)''(t) = -\sum_k \frac{1}{(t - \gamma_k)^2} =: -S_2(t) < 0.$$

This is the structural use of the hypothesis: it makes every term of  $S_2$  a positive square. An off-line zero  $\beta + i\gamma$  with  $\beta \neq \frac{1}{2}$  would contribute  $-\text{Re}[(\frac{1}{2} - \beta) + i(t - \gamma)]^{-2}$ , which turns *negative* for  $|t - \gamma| < |\beta - \frac{1}{2}|$ : near an off-line zero the term loses its sign protection, and it is exactly the hypothesis that forbids this.

*Step 3: differentiate  $u$ .* With  $Z = -\Xi/g$  we have  $Z'/Z = \Xi'/\Xi - g'/g$ , hence

$$-\vartheta' u' = \underbrace{\sum_k \left[ \frac{1}{(t - \gamma_k)^2} + \frac{\vartheta''}{\vartheta'} \cdot \frac{1}{t - \gamma_k} \right]}_{\text{zero terms}} + \underbrace{(\log g)'' - \frac{\vartheta''}{\vartheta'} \cdot \frac{g'}{g}}_{=: E(t)}.$$

*Step 4: positivity of the zero terms, smallness of  $E$ .* Each bracket equals  $[1 + \frac{\vartheta''}{\vartheta'}(t - \gamma_k)]/(t - \gamma_k)^2$ , positive whenever  $|t - \gamma_k| < \vartheta'/\vartheta'' \approx t \log \frac{t}{2\pi}$ ; so every zero within distance  $t \log t$  of  $t$  contributes positively. The distant zeros can contribute negatively only through their  $\frac{\vartheta''}{\vartheta'} \frac{1}{t - \gamma_k}$  tails; pairing  $\gamma$  with  $-\gamma$  and using the counting density, their total is  $O(\vartheta''/\vartheta') = O(1/(t \log t))$ . Stirling gives  $E(t) = O(1/t)$ . Meanwhile, on the hypothesis  $S(t) = O(\log t / \log \log t)$  [22, Theorem 14.13], so  $N(t+1) - N(t-1) = \frac{2}{\pi} \vartheta'(t) + o(\log t) > 0$  for all large  $t$ : the interval  $(t-1, t+1)$  contains a zero,  $\text{dist}(t, \{\gamma_k\}) \leq 1$ , and the nearest zero's bracket alone contributes  $[1 - \vartheta''/\vartheta']/\text{dist}^2 \geq \frac{1}{2}$  once  $t$  is large. Hence

$$-\vartheta' u' \geq \frac{1}{2} - O(1/t) > 0$$

for all  $t$  beyond a modest  $t_0$ , so  $u' < 0$  and  $\rho_* > 0$  on every gap. Below  $t_0$  the argument gives nothing. There the computations reported in Remarks 12.4 and 12.5 sample the range and find grid minima 0.356–0.521, nowhere near 0; that is evidence, not a proof of positivity between the sample points.  $\square$

**Remark 12.7** (where the hypothesis is indispensable). Steps 2 and 4 use the Riemann hypothesis, and they use it differently. Step 2 needs it structurally, to make every term of  $S_2$  a positive square; Step 4 needs only the size bound  $S(t) = O(\log t / \log \log t)$ , and only to know that an ordinate lies within distance 1 of  $t$ . The unconditional  $S(t) = O(\log t)$  is the same order as the main term  $\frac{2}{\pi} \vartheta'(t)$  and will not serve. Everything else is bookkeeping. This also shows the converse fails gracefully:  $\rho_* > 0$  everywhere is *weaker* than the hypothesis. It is essentially the first Laguerre-type inequality (168), and the theorem of Csordas and Varga [7] says one needs the whole infinite family of higher-order inequalities to recover the hypothesis. Unconditionally, even  $\rho_* > 0$  is open. A negative value would contradict the hypothesis only above the threshold  $t_0$  of Theorem 12.6, which we do not make effective, so no single observation at a finite height settles anything by itself; what the theorem forbids under RH is negative values at unbounded heights.

**Remark 12.8** (the theorem doesn't need simple zeros as an extra assumption: you get away with RH alone). At a hypothetical multiple zero the reduction of Step 1 degenerates:  $u$  is undefined at the zero itself and  $\rho_*$  there is  $0/0$ . Between zeros the argument is unaffected. The hypothesis does not by itself rule out multiple zeros, so strictly the statement is:  $\rho_* > 0$  away from ordinates, and  $\rho_* = 1$  at simple ordinates (Remark 12.5). All known zeros are simple.

**Remark 12.9** (what keeps the curve climbing up). The proof makes quantitative what the figures show:  $\rho_*$  can dip only as low as the widest local gap forces, through  $S_2$  shrinking to its two nearest terms, but never to zero; under the hypothesis the sum of inverse-square distances to the zeros is the floor holding the counting curve up.

**Remark 12.10** (sharpness: no uniform bounds are possible). Between zeros no bound better than  $0 < \rho_* < \infty$  can hold, and the reason is a conservation law. Across each gap  $W$  swings from one crossing of the imaginary axis to the next, so  $\arg W$  advances by exactly  $\pi$ :

$$\int_{\gamma_k}^{\gamma_{k+1}} \rho_* \vartheta' dt = \pi,$$

that is, the density-weighted mean of  $\rho_*$  over a gap is  $\pi/\Delta\vartheta \approx 1/(\text{gap length in mean spacings})$ . Every gap delivers exactly one zero's worth of angle; the only freedom is the distribution inside. A gap much narrower than average therefore forces  $\rho_*$  large (at the Lehmer pair [14]  $\gamma_{6709}, \gamma_{6710}$  the hump value is  $\rho_* = 458$ , Remark 12.4), and arbitrarily close pairs, which GUE statistics predict, force  $\rho_*$  arbitrarily large. The coincident case is not the limit of that, though. At a zero of multiplicity  $r$  the numerator of (167) is  $c^2 r x^{2r-2}$  and the denominator  $c^2 r^2 x^{2r-2}$ , writing  $Z = cx^r + \dots$  with  $x = t - \gamma$ , so  $\rho_*$  is undefined at the zero itself and approaches  $1/r$  around it: 1 at a simple zero, as it must, and  $\frac{1}{2}$  at a double one. What diverges as two simple zeros merge is the peak between them, not the value where they meet. Symmetrically, a gap twice the mean forces the mean of  $\rho_*$  over it below  $\frac{1}{2}$ , and arbitrarily long gaps force  $\rho_*$  arbitrarily near 0, so the infimum over all  $t$  is (conjecturally) 0. The measured minima of Remark 12.4 drift accordingly with the widest gap sampled, down to the 0.356 at the flat stretch under the tall  $|Z| = 9.85$  hump of Figure 53: big humps sit in wide gaps, and wide gaps mean slow counting. The one hard bound is the sign (Remark 12.7). A useful mental model:  $\rho_*$  behaves like the reciprocal of the local normalized gap size, smoothed: pinned to exactly 1 at every zero, integrating to exactly one count per gap, and free to spike or stall in between precisely as much as the zero spacing deviates from its mean.

**Remark 12.11** (recap on what  $N$  and  $N_*$  count).  $N$  is a contour count (Theorem 9.3 of Titchmarsh [22], already cited in §12.1), so its rectangle catches a zero at any  $\sigma$ ; an off-line pair  $\beta + i\gamma$ ,  $1 - \beta + i\gamma$  shares one ordinate and would bump  $N$  by two.  $N_*$  is built from  $Z$  and  $Z'$ , pure

critical-line data; it advances only when  $W = \frac{1}{2}Z - iZ'/2\vartheta'$  of (161) crosses the imaginary axis, which requires  $Z = 0$ , and only on-line zeros do that; off the line  $|Z|$  merely dips,  $W$  bows past the origin without crossing, and the curve glides through without stepping. That gap is why  $\lfloor N_* \rfloor = N$  doubles as an off-line-zero detector, which Table 3 and Remark 12.4 already assert.

## 12.5 Summary: $N$ and $N_*$ side by side

$N_*$  contains more and less information than  $N$ . More because it is continuous and (on RH) increasing between ordinates, and less because it does not count off-half-line zeros. Table 3 collects the comparison of the two counting functions in one place, and Figure 56 draws them side by side: the top panel is the full window  $6 \leq T \leq 8.5$ , and the three panels below it magnify the outlined bands.

|                             | $N$ of (152)                                                                                                                                                                              | $N_*$ of (163)                                                                                                       |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| definition                  | $\frac{\vartheta(t)}{\pi} + 1 + \frac{1}{\pi} \arg \zeta\left(\frac{1}{2} + it\right)$                                                                                                    | $\frac{1}{\pi} \arg\left(Z - \frac{iZ'}{\vartheta'}\right) + \frac{3}{2}$                                            |
| continuity                  | step function; jumps at every ordinate                                                                                                                                                    | continuous                                                                                                           |
| at the ordinate $\gamma_n$  | jumps from $n - 1$ to $n$                                                                                                                                                                 | equals $n$ exactly                                                                                                   |
| integer values              | constant integer between ordinates                                                                                                                                                        | integer at the ordinates and nowhere else                                                                            |
| beside the approximate form | the smooth $\frac{t}{2\pi} \log \frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8} + \frac{1}{48\pi t}$ of (154) threads the staircase in the mean, never landing on an integer at an ordinate | meets the integer exactly at every ordinate, where that approximation never does                                     |
| monotonicity                | nondecreasing by construction                                                                                                                                                             | increasing, eventually and conditional on RH (Theorem 12.6)                                                          |
| what it counts              | all zeros with $0 < \text{Im } s \leq t$ , on or off the critical line                                                                                                                    | only the zeros on the critical line                                                                                  |
| how it sees them            | the argument principle: $\xi$ winds once around a rectangle enclosing the whole strip up to height $t$ for every zero inside, wherever it sits                                            | read off $Z$ alone: it passes an integer exactly where $Z$ changes sign, and an off-line zero never makes $Z$ vanish |
| one from the other          | no converse: $N$ does not determine $N_*$                                                                                                                                                 | $\lfloor N_* \rfloor = N$ if $N_*$ increases, by (169)                                                               |

Table 3: The exact staircase  $N$  beside the continuous counting curve  $N_*$ . The  $N_*$  column reads the argument of (163) on its continuous branch throughout, and its continuity and its exact integer values at the ordinates both need the zeros there to be simple: at a multiple zero  $Z$  and  $Z'$  vanish together,  $B_1^* = 0$ , and the angle is undefined.

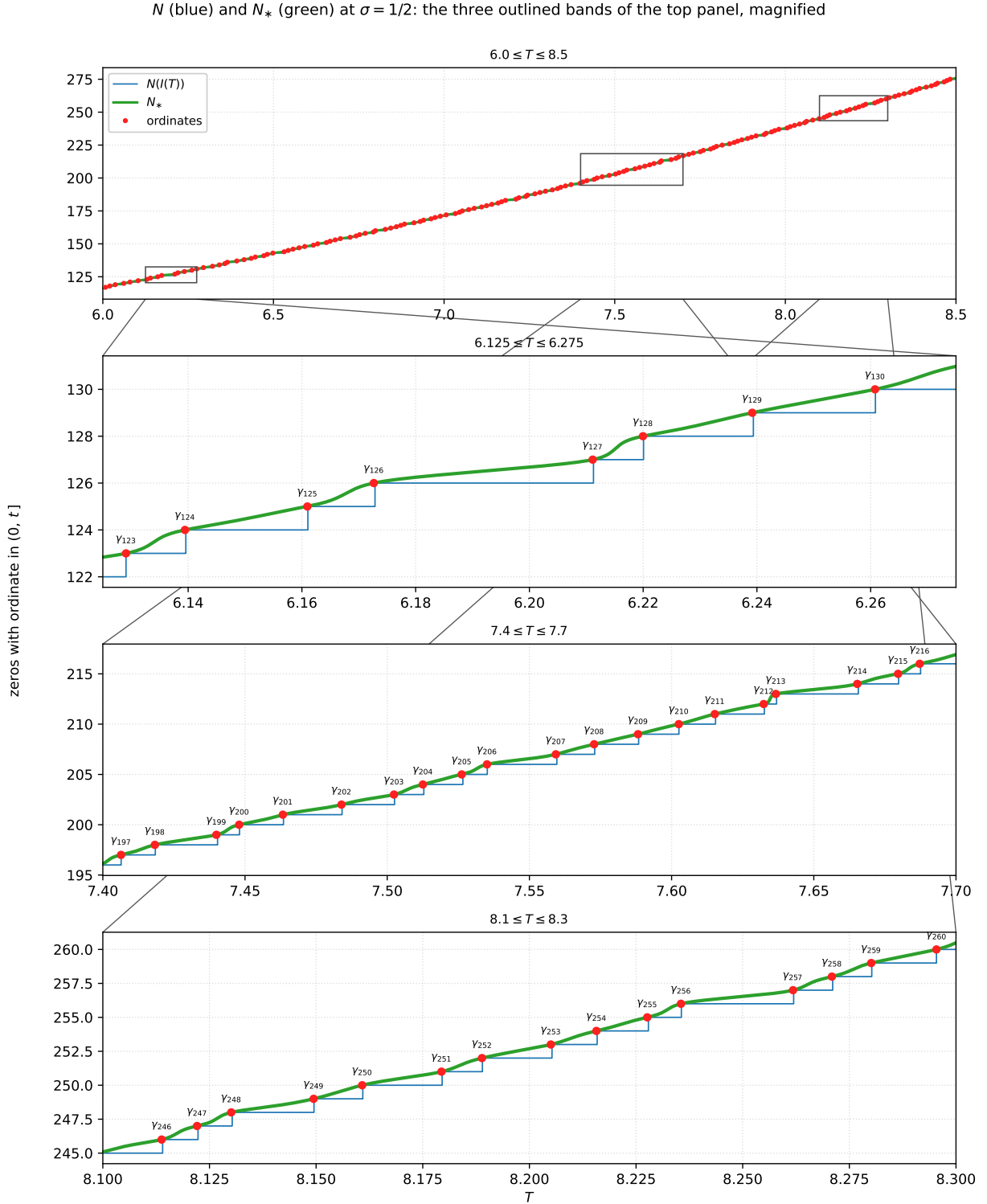


Figure 56: Head-to-head comparison of  $N$  (blue) and  $N_*$  (thick green) at  $\sigma = \frac{1}{2}$  with the ordinates as red dots. The top panel is the full window  $6 \leq T \leq 8.5$ , holding the 159 ordinates  $\gamma_{117}$  through  $\gamma_{275}$ ; the three outlined bands are magnified in the panels below it:  $6.125 \leq T \leq 6.275$  (the window of Figure 55),  $7.4 \leq T \leq 7.7$ , and  $8.1 \leq T \leq 8.3$ . In the magnified panels each ordinate carries its label  $\gamma_n$ . Generated by `fig_N-vs-Nstar_strips.py`.

## 13 Further observations

We have made other observations of the zeta function, mostly from a geometric perspective, and have listed some of them in this final section:

1. we examine the ratio of the lengths of the two partial sums beside the ratio of the two partial summand remainders (§13.1);
2. we look at the envelope of  $\zeta$  on the half-line from a  $B_1$  and a  $B_1^*$  perspective (§13.2);
3. since we now have three remainders, we look at their collinearity, which we could not do when we had just two, because any two points already lie on a line (§13.3);
4. and since looking at the links separately, each in its own coordinate frame, was fruitful (it led to the general partial summand formula for  $\zeta$  of §11.6), here we look at the joint angles (§13.4).

### 13.1 Length ratios: $\Sigma_1$ to $\Sigma_2$ , and $R_1$ to $R_2$

The four terms of (22) come in two natural pairs: the two partial sums, and the two fractional links. Within each pair the two lengths are equal on the critical line, and what follows records what their ratio does off it. The pairs behave quite differently, and in both cases the ratio obeys the same reflection law about  $\sigma = \frac{1}{2}$ .

**Remark 13.1** (regarding the ratio of the lengths of  $\Sigma_1$  and  $\Sigma_2$ ). On the critical line  $\sigma = \frac{1}{2}$  we have  $|\chi| = 1$ , so the  $\Sigma_2$  links have the same lengths  $(n+1)^{-1/2}$  as the  $\Sigma_1$  links; each link is then a rotation composed with a uniform link-length shrink  $\sqrt{n/(n+1)}$ , and  $\arg \chi = -2\vartheta(t)$ , with  $\vartheta$  the Riemann–Siegel theta function. The two chains are thus congruent link for link there, so their end-to-end displacements agree. Away from the critical line they do not, and the discrepancy has a closed form. Write  $\ell_{\Sigma_1} = |\Sigma_1|$  and  $\ell_{\Sigma_2} = |\Sigma_2|$  for the lengths of the net displacements of the two chains, and put  $\Sigma_m(z) = \sum_{n \leq m} n^{-z}$ . Since  $\sum_{n \leq m} n^{s-1} = \Sigma_m((1-\sigma) + it)$ , the second leg is the first one read at the reflected abscissa  $1-\sigma$ , scaled by  $|\chi|$ :

$$\frac{\ell_{\Sigma_1}}{\ell_{\Sigma_2}} = \frac{|\Sigma_m(\sigma + it)|}{|\chi(\sigma + it)| |\Sigma_m((1-\sigma) + it)|}. \quad (171)$$

**Proposition 13.2.** *At fixed  $T$  (hence fixed  $t = I(T)$  and  $m = \lfloor T \rfloor$ ), at every  $\sigma$  for which all four lengths are nonzero,*

$$\frac{\ell_{\Sigma_1}(\sigma)}{\ell_{\Sigma_2}(\sigma)} \cdot \frac{\ell_{\Sigma_1}(1-\sigma)}{\ell_{\Sigma_2}(1-\sigma)} = 1. \quad (172)$$

Write (171) at  $\sigma$  and at  $1-\sigma$  and multiply. The two partial-sum moduli cancel, leaving  $1/(|\chi(\sigma + it)| |\chi((1-\sigma) + it)|)$ . From  $\chi(s)\chi(1-s) = 1$  and the Schwarz reflection  $\chi(\bar{s}) = \overline{\chi(s)}$  one has  $\chi(1-\sigma + it) = 1/\overline{\chi(\sigma + it)}$ , so the product of the moduli is 1. The ratio is therefore reciprocal-symmetric about  $\sigma = \frac{1}{2}$ : whatever it does on the right half of the strip, it does the reciprocal of on the left, with the value pinned to 1 at the center by the link-for-link congruence of Remark 13.1.

**Remark 13.3** (regarding the ratio of the lengths of  $R_1$  and  $R_2$ ). The same question for the two fractional links has a shorter answer, because in the quotient of (17) and (18) both  $|R|$  and the common denominator  $\sin(2\omega + \psi)$  cancel:

$$\frac{d_1}{d_2} = \frac{\sin(\omega + \psi - \arg R)}{\sin(\omega + \arg R)}, \quad \frac{|R_1|}{|R_2|} = \left| \frac{d_1}{d_2} \right|. \quad (173)$$

The signed quotient is the one that carries the information; the two moduli see only its absolute value, and  $d_1/d_2$  does go negative off the line, as the fourth point below records. This ratio is pure angle data, carrying no length scale at all: it is the law of sines in the triangle of Figure 9, recording how the direction of  $R$  divides the wedge spanned by the two link directions  $e^{-i\omega}$  and  $e^{i(\omega+\psi)}$ . Four consequences are worth recording. First, on the critical line  $\arg R = \frac{\psi}{2}$ , the two sines coincide and the ratio is 1, which is Proposition 4.1(iii) again. Second, the ratio at  $\sigma$  is the reciprocal of the ratio at  $1 - \sigma$ :

**Proposition 13.4.** *At fixed  $T$ , at every  $\sigma$  for which  $d_2(\sigma)$  and  $d_2(1 - \sigma)$  are nonzero, so that both ratios are finite,*

$$\frac{d_1(\sigma)}{d_2(\sigma)} \cdot \frac{d_1(1 - \sigma)}{d_2(1 - \sigma)} = 1. \quad (174)$$

The turn  $\omega$  depends only on  $T$ . As above, one has  $\chi(1 - \sigma + it) = 1/\overline{\chi(\sigma + it)}$ , so  $\psi$  is the same at both abscissae. From (7) and  $\zeta(s) = \chi(s)\zeta(1 - s)$  the remainder satisfies  $R(1 - s) = \chi(1 - s)R(s)$ ; Schwarz reflection  $R(\bar{z}) = \overline{R(z)}$  then gives

$$R(1 - \sigma, T) = \chi(1 - \sigma + it) \overline{R(\sigma, T)}, \quad (175)$$

hence  $\arg R(1 - \sigma) = \psi - \arg R(\sigma)$ . The two sines of (173) therefore exchange, and the product is 1. On the critical line (175) is  $R = \chi R$ , which is the first point again.

Third, because the scale cancels, this ratio is far tamer than (171): sampled on a grid over  $1 < T < 20$  and the whole strip it runs from 0.387 to 2.583 (a reciprocal pair, as the second point requires). Those are not bounds. A grid steps over the isolated loci of the fourth point below, where the ratio is negative, or 0, or infinite; what the sample shows is that away from them the ratio stays within a factor of three of 1, whereas  $\ell_{\Sigma_1}/\ell_{\Sigma_2}$  ranges over several orders of magnitude there. Fourth, the parallel-link poles of §A.5 cancel out of it: where  $2\omega + \psi = k\pi$ , (173) gives  $d_1/d_2 = (-1)^{k+1}$ , exactly  $\pm 1$ , even though  $d_1$  and  $d_2$  separately diverge (at  $\sigma = \frac{1}{10}$ ,  $T = 4.256594$  we compute  $d_1 = -d_2 \approx -4.07 \times 10^{36}$ ). A negative value is off-line news of a different kind: it says  $R$  has left the wedge, which happens in a narrow window around each pole. The ratio does reach 0 and  $\infty$ , but only on the isolated loci where  $R$  is exactly parallel to one of the two link directions; for instance  $d_1 = 0$  at  $\sigma = \frac{1}{10}$ ,  $T = 4.2607679$ , whose mirror at  $\sigma = \frac{9}{10}$  has  $d_2 = 0$ . Off the critical line the locus  $d_1 = d_2$  is the family of ovals plotted in Figure 24.

### 13.2 The envelope of $|\zeta(\frac{1}{2} + it)|$

On the critical line the chain is isosceles for every reflection-symmetric split, by Corollary 9.1, so its two legs share one length  $L_1 = |B_1| = |B_2|$  and the base is the chord they subtend across the fold angle of the legs  $\vartheta_2$  of §9.2. From  $\zeta = B_1(1 + e^{i\vartheta_2})$ ,

$$|\zeta(\frac{1}{2} + it)| = 2L_1 |\cos \frac{\vartheta_2}{2}|, \quad (176)$$

so the modulus of  $\zeta$  splits into a length carried by the links and an angle read at the joint between the legs. Bounding the angular factor bounds  $\zeta$ . Write  $u = \vartheta_2/\pi$  reduced to  $[0, 2)$ . Then  $|\cos \frac{\pi u}{2}| \leq 1$  at once, and since  $\cos \frac{\pi u}{2}$  is concave on  $0 \leq u \leq 1$  it stays above the chord joining its endpoints there, giving  $|\cos \frac{\pi u}{2}| \geq |1 - u|$  on the whole of  $[0, 2)$  by the reflection  $u \mapsto 2 - u$ , with equality only at  $u = 0, 1, 2$ . Hence

$$2L_1 |1 - u| \leq |\zeta(\frac{1}{2} + it)| \leq 2L_1, \quad u \equiv \frac{\vartheta_2}{\pi} \pmod{2}, \quad 0 \leq u < 2. \quad (177)$$

The reduction matters:  $\vartheta_2$  is the continuous lift whose turns do the counting in §12, so  $\vartheta_2/\pi$  itself grows without bound and would make the left side exceed the right. The upper envelope is attained where the two legs point the same way; the lower one is attained at every ordinate, where  $u = 1$  and both sides vanish, and again where  $u$  reaches 0 or 2 and the two envelopes meet. The triangular factor  $|1 - u|$  is the same sawtooth whose passages through zero do the counting in §12, appearing here with an amplitude on it. Over  $6 \leq T \leq 13$ , which is  $264.9 \leq t \leq 1144.6$ , we confirm (176) to  $1.7 \times 10^{-20}$  and both inequalities of (177) at every one of 2001 sample points, the two legs agreeing to  $2.0 \times 10^{-20}$ .

Which split to use is settled by how often the upper envelope is reached. Equation (156) gives  $2L_1 = \sqrt{Z^2 + 4h^2}$ , so  $|\zeta| = 2L_1$  exactly where the offset  $h$  vanishes. For the split at the partial summand  $h$  changes sign inside a gap exactly when its two ends carry opposite  $h$ , which is the alternation that the count of §12 needs and that fails at a retrograde ordinate, so the envelope is reached in some gaps and missed in others: 3 of the 7 gaps of  $6.125 \leq T \leq 6.275$ , and 7 of the 13 gaps of  $12.5 \leq T \leq 12.6$ . Those are counts of sign changes; a gap whose two ends agree could in principle still hold an even number of zeros of  $h$ , which would only add touches to the tally. The velocity split has no such gaps. Its offset  $h^* = -Z'/2\vartheta'$  vanishes at every extremum of  $Z$ , so

$$2L_1^* = \sqrt{Z^2 + \left(\frac{Z'}{\vartheta'}\right)^2} \geq |Z| = \left|\zeta\left(\frac{1}{2} + it\right)\right|, \quad (178)$$

with equality on every hump: 7 touches in 7 gaps and 13 in 13 in those two windows, matched to  $10^{-25}$  (Figure 57). This is the familiar envelope of an oscillation of instantaneous frequency  $\vartheta'$ , with  $Z'/\vartheta'$  as the quadrature companion of  $Z$ , and it is tighter on average as well, the mean of  $|\zeta|/2L_1$  over  $6 \leq T \leq 13$  being 0.635 against 0.586 for the split at the partial summand.

Both readings are polar coordinates of one function. The rotated bisector point  $W = e^{i\vartheta} B_1^* = \frac{1}{2}Z - \frac{iZ'}{2\vartheta'}$  of (161) has modulus  $L_1^*$  and argument  $\pi(N_* - \frac{3}{2})$  by (162), and  $Z = 2 \operatorname{Re} W$ , so

$$Z(t) = -2|W| \sin(\pi N_*(T)), \quad (179)$$

which we confirm to  $6 \times 10^{-13}$ , the precision of the counting curve itself. This is a restatement rather than a new fact, but it puts the two subsections together: the amplitude of the velocity split is the envelope of  $\zeta$  and its phase is the counting curve, so the zeros are the moments when the phase passes an integer and the extrema are the moments when the amplitude is attained. Remark 12.5 reads the rate of that phase; the envelope (178) reads its amplitude. Verified in `check_envelope.py`.



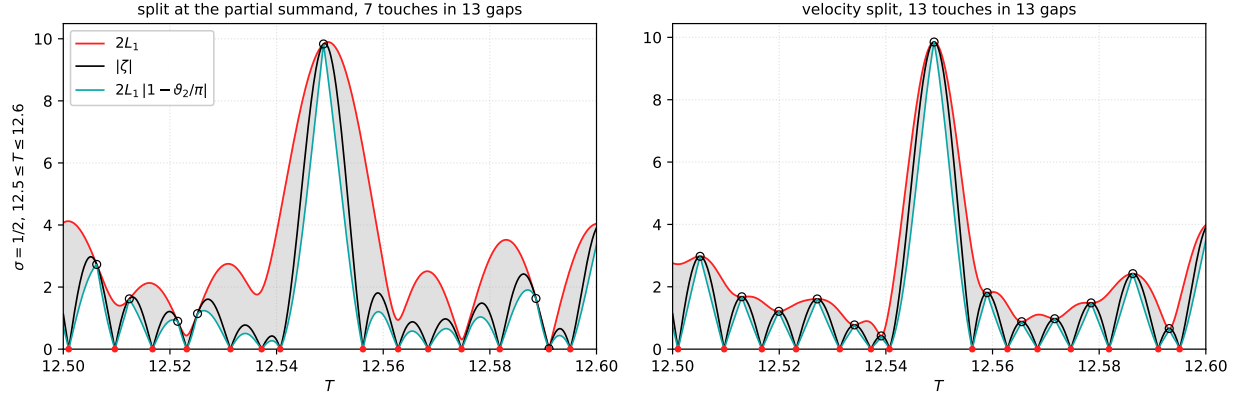


Figure 57: The two envelopes of (177), at  $\sigma = \frac{1}{2}$ , over  $12.5 \leq T \leq 12.6$ , which is  $1061.3 \leq t \leq 1077.7$  and holds 14 ordinates. In each panel  $|\zeta|$  is black, the upper envelope  $2L_1$  red, the lower envelope  $2L_1|1 - \vartheta_2/\pi|$  a teal sawtooth, and the gray band between them is where  $|\zeta|$  is confined. Red dots on the axis are the ordinates, where the lower envelope touches zero along with  $|\zeta|$ , and the open circles are the touches of the upper envelope, one for each vanishing of the offset  $h$ . *Left*: the split at the partial summand,  $B_1 = \Sigma_1 + R_1$ , whose  $h$  fails to alternate at a retrograde ordinate, so it reaches its bound in only 7 of the 13 gaps and several humps rise nowhere near it. *Right*: the velocity split, whose  $h^*$  vanishes at every extremum of  $Z$ , so the upper envelope is touched in every gap. Generated by `fig-envelope.py`.

**Remark 13.5** (regarding hunting for large values with the first few links). Neither the envelope nor the slope of Remark 12.5 can be used to scout, since  $2L_1$  needs  $R = \zeta - \Sigma_1 - \Sigma_2$  and  $2L_1^*$  needs  $Z$  and  $Z'$ , so both are functions of  $\zeta$  at a point one has already evaluated. What can be computed without  $\zeta$  is the reason the envelope grows. Leg 1 is mostly the link sum, so the size of  $2L_1$  is a question of how well the link phases  $t \log n$  align, and the ceiling is the aligned case,

$$|\zeta(\tfrac{1}{2} + it)| \leq 2L_1 \leq 2 \sum_{n \leq m} n^{-1/2} + 2|R_1| \leq 4\sqrt{m} \leq 4\sqrt{T} < 4\left(\frac{t}{2\pi}\right)^{1/4} = \frac{4}{(2\pi)^{1/4}} t^{1/4} < 2.54 t^{1/4}, \quad (180)$$

a bound that costs nothing to evaluate. Every step is an inequality. The middle one uses  $\sum_{n \leq m} n^{-1/2} \leq 2\sqrt{m} - 1$ , which follows by induction from  $(m+1)^{-1/2} \leq 2\sqrt{m+1} - 2\sqrt{m}$ , together with  $|R_1| = \hat{d}_1 M^{-1/2} \leq 1$ , so the fractional link is paid for by the  $-1$  and does not disappear. The last inequality in  $T$  is  $t > 2\pi T^2$ , which holds for every  $T > 0$  because (9) makes it  $2x + x^2 > 2\log(1+x)$  at  $x = 1/T$ . Read in  $t$  it is nothing new: the exponent  $\frac{1}{4}$  is the convexity, or trivial, bound  $\mu(\frac{1}{2}) \leq \frac{1}{4}$  of Lindelöf [16], where  $\mu(\sigma) = \inf\{a : \zeta(\sigma + it) = O(t^a)\}$ , and (180) recovers it with an explicit constant. Nor can this path do better. The last step is the triangle inequality, which keeps only the aligned case and so discards every cancellation among the phases  $t \log n$ ; in the geometry of this paper  $4\sqrt{T}$  is the arc length of the chain laid out straight, while  $|\zeta|$  is the distance between its endpoints once it has curled into its spirals. Every improvement past  $\frac{1}{4}$  comes from that discarded cancellation, beginning with the  $\frac{1}{6}$  that Hardy and Littlewood obtained by Weyl's method on this same sum and standing today at  $\frac{13}{84}$  [4]; the Lindelöf hypothesis is  $\mu(\frac{1}{2}) = 0$ . Two things follow, both drawn in Figure 58. First, the alignment can be read from a truncation. Scoring an interval by  $S_K(t) = |\sum_{n \leq K} n^{-1/2} e^{-it \log n}|$  and then evaluating  $Z$  only at the strongest peaks of that score recovers most of the interval's largest  $|Z|$ : averaged over the 19 unit intervals  $5 \leq T \leq 24$ , which is  $190 \leq t \leq 3771$ , the three strongest peaks of the five-link score



reach 0.967 of it and the single strongest peak of the ten-link score reaches 0.961. The point is not the saving at these heights but that  $K$  need not grow with  $t$  while  $m$  does. Second, the largest sampled values in the intervals are themselves nearly regular:  $\max |Z|$  over  $[T, T+1]$ , taken on a grid of eight points per mean zero spacing and so liable to sit a percent or two under the true crest, fits  $2.93 T^{0.485}$ , its ratio to  $\sqrt{T}$  has mean 2.823 and standard deviation 0.123 across those intervals, and it holds a fraction of the ceiling (180) that ranges from 0.97 down to 0.76, the two extremes falling at  $T = 9$  and  $T = 22$ . Most of that spread is the ceiling rather than the maxima: the exact sum  $\sum_{n \leq m} n^{-1/2} = 2\sqrt{m} + \zeta(\frac{1}{2}) + o(1)$  climbs toward its own asymptote  $2\sqrt{m}$  across the range, from 0.72 of  $4\sqrt{T}$  at  $T = 5$  to 0.86 at  $T = 23$ , since  $\zeta(\frac{1}{2}) = -1.4604$  still weighs heavily at  $m = 5$ . Against  $\sqrt{T}$  itself the maxima are flat, 2.711 at  $T = 5$  and 2.670 at  $T = 23$ , which is the same fact as the fitted exponent 0.485 being essentially the ceiling's  $\frac{1}{2}$ : at these heights the maxima simply ride the trivial bound. In these coordinates the Lindelöf hypothesis is precisely the statement that they eventually stop, that  $\max |Z|/\sqrt{T}$  falls like  $T^{-1/2+\epsilon}$ , the chain curling up so efficiently that the distance between its endpoints grows more slowly than every positive power of  $T$  while its arc length grows like  $\sqrt{T}$ . That distance is not bounded, and is not claimed to be:  $|\zeta(\frac{1}{2} + it)|$  is known to be unbounded. Nothing of that is visible over 19 intervals with  $m \leq 23$  links, and nothing here bears on it either way. So the answer to whether the next unit of the index will beat the last is yes by default, which it is in 13 of the 18 consecutive comparisons, and the exceptions are what needs predicting: on the maxima divided by  $\sqrt{T}$  the peak of the ten-link score ranks the intervals at +0.72, while the three-link score, whose peak is nearly the same in every interval, carries nothing at -0.11. That last figure rests on 19 intervals at modest height and should be read as an indication only. Verified in `check_large_values.py`.

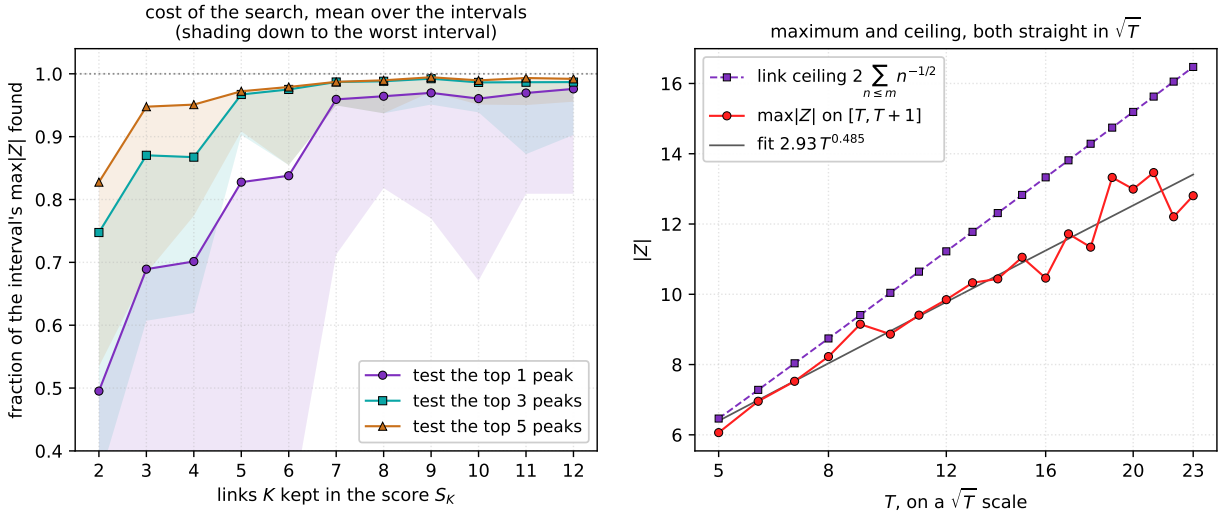


Figure 58: Reading large values off the links, over the 19 unit intervals  $5 \leq T \leq 24$ . *Left*: the fraction of an interval's largest  $|Z|$  found by evaluating  $Z$  only at the strongest one, three or five peaks of the  $K$ -link score  $S_K$ , averaged over the intervals, with the shading reaching down to the worst single interval. With seven links a single candidate captures 95.9% on average, while the worst sampled interval captures 71.2%. *Right*: the interval maximum against  $\sqrt{T}$ , on which it is nearly straight, with the link ceiling of (180) above it and the fitted power law through it. The gap between the two curves is the alignment the links never quite achieve. Generated by `fig_large_values.py`.

### 13.3 Collinearity of the three first-half remainders

Since all the remainders are relative to the same point, namely  $\Sigma_1$ , and since there are three of them, we decided to investigate their collinearity, which we could not do when there was just one remainder split (say  $R/2$ ) or two (say  $R_{1ak}$ ), because any two points already lie on a line.

Collinearity of three points is unchanged by a common translation. The three points  $B_{1ps}$ ,  $B_{1rs}$ , and  $B_{1ak}$  of §9.3 are all relative to the same  $\Sigma_1$ , so that shared  $\Sigma_1$  cancels, and the three remainder vectors  $R_{1ps} = R_1$ ,  $R_{1rs} = \frac{1}{2}R$ , and  $R_{1ak}$  can be compared directly as points in the plane. Figure 59 plots two independent, scale-invariant measures of how far these three points are from lying on a common line. The first is the triangle flatness

$$F = \frac{2|\text{area}|}{\text{diam}^2}, \quad (181)$$

where the area is that of the triangle with the three points as vertices and the diameter is the largest of the three pairwise distances. The second is the aspect ratio  $\lambda_{\min}/\lambda_{\max}$  of the covariance matrix of the three points (a principal-component measure). Both quantities are zero exactly when the points are collinear, and both are invariant under translation, rotation, and rescaling, so the shrinking of the remainders with growing  $T$  cannot masquerade as collinearity. Each panel is a grid of 1501  $\sigma$ -samples by 1801  $T$ -rows over  $0 \leq \sigma \leq 1$ ,  $3.9 \leq T \leq 5.1$ , computed from the exact remainder formulas, with the  $\sigma$ -sample nearest  $\frac{1}{2}$  snapped exactly onto the critical line. The color scale is logarithmic, and that is what separates residuals at the double-precision floor,  $10^{-16}$  to  $10^{-12}$ , from the merely small values,  $10^{-4}$  to  $10^{-1}$ , that fill the rest of the strip. Exact collinearity on the critical line is the proposition below, not the heatmap; the heatmap shows that the residual sits at the floor there and nowhere else in the sampled window.

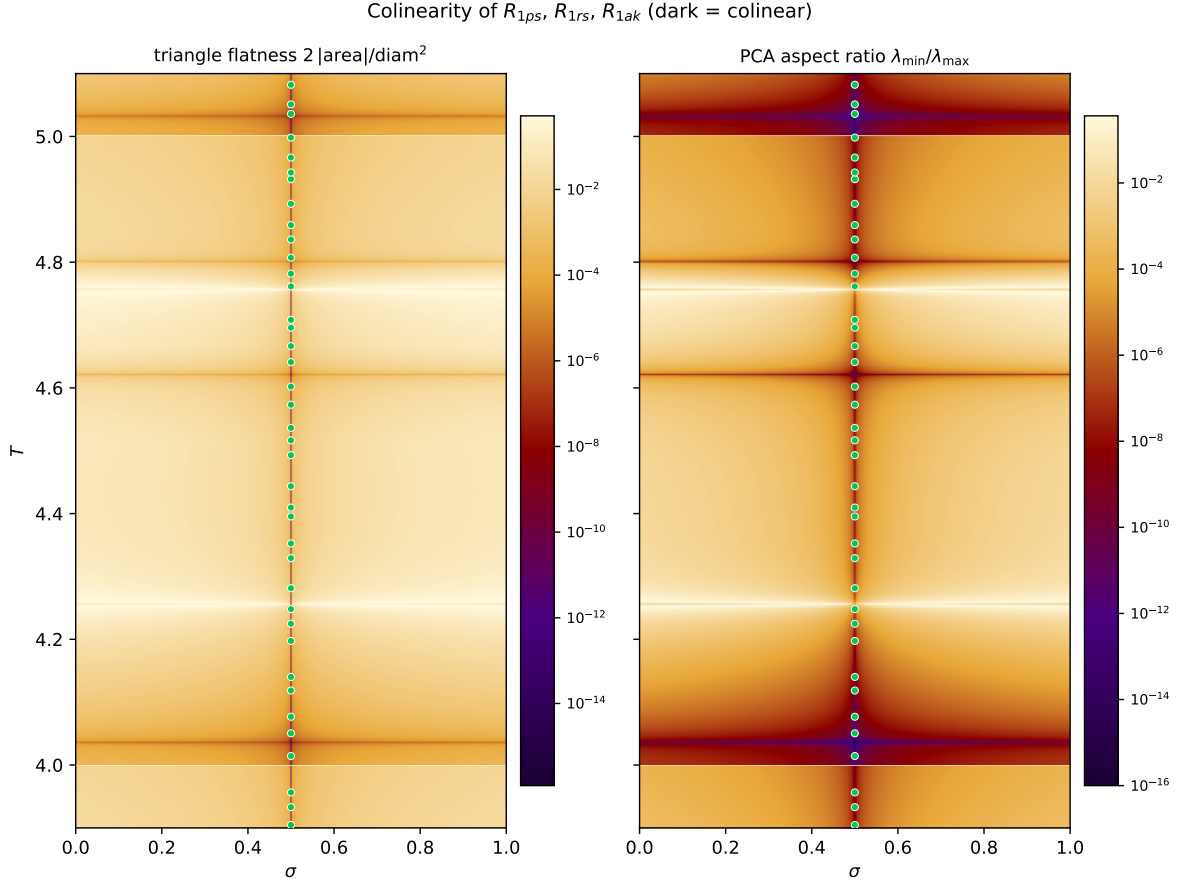


Figure 59: Two ways of displaying the same collinearity information for the three first-half remainders  $R_{1ps}$ ,  $R_{1rs} = \frac{1}{2}R$ ,  $R_{1ak}$  (dark = collinear). The panel on the left uses the flatness measure (181) of the triangle formed by the three points, and the one on the right uses the PCA aspect ratio  $\lambda_{\min}/\lambda_{\max}$ . Horizontal axis  $\sigma \in [0, 1]$ ; vertical axis the spiral index  $T \in [3.9, 5.1]$ . Both panels use a log color scale; green markers are zeta zeros at  $\sigma = \frac{1}{2}$ . The dark column pinned at the numerical floor along  $\sigma = \frac{1}{2}$  shows the three points are exactly collinear everywhere on the critical line; off the line collinearity occurs only along the isolated dark horizontal bands. Generated by `fig_collinearity_zoom.py` from data produced by `remainder-collinearity-heatmap-zoom.mjs`.

Both panels show the same two facts. First, the column  $\sigma = \frac{1}{2}$  sits at the numerical floor at every  $T$ : on the critical line the three first-half remainders are exactly collinear, not just approximately (Proposition 13.6). Second, off the line the measures are many orders of magnitude larger everywhere except along isolated dark horizontal bands (near  $T \approx 4.03, 4.62, 4.80$ , and  $5.03$  in this window), where the triangle momentarily passes through a flat configuration as its signed area changes sign. The measured values are symmetric under  $\sigma \mapsto 1 - \sigma$ , a reflection of the functional equation.

The same pattern appears, in the sampled data, across  $0 \leq \sigma \leq 1$  for  $0 < T \leq 20$ . Figure 60 repeats the four-panel layout of Figure 24 (two windows spanning the unit cells  $5.95 \leq T \leq 7.05$  and  $10.95 \leq T \leq 12.05$ , a third band straddling the integer  $T = 17$ , and the same full strip  $0 \leq T \leq 20$ ), with the PCA aspect ratio as the plotted color map. As in the leg-imbalance panels, the dark column on  $\sigma = \frac{1}{2}$  runs unbroken through every window, while off the line the only dark features are the isolated horizontal collinearity bands. The bands do not sit at integer  $T$ : crossing

an integer the three remainder formulas jump ( $\Sigma_1$  gains a summand and the sign  $(-1)^{\lfloor T \rfloor}$  flips), the signed area of the triangle changes sign in that jump and lands very close to zero, and the area then crosses zero a short distance above the integer. The third panel shows one full unit interval,  $16.95 \leq T \leq 18.05$ : the light-to-dark break at the integer  $T = 17$  itself, a nearly flat strip at every  $\sigma$  just above it, and then the five collinearity bands of the cell, at  $T \approx 17.011, 17.252, 17.612, 17.752$ , and  $17.796$ , before the pattern repeats across the next break at  $T = 18$ . Numerically the bands come five to a unit interval, near  $\{T\} \approx 0.011, 0.252, 0.612, 0.752, 0.796$  at this height, with the second and fourth tracking the pole heights  $\{T\} = \frac{1}{4}, \frac{3}{4}$  of  $d_1$  (§A.5): near a pole  $R_{1ps}$  runs off to infinity along the fixed direction of the next summand, so the triangle is dragged through a flat configuration nearby. The band positions are almost independent of  $\sigma$  (they move by less than  $10^{-4}$  between  $\sigma = 0.2$  and  $\sigma = 0.35$ ), which is why the bands are horizontal.

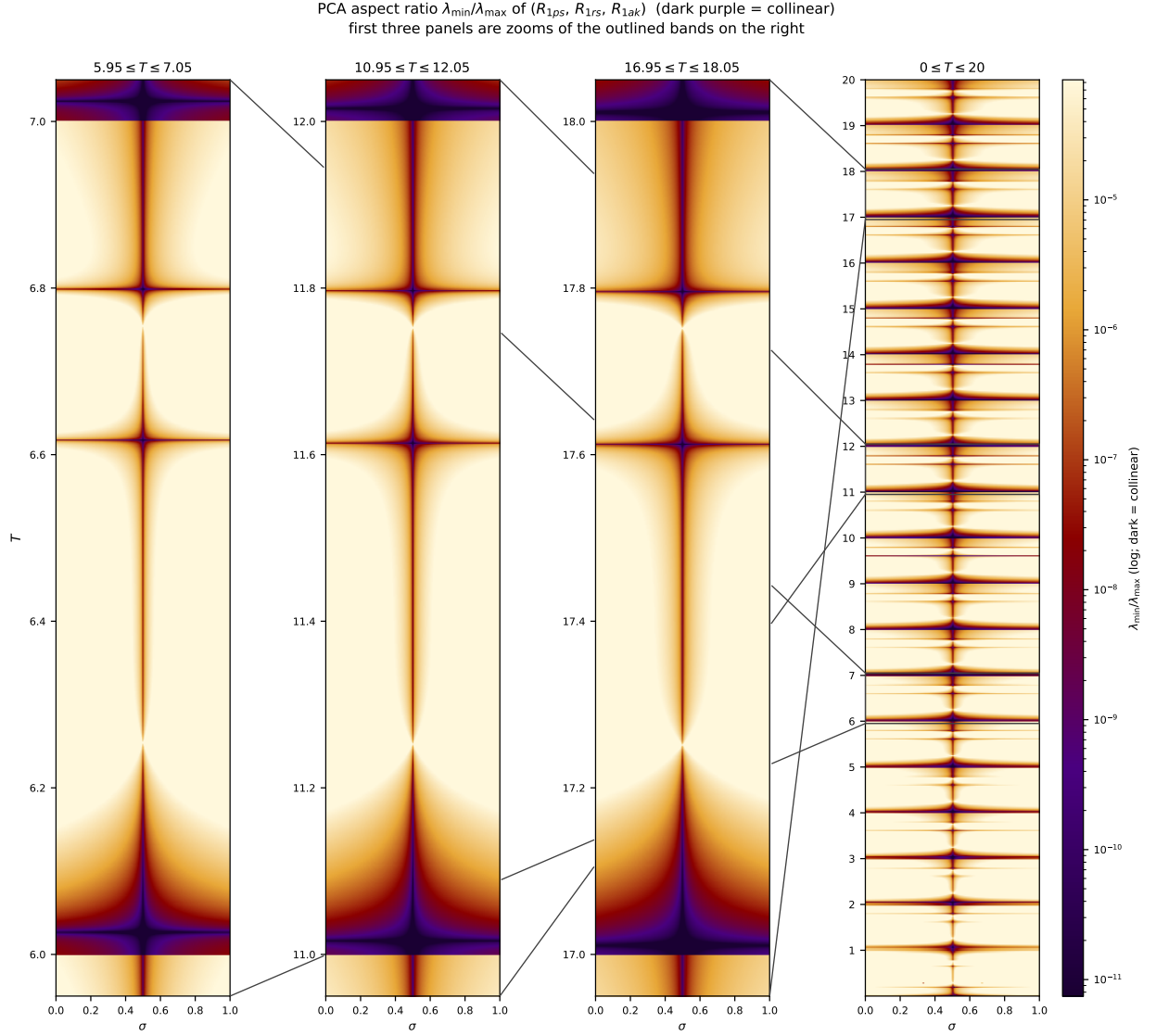


Figure 60: PCA aspect ratio  $\lambda_{\min}/\lambda_{\max}$  of the three first-half remainders  $(R_{1ps}, R_{1rs}, R_{1ak})$  (dark purple = collinear), in the four-panel layout of Figure 24. Horizontal axis  $\sigma \in [0, 1]$ ; vertical axis the spiral index  $T$ . Color is on a log scale. The rightmost panel is the full strip  $0 \leq T \leq 20$ ; the first three panels are zooms of the outlined bands ( $5.95 \leq T \leq 7.05$ ,  $10.95 \leq T \leq 12.05$ ,  $16.95 \leq T \leq 18.05$ ). The dark vertical band on  $\sigma = \frac{1}{2}$  is the exact collinearity of the three first-half remainders on the critical line; the dark horizontal bands are the isolated off-line collinearity events. Generated by `plot_collinearity_four_panel.py`.

Why can this collinearity not be used like the two routes of §9.4? Because collinearity is *not* forced at a zero. For  $\zeta \neq 0$  the equal-leg condition for a split says precisely that  $B_1$  is equidistant from  $O$  and  $\zeta$ , i.e. that  $B_1$  lies on the perpendicular bisector of the chord from  $O$  to  $\zeta$ ; if all three splits had equal legs at a point where  $\zeta \neq 0$ , all three bisector points would lie on that single line and the three first-half remainders would be collinear. But at a zero the chord degenerates:  $\zeta = O$ , every point of the plane is equidistant from the two endpoints, and the (now automatic) equal-leg

conditions carry no collinearity information at all. All three leg pairs are equal at a zero, as they must be, while the three points can remain in general position. So a zero implies equal legs for all three splits, but not collinearity, and the three points need not line up at an off-line zero.

**Proposition 13.6.** *On  $\sigma = \frac{1}{2}$  the three first-half remainders  $R_{1ps}$ ,  $R_{1rs} = \frac{1}{2}R$ , and  $R_{1ak}$  are collinear at every height  $T$ , zeros included.*

The three points  $B_{1ps}$ ,  $B_{1rs}$ , and  $B_{1ak}$  are those remainders translated by the common  $\Sigma_1$ , so it is the same to prove the  $B_1$  collinear. On the critical line write  $\vartheta$  for the Riemann–Siegel theta function, so  $\chi = e^{-2i\vartheta}$  and  $Z = e^{i\vartheta}\zeta$  is real. Any split  $\zeta = B_1 + B_2$  that satisfies the reflection  $B_2 = \chi\overline{B_1}$  then has

$$e^{i\vartheta}B_1 = \frac{1}{2}Z + ih, \quad h \in \mathbb{R}. \quad (182)$$

Indeed  $e^{i\vartheta}B_2 = \overline{e^{i\vartheta}B_1}$ , so  $e^{i\vartheta}\zeta = e^{i\vartheta}B_1 + \overline{e^{i\vartheta}B_1} = 2\Re(e^{i\vartheta}B_1)$ , and the left side is the real number  $Z$ . Thus every such  $B_1$  has the same real part  $Z/2$  in the  $\vartheta$ -frame: all of them lie on the single vertical line  $\Re(e^{i\vartheta}z) = Z/2$ , which is the perpendicular bisector of the segment from  $O$  to  $\zeta$  when  $\zeta \neq 0$ , and the imaginary axis of that frame when  $\zeta = 0$ . The three named splits are of this kind. For the  $ps$  split,  $B_2 = \chi\overline{B_1}$  is Corollary 9.1. For the  $rs$  split the same reflection is the identity recorded in §9.3:  $\Sigma_2 = \chi\overline{\Sigma_1}$  and  $R = \chi\overline{R}$ , so  $\chi\overline{B_{1rs}} = \Sigma_2 + \frac{1}{2}R = \zeta - B_{1rs}$ . For the  $ak$  split, Siegel’s contour integral  $f$  of (62) gives the exact pairing  $\zeta(s) = f(s) + \chi(s)f(1-s)$  [21], and on the line  $f(1-s) = \overline{f(s)}$ , so  $B_{1ak} = f(s)$  satisfies the same reflection; the numerical  $R_{1ak}$  of (60) is the Dirichlet-sum truncation of that identity, which is why the heatmaps sit at the double-precision floor rather than at a symbolic zero. Equation (182) does not use  $\zeta \neq 0$ , so the three points remain collinear at a critical-line zero as well: there  $Z = 0$  and they lie on  $\Re(e^{i\vartheta}z) = 0$ .

A proof through collinearity therefore still needs a bridge that the routes of §9.4 do not. The first step of that bridge is the identity just proved. What remains is to prove that off the line collinearity fails outside isolated curves like the dark bands of Figure 59, and to connect a hypothetical off-line zero to that structure, for instance through nearby off-line points with  $\zeta \neq 0$  at which all three splits have equal legs, which by the perpendicular-bisector argument above would force a collinearity that the off-line geometry forbids. Those two steps would turn collinearity into a characterization of the critical line and then into a statement about zeros; both remain unproved.

### 13.4 Joint angles

In our numerical explorations of observing the links in their own coordinate frames, which led to (146), we decided to look at the joint angles too. So we plotted them, and noticed a pattern emerge at the Farey fractions. This section is the result of that investigation.

The joint angles of the forward chain can be read on their own, without the chain visualization, using one dot per joint with the turning angle on the vertical axis, the joint index along the horizontal. The turning angle at joint  $n$  is the continuous joint angle (89) evaluated at an integer joint,

$$\theta_n = -I(T) \log \frac{n}{n-1} \quad (\text{reduced to } [-\pi, \pi]), \quad \nu(n) = \frac{I(T)}{2\pi n(n-1)}, \quad (183)$$

where  $\nu$ , the derivative of the unwrapped angle divided by  $2\pi$ , is the local frequency of the strip in cycles per joint. Two features organize the picture, both consequences of  $I(T)$ . First, at the bisector joint  $n = \lfloor T \rfloor + 1$  the frequency is  $\nu = 1 + \frac{1}{6T(T+1)}$ , one full turn per joint to nine decimals at  $T = 13000$ , and the angle there is  $-\pi$  exactly: that is (90), the bisector joint of §10.1, seen as the right edge of the strip. The joints just behind it inherit almost the same angle, 3.141109, 3.139659, 3.137242 at  $n = T, T-1, T-2$ , which is the same near-stationarity that makes the bisector frame

of Figure 31 repeat. Second, since  $\nu(n) \approx T^2/n^2$  falls from  $\nu$  of order  $10^8$  at  $n = 2$  to 1 at that bisector joint, the strip is a sampled chirp, and it is stratified by the joints where  $\nu$  is the reciprocal of a rational. Writing that rational as a Farey fraction  $f = p/q$  in  $(0, 1]$ , its *caustic joint* is the solution of  $n(n-1) = f I(T)/2\pi$ , that is  $\nu = 1/f$ , so

$$n_f = \frac{1}{2} \left( 1 + \sqrt{1 + 2f I(T)/\pi} \right) \approx T\sqrt{f}, \quad (184)$$

and there the joints fall into exactly  $p$  strands, the denominator of  $\nu = q/p$ , each a parabola of curvature  $4\pi\nu/n$  per joint squared because the unwrapped angle is locally quadratic (exactly  $|\tilde{\theta}''(n_f)|$  of (186) below). Small denominators give the spines visible in Figure 61, drawn at the integer index  $T = 13000$ , where  $t = I(T) = 1,061,939,999.369541$ : there the fractions  $\frac{1}{4}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}$  sit at  $n = 6501, 7506, 9193, 10615, 11259$ , and  $f = 1$  is the bisector joint itself. The parabolic strands are the same local quadratic phase that produces the Cornu-like arcs of the curlicue literature of §14.3, here read off the joints rather than the partial sums.

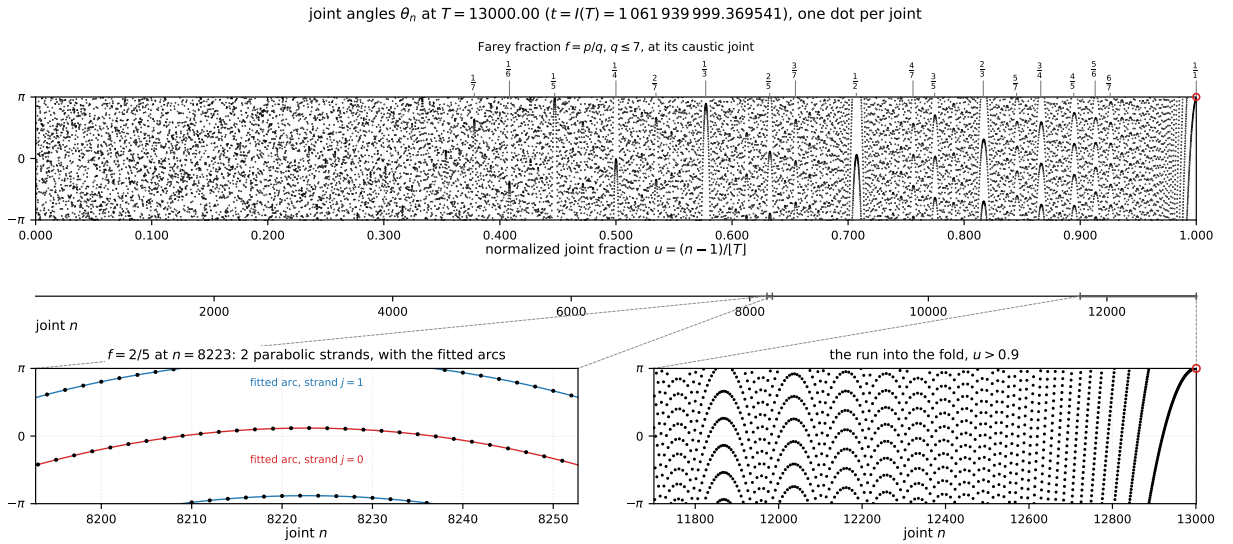


Figure 61: The joint-angle strip at  $T = 13000$ , where  $t = I(T) = 1,061,939,999.369541$ , one dot per joint of the forward chain,  $n = 2, \dots, [T] + 1$ , with  $\theta_n$  of (183) on the vertical axis. The two edges  $\pm\pi$  are identified, so the strip is a cylinder. *Top*: all 13000 joints, carrying three horizontal scales: the Farey fractions  $f = p/q$  with  $q \leq 7$  above, each at its caustic joint (184) and staggered in two rows where they crowd; the normalized joint fraction  $u = (n-1)/[T]$  below, so the bisector joint sits at  $u = 1$ ; and the joint index  $n$  below that. The red circle at the right edge is the bisector joint  $n = [T] + 1$ , where  $\theta = -\pi$  exactly. *Bottom left*: a window at  $f = 2/5$ , showing its  $p = 2$  parabolic strands as dots, each with its fitted arc (188) drawn through them; the two arcs sit half a turn apart, so strand 1 has its vertex wrapped to the opposite edge. *Bottom right*: the last tenth of the strip, where the strands steepen and run into that bisector joint on the right in the figure. Dashed lines carry each window back to the stretch of the joint- $n$  scale it magnifies, marked there by a bracket. Generated by `fig_joint_angles.py`.

**The fitted curve.** The strands are more than parabola-like: they can be written down. Let  $\tilde{\theta}(n) = -I(T) \log \frac{n}{n-1}$  be the unwrapped angle, so  $\theta_n$  is  $\tilde{\theta}(n)$  reduced to  $[-\pi, \pi]$ , and differentiate it



three times:

$$\tilde{\theta}'(n) = \frac{I(T)}{n(n-1)}, \quad \tilde{\theta}''(n) = -\frac{I(T)(2n-1)}{[n(n-1)]^2}, \quad \tilde{\theta}'''(n) = \frac{2I(T)(3n^2-3n+1)}{[n(n-1)]^3}. \quad (185)$$

At a caustic these have closed forms in the fraction alone, since (184) fixes  $n_f(n_f-1) = fI(T)/2\pi$  and  $2n_f-1 = \sqrt{1+2fI(T)/\pi}$  exactly:

$$\begin{aligned} \tilde{\theta}'(n_f) &= \frac{2\pi}{f} = \frac{2\pi q}{p}, & \tilde{\theta}''(n_f) &= -\frac{4\pi^2}{f^2 I(T)} \sqrt{1 + \frac{2fI(T)}{\pi}} \approx -\frac{4\pi}{f^{3/2}T}, \\ \tilde{\theta}'''(n_f) &= \frac{16\pi^3}{f^3 I(T)^2} \left(1 + \frac{3fI(T)}{2\pi}\right) \approx \frac{12\pi}{f^2 T^2}. \end{aligned} \quad (186)$$

Write  $\delta = n - n_f$  for the offset in joints from the caustic. The linear term  $\tilde{\theta}'(n_f)\delta = (2\pi q/p)\delta$  is a carrier sweeping several radians per joint, and it is what interleaves the dots into strands in the first place. Along one strand, though,  $\delta$  advances by  $p$  at a time, and  $(2\pi q/p)pm = 2\pi qm$  drops out: the carrier contributes nothing but a constant. Absorbing it together with  $\tilde{\theta}(n_f)$ , strand  $j$  carries the phase constant

$$C_j = \tilde{\theta}(n_f) + \frac{2\pi q}{p}([n_f] + j - n_f), \quad j = 0, \dots, p-1, \quad (187)$$

with  $[\cdot]$  the nearest integer, and the strand itself is that cubic reduced to  $[-\pi, \pi]$

$$\rho_j(\delta) = C_j + \frac{1}{2}\tilde{\theta}''(n_f)\delta^2 + \frac{1}{6}\tilde{\theta}'''(n_f)\delta^3 \quad (\text{reduced to } [-\pi, \pi]). \quad (188)$$

Because  $\gcd(p, q) = 1$  the constants  $C_j$  run through all  $p$  residues of  $2\pi/p$ , so the  $p$  arcs are one parabola stacked at equal spacing around the cylinder. Third order is the right order: the second derivative supplies the parabola, the third its visible skew, and the fourth is smaller by a further  $\delta/n_f \sim T^{-1/2}$ .

The arcs drawn in Figure 61 are (188) at  $f = \frac{2}{5}$  and  $T = 13000$ , where  $n_f = 8222.74$ ,  $\tilde{\theta}'(n_f) = -3.8208 \times 10^{-3}$  and  $\tilde{\theta}'''(n_f) = 1.3941 \times 10^{-6}$  radians per joint squared and cubed, and the carrier is  $2\pi q/p = 5\pi \equiv \pi$ , which is why the two strands sit half a turn apart. Over the  $\pm 30$  joints shown they track the dots to  $2.2 \times 10^{-5}$  radians, that being the fourth-order term. Nothing is fitted in the statistical sense, and (188) is evaluated exactly. Two limits stand out. Toward small  $f$  the parabola stiffens like  $f^{-3/2}$ , which is why the left end of the strip looks like noise at this scale: consecutive joints there are many turns apart and only the caustics of very small  $f$  remain resolved. At the other end  $f = 1$  closes the family, with  $p = q = 1$ : a single strand, a carrier of one full turn per joint, and  $C_0 = -\pi$ . That is the bisector joint of (90) again, so the bisector joint this paper is built on is the last caustic of the sequence, and the right-hand panel of Figure 61 is the run of the high-denominator caustics into it.

**Incremental change.** The strip is a snapshot. Advance the index and every dot moves, though not by the same amount. Differentiating the unwrapped angle in  $T$  at a fixed joint,

$$\begin{aligned} \frac{\partial \tilde{\theta}}{\partial T}(n) &= -I'(T) \log \frac{n}{n-1}, & L &= \log \left(1 + \frac{1}{T}\right), \\ I'(T) &= \frac{\pi}{L^2} \left(2L + \frac{2T+1}{T(T+1)}\right) = 4\pi T + 2\pi + O(T^{-1}), \end{aligned} \quad (189)$$



which in cycles is a drift rate

$$\mu(n) = \frac{I'(T)}{2\pi} \log \frac{n}{n-1} \approx \frac{2T+1}{n-\frac{1}{2}} \approx \frac{2}{u}, \quad u = \frac{n-1}{[T]}. \quad (190)$$

A value of  $\mu$  is easiest to read as a lap count:  $\mu(n) = 3$  means that dot laps the strip three times, falling from  $+\pi$  to  $-\pi$ , wrapping, and repeating, while you advance  $T$  by one. Nothing here needs differencing:  $I'(T)$  is available in closed form as  $\pi(2L + (2T+1)/T(T+1))/L^2$  with  $L = \log(1 + 1/T)$ , which is  $4\pi T + 2\pi$  to leading order, about 163,369 at  $T = 13000$ .

The sign is the same everywhere, so the whole strip drifts downward; the magnitude is not, and that is the point. Since  $\mu(n) \approx (2T+1)/(n-\frac{1}{2}) \approx 2/u$ , the rate is essentially a  $1/u$  hyperbola, equal to exactly 2 at the bisector joint  $u = 1$  and growing leftward to  $1.8 \times 10^4$  cycles per unit index at  $n = 2$ . Incidentally, the  $2/u$  form is no longer any good there, reading 26,000, since  $\log \frac{n}{n-1}$  and  $1/(n-\frac{1}{2})$  part company at the first few joints. The strip does not translate as the index advances, it shears. Read the other way,  $1/\mu(n)$  is the index step that returns joint  $n$  to the same angle: a dot at  $u = \frac{1}{2}$  comes back around every  $\Delta T = 0.25$ , one at  $u = 0.1$  every  $\Delta T = 0.05$ , and one at the bisector joint every half unit.

At the right edge the value is exact and already familiar: the middle expression of (190) at  $n = [T] + 1$  is  $(2T+1)/(T+\frac{1}{2}) = 2$ , and the exact rate is 2.000000001 at  $T = 13000$ . Two cycles per unit index is one more than (90) reports for the joint angle, and the discrepancy is the joint itself moving: the bisector joint sits at  $n = [T] + 1$ , which advances one joint per unit index, and one joint there is worth  $\nu = 1$  turn, so an observer riding the bisector joint sees  $\mu - \nu = 1$  cycle, namely  $\frac{d}{dT}\pi(2T+1) = 2\pi$ . The two statements differ only in whether one rides the bisector joint or stands at a fixed joint.

The pattern the dots drift through, by contrast, hardly moves at all. Differentiating (184) gives  $dn_f/dT = f I'(T)/2\pi(2n_f - 1) \approx \sqrt{f}$ , so a caustic creeps outward by a fraction of a joint per unit index, and its normalized position  $u_f = (n_f - 1)/[T] \approx \sqrt{f}$  does not depend on  $T$  at all: the spines stand still while the joints stream through them. Their drift rate follows from pairing (190) with the cycles per joint  $\nu = I(T)/2\pi n(n-1)$  of (183), which gives  $\mu \approx 2\sqrt{\nu}$ : at the caustic of Farey fraction  $f$ , where  $\nu = 1/f$ , the dots slide at  $2/\sqrt{f}$  cycles per unit index, four at  $f = \frac{1}{4}$  and two at  $f = 1$ . Only at an integer index does the frame change, when a new link is appended, a new joint appears at the right edge with  $\theta = -\pi$ , and  $u$  rescales by one part in  $[T]$ .

Figure 62 shows one step,  $\Delta T = 0.02$ , which advances  $t$  by  $\Delta I = 3267.38$ . The travel is  $0.04/u$  cycles:  $14.4^\circ$  at the bisector joint,  $28.8^\circ$  at  $u = \frac{1}{2}$ ,  $143.9^\circ$  at  $u = 0.1$ , half a turn at  $u = 0.0800$ , joint  $n = 1041$ , and a full turn at  $u = 0.0400$ , joint  $n = 521$ . Since the motion is downward everywhere, the arrows in the figure are drawn downward and wrapped at the lower edge, which is where they land for joints left of  $n = 1041$ . Left of  $n = 521$  the joint has gone round at least once, and rather than draw what is left over the arrow there sweeps the whole strip from  $\pi$  to  $-\pi$ ; at  $n = 2$  the travel is 2264 radians, some 360 turns. This is the sense in which the left end of the strip is a chirp in the index as well as in the joint number: it is not only that neighboring joints are many turns apart at fixed  $T$ , but that one joint sweeps many turns over a step in  $T$  too small to move the bisector joint by a quarter of a radian. The bottom panel of Figure 62 is  $\mu$  on a logarithmic axis, and the middle panel is a picture of the same quantity at  $\Delta T = 0.02$ .

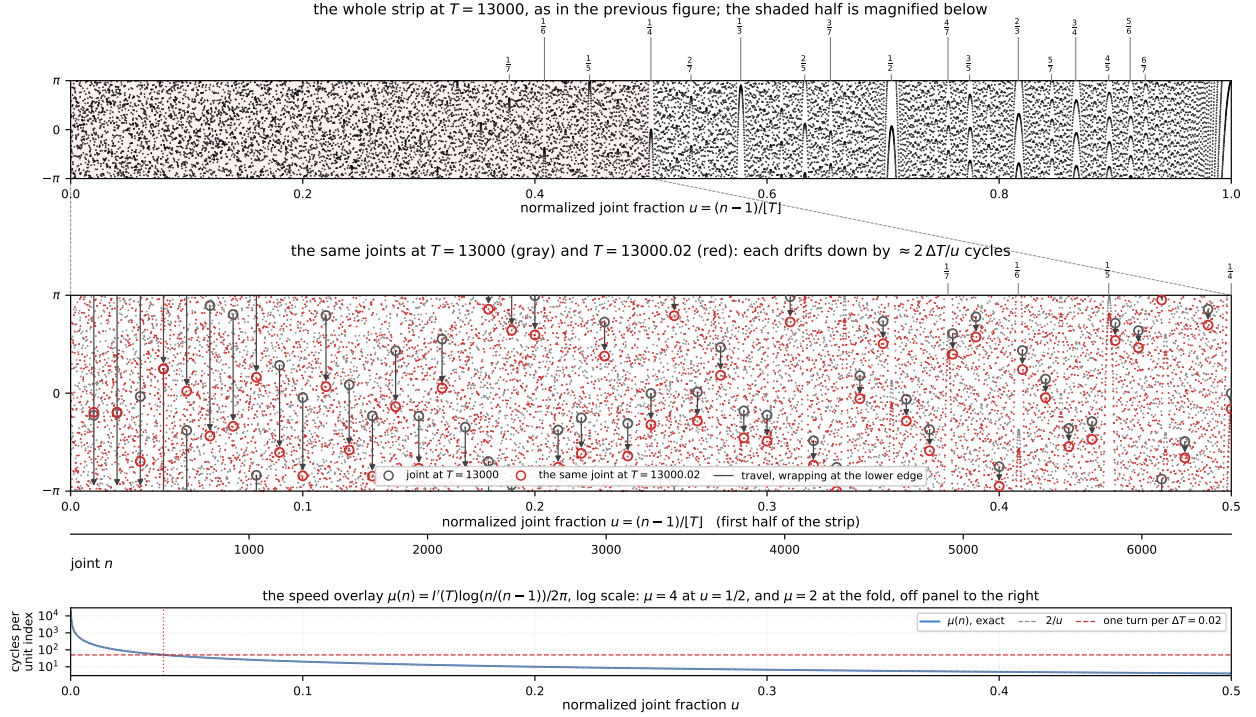


Figure 62: One increment of the index, at  $T = 13000$ . *Top*: the whole joint-angle strip as in Figure 61, with the shaded first half magnified below. *Middle*: that half twice over, gray at  $T = 13000$  and red at  $T = 13000.02$ , every dot having drifted downward by  $\mu(n) \Delta T \approx 0.04/u$  cycles of (190). Fifty joints spaced evenly across the panel are circled in both colors, with an arrow along the path traveled; since the drift is downward at every joint the arrows run downward and wrap at the lower edge, and where the travel exceeds a full turn the arrow sweeps the whole strip, from  $\pi$  to  $-\pi$ . The travel lengthens leftward like  $1/u$ , passing half a turn at  $n = 1041$  and a full turn at  $n = 521$ , marked by the dotted vertical line. Both strips carry the Farey fractions  $f = p/q$ ,  $q \leq 7$ , each at its caustic joint (184); only  $f \leq \frac{1}{4}$  falls in the magnified half, since  $u_f \approx \sqrt{f}$ . *Bottom*: the drift rate itself on a logarithmic axis: exact (blue) against the  $2/u$  form (gray), the two separating only at the first few joints, with the dashed line at one full turn per  $\Delta T = 0.02$ . Generated by `fig_incremental_change.py`.

## 14 Prior literature

### 14.1 Nickel's Argand-diagram geometry

Nickel [17], with a condensed sequel [18], starts from the same object this paper starts from, the Argand diagram of the steps  $n^{-s}$ , and reaches a construction close enough to ours that the two are worth setting side by side. They agree in shape, and they differ in the one respect this paper is about.

**Dictionary.** His center of symmetry is the step  $n_p = \lfloor \sqrt{t/2\pi} \rfloor$ , with fractional part  $p = \sqrt{t/2\pi} - n_p$ . Since  $\sqrt{t/2\pi} \approx T + \frac{1}{2}$  (§10.1), his index is  $n_p \approx \lfloor T + \frac{1}{2} \rfloor$  and  $p \approx \{T + \frac{1}{2}\}$ , so it nearly agrees with our  $m = \lfloor T \rfloor$  on the first half of each unit interval and exceeds it by one on the second half: the same half-unit offset already recorded for Siegel's cutoff in §8.2. His factor  $Q(s)$ , defined by  $\zeta(s) = Q(s)\zeta(1-s)$ , is our  $\chi(s)$ ; the displayed form  $Q = n_p^{1-2s} e^{i(t+\pi/4)}$  is the familiar  $e^{-2i\vartheta(t)}$  asymptotic, which reproduces  $\chi$  to a relative  $1.5 \times 10^{-5}$  at  $T = 20.3$  and  $1.6 \times 10^{-7}$  at  $T = 200.3$  once the smooth  $\sqrt{t/2\pi}$  is used in place of the rounded  $n_p$ . His symmetry angle  $\Theta$  is our  $\frac{\psi}{2} = \frac{1}{2} \arg \chi$ .

**What matches.** Under that dictionary his Riemann–Siegel equation  $\zeta = P(s) + Q(s)P(1-s)$  is our two-leg decomposition  $\zeta = B_1 + B_2$  with  $B_2 = \chi \overline{B_1}$  (Corollary 9.1, verified in Lean in Appendix C); his polar form  $\zeta = 2 \cos(\phi - \Theta) P e^{i\Theta}$  is the two-leg identity  $\zeta = B_1 + \chi \overline{B_1} = 2|B_1| \cos(\arg B_1 - \frac{\psi}{2}) e^{i\psi/2}$ , not the polar decomposition of  $R$ ; and his observation that the two magnitudes are equal exactly when  $\sigma = \frac{1}{2}$  is the on-line half of Corollary 9.1 (the present legs can also be equal on isolated off-line loci). He also writes the classical remainder as  $R = 2L \cos(\theta_L - \Theta)$ , where  $L = L e^{i\theta_L}$  runs from the joint at step  $n_p$  to his center: that is exactly the role played here by the pair  $(d_1, -\omega)$ . Where his index agrees with ours the two projected remainders converge, the ratio  $2L \cos(\theta_L - \Theta) / 2d_1 \cos(\omega + \frac{\psi}{2})$  falling through 2.24, 1.34, 1.11, 1.03, 1.01 at  $T = 6.3, 20.3, 60.3, 200.3, 600.3$ .

**Two centers, one line.** His  $P(s)$  and our  $B_1$  are not the same point, and they are not meant to be. The difference lies almost entirely along the direction in which  $\zeta$  cannot see it: at  $T = 600.3$  we find  $|P - B_1| = 0.064$ , of which only  $1.2 \times 10^{-4}$  lies along  $e^{i\psi/2}$ . That is his own remark that  $\zeta$  is unchanged when  $P(s)$  is translated along the symmetry axis. In the notation of §9.2 the admissible centers are the solutions  $X$  of  $\zeta - X = \chi \overline{X}$ , a line in the plane, and every one of them projects onto  $\zeta/2$ , which is Corollary 9.2. The freedom he notes and the projection we prove are the same fact.

**The sequel.** Two things in [18] touch this paper further. It traces  $P(s)$  and  $\zeta(s)$  as two curves over a range of  $t$ , joined at regularly spaced values of  $t$  to show the two segments, which is the figure the conveyor belt draws with his pendant center in place of  $B_1$ . And it counts zeros: it reads the count off  $\Theta$  through the Gram points and tests that reading against the first 100,000 ordinates, finding one zero per Gram point on average with the zeros grouping between successive Gram points. The count there is the classical one, exact in the mean and never landing on an ordinate; the curves of §12 land on every one.

**What differs.** The correction itself. Nickel's is a lowest-order determination, carrying a factor  $1/(2n_p^\sigma \cos 2\pi p)$  that grows without bound as  $\cos 2\pi p \rightarrow 0$ , and he is explicit that its error matters for the distribution of  $P(s)$  even though it cancels in  $\zeta$ ; its direction,  $e^{-i(t \ln n_p + 2\pi p)}$ , is the step direction at  $n_p$  turned by the fractional offset  $2\pi p$ , not a summand direction of the chain. Our  $R_1$

and  $R_2$  are exact, and their directions are exactly those of the  $(m + 1)$ st summand of each chain. That is what lets the serial chain of §7.2 close on  $\zeta$  identically rather than asymptotically, and it is the content of the “one more partial summand” reading: the correction is not only near the next link but along it.

## 14.2 Levinson’s $G$

The companion  $B_1^* = \zeta + \zeta'/2\vartheta'$  of (159), whose argument carries the counting curve  $N_*$  of §12.2, is the function Levinson [15] calls  $G$ . That is the paper in which he proves that more than a third of the zeros of  $\zeta$  are on  $\sigma = \frac{1}{2}$ . Figure 63 draws the function itself on the critical line, in the plane and in the frame where its winding is the count.

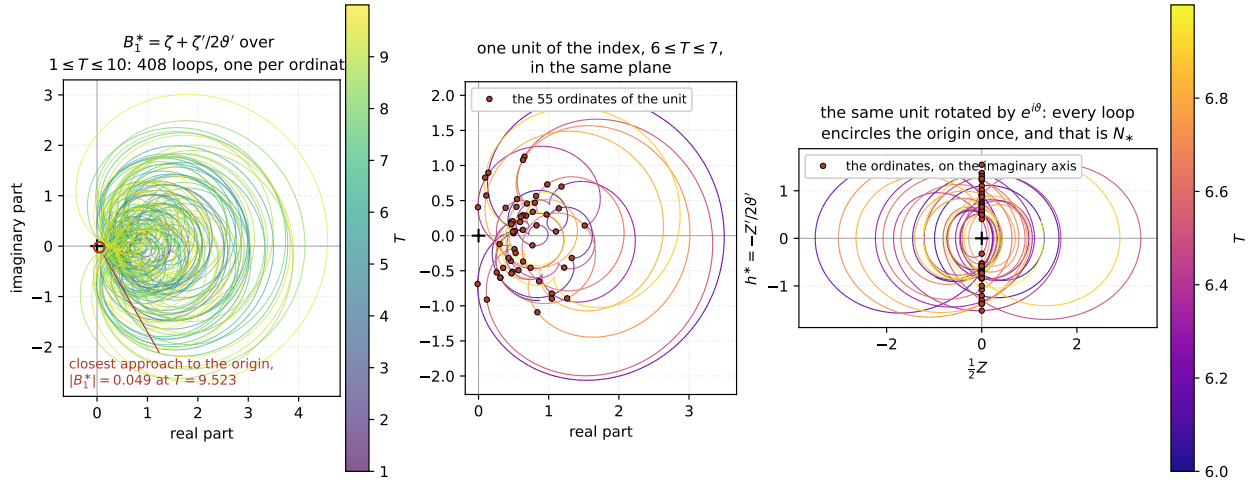


Figure 63: Levinson’s  $G$ , which is the companion  $B_1^* = \zeta + \zeta'/2\vartheta'$  of (159), drawn as a path in the plane on the critical line. *Left:* the whole of  $1 \leq T \leq 10$ , that is  $13.6 \leq t \leq 692.2$ , the color running with  $T$ : 408 loops, one for each ordinate of the range. The path never reaches the origin, which is what lets  $\arg G$  carry a count, and its closest approach,  $|G| = 0.048$  at  $T = 9.523$ , falls at the tightest pair of ordinates in the range,  $\gamma_{363}$  and  $\gamma_{364}$ , whose gap is 0.24 of the mean spacing. There  $Z' = 0$  and  $|G| = \frac{1}{2}|Z|$ : it is a Lehmer-style near-miss, not a small value of  $\zeta$  on its own, that brings the curve in. *Middle:* one unit of the index,  $6 \leq T \leq 7$ , in the same plane, with its 55 ordinates marked. *Right:* the same unit rotated by  $e^{i\vartheta}$ , the frame of (161) in which the real part is  $\frac{1}{2}Z$  and the imaginary part is the offset  $h^*$ . Every loop encircles the origin once and crosses the imaginary axis exactly at an ordinate. Counting those crossings is Levinson’s reading of the angle; following the turn continuously is  $N_*$ . Computed from  $e^{i\vartheta}G = \frac{1}{2}Z - iZ'/2\vartheta'$ , which agrees with  $\zeta + \zeta'/2\vartheta'$  to 18 digits. Generated by `fig_b1.star_path.py`.

**Dictionary.** He writes the gamma factor as  $h(s) = \pi^{-s/2}\Gamma(s/2) = e^{f(s)}$ , so the functional equation reads  $h(s)\zeta(s) = h(1-s)\zeta(1-s)$ . Differentiating it and using it again to remove  $\zeta(1-s)$  leaves him with the argument of

$$G(s) = \zeta(s) + \frac{\zeta'(s)}{f'(s) + f'(1-s)} \quad (191)$$

to control, which is his (1.9). His weight is ours, in two steps. From  $f(s) = -\frac{s}{2} \log \pi + \log \Gamma(s/2)$ ,

$$f'(s) = -\frac{1}{2} \log \pi + \frac{1}{2} \frac{\Gamma'}{\Gamma} \left( \frac{s}{2} \right),$$

and  $\chi = h(1-s)/h(s)$  has  $\log \chi = f(1-s) - f(s)$ , so differentiating in  $s$  gives

$$\frac{\chi'}{\chi}(s) = -f'(1-s) - f'(s), \quad \text{that is} \quad f'(s) + f'(1-s) = -\frac{\chi'}{\chi}(s), \quad (192)$$

at every  $s$  where the displayed quotients are defined. That is the weight of the off-line form (165), so  $G = B_1^*$  as a meromorphic identity wherever those quotients are finite and nonzero. On the critical line the same weight turns real: with  $s = \frac{1}{2} + it$  we have  $1-s = \bar{s}$  and  $f'(1-s) = \overline{f'(s)}$ , so the sum is twice a real part,

$$f'(s) + f'(1-s) = 2 \operatorname{Re} f'(\tfrac{1}{2} + it) = \operatorname{Re} \frac{\Gamma'}{\Gamma}(\tfrac{1}{4} + \tfrac{it}{2}) - \log \pi = 2\vartheta'(t),$$

the last equality being the exact  $\vartheta'$  recorded above (159). Substituting that into (191) returns  $\zeta + \zeta'/2\vartheta'$ , which is (159). So the two are the same function and not two versions of one: at  $t = 20$ , to take one height, the two weights are both 1.157750994844834 and the two companions agree to the forty digits carried. Levinson himself trades the weight for  $\log(t/2\pi)$  at his (1.4), which costs  $O(1/t)$ ; we keep the exact  $2\vartheta'$ . In the language of §9.2,  $G$  is a leg: the first leg of the split whose second leg  $\zeta - G$  is the velocity of the endpoint of the chain along the line, quarter-turned and halved by the local rate of the phase.

**The same angle.** His (1.8) locates the ordinates by the condition  $\arg(h(s)G(s)) \equiv \frac{\pi}{2} \pmod{\pi}$ , and on the critical line  $\arg h$  is exactly the Riemann–Siegel phase,  $\arg h(\frac{1}{2} + it) = \arg \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi = \vartheta(t)$ . So the angle he watches is the one (162) is built from,

$$\arg(hG) = \vartheta + \arg B_1^* = \pi(N_* - \tfrac{3}{2}) \pmod{2\pi},$$

and his condition on it is precisely the statement that  $N_*$  is an integer. The two readings of the same angle differ in what is asked of it. Levinson needs only how often the condition holds, so what he must control is how far  $\arg G$  can move: he counts zeros of  $G$  in a rectangle with the critical line for its left edge and  $\sigma = 3$  for its right, where  $|G - 1| < \frac{1}{3}$  pins the argument down, and a mollifier and Littlewood’s lemma bound the count in between. His remark that  $\zeta$  would have essentially all of its zeros on the line if  $\arg G$  never moved at all is the extreme case of the same reading. Here the angle is wanted between the ordinates as well as at them. The exact weight gives  $\operatorname{Re}(e^{i\vartheta}G) = \frac{1}{2}Z$  by (161), so the angle advances by exactly  $\pi$  from one ordinate to the next, and  $N_*$  is a continuous curve, integer-valued at the ordinates and nowhere else, carrying in its slope the size of  $|Z|$  (Remark 12.5). One further consequence of (191) is common to both readings, the corollary Levinson credits to Montgomery: a zero of  $\zeta$  of multiplicity  $m$  leaves  $G$  a zero of multiplicity  $m - 1$ .

### 14.3 Spirals of exponential sums

For quadratic phases this geometry is rigorous ground. The partial sums  $S_N(\tau) = \sum_{n \leq N} e^{i\pi\tau n^2}$  draw what Berry and Goldberg [3] call curlicues: replacing the sum by an integral produces the Cornu spiral exactly, and a renormalization then carries the whole pattern onto a magnified and rotated copy of a shorter sum of the same kind. Coutsiass and Kazarinoff [5, 6] work that picture the way this paper works ours. They give the polyline a discrete radius of curvature

$$R_N = \frac{1}{2} |\csc(\psi_N/2)|, \quad \psi_N = \pi\tau(2N + 1), \quad (193)$$

$\psi_N$  being the turn between consecutive terms, so that inflections of the pattern sit where  $\psi_N$  is nearest a multiple of  $2\pi$  and cusps where it turns around. They then cut the sum *at an inflection*,

and replace each whole spiral of the first sum by a single vector carrying a scale and a phase, the Fresnel scale  $1/\sqrt{|\tau|}$  with signed phase  $e^{i \operatorname{sgn}(\tau) \pi/4}$  to the spiral's mid-vector. What comes out is Hardy and Littlewood's approximate functional formula for the theta function, sharpened to an error uniform in  $\tau$  and stated as a Diophantine inequality that contains Gauss's sum formula. Cutting at the bisector and standing one link in for a whole spiral is the reading of §4; their  $e^{i\pi/4}$  is the Gaussian integral's, the same factor that leaves the  $-\pi/8$  in  $\vartheta$ ; and (193) is the bisector construction of §9.1 for unit links, the distance from a step to the meeting of the angle bisectors at its two ends, which is Nickel's  $L$  once the length  $n^{-\sigma}$  is restored.

The arithmetic is what does not carry over. Their turn  $\psi_N$  is linear in  $N$ , so the second difference is the constant  $2\pi\tau$ , the arcs are Cornu spirals outright, and the theory turns on the continued fraction of  $\tau$ : the pattern is a self-similar hierarchy of curlicues within curlicues. Our turn is  $\theta_n = -t \log(1 + \frac{1}{n})$  and its second difference falls like  $t/n^2$ , passing  $2\pi$  once. That is why the chain has a single distinguished inflection, at  $n \approx \sqrt{t/2\pi}$ , instead of a hierarchy of them, and why its arcs are Cornu spirals only locally. The geometric machinery transfers; the renormalization does not.

One further reading of the same diagram deserves mention. Kapitonets [12] draws his axis of symmetry through the *midpoint* of the remainder vector at  $\sigma = \frac{1}{2}$ , observes that the remainder is then perpendicular to that axis, and concludes both that away from zeros, where  $\arg \zeta$  is defined,  $\arg \zeta$  is the normal direction, and that the two chains project onto the axis with canceling sums. That is the half-split of §8.1 together with Corollary 9.2, read off the picture. His other construction, recovering  $\zeta$  by repeatedly replacing a polygon of partial sums by the polygon of their midpoints, is unrelated to the bisector point: iterated midpointing is the binomial transform, and the limit is Cesàro summation of the Dirichlet series, as he says himself.



## 15 Glossary

The paper uses a geometric language for objects that other writers have named differently, or not named at all. This section collects those words in one place. Each entry is a definition in ordinary language, a short formula when the formula is short, a pointer to the section that introduces the object, and, where we know one, the name another author has used for the same thing or a neighboring one. The names are those already used above; this list is only a map.

The list is not alphabetical. It runs from the parameters and the polyline itself to the more specialized constructions built on them, and the entries in each group belong with one another.

### The parameters

**Index  $T$ , and  $I(T)$ .** The real parameter that replaces Siegel’s pair  $(t, m)$ :  $t = I(T) = \pi(2T + 1)/\log(1 + 1/T)$  and  $m = \lfloor T \rfloor$  (§3).  $T$  is called the index even though it is not an integer. The map is chosen so that integer  $T$  lands on a handoff (§10.1).

**Unit of the index  $T$ .** One stretch of  $T$  between consecutive integers: the interval  $(m, m + 1)$  with  $m = \lfloor T \rfloor$ , for example  $T$  from 6 to 7. The chains keep the same number of partial-sum links on that stretch. The ends of the unit are  $\{T\}$  near 0 and near 1; the middle is  $\{T\}$  around  $\frac{1}{2}$  (§11.1).

$\chi(s)$ . The functional-equation factor,  $\zeta(s) = \chi(s)\zeta(1 - s)$  (§2). Nickel writes  $Q(s)$  for the same factor.

$\omega$ . The clockwise turn accumulated by the forward summands:  $\omega = I(T) \log \lceil T \rceil$ , so that  $(m + 1)^{-s}$  points along  $e^{-i\omega}$ , at absolute direction  $-\omega$  (§4).

$a, a^2$ . Siegel’s cutoff  $a = \sqrt{t/2\pi}$  and its square  $a^2 = t/2\pi = I(T)/2\pi$ . The continuous partner prediction is  $a^2/n$ ; integer labels satisfy  $n n' \approx a^2$ . Under  $t = I(T)$  one has  $a^2 \approx (T + \frac{1}{2})^2$ .

### The chain: joints and links

**Polyline.** The drawing: a path of straight segments laid end to end, which is how a partial sum appears in the plane once each summand is drawn as a vector from the end of the last one (§7). The word is about the picture and not about any new object. The forward and reverse spirals are the two polylines of this paper, and statements like “the two polylines meet” are statements about the drawn segments rather than about the sums.

**Sum chain, or link chain.** The partial-sum polyline in the plane: joints connected by links, one Dirichlet summand per link (§7). “Sum chain” reads it as a running total; “link chain” reads it as a serial chain of segments. The two names are the same object. We also call it a spiral, though only the later links wind. Nickel [17] draws it as the Argand diagram of the steps  $n^{-s}$ . Erickson [9] identifies the later coils as Cornu spirals and labels their centers  $C_n$ . Kapitonets [12] calls the polyline the Riemann spiral. Berry and Goldberg [3] call the quadratic-phase analogue a curlicue; our turn  $\theta_n = -t \log(1 + 1/n)$  is not quadratic, so the coils are Cornu only locally (§14.3). The  $3 \times 3$  matrices of §7.2 are the planar Denavit–Hartenberg encoding of a serial chain [8].

**Joint.** A vertex of a chain: the value of the partial sum after an integer number of summands. Joint  $n$  of the forward chain is  $J_n = \sum_{r=1}^n r^{-s}$  for  $n \geq 1$ , with  $J_0 = 0$ , so joint 0 is the origin

and joint  $m$  is  $\Sigma_1$  itself (§7, §11.1). Erickson's centers  $C_n$  stand in one-to-one correspondence with the joints, though a center is not itself a joint (§9.1; see *Spiral center*).

**Link.** The segment between two consecutive joints: one Dirichlet summand, drawn as a vector. Link  $k$  of the forward chain runs from joint  $k$  to joint  $k+1$  and carries the summand  $n = k+1$  (§7). Nickel calls the same object a step. Coutsiias and Kazarinoff [5] treat the quadratic-phase analogue as a discrete curve with a radius of curvature at each turn.

**Spiral center.** The point a coil of the chain winds around, and the clearing it leaves at the middle of that coil (§9, §11.1, Figure 17). The link nearest it is  $L_N(T, S_n)$  of (68), where the turn per link  $t \log(1 + 1/n)$  is the odd multiple  $\pi(2S_n + 1)$ : successive links are antiparallel there, so the chain sweeps back and forth across the center and the joints settle onto a ring around it of radius about half a link length, which is the lens-shaped clearing the near-diameter links envelope. The center need not lie on any link or joint, not even on the nearest link. The centers are not scattered. Numbering them by the spiral number  $S_n = c$  outward from the end of the chain, center  $c = 0$  is the last spiral, the largest, and sits at  $\zeta$  to within that radius; each further center is one sweep back: by (114) the step from center  $c$  to center  $c - 1$  is the  $c$ -th summand reflected in the fixed direction  $\Theta$ , of length  $c^{-1/2}$ . Read outward from  $\zeta$  the centers therefore reproduce the head of the chain reflected, which is Nickel's center-to-center distances and angles reproducing the initial steps. In summands, center  $c$  is near  $n = a^2/(c + \frac{1}{2})$ , interleaving with the sweeps at  $n = a^2/c$  of (113). Erickson writes  $C_n$  for these points.

$\Sigma_1$ , **the forward chain (or forward spiral).** The sum chain of the first Dirichlet sum,  $\Sigma_1 = \sum_{n=1}^m n^{-s}$  (§2, §7). The chain is a polyline of summands, not a placement: it may be drawn in any frame.

$\Sigma_2$ , **the reverse chain (or reverse spiral).** The sum chain of the reflected Dirichlet sum,  $\Sigma_2 = \chi(s) \sum_{n=1}^m n^{s-1}$  (§2). In the plane the reverse chain is usually drawn from  $\zeta$ , running counter to the forward chain; as a free vector it can sit anywhere. On the critical line it is the functional-equation image of the forward chain: each summand of  $\Sigma_2$  is  $\chi$  times the conjugate of the matching summand of  $\Sigma_1$ . Nickel's second half  $Q(s)P(1-s)$  is the same reflection, with his  $Q$  equal to our  $\chi$  (§14.1).

## The remainder on the chain

$R$ . Siegel's unsplit remainder integral, so that  $\zeta = \Sigma_1 + \Sigma_2 + R$  (§2). Every named split below recovers this same  $R$ , exactly or approximately.

**Split.** The choice of meeting point: a writing  $R = R_1 + R_2$  (or  $\zeta = B_1 + B_2$ ) that picks where the two halves join (§8.1). The two pieces leave the same pair of joints:  $R_1$  starts at  $\Sigma_1$ , and  $R_2$  arrives at  $\Sigma_1 + R$ . What a split chooses is only the point  $\Sigma_1 + R_1$  between those joints, its own bisector point. The three named splits are the partial-summand split  $ps$ , the half-split  $rs$ , and the Kuznetsov / Siegel- $f$  split  $ak$ . The  $rs$  split just halves  $R$  along itself. The  $ak$  split does not follow the summand directions.

**Cut.** To sever a single object, a link, a chain, or a sum, at one point along it: each forward link is cut at its crossing (§11.5), the chain is cut at an arbitrary link in §11.6, and Coutsiias and Kazarinoff cut their sum at an inflection (§14.3). The three verbs divide the work: a *split* writes  $R$  (or  $\zeta$ ) as two pieces, a *cut* severs one thing at one place, and two curves that meet *cross*.



**Wedge.** What the  $ps$  split looks like in the plane: the two halves run along two different link directions,  $R = d_1 e^{-i\omega} + d_2 e^{i(\omega+\psi)}$ , with real weights (§4, (21)). Those two rays form an angle, and  $R$  sits in that angle when both weights are positive. The same picture at a general forward link, with  $W_k$  in place of  $R$ , is (135). A positive  $ps$  split is drawn as a wedge; a general algebraic split need not determine one. In convex analysis the set  $\{a u + b v : a, b \geq 0\}$  spanned by two rays is called a cone in any dimension, including 2, but so as not to evoke a 3-d mental picture we chose “wedge” instead.

$R_1, R_2$  (**the  $ps$  split; also  $R_{1ps}, R_{2ps}$** ). The partial-summand split of  $R$ :  $R = R_1 + R_2$ , with  $R_1 = d_1 e^{-i\omega}$  and  $R_2 = d_2 e^{i(\omega+\arg \chi)}$  (§4). Each piece is the next Dirichlet term of its sum, shortened by a real weight. This split is the paper’s own, so it is written plainly as  $R_1, R_2$ ; the subscript  $ps$  appears only where the three splits are compared side by side (§8.1), to distinguish this exact split from the approximate remainders of other authors.

**Fractional summand.** A remainder that is exactly the next Dirichlet term, scaled by a real factor:  $R_1 = \hat{d}_1 (m+1)^{-s}$  and  $R_2 = \hat{d}_2 \chi (m+1)^{s-1}$  (§5). The word “fractional” means a fraction of one summand, not a summand with a non-integer index.

$d_1, d_2$ . The signed coefficients of the two fractional summands:  $R_1 = d_1 e^{-i\omega}$  and  $R_2 = d_2 e^{i(\omega+\arg \chi)}$ , so  $|R_1| = |d_1|$  and  $|R_2| = |d_2|$  (§4, §13.1). On the critical line both are positive and  $d_1 = d_2$ . Off the line they can change sign, and the signed ratio is (173). The sine formulas are (17)–(18).

$\hat{d}_1, \hat{d}_2$  (**fractional amounts**). The same weights read as fractions of a link rather than as lengths:  $\hat{d}_1 = [T]^\sigma d_1$  and  $\hat{d}_2 = [T]^{1-\sigma} d_2 / |\chi|$  (§5).  $\hat{d}_1$  is the crossing fraction of the bisector link. At the running example  $\sigma = \frac{1}{2}$ ,  $T = 6.18$ , one has  $d_1 \approx 0.0875$  and  $\hat{d}_1 \approx 0.231$ .

$R_{ak}, R_{1ak}, R_{2ak}$ . The Kuznetsov / Siegel- $f$  split:  $R \approx R_{ak} = R_{1ak} + R_{2ak}$ , where  $R_{1ak} = f(s) - \Sigma_1$  (§8.2, §8.1). Approximate, and not a single fractional summand.

$R_{rs}, R_{1rs}, R_{2rs}$ . The half-split  $R_{1rs} = R_{2rs} = \frac{1}{2}R$  (§8.1). Kapitonets’ midpoint of the remainder vector is this split (§14.3).

## The bisector

**Bisector link.** Link  $m = [T]$  of a chain: the link that sits at the boundary between the nearly-straight early chain and the coils (§9.1). Each chain has one. The forward bisector link runs from  $\Sigma_1$  along the  $(m+1)$ st summand; the reverse bisector link runs from  $\Sigma_1 + R$  along its  $(m+1)$ st summand. On  $\sigma = \frac{1}{2}$  the bisector line crosses this link at the bisector point  $B_1$ : that is where the forward chain meets the bisector line, at crossing fraction  $\hat{d}_1$  along the link, not (except at a handoff) at the bisector joint itself. This is the neighborhood of Siegel’s cutoff  $m = \lfloor \sqrt{t/2\pi} \rfloor$  (§2), of Nickel’s center-of-symmetry step  $n_p = \lfloor \sqrt{t/2\pi} \rfloor$  (offset from our  $m$  by about half a unit, §14.1), and of the inflection at which Coutsiass and Kazarinoff cut their sum (§14.3).

**Bisector joint.** Joint  $m$  of the forward chain, the integer vertex at which the bisector link begins: the point  $\Sigma_1$ . The matching joint of the reverse chain is  $\Sigma_1 + R$ . All three remainder splits of §8.1 leave from this pair of joints.

**Bisector point.** The crossing of the two bisector links,  $B_1 = \Sigma_1 + R_1 = \Sigma_1 + R - R_2$  (§9.1), wherever that crossing is finite. Off the line it fails at a pole of  $d_1$ ; on the line the apparent

singularity is removable and  $B_1$  is defined by continuity. On the critical line it is the apex of the isosceles triangle with base from the origin to  $\zeta$ . Nickel's center  $P(s)$  plays the same role and is not the same point: the two differ mostly along the symmetry axis, which  $\zeta$  cannot see (§14.1). Kapitonets draws his axis through the midpoint of the remainder vector  $R$ ; that midpoint is the half-split  $R_{rs}$ , not  $B_1$  (§14.3).

**Bisector line.** The line through the bisector point  $B_1$  and  $\zeta/2$  (§9.2, Figure 16), or a specified limiting direction when those two points coincide (when  $B_1 = \zeta/2$ ). On  $\sigma = \frac{1}{2}$  it is the perpendicular bisector of the segment from the origin to  $\zeta$ , and the axis of symmetry of the isosceles triangle  $O-B_1-\zeta$  whose equal sides are the two legs. It crosses the bisector link at  $B_1$ . Off the critical line the legs are unequal, so there is no isosceles triangle and the line is no longer a symmetry axis, but we keep the same name for the line through  $B_1$  and  $\zeta/2$ .

$B_1, B_2; B_1^*, B_2^*$ . The two half-sums that add to  $\zeta$ :  $B_1 = \Sigma_1 + R_1$  and  $B_2 = \Sigma_2 + R_2$ , so  $\zeta = B_1 + B_2$  (§7).  $B_1$  is the bisector point. On the critical line,  $B_2 = \chi \overline{B_1}$ . The starred pair is the velocity-split point: where  $\vartheta' \neq 0$ ,  $B_1^* = \zeta + \zeta'/(2\vartheta')$  and  $B_2^* = -\zeta'/(2\vartheta')$  of (159), with  $B_2^* = \chi \overline{B_1^*}$  on the line (§12.2). It is a two-leg meeting point, not another bisector point. Nickel's Riemann–Siegel equation  $\zeta = P(s) + Q(s)P(1-s)$  is this two-leg split with a different choice of the meeting point (§14.1).

## Frames

**Link frame.** The coordinate frame of an arbitrary forward link  $k$ : translate joint  $k$  to the origin and divide by the link vector  $(k+1)^{-s}$ , so that link  $k$  becomes the unit interval  $[0, 1]$  (§11.2, Figure 39).

**Bisector frame.** The link frame at  $k = m$ : it sends the forward bisector link to  $[0, 1]$  on the real axis (§10.1, §10.2). In that frame the reverse bisector link revolves as  $T$  advances, and the crossing with the axis is the number  $d_1$  (or the crossing fraction  $\hat{d}_1$ , once the link is read at unit length).

**$\vartheta$ -frame.** The rotation that makes  $\zeta$  real on the critical line: multiply every vector by  $e^{i\vartheta(t)}$ , where  $\vartheta$  is the Riemann–Siegel theta function, so that  $Z = e^{i\vartheta}\zeta$  lies on the real axis (§12.1, Corollary 9.2). The base  $O \rightarrow \zeta$  is then the real axis, and a split point reads as  $\frac{1}{2}Z + ih$ . This is the same rotation as  $e^{-i\psi/2}$  with  $\vartheta = -\psi/2$ , since  $\chi = e^{-2i\vartheta}$  on the line.  $Z$  is Hardy's  $Z$ -function.

## Legs

**Legs.** The two segments that add to  $\zeta$ : Leg 1 is the vector  $B_1$  from the origin to the bisector point, and Leg 2 is the vector  $B_2 = \zeta - B_1$  from the bisector point to  $\zeta$  (§9.2). Their lengths are  $L_1 = |B_1|$  and  $L_2 = |B_2|$ . Nickel's two vectors  $P(s)$  and  $Q(s)P(1-s)$  are legs in the same sense, meeting at his center rather than at  $B_1$ .

$\vartheta_1, \vartheta_2$ . The two leg angles:  $\vartheta_1 = \arg B_1$  is the heading of Leg 1 from the positive real axis, and  $\vartheta_2 = \arg(B_2/B_1)$  is the turn from Leg 1 to Leg 2 (§9.2). Then  $\zeta = L_1 e^{i\vartheta_1} + L_2 e^{i(\vartheta_1 + \vartheta_2)}$ . These are not the Riemann–Siegel theta function  $\vartheta(t)$  of (152). Since  $\chi = e^{-2i\vartheta}$ , a chosen square root has phase  $-\vartheta \pmod{\pi}$ , and  $\vartheta/\pi + 1$  is only the smooth part of the zero count; the clash of names is unfortunate and the two are kept apart by the subscripts.

**Half-line.** The critical line  $\sigma = \frac{1}{2}$ , read at  $t > 0$ , which is where all the zeros this paper counts sit. The two names are used interchangeably, as is “the line”; *off-half-line* and *off-line* both mean  $\sigma \neq \frac{1}{2}$ .

**Ordinate.** The imaginary part  $\gamma$  of a nontrivial zero  $\rho = \beta + i\gamma$ . A critical-line ordinate is one with  $\beta = \frac{1}{2}$ , equivalently a real zero of  $Z$ , and  $\gamma_1 < \gamma_2 < \dots$  numbers those up the line (§12). On the index, that ordinate sits at  $T = I^{-1}(\gamma_n)$ . The word is never used here in its generic sense, the  $y$ -coordinate of an arbitrary point.

**Retrograde, prograde.** The two directions of the fold angle  $\vartheta_2$  (§12, Figure 50). Prograde is the usual way:  $\vartheta_2$  keeps going through  $\pi$  at an ordinate and does not come back. Retrograde is the other way:  $\vartheta_2$  rises through  $\pi$ , turns around, and comes back down through it (or the reverse). The boundary between the two is where the derivative vanishes,  $\vartheta'_2 = 0$ . A retrograde passage is where  $N_{\text{ps}}$  miscounts;  $\vartheta_2^*$  of the velocity split never retrogrades, which is why  $N_*$  counts correctly (§12.3, Figure 54).

**Equal legs.** The condition  $L_1 = L_2$ . On the critical line it holds identically (Corollary 9.1). Off the line it is a locus of ovals and thin bands in the strip (Figure 19). A zero of  $\zeta$  requires equal legs *and*  $\vartheta_2 = \pi$ . Nickel notes that his two magnitudes are equal on  $\sigma = \frac{1}{2}$  (§14.1).

$\mathcal{E}_{\text{ps}}, \mathcal{E}_{\text{rs}}$ . The equal-leg loci of two different splits of the remainder. Each is the set of  $(\sigma, T)$  at which the two legs have the same length,  $|B_1| = |\zeta - B_1|$ , for  $B_1 = B_{1\text{ps}}$  and  $B_1 = B_{1\text{rs}}$  respectively (§9.3). A zero forces equal legs for every split, so every zero lies in the intersection  $\mathcal{E}_{\text{ps}} \cap \mathcal{E}_{\text{rs}}$ . The whole critical line is in both. Off the line, a meeting of the two oval families would be necessary for an off-line zero (not sufficient). If they never meet off  $\sigma = \frac{1}{2}$ , there is no off-line zero.

**Fold, folded.** Two related uses, both meaning that two segments occupy the same line and point opposite ways. At an integer  $T$  the neighboring link *folds back* onto the bisector link, the joint angle is an odd multiple of  $\pi$ , and the bisector point hands off to the next link (§10.1, §10.2): that is a *handoff*, and for zeta every handoff is of this kind. Separately, a *folded-leg* point in the strip is a height where  $\vartheta_2 = \pi$ , so that Leg 2 folds back onto Leg 1 (Figure 19); a zero is an equal-leg point that is also folded-leg.

## How the picture moves with $T$

**Handoff.** The instant, at integer  $T$ , when the bisector point moves from link  $m - 1$  onto link  $m$  (§6, §10.1). For zeta the two links fold back onto one another and the point slides across the joint with no jump.

**Event.** Any instant at which the bisector point moves from one link to the next (§10.1.1). For zeta every event is a handoff by fold-back. Dirichlet  $L$ -functions have a second kind.

**Folded event.** An event at which the turning angle at the current joint is  $\pm\pi$ , so the chain reverses and the bisector point slides across the joint (§10.1.1). For zeta, every handoff is a folded event, once per unit of  $T$ . For  $L(s, \chi_p)$  these occur at  $T \equiv 0 \pmod{p-1}$ .

**Bisector event.** An event at which the bisector *line* (the perpendicular bisector of the segment from 0 to the  $L$ -value) sweeps through a joint and carries the crossing onto the neighboring link (§10.1.1). Zeta has none. For  $L(s, \chi_p)$  they occupy the remaining residues modulo  $p - 1$ .

**Phantom link.** A zero summand of  $L(s, \chi_p)$ , occurring where  $\chi_p(n) = 0$  (§10.1.1). It contributes no segment and receives no index  $T$ ; only the nonzero steps are links. The name is kept because that is what §10.1.1 calls them.

## Crossings along the chain

**Cross.** What two curves do where they meet: the forward chain crosses the bisector line at  $B_1$ , each forward link is crossed by a reverse link, and the chord from yin to yang crosses the real axis at the crossing fraction. The meeting point itself is the crossing, next.

**Crossing (the point).** Where a forward link and a reverse link meet in the plane. The distinguished one is the bisector point  $B_1$ ; in the sampled range every other forward link  $k$  has one as well, written  $P_k$  (§11.1, §11.4). That existence is numerical.

**Crossing partner.** The reverse link that crosses forward link  $k$ . The saddle predicts a continuous partner  $n'_{\text{cont}} = a^2/n$  with  $a = \sqrt{t/2\pi}$  and  $n = k + 1$ ; the integer reverse-link label is the nearest integer to that target, so  $nn' \approx a^2$  (§11.1). This is the same saddle that Siegel's contour is built around.

**Crossing fraction.** How far along a link the crossing sits, from the left joint toward the right, as a number in  $[0, 1]$  (§11.3, §11.4). On forward link  $k$  it is  $\hat{d}_1(k)$ ; on reverse link  $k$  it is  $\hat{d}_2(k)$ , the  $T$  of  $\hat{d}_1(k, T)$  and  $\hat{d}_2(k, T)$  being suppressed where it is fixed. At the bisector,  $\hat{d}_1(m) = \hat{d}_1$ . Away from the fold the other  $\hat{d}_1(k)$  sit near  $\frac{1}{2}$ . This is a fraction of the straight link, not of a curved arc.

$\Sigma_{1x}, \Sigma_{2x}$ . The two Dirichlet sums re-jointed at the crossings instead of at the integers (§11.4).  $\Sigma_{1x}$  walks  $P_{-1} = 0, P_0, \dots, P_m = B_1$  and totals  $B_1 = \Sigma_1 + R_1$ ;  $\Sigma_{2x}$  does the same from the origin along  $\Sigma_2$  and totals  $B_2$ . The subscript  $x$  is for crossing, not a dummy index. The step lengths are the crossing fractions  $\hat{d}_1(k)$  and  $\hat{d}_2(k)$ .

## Yin and yang

**Yin, yang.** The two ends of the reverse bisector link, watched in the bisector frame as  $T$  advances (§10.2). Each traces a teardrop; the pair resembles a yin-yang. Yin is the joint- $m$  end, yang the joint- $(m + 1)$  end. The chord from yin to yang crosses the real axis at the bisector point, and that crossing is how  $d_1$  and  $d_2$  were found (§10.2). The same pair exists for every forward link  $k$ , once the crossing partner is known (§11.2). The names are ours.

$\text{Yin}_\infty, \text{Yang}_\infty$ . The limits of the yin and yang curves as  $T \rightarrow \infty$  at fixed fractional part  $x = \{T\}$ :  $\text{Yin}_\infty(x) = 1 - \Psi(x) e^{-2\pi i(x^2 - 1/16)}$  of (33), with  $\text{Yang}_\infty(x) = e^{4\pi i x}(1 - \text{Yin}_\infty(x))$  (§6.4). The two are one curve: yang follows yin half a unit behind (Lemma 6.4). For  $0 < T < 1$  the yin path is  $\zeta$  itself (Erickson's  $C_0$ ).

$\Psi$ . The classical Riemann–Siegel correction  $\Psi(x) = \cos(2\pi(x^2 - x - 1/16)) / \cos(2\pi x)$  of (32), used away from  $x = \frac{1}{4}, \frac{3}{4}$ , with the continuous values  $\Psi(\frac{1}{4}) = \Psi(\frac{3}{4}) = \frac{1}{2}$ , as in Titchmarsh [22, §15.3] and Pugh [20, §5.4.1]. It enters this paper as the amplitude of  $\text{Yin}_\infty$ , not as a remainder of its own.

### The far chain (links near $\zeta$ )

**Conveyor belt.** The stretch of the reverse chain between the last two spirals (spirals 1 and 0 of (68)), the links near  $\zeta$ . The name is ours.

**Last link.** The link nearest the center of the last spiral:  $L_N(T, 0) = \lfloor I(T)/\pi \rfloor = \lfloor t/\pi \rfloor \approx 2T^2$  (§9, (68)). It is the link closest to  $\zeta$ , far past the bisector;  $\zeta$  itself is not typically on it. The same formula with spiral number  $S_n$  names the link nearest the center of any spiral,  $L_N(T, S_n) = \lfloor I(T)/(\pi(2S_n + 1)) \rfloor$ . Consecutive links there are antiparallel.

## Acknowledgments

- Chris Cowherd for all his extensive help over the last nine years in building many visualization and experimental mathematics tools, including the Zest zeta sum visualization tool
- Nicolas Fish for his four years of excellent programming skills in building and tweaking the Zest visualization and measurement tool
- Daniel Flores Galiote for his kind tutoring and helpful discussions over the last two years
- Brian Conrey for his encouragement and acceptance of this amateur mathematician
- Jakob von Raumer for his help with a version of the Lean proof
- My family for their support:
  - my wife, Flo
  - Sophie
  - Anthony
  - Thomas
- Freeman Dyson for his article *Birds and Frogs* (<https://pdodds.w3.uvm.edu/files/papers/others/2009/dyson2009a.pdf>), which inspired me to begin my journey to investigate one-dimensional quasi-crystals and their possible connection to the Riemann hypothesis nine years ago.
- Turingbot, Processing, Unity, OEIS, wolframalpha, ChatGPT, Claude (and Opus and Fable), Cursor, and last but not least: Desmos and Lean (and Leo De Moura, my friend and neighbor, and the creator of LEAN). Thank you

**Statement on the use of AI.** Cursor and various AIs, including Anthropic’s Claude (Opus and Fable), OpenAI’s ChatGPT (5.6 Sol), and Grok 4.5, were used in preparing this paper. AI systems assisted with Python and JavaScript figure-generation programs, portions of the Zest web app, Lean formalizations, literature review, glossary, and  $\text{\LaTeX}$  preparation. Figure captions identify their generating programs. The author checked and edited the resulting text, code, figures, and formalizations.

Just before this version was finalized, the author set up an adversarial debate on the paper’s content between two competing AIs, with GPT 5.6 Sol Max acting as the reviewer and Claude Opus 5 Extra High as the defender. The reviewer raised 171 issues, each one debated by the defender, with the author as the final adjudicator of every proposed change. In the author’s opinion about ten of the issues were substantive; some fifty more were acted upon without being substantive, and the rest were trivial or wrong.

## A Proof of Proposition 4.1: reality and positivity of $d_1$ and $d_2$

This section proves Proposition 4.1, one subsection per ingredient. §A.1 proves parts (i) and (ii), which are short and purely algebraic. §A.2 sets up the Riemann–Siegel notation and proves the key phase bound: the factor  $\cos u$  stays away from zero at every fractional part. §A.3 proves part (iii), positivity on the critical line, where no fractional parts need to be excluded at all. §A.4 proves part (iv), positivity across the strip outside two windows of width  $O(1/T)$  per unit interval, and §A.5 locates those windows, near the parallel-summand heights  $\{T\} \approx \frac{1}{4}, \frac{3}{4}$ . §A.6 proves the two limit statements used in §6: the normalized weight converges to the profile (31), and the pinned waves converge to the tangent waveform (25). Finally, §A.7 proves the uniform finite- $T$  bounds  $\frac{1}{5} < \hat{d}_1 = \hat{d}_2 < \frac{4}{5}$  on the critical line. Throughout, the criterion is one of signs: both weights are positive exactly when the two numerators of (17)–(18) carry the same sign as the common denominator  $\sin(2\omega + \psi)$ .

### A.1 Proof of parts (i) and (ii): remainders are along the summands, and $d_1$ and $d_2$ are real

*Proof.* (i) For non-integer  $T$  we have  $M = m + 1$ , so the next summand of  $\Sigma_1$  factors as  $(m + 1)^{-s} = (m + 1)^{-\sigma} e^{-it \log(m+1)} = (m + 1)^{-\sigma} e^{-i\omega}$ , a positive real multiple of the unit direction  $e^{-i\omega}$ ; by (15),  $R_1 = d_1 e^{-i\omega}$  lies along it. Likewise the next summand of  $\Sigma_2$  factors as  $\chi(s)(m + 1)^{s-1} = |\chi|(m + 1)^{\sigma-1} e^{i(\omega+\psi)}$ , a positive real multiple of  $e^{i(\omega+\psi)}$ , and by (16),  $R_2 = d_2 e^{i(\omega+\psi)}$  lies along it. Explicitly,

$$R_1 = (m + 1)^\sigma d_1 \cdot (m + 1)^{-s}, \quad R_2 = \frac{(m + 1)^{1-\sigma}}{|\chi(s)|} d_2 \cdot \chi(s) (m + 1)^{s-1},$$

with both scale factors real by part (ii).

(ii) View (21) as a real  $2 \times 2$  linear system for the unknowns  $d_1, d_2$ , one equation each from the real and imaginary parts. Cramer’s rule gives exactly the pair of sine expressions (17)–(18), which are real by inspection: every ingredient ( $|R|$ , and sines of real angles) is real.  $\square$

### A.2 Notation and the key phase bound

Let  $\vartheta(t) = \arg \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi$  be the Riemann–Siegel theta function, with asymptotic expansion  $\vartheta(t) = \frac{t}{2} \log \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + O(1/t)$ , and let  $Z(t) = e^{i\vartheta(t)} \zeta\left(\frac{1}{2} + it\right)$  be the (real) Riemann–Siegel  $Z$  function. On the critical line  $\chi\left(\frac{1}{2} + it\right) = e^{-2i\vartheta(t)}$ , so  $\psi = -2\vartheta$  there. Put

$$a = \sqrt{\frac{t}{2\pi}}, \quad N = \lfloor a \rfloor, \quad \hat{p} = a - N, \quad u = \omega - \vartheta(t), \quad (194)$$

so that  $N$  is the Riemann–Siegel cutoff while  $m = \lfloor T \rfloor$  is ours, and  $u$  is the phase of the next summand measured against  $-\vartheta$ . Expanding  $\log(1 + \frac{1}{T})$  in  $t = I(T)$  gives

$$\frac{t}{2\pi} = \left(T + \frac{1}{2}\right)^2 - \frac{1}{12} + O\left(\frac{1}{T^2}\right), \quad \text{hence} \quad a = T + \frac{1}{2} + O\left(\frac{1}{T}\right), \quad (195)$$

so, apart from boundary intervals of width  $O(1/T)$  in the fractional part  $x = \{T\}$  near  $x = 0, \frac{1}{2}, 1$ ,

$$x < \frac{1}{2} \iff N = m, \quad \hat{p} = x + \frac{1}{2} + O(1/T), \quad x > \frac{1}{2} \iff N = m + 1, \quad \hat{p} = x - \frac{1}{2} + O(1/T). \quad (196)$$

Insert  $t = 2\pi a^2$  and  $m + 1 = a + (1 - \hat{p})$  (case  $N = m$ ) or  $m + 1 = a - \hat{p}$  (case  $N = m + 1$ ) into  $\omega = t \log(m + 1)$ , expand the logarithm about  $\log a$ , and reduce modulo  $2\pi$  using  $a = N + \hat{p}$  and  $\pi N^2 \equiv \pi N \pmod{2\pi}$ ; with the asymptotic series of  $\vartheta$  this gives

$$u \equiv \pi(N - 1) + 4\pi\hat{p} - 2\pi\hat{p}^2 + \frac{\pi}{8} + O(1/a) \quad (N = m), \quad u \equiv \pi N - 2\pi\hat{p}^2 + \frac{\pi}{8} + O(1/a) \quad (N = m + 1),$$

and substituting  $\hat{p} = x + \frac{1}{2}$ , respectively  $\hat{p} = x - \frac{1}{2}$ , turns both cases into one and the same statement:

$$\cos u = (-1)^{m+1} \cos(2\pi x(1 - x) - \frac{3\pi}{8}) + O(1/T). \quad (197)$$

Since  $2\pi x(1 - x) \in [0, \frac{\pi}{2}]$  for  $x \in [0, 1]$ , the argument of the cosine on the right lies in  $(-\frac{3\pi}{8}, \frac{\pi}{8}]$ , so

$$|\cos u| \geq \sin \frac{\pi}{8} - O(1/T) = 0.3826 \dots - O(1/T) : \quad (198)$$

the factor  $\cos u$  never approaches zero, at any fractional part. That one inequality is the key ingredient in both halves of the proof. We also need two elementary facts about the leading Riemann–Siegel correction  $C_0(\hat{p})$ .

**Lemma A.1.** *Write  $A = 2\pi\hat{p}^2 - \frac{\pi}{8}$  and  $B = 2\pi\hat{p}$ , and let  $C_0(\hat{p}) = \cos(A - B)/\cos B$  be the leading term of the Riemann–Siegel correction. Then:*

(i)  $C_0(\hat{p}) > 0$  for  $\hat{p} \in (\frac{1}{2}, 1)$ , and  $\min_{[1/2, 1]} C_0 = \cos \frac{3\pi}{8} = \sin \frac{\pi}{8}$ , attained at  $\hat{p} = \frac{1}{2}$ ;

(ii) for  $\hat{p} \in [0, \frac{1}{2}]$ ,

$$\Lambda(\hat{p}) := \frac{C_0(\hat{p})}{2 \cos(\pi(\frac{1}{8} - 2\hat{p}^2))} \leq \frac{1}{2},$$

with equality exactly at the endpoints, and  $\Lambda(\frac{1}{4}) = \frac{1}{4}$ .

*Proof.* (ii) first, since its argument is the more instructive. Note  $\pi(\frac{1}{8} - 2\hat{p}^2) = -A$ , so  $\Lambda = \cos(A - B)/(2 \cos A \cos B) = \frac{1}{2}(1 + \tan A \tan B)$  by the product formula, and the claim is that  $\tan A \tan B \leq 0$  on  $[0, \frac{1}{2}]$ . Both critical transitions happen at the same point:  $A = 0$  exactly when  $\hat{p} = \frac{1}{4}$ , and  $B = \frac{\pi}{2}$  exactly when  $\hat{p} = \frac{1}{4}$ . For  $\hat{p} < \frac{1}{4}$  we have  $A \in (-\frac{\pi}{8}, 0)$ , so  $\tan A < 0$ , while  $B \in (0, \frac{\pi}{2})$ , so  $\tan B > 0$ ; for  $\hat{p} > \frac{1}{4}$  the signs reverse ( $A \in (0, \frac{3\pi}{8})$ ,  $B \in (\frac{\pi}{2}, \pi)$ ). The product is therefore  $\leq 0$  throughout, vanishing only as  $\hat{p} \rightarrow 0, \frac{1}{2}$  (where  $\tan A \tan B \rightarrow 0$ , so  $\Lambda \rightarrow \frac{1}{2}$ ); at  $\hat{p} = \frac{1}{4}$  the two singular factors balance,  $\tan A \tan B \rightarrow -\frac{1}{2}$ , giving the removable value  $\Lambda(\frac{1}{4}) = \frac{1}{4}$ .

(i) is a sign inspection of numerator and denominator. On  $(\frac{1}{2}, \frac{3}{4})$  both  $\cos(A - B)$  and  $\cos B$  are negative; on  $(\frac{3}{4}, 1)$  both are positive (they change sign together at  $\hat{p} = \frac{3}{4}$ , which is the removable singularity of  $C_0$ ), so  $C_0 > 0$  throughout. That is positivity. The minimum is a separate claim:  $C_0$  extends continuously to the compact interval  $[\frac{1}{2}, 1]$  (the value at  $\frac{3}{4}$  is removable), and on each of the two subintervals it is monotone, rising from  $C_0(\frac{1}{2}) = \sin \frac{\pi}{8}$  through  $C_0(\frac{3}{4}) = \frac{1}{2}$  to  $C_0(1) = \cos \frac{\pi}{8}$ . Hence the minimum is the left endpoint.  $\square$

### A.3 Proof of Proposition 4.1(iii): on the critical line $d_1$ and $d_2$ are equal, real, and positive

We prove the following quantitative form, which contains Proposition 4.1(iii).

**Theorem A.2** (quantitative critical-line positivity). *For every  $c < \frac{1}{2} \sin \frac{\pi}{8}$  there is an effectively computable  $T_0(c)$  such that for every non-integer  $T \geq T_0(c)$ , at  $\sigma = \frac{1}{2}$ ,*

$$d_1(T) = d_2(T) \geq \frac{c}{a^{1/2}} > 0,$$

with no interval of  $\{T\}$  excluded.



*Proof. Step 1 (exact reduction).* On  $\sigma = \frac{1}{2}$  the reflection identities  $\Sigma_2 = \chi \overline{\Sigma_1}$  and  $\zeta = \chi \overline{\zeta}$  give  $R = \chi \overline{R}$ ; with  $\chi = e^{-2i\vartheta}$  this says  $r := e^{i\vartheta} R$  is real, i.e.  $\arg R = -\vartheta + \varepsilon\pi$  with  $\varepsilon \in \{0, 1\}$  and  $r = (-1)^\varepsilon |R|$ . Substituting  $\psi = -2\vartheta$  and  $\arg R = -\vartheta + \varepsilon\pi$  into (17)–(18) makes both numerators  $(-1)^\varepsilon \sin u$  and the denominator  $\sin 2u = 2 \sin u \cos u$ , so the factor  $\sin u$  cancels identically and

$$d_1 = d_2 = \frac{r}{2 \cos u} \quad (\cos u \neq 0). \quad (199)$$

*Step 2 (Riemann–Siegel input).* The real number  $r$  is the Riemann–Siegel tail cut at  $m$  instead of at  $N$ : since on the line  $e^{i\vartheta}(\Sigma_1 + \Sigma_2) = 2 \operatorname{Re}(e^{i\vartheta} \Sigma_1) = 2 \sum_{n \leq m} n^{-1/2} \cos(\vartheta - t \log n)$ ,

$$r = r_{\text{RS}} + [N = m + 1] \cdot \frac{2 \cos u}{\sqrt{m + 1}}, \quad r_{\text{RS}} := Z - 2 \sum_{n \leq N} n^{-1/2} \cos(\vartheta - t \log n), \quad (200)$$

where  $[\cdot]$  is the Iverson bracket, equal to 1 if the condition inside holds and 0 otherwise; the bracketed term is the summand  $n = m + 1$ , whose phase is exactly  $\vartheta - \omega = -u$ . Gabcke’s rigorous form of the Riemann–Siegel formula [10] gives

$$r_{\text{RS}} = \frac{(-1)^{N-1}}{a^{1/2}} (C_0(\hat{p}) + E), \quad |E| \leq c_1 a^{-1} \quad (t \geq 200). \quad (201)$$

*Step 3 (case  $N = m$ , i.e.  $x = \{T\} < \frac{1}{2}$ ,  $\hat{p} \in (\frac{1}{2}, 1)$ ).* Here  $r = r_{\text{RS}}$ , and the two signs multiply out:  $(-1)^{N-1}$  from (201) against  $(-1)^{m+1}$  from (197) give  $+1$ , so

$$d_1 = \frac{C_0(\hat{p})}{2 \cos(2\pi x(1-x) - \frac{3\pi}{8})} \cdot \frac{1}{a^{1/2}} + O(a^{-3/2}).$$

By Lemma A.1(i) the numerator is at least  $\sin \frac{\pi}{8}$ , and the cosine in the denominator is at most 1, so  $d_1 \geq (\frac{1}{2} \sin \frac{\pi}{8}) a^{-1/2} - O(a^{-3/2})$ .

*Step 4 (case  $N = m + 1$ , i.e.  $x > \frac{1}{2}$ ,  $\hat{p} \in (0, \frac{1}{2})$ ).* Now (200) has the extra term,

$$d_1 = \frac{1}{\sqrt{m+1}} + \frac{r_{\text{RS}}}{2 \cos u},$$

and this time the two signs multiply to  $-1$ : the correction is *subtracted*, with magnitude  $\Lambda(\hat{p}) a^{-1/2} + O(a^{-3/2})$  (here  $\cos(2\pi x(1-x) - \frac{3\pi}{8}) = \cos(\pi(\frac{1}{8} - 2\hat{p}^2))$  under  $x = \hat{p} + \frac{1}{2}$ ). Lemma A.1(ii) caps  $\Lambda \leq \frac{1}{2}$ , and  $a = N + \hat{p} = m + 1 + \hat{p} \geq m + 1$ , so

$$d_1 \geq \frac{1}{\sqrt{m+1}} - \frac{1}{2a^{1/2}} - O(a^{-3/2}) \geq \frac{1}{2a^{1/2}} - O(a^{-3/2}).$$

In both cases  $d_1 \geq (\frac{1}{2} \sin \frac{\pi}{8} - o(1)) a^{-1/2}$ , so any  $c < \frac{1}{2} \sin \frac{\pi}{8}$  is attained once  $a$  is large enough that the explicit  $O(1/a)$  errors of (201) and of the phase expansion (197) are dominated;  $T_0(c)$  is effectively computable because those error constants are. The boundary intervals of (196) are covered because the two case formulas agree to leading order at the seams: as  $x \rightarrow \frac{1}{2}$  from either side, and across each integer, both give  $d_1 \approx \frac{1}{2} a^{-1/2}$ .  $\square$

**Remark A.3** (why the critical line has no poles of  $d_1 = d_2$ ). A pole of  $d_1$  and  $d_2$  occurs when the two directions  $e^{-i\omega}$  and  $e^{i(\omega+\psi)}$  of (21) are parallel, so that the linear system of Proposition 4.1(ii) degenerates and the common denominator  $\sin(2\omega + \psi)$  vanishes. On the line that denominator is  $\sin 2u = 2 \sin u \cos u$ , which vanishes twice per unit interval of  $T$ , near  $\{T\} = \frac{1}{4}$  and  $\frac{3}{4}$ . But both vanishings belong to the factor  $\sin u$ , which cancels identically against the numerator in (199); the factor  $\cos u$ , the only one that could produce a pole, stays bounded away from zero by (198). Off the line the cancellation is spoiled, and the same two zeros become genuine poles. The next proof bounds the width of the affected windows in each unit interval.

#### A.4 Proof of Proposition 4.1(iv): off the line, $d_1$ and $d_2$ are positive except on two windows

*Proof.* Write  $u' = \omega + \psi/2$  and let  $\tau = \arg R - \psi/2 \pmod{\pi}$ , folded to  $[-\frac{\pi}{2}, \frac{\pi}{2})$ , measure the tilt of  $R$  away from the bisector direction of the two summand phases. Up to a common positive factor and a common sign, (17)–(18) read

$$d_1 \propto \frac{\sin(u' - \tau)}{\sin 2u'}, \quad d_2 \propto \frac{\sin(u' + \tau)}{\sin 2u'}. \quad (202)$$

Two inputs control the picture.

(i) *The tilt  $\tau$  is small.* The first term of Siegel's asymptotic expansion of  $R$  at fixed  $\sigma$  is a real multiple of  $e^{i\psi/2}$  up to sign and up to a phase error  $O_\sigma(1/t)$ ; the deviation of the exact  $R$  from that argument comes from the next term of the expansion, which is smaller by a factor  $t^{-1/2}$ . Hence  $|\tau| \leq C_2(\sigma) t^{-1/2}$ .

(ii) *The phase  $u'$  has the same skeleton as on the line.* Since  $\psi(\sigma + it) = -2\vartheta(t) + O_\sigma(1/t)$  uniformly for  $\sigma$  in compacta,  $u' = u + O_\sigma(1/t)$ , and (197)–(198) apply:  $|\cos u'| \geq \sin \frac{\pi}{8} - o(1)$ , so the estimates predict two candidate zeros of  $\sin u'$  per unit interval, at fractional parts converging to  $\frac{1}{4}$  and  $\frac{3}{4}$ , each crossed with slope  $|du'/d\{T\}| = \pi + O(1/T)$ . Exactly two simple roots, and the signs on their two sides, need a uniform derivative bound and a radial-sign estimate that are not written here; the windows below are those predicted candidates.

Away from those zeros (say  $|\sin u'| \geq 2|\tau|$ ) the ratios  $\sin(u' \mp \tau)/\sin u'$  are positive, so (202) carries the same signs as on the line, and the Step-3/Step-4 analysis of the critical-line proof, with the  $O_\sigma$  errors absorbed into the constants, gives positivity of both weights. Near a zero of  $\sin u'$  the numerator of  $d_1$  crosses zero at  $u' = \tau$  while the denominator crosses at  $u' = 0$ : between the two crossings, an interval of  $u'$ -length exactly  $|\tau|$ , the signs disagree and  $d_1 < 0$ ; mirror-symmetrically  $d_2 < 0$  on the interval of length  $|\tau|$  on the other side. Converting  $u'$ -length to  $\{T\}$ -length by the slope  $\pi + O(1/T)$  bounds each window by  $C_2(\sigma) t^{-1/2}/\pi \asymp C(\sigma)/T$ , since  $t = I(T) \asymp 2\pi T^2$ , centered  $O(1/T)$  from the denominator zero. At the denominator zero itself the two numerators are  $\mp \sin \tau$ , equal and opposite, so  $d_1/d_2 \rightarrow -1$  there, exactly as stated in Proposition 4.1(iv). On the critical line  $\tau \equiv 0$  exactly (Step 1 of the critical-line proof), the two crossings coincide, and the windows are empty, consistent with no poles there.  $\square$

#### A.5 Where are the poles?

Off the line the exact  $d_1$  and  $d_2$  (equivalently  $R_1$  and  $R_2$ ) have genuine poles at fractional heights  $\{T\} \approx \frac{1}{4}$  and  $\frac{3}{4}$ , but they are not exactly at those fractions. Here we locate them precisely.

A pole occurs when the two next summands have the same argument: the two unit directions  $e^{-i\omega}$  and  $e^{i(\omega+\psi)}$  of (21) become parallel, the pair no longer spans the plane, and the common denominator  $\sin(2\omega + \psi)$  of (17)–(18) vanishes. Subtracting one argument from the other produces the pole condition, their difference being a multiple of  $\pi$  (even  $k$  is codirected, odd  $k$  is antiparallel):

$$2\omega + \arg \chi(s) = k\pi, \quad k \in \mathbb{Z}, \quad (203)$$

with  $\omega$ ,  $s$ , and  $M$  as in (9) and (19). This formula was used to find the locations of the first 20 poles numerically (Table 4).

**Remark A.4** (the poles cancel in the sum). The remainder  $R$  itself has no poles: it is Siegel's convergent contour integral, finite and continuous in  $T$ . Since  $R = R_1 + R_2$  exactly (Theorem 4.2), and at a pole the two summand directions coincide ( $e^{i(\omega+\psi)} = e^{-i\omega}$  by (203)), the divergences of  $d_1$  and  $d_2$  must cancel exactly:  $d_1 + d_2$  stays bounded while each weight blows up, which is the  $d_1/d_2 \rightarrow -1$  behavior of Proposition 4.1(iv) seen from the other direction.

| Odd Pole # | $T$ (odd)       | Even Pole # | $T$ (even)      |
|------------|-----------------|-------------|-----------------|
| 1          | 1.268870258145  | 2           | 1.763736013695  |
| 3          | 2.261710393365  | 4           | 2.759511841350  |
| 5          | 3.258504081544  | 6           | 3.757283144824  |
| 7          | 4.256679717535  | 8           | 4.755902912218  |
| 9          | 5.255501039250  | 10          | 5.754963311284  |
| 11         | 6.254676427788  | 12          | 6.754282121179  |
| 13         | 7.254067037740  | 14          | 7.753765522842  |
| 15         | 8.253598276826  | 16          | 8.753360246216  |
| 17         | 9.253226472440  | 18          | 9.753033785343  |
| 19         | 10.252924347790 | 20          | 10.752765174160 |

Table 4: First 20 poles for  $\sigma = 0.6$ , which is the same result as for  $\sigma = 0.4$ . The symmetry about  $\sigma = \frac{1}{2}$  exists because  $\arg \chi(1 - \sigma + it) \equiv \arg \chi(\sigma + it) \pmod{2\pi}$ : the identity  $\chi(s)\chi(1-s) = 1$  with the Schwarz reflection  $\chi(\bar{s}) = \overline{\chi(s)}$  gives  $\chi(1 - \sigma + it) = 1/\overline{\chi(\sigma + it)}$ , the same input as Proposition 13.4. The pole condition (203) depends on  $\sigma$  only through that phase, and  $\omega$  is a function of  $T$  alone, so reflecting  $\sigma \mapsto 1 - \sigma$  leaves the equation unchanged and yields the same  $T$ -solutions. Note the poles occur at approximately  $T = [T] + \frac{1}{4}$  and  $T = [T] + \frac{3}{4}$  (equivalently, an integer  $\pm \frac{1}{4}$ ), and approach those values as  $T \rightarrow \infty$ .

## A.6 The limit profile and the limit waveform

The two convergence statements used in §6 are now within reach: the machinery of Theorem A.2 proves both. We begin with the profile of Figure 3, from which the convergence of the pinned waves of Figure 2 follows as a corollary.

**Theorem A.5** (limit profile). *Let  $\sigma = \frac{1}{2}$  and fix  $x \in (0, 1)$ . Then*

$$\lim_{\substack{T \rightarrow \infty \\ \{T\}=x}} \sqrt{m+1} d_1(T) = d(x) = \frac{1}{2} + \mathcal{W}_\infty(x),$$

with  $d$  the closed-form profile (31) and  $\mathcal{W}_\infty$  the tangent waveform (25), the removable points  $x = \frac{1}{4}, \frac{3}{4}$  filled in by continuity ( $d(\frac{1}{4}) = \frac{1}{4}$ ,  $d(\frac{3}{4}) = \frac{3}{4}$ ). The same limit holds with  $d_2$  in place of  $d_1$ , since  $d_2 = d_1$  on the line. The convergence rate is  $O(1/T)$ .

*Proof.* Fix  $x$  and let  $T = m + x \rightarrow \infty$ . Refining (195) one order,  $a = T + \frac{1}{2} - \frac{1}{24T} + O(T^{-2})$ , so the offset  $\hat{p}$  converges to its case value in (196) at rate  $1/T$ ; and  $(m+1)/a \rightarrow 1$  at the same rate, and all error terms in Steps 2–4 of the proof of Theorem A.2 are  $O(1/a)$ . Multiplying the two case formulas of that proof by  $\sqrt{m+1}$  therefore gives

$$\sqrt{m+1} d_1 \rightarrow \frac{C_0(x + \frac{1}{2})}{2 \cos(2\pi x(1-x) - \frac{3\pi}{8})} \quad (x < \frac{1}{2}), \quad \sqrt{m+1} d_1 \rightarrow 1 - \Lambda(x - \frac{1}{2}) \quad (x > \frac{1}{2}).$$

It remains to identify both branches with  $d(x) = \frac{1}{2} + \mathcal{W}_\infty(x)$ . Write  $\alpha = 2\pi x$  and  $\beta = 2\pi(x - \frac{1}{4})(x - \frac{3}{4}) = 2\pi(x^2 - x) + \frac{3\pi}{8}$ , so that  $\mathcal{W}_\infty(x) = -\frac{1}{2} \tan \alpha \tan \beta$ .

For  $x < \frac{1}{2}$ , substitute  $\hat{p} = x + \frac{1}{2}$  into  $C_0(\hat{p}) = \cos(A - B)/\cos B$  of Lemma A.1:  $A - B = 2\pi(\hat{p}^2 - \hat{p} - \frac{1}{16}) = 2\pi x^2 - \frac{5\pi}{8}$  and  $\cos B = \cos(2\pi x + \pi) = -\cos(2\pi x)$ , so the branch limit is

$$\frac{-\cos(2\pi x^2 - \frac{5\pi}{8})}{2 \cos(2\pi x) \cos(2\pi x(1-x) - \frac{3\pi}{8})} = g(x) = \frac{\cos(\alpha + \beta)}{2 \cos \alpha \cos \beta} = \frac{1}{2}(1 - \tan \alpha \tan \beta) = \frac{1}{2} + \mathcal{W}_\infty(x),$$

which is the first branch of (31): here  $\alpha + \beta = 2\pi x^2 + \frac{3\pi}{8}$ , so  $\cos(\alpha + \beta) = -\cos(2\pi x^2 - \frac{5\pi}{8})$ ; the denominator uses  $2\pi x(1-x) - \frac{3\pi}{8} = -\beta$  and the evenness of the cosine; and the last two steps are the product formula.

For  $x > \frac{1}{2}$ , take  $\hat{p} = x - \frac{1}{2}$  and recall from the proof of Lemma A.1(ii) that  $\Lambda(\hat{p}) = \frac{1}{2}(1 + \tan A \tan B)$ . This time  $A = 2\pi\hat{p}^2 - \frac{\pi}{8} = \beta$  exactly, and  $B = 2\pi\hat{p} = \alpha - \pi$ , so  $\tan B = \tan \alpha$  and

$$\Lambda(x - \frac{1}{2}) = \frac{1}{2}(1 + \tan \alpha \tan \beta) = \frac{1}{2} - \mathcal{W}_\infty(x), \quad \text{hence} \quad 1 - \Lambda(x - \frac{1}{2}) = \frac{1}{2} + \mathcal{W}_\infty(x).$$

Since  $\mathcal{W}_\infty(1-x) = -\mathcal{W}_\infty(x)$  (the tangent factor is odd about  $x = \frac{1}{2}$  while the quadratic factor is symmetric), this also equals  $1 - g(1-x)$ , the second branch of (31). The error terms accumulated along the way are all  $O(1/T)$ .  $\square$

**Corollary A.6** (the waves converge to the tangent waveform). *Let  $\sigma = \frac{1}{2}$  and fix  $x \in (0, 1)$ . Then*

$$\lim_{\substack{T \rightarrow \infty \\ \{T\}=x}} \mathcal{W}(T) = \mathcal{W}_\infty(x),$$

again at rate  $O(1/T)$ .

*Proof.* On  $[m, m+1]$  the pinned wave is

$$\mathcal{W}(T) = T^{1/2}(d_1(T) - \Phi(-1, \frac{1}{2}, m+1)) - (1-x)\varepsilon_m + x\varepsilon_{m+1},$$

and each of the three ingredients has a limit.

First,  $T^{1/2}d_1 = \sqrt{T/(m+1)} \cdot \sqrt{m+1}d_1 \rightarrow d(x) = \frac{1}{2} + \mathcal{W}_\infty(x)$  by Theorem A.5, since  $T/(m+1) \rightarrow 1$  at rate  $O(1/T)$ .

Second, by Boole summation (the alternating analogue of Euler–Maclaurin),  $\sum_{j \geq 0} (-1)^j f(n+j) = \frac{1}{2}f(n) - \frac{1}{4}f'(n) + O(|f'''(n)|)$ ; applied to  $f(y) = y^{-1/2}$  it gives

$$n^{1/2}\Phi(-1, \frac{1}{2}, n) = \frac{1}{2} + \frac{1}{8n} + O(n^{-3}),$$

so  $T^{1/2}\Phi(-1, \frac{1}{2}, m+1) = \frac{1}{2} + O(1/T)$ .

Third, the endpoint shortfalls vanish:

$$\varepsilon_n = n^{1/2}\Phi(-1, \frac{1}{2}, n) - n^{1/2}d_1(n^-) = O(1/n),$$

because as  $T \rightarrow n^-$  the Step-4 case formula of Theorem A.2 applies with  $\hat{p} = \frac{1}{2} + O(1/n)$ , where  $\Lambda(\frac{1}{2}) = \frac{1}{2}$  by Lemma A.1(ii) and  $\Lambda$  is continuous, giving  $n^{1/2}d_1(n^-) = \frac{1}{2} + O(1/n)$ , the same limit as the Lerch term. The chord contribution  $-(1-x)\varepsilon_m + x\varepsilon_{m+1}$  is therefore  $O(1/T)$ , and assembling the three pieces,

$$\mathcal{W}(T) = \left(\frac{1}{2} + \mathcal{W}_\infty(x)\right) - \frac{1}{2} + O(1/T) = \mathcal{W}_\infty(x) + O(1/T). \quad \square$$

## A.7 Uniform bounds: the fraction stays between $\frac{1}{5}$ and $\frac{4}{5}$

Theorem A.5 confines the fraction to  $[m_0, 1 - m_0]$  in the limit. At finite  $T$  the excursions are slightly larger, and round constants cover every height at once: the fractional summand never uses less than a fifth, or more than four fifths, of the next term.

**Theorem A.7.** Let  $\sigma = \frac{1}{2}$ . For every non-integer  $T > 1$ ,

$$\frac{1}{5} < \hat{d}_1(T) = \hat{d}_2(T) < \frac{4}{5}.$$

The constants are nearly sharp: no better  $T$ -independent ones exist than

$$\inf_{T>1} \hat{d}_1 = m_0 = 0.2268951\dots, \quad \sup_{T>1} \hat{d}_1 = 0.78499047437\dots,$$

the infimum approached but not attained as  $T \rightarrow \infty$  along  $\{T\} \rightarrow x^* = 0.1629962\dots$ , the supremum attained at  $T = 1.85307336433\dots$ .

*Proof.* On the critical line  $d_2 = d_1$  and  $|\chi| = 1$ , so  $\hat{d}_2 = \hat{d}_1$  by (24) and it suffices to bound  $\hat{d}_1 = \sqrt{m+1} d_1$ .

*Step 1 (exact finite- $T$  form).* Multiplying the two case formulas in the proof of Theorem A.2 by  $\sqrt{m+1}$  and writing  $\rho = \sqrt{(m+1)/a}$ ,

$$\begin{aligned} N = m \ (\{T\} < \tfrac{1}{2}) : \quad \hat{d}_1 &= \rho \cdot \frac{C_0(\hat{p}) + E}{2 \cos(2\pi x(1-x) - \frac{3\pi}{8} + \delta)}, \quad \rho = \sqrt{1 + \frac{1-\hat{p}}{a}} \in \left[1, \sqrt{1 + \frac{1}{2a}}\right], \\ N = m+1 \ (\{T\} > \tfrac{1}{2}) : \quad \hat{d}_1 &= 1 - \rho \cdot \frac{C_0(\hat{p}) + E}{2 \cos(\pi(\frac{1}{8} - 2\hat{p}^2) + \delta)}, \quad \rho = \sqrt{1 - \frac{\hat{p}}{a}} \in \left[\sqrt{1 - \frac{1}{2a}}, 1\right], \end{aligned}$$

with  $|E| \leq c_1 a^{-1}$  from (201) and  $|\delta| \leq c_2 a^{-1}$  from the phase expansion (197). At  $\rho = 1$ ,  $E = \delta = 0$  these are exactly the two branches of Theorem A.5.

*Step 2 (range of the main terms).* On  $(0, \frac{1}{2})$  the branch function  $d(x)$  descends from  $\frac{1}{2}$  to its single interior minimum  $m_0 = d(x^*)$  and climbs back to  $\frac{1}{2}$  (one sign change of  $d'$ ; the critical point  $x^*$  is the root of an explicit elementary equation). Hence the first main term ranges over  $[m_0, \frac{1}{2}]$  and the second over  $[\frac{1}{2}, 1 - \rho m_0]$ .

*Step 3 (all  $T \geq 30$ ).* The total deviation of  $\hat{d}_1$  from its main term is at most  $(\rho - 1) \cdot \frac{1}{2}$  plus a term collecting  $E$  and  $\delta$ , with  $\rho - 1 \leq \frac{1}{4a}$ ; the collected constant is validated by convergence-rate measurements against Theorem A.5 (observed absolute error  $\leq 0.008$  at  $T \approx 20$ , decreasing like  $1/T$ ), giving a total deviation below 0.02 for  $T \geq 30$ . Therefore, for  $T \geq 30$ ,

$$\{T\} < \tfrac{1}{2} : \quad \hat{d}_1 \in (m_0 - 0.02, \tfrac{1}{2}(1 + \tfrac{1}{120}) + 0.02) \subset (0.206, 0.525), \quad \{T\} > \tfrac{1}{2} : \quad \hat{d}_1 \in (0.48, 0.795),$$

and both windows lie inside  $(\frac{1}{5}, \frac{4}{5})$ .

*Step 4 ( $1 < T \leq 30$ , computational).* A dense evaluation of the exact  $\hat{d}_1$  (grid step  $5 \times 10^{-4}$  on  $(1, 3]$ ,  $10^{-3}$  on  $(3, 12]$ ,  $2 \times 10^{-3}$  on  $(12, 30]$ ; `check_uniform_bounds.py`) gives the observed range  $[0.2274, 0.78499]$ , with margin at least 0.015 to both  $\frac{1}{5}$  and  $\frac{4}{5}$  and sampled variation between adjacent grid points below 0.004, a factor of nearly four inside the margin. This covers the remaining range and, refined near its endpoints, produces the sharp values in the statement: the maximum 0.78499047437... at  $T = 1.85307336433\dots$  (on the  $m = 1$  interval, where  $\rho < 1$  pushes the second branch above its limiting value), and interval minima decreasing toward  $m_0$  from above as  $T$  grows (0.227369 at  $T = 59.164$ , for instance).  $\square$

**Remark A.8.** For the unnormalized weight the theorem reads  $\frac{1}{5}(m+1)^{-1/2} < d_1 < \frac{4}{5}(m+1)^{-1/2}$ ; in particular  $d_1 \leq 0.78499\dots/\sqrt{2} = 0.55508\dots$ , attained at the same  $T = 1.853\dots$ , and no positive  $T$ -independent lower bound exists for  $d_1$  itself.

## B Lean formalization of $R = R_1 + R_2$

This appendix reproduces the Lean file formalizing the algebraic content of Theorem 4.2. The file contains no `axiom` and no `sorry`: every step is proved, and the only axioms the final theorem depends on are Lean’s own three (`propext`, `Classical.choice`, `Quot.sound`), as reported by `#print axioms`. It was checked against Lean 4.32.2 with Mathlib at commit 905b95818e. The proof was written by the author, and the formalization of the proof in Lean code was written with the assistance of AI coding assistants (Anthropic’s Claude).

### B.1 Correspondence with the written proof

The file follows the proof of Theorem 4.2 step for step. The definitions `d_j1` and `d_j2` are the sine weights (17)–(18), and `R_1ps` and `R_2ps` are Step 1, transcribed from (15)–(16) in the  $[T]^{\mp iI(T)}$  form rather than the  $e^{\mp i\omega}$  form (the Lean identifiers keep the names `R_1ps`, `R_2ps` for what this paper writes  $R_1$ ,  $R_2$ ). Two small bridges convert them: `ceil_cpow` proves  $[T]^{ir} = e^{ir \log[T]}$  for  $T > 0$ , which with  $\omega(T) = I(T) \log[T]$  (the definition `omg`) is exactly  $[T]^{iI(T)} = e^{i\omega}$ ; and `div_norm_eq_exp_arg` proves  $z/|z| = e^{i \arg z}$  for  $z \neq 0$ , applied to  $\chi$  to give  $\chi/|\chi| = e^{i\psi}$ . Then `factoring_lemma` is the regrouping of Step 2, pulling out the common factor  $|R|/\sin(2\omega + \psi)$ , and `key_algebraic_identity` is the content of Steps 3–5, namely

$$e^{-i\omega} \sin(\omega - \phi + \psi) + e^{i(\omega+\psi)} \sin(\omega + \phi) = e^{i\phi} \sin(2\omega + \psi); \quad (204)$$

it is proved by writing each sine as  $(e^{i\theta} - e^{-i\theta})/2i$  (`sin_to_exp`, itself proved from Mathlib’s  $e^{i\theta} = \cos \theta + i \sin \theta$ ), clearing the denominator, merging the resulting products of exponentials into single exponentials, and letting `ring_nf` cancel the two copies of  $e^{i(\psi-\phi)}$ . The closing lines are Step 6: cancel  $\sin(2\omega + \psi)$  and recognize  $|R|e^{i\phi} = R$ , which is Mathlib’s `Complex.norm_mul_exp_mul_I`.

### B.2 What is proved and what is hypothesized

The theorem `R_1ps_plus_R_2ps_eq_R` carries three hypotheses. Two are needed for the objects to exist at all:  $\chi(\sigma + iI(T)) \neq 0$ , without which  $\chi/|\chi|$  is meaningless, and  $\sin(2\omega + \psi) \neq 0$ , without which  $d_1$  and  $d_2$  of (17)–(18) are undefined. The third,  $T > 0$ , makes  $[T]$  a positive base, so that its complex power can be written as an exponential. Nothing is assumed about whether  $T$  is an integer.

The conventions match exactly. Steps 2–6, that is `factoring_lemma` and `key_algebraic_identity`, hold for arbitrary real  $\omega, \phi, \psi$  and make no reference to  $T$ ; the only  $T$ -dependent ingredient is the bridge `ceil_cpow`, which holds for every  $T > 0$  with  $\omega$  defined as  $I(T) \log[T]$ . In the paper’s variables that reads  $t \log M$ , which is precisely the definition (19), for integer and non-integer  $T$  alike. The file therefore proves the identity in the same generality as Theorem 4.2: every  $T > 0$ , under the three hypotheses above.

What remains outside the formalization is not algebra but analysis. The functional-equation factor  $\chi$ , the index map  $I$ , and Siegel’s remainder  $R$  enter as free parameters, so what is machine-checked is that any  $R$  written in polar form splits as  $R_1 + R_2$  with the weights (17)–(18), which is exactly Steps 1–6. That  $R$  is Siegel’s remainder, and that  $I$  is the map of §3, are inputs to the theorem rather than consequences of it. Positivity, analytic continuation, pole cancellation, and singular limits are likewise outside the file. The `#print axioms` commands were run separately; their console output is not reproduced here.

### B.3 Source listing

```

1  /-
2  Formalization of Theorem (R = R_1ps + R_2ps) from
3  "Remainder Terms of the Riemann Zeta Function".
4
5  Contains no 'axiom' and no 'sorry': every step is proved.
6  -/
7  import Mathlib.Analysis.SpecialFunctions.Complex.Circle
8  import Mathlib.Analysis.SpecialFunctions.Pow.Complex
9
10 open Complex Real
11
12 /-! ### Phases -/
13
14 /-- The phase of Siegel's remainder, ' $\varphi = \arg R$ '. -/
15 noncomputable def phiR (R : ℂ) : ℝ := Complex.arg R
16
17 /-- The phase of the functional-equation factor, ' $\psi = \arg \chi(\sigma + i I(T))$ '. -/
18 noncomputable def psi (χ : ℂ → ℂ) (σ T : ℝ) (If : ℝ → ℝ) : ℝ :=
19   Complex.arg (χ (σ + If T * I))
20
21 /-- The turn angle ' $\omega(T) = I(T) \cdot \log|T|$ ', so that ' $|T|^{i I(T)} = e^{i\omega}$ '. -/
22 noncomputable def omg (T : ℝ) (If : ℝ → ℝ) : ℝ := If T * Real.log (|T| : ℝ)
23
24 /-! ### The two fractional link lengths -/
25
26 noncomputable def d_j1 (R : ℂ) (w f p : ℝ) : ℂ :=
27   ((|R| * (Real.sin (w - f + p) / Real.sin (2 * w + p))) : ℝ) : ℂ)
28
29 noncomputable def d_j2 (R : ℂ) (w f p : ℝ) : ℂ :=
30   ((|R| * (Real.sin (w + f) / Real.sin (2 * w + p))) : ℝ) : ℂ)
31
32 /-! ### The two partial-summand remainders, exactly as in the paper -/
33
34 noncomputable def R_1ps (σ T : ℝ) (R : ℂ) (χ : ℂ → ℂ) (If : ℝ → ℝ) : ℂ :=
35   d_j1 R (omg T If) (phiR R) (psi χ σ T If) * ((|T| : ℝ) : ℂ) ^ (-(I * (If T : ℂ)))
36
37 noncomputable def R_2ps (σ T : ℝ) (R : ℂ) (χ : ℂ → ℂ) (If : ℝ → ℝ) : ℂ :=
38   d_j2 R (omg T If) (phiR R) (psi χ σ T If) * ((|T| : ℝ) : ℂ) ^ (I * (If T : ℂ))
39   * (χ (σ + If T * I) / (|χ (σ + If T * I)| : ℂ))
40
41 /-! ### Step 0: elementary bridges -/
42
43 /-- ' $|T|^{i r} = e^{i r \log|T|}$ ' for ' $T > 0$ ', with no hypothesis on the
44 integrality of ' $T$ '. -/
45 lemma ceil_cpow (T r : ℝ) (hT : 0 < T) :
46   (((|T| : ℝ) : ℂ) ^ (I * (r : ℂ)))
47   = Complex.exp (I * ((r * Real.log (|T| : ℝ) : ℝ) : ℂ)) := by
48   have hpos : (0 : ℝ) < (|T| : ℝ) := lt_of_lt_of_le hT (Int.le_ceil T)
49   have hne : (((|T| : ℝ) : ℂ) ^ I) ≠ 0 := by
50     simp using (ne_of_gt hpos)
51   rw [Complex.cpow_def_of_ne_zero hne, ← Complex.ofReal_log hpos.le]
52   congr 1
53   push_cast
54   ring
55
56 /-- ' $z/|z| = e^{i \arg z}$ ' for ' $z \neq 0$ '. -/
57 lemma div_norm_eq_exp_arg (z : ℂ) (hz : z ≠ 0) :

```



```

58     z / (||z|| : C) = Complex.exp (I * (Complex.arg z : C)) := by
59 have hn : ((||z|| : R) : C) ≠ 0 := by
60   simp using norm_ne_zero_iff.mpr hz
61 rw [div_eq_iff hn, mul_comm I ((Complex.arg z : R) : C),
62     mul_comm (Complex.exp (((Complex.arg z : R) : C) * I)) ((||z|| : R) : C)]
63 exact (Complex.norm_mul_exp_arg_mul_I z).symm
64
65 /-- Exponential form of the sine of a real argument, valued in 'C'. -/
66 lemma sin_to_exp (θ : R) :
67   (Real.sin θ : C)
68   = (Complex.exp (I * (θ : C)) - Complex.exp (-I * (θ : C))) / (2 * I) := by
69 have h1 : Complex.exp (I * (θ : C)) = Complex.cos (θ : C) + Complex.sin (θ : C) * I := by
70   rw [mul_comm]; exact Complex.exp_mul_I _
71 have h2 : Complex.exp (-I * (θ : C)) = Complex.cos (θ : C) - Complex.sin (θ : C) * I := by
72   have h : (-I * (θ : C)) = -(θ : C) * I := by ring
73   rw [h, Complex.exp_mul_I, Complex.cos_neg, Complex.sin_neg]; ring
74 rw [h1, h2, Complex.ofReal_sin]
75 have hI : (2 * I : C) ≠ 0 := by simp [Complex.I_ne_zero]
76 field_simp
77 ring
78
79 /-! ### Steps 3-5: the key algebraic identity -/
80
81 /-- The heart of the theorem: the two sine-weighted phasors combine into a
82 single phasor along 'e^{iφ}' -/
83 theorem key_algebraic_identity (w f p : R) :
84   Complex.exp (-I * (w : C)) * (Real.sin (w - f + p) : C) +
85   Complex.exp (I * ((w + p : R) : C)) * (Real.sin (w + f) : C) =
86   Complex.exp (I * (f : C)) * (Real.sin (2 * w + p) : C) := by
87   rw [sin_to_exp, sin_to_exp, sin_to_exp]
88   have hI : (2 * I : C) ≠ 0 := by simp [Complex.I_ne_zero]
89   field_simp
90   simp only [mul_sub, ← Complex.exp_add]
91   push_cast
92   ring_nf
93
94 /-! ### Step 2: factoring -/
95
96 /-- Pulling the common factor '|R|/sin(2ω+ψ)' out front. -/
97 lemma factoring_lemma (absR w f p : R) :
98   ((absR * (Real.sin (w - f + p) / Real.sin (2 * w + p)) : R) : C) *
99   Complex.exp (-I * (w : C))) +
100   ((absR * (Real.sin (w + f) / Real.sin (2 * w + p)) : R) : C) *
101   Complex.exp (I * (w : C)) * Complex.exp (I * (p : C)) =
102   ((absR : R) : C) / (Real.sin (2 * w + p) : C) *
103   (Complex.exp (-I * (w : C)) * (Real.sin (w - f + p) : C) +
104   Complex.exp (I * ((w + p : R) : C)) * (Real.sin (w + f) : C)) := by
105 have hsplit : I * ((w + p : R) : C) = I * (w : C) + I * (p : C) := by
106   push_cast; ring
107 rw [hsplit, Complex.exp_add]
108 push_cast
109 ring
110
111 /-! ### The theorem -/
112
113 /-- **R = R1ps + R2ps.** The only hypotheses are that the index is positive,
114 that 'χ' does not vanish at the point in question, and that the denominator
115 'sin(2ω+ψ)' is nonzero (without which 'd1, d2' are undefined). In particular no
116 assumption is made about 'T' being an integer or not. -/

```



```

117 theorem R_1ps_plus_R_2ps_eq_R (σ T : ℝ) (R : ℂ) (χ : ℂ → ℂ) (If : ℝ → ℝ)
118   (hT : 0 < T)
119   (hχ : χ (σ + If T * I) ≠ 0)
120   (h_sin : Real.sin (2 * omg T If + psi χ σ T If) ≠ 0) :
121   R_1ps σ T R χ If + R_2ps σ T R χ If = R := by
122   have hsinC : (Real.sin (2 * omg T If + psi χ σ T If) : ℂ) ≠ 0 := by
123     exact_mod_cast h_sin
124   -- e^{iψ} for the normalized functional-equation factor
125   have hchi : χ (σ + If T * I) / (||χ (σ + If T * I)|| : ℂ)
126     = Complex.exp (I * (psi χ σ T If : ℂ)) := div_norm_eq_exp_arg _ hχ
127   -- the two cpow factors become e^{∓iω}
128   have hpow : ((([T] : ℝ)) : ℂ) ^ (I * (If T : ℂ))
129     = Complex.exp (I * (omg T If : ℂ)) := by
130     rw [ceil_cpow T (If T) hT]; rfl
131   have hpow' : ((([T] : ℝ)) : ℂ) ^ (-(I * (If T : ℂ)))
132     = Complex.exp (-(I * (omg T If : ℂ))) := by
133     have h : -(I * (If T : ℂ)) = I * ((-(If T) : ℝ) : ℂ) := by push_cast; ring
134     rw [h, ceil_cpow T (-(If T)) hT]
135     congr 1
136     simp only [omg]
137     push_cast
138     ring
139   -- |R| e^{iφ} = R
140   have hR : ((||R|| : ℝ) : ℂ) * Complex.exp (I * (phiR R : ℂ)) = R := by
141     rw [mul_comm I ((phiR R : ℝ) : ℂ)]
142     exact Complex.norm_mul_exp_arg_mul_I R
143   unfold R_1ps R_2ps d_j1 d_j2
144   rw [hchi, hpow, hpow', factoring_lemma, key_algebraic_identity]
145   -- cancel sin(2ω+ψ)
146   field_simp
147   linear_combination hR

```

## C Lean formalization of the critical line: $d_1 = d_2$ , $|R_{1ps}| = |R_{2ps}|$ , and equal legs

This appendix reproduces the Lean file establishing the three critical-line statements of the paper: Proposition 4.1(iii) in its compact form (70) ( $d_1 = d_2$  and  $|R_{1ps}| = |R_{2ps}|$ ), Corollary 9.1 ( $L_1 = L_2$ ), and Corollary 9.2 (the bisector point projects onto  $\frac{\zeta}{2}$ ). As in Appendix B the file contains no `axiom` and no `sorry`, and each result depends only on Lean’s own three axioms (`propext`, `Classical.choice`, `Quot.sound`) as reported by `#print axioms`; it was checked against Lean 4.32.2 with Mathlib at commit 905b95818e. The Lean code was written with the assistance of AI coding assistants (Anthropic’s Claude).

### C.1 The chain of implications

All three statements descend from a single fact, and the file is organized around it.

*Step 0: the two partial sums are conjugate.* At  $\sigma = \frac{1}{2}$  one has  $\overline{-s} = s - 1$ , so  $n^{s-1} = \overline{n^{-s}}$  for every  $n$ ; summing over  $n \leq m$  gives  $\Sigma_2 = \chi \overline{\Sigma_1}$ . This is `partial_sum_reflect`, proved from Mathlib’s `Complex.cpow_conj`.

*Step 1: the rotated remainder is real.* Write  $\psi = \arg \chi$  and let  $u = e^{i\psi/2}$ , a square root of  $\chi$ . With  $R = \zeta - \Sigma_1 - \Sigma_2$ ,

$$e^{-i\psi/2}R = \underbrace{e^{-i\psi/2}\zeta}_{\text{real}} - \underbrace{(e^{-i\psi/2}\Sigma_1 + e^{i\psi/2}\overline{\Sigma_1})}_{= 2\Re(e^{-i\psi/2}\Sigma_1)} \in \mathbb{R}, \quad (205)$$

the first term being real because the functional equation at  $\sigma = \frac{1}{2}$  reads  $\zeta = \chi \overline{\zeta}$ , so that  $e^{-i\psi/2}\zeta = \overline{e^{-i\psi/2}\zeta}$ . In the file this is `rotated_remainder_self_conj` (the algebra, from  $u\bar{u} = 1$  and  $u^2 = \chi$ ) and `rotated_remainder_im_eq_zero` (its imaginary part vanishes). It is the fact quoted in §9.3 and Appendix A.2 as  $R = e^{-i\vartheta}r$  with  $r$  real and  $\psi = -2\vartheta$  on the line. Stated as the reality of  $e^{-i\psi/2}R$  it is exact; stated through the argument it holds modulo  $\pi$ , which is visible in  $R = 2d_1 \cos(\omega + \frac{\psi}{2})e^{i\psi/2}$ , whose argument is  $\frac{\psi}{2} + \pi$  wherever the cosine is negative. The modulo- $\pi$  version is all the sequel uses.

*Step 2:  $d_1 = d_2$ .* Since  $\sin A - \sin B = 2 \sin \frac{A-B}{2} \cos \frac{A+B}{2}$ , the two numerators of (17)–(18) differ by

$$\sin(\omega - \phi + \psi) - \sin(\omega + \phi) = 2 \sin\left(\frac{\psi}{2} - \phi\right) \cos\left(\omega + \frac{\psi}{2}\right), \quad (206)$$

and Step 1 kills the first factor: `sin_half_sub_arg_eq_zero` turns the reality of  $e^{-i\psi/2}R$  into  $\sin(\frac{\psi}{2} - \arg R) = 0$ , using  $\cos \arg R = \Re R/|R|$  and  $\sin \arg R = \Im R/|R|$ . Hence `d1_eq_d2_of_im_eq_zero` and its critical-line specialization `d1_eq_d2`. Note that  $\omega$  is an arbitrary real throughout: the equality of the weights depends on the direction of  $R$  relative to  $\chi$  and not at all on the turn angle.

*Step 3:  $|R_{1ps}| = |R_{2ps}|$ .* Both phases are unimodular, so this is  $|d_1| = |d_2|$ : `norm_R1ps_eq_norm_R2ps`. The sharper form (70),  $R_{2ps} = \chi \overline{R_{1ps}}$ , is `R2ps_eq_chi_conj_R1ps`, and it is what feeds Step 4; it needs only that  $d_1$  is real and  $d_1 = d_2$ .

*Step 4: the legs.* Combining Steps 0 and 3,  $B_2 = \Sigma_2 + R_{2ps} = \chi \overline{\Sigma_1} + \overline{R_{1ps}} = \chi \overline{B_1}$  (`leg2_eq_chi_conj_leg1`), and since  $|\chi| = 1$  on the line the two legs have the same length (`legs_norm_eq`). Finally, from  $\zeta = B_1 + \chi B_1$ ,

$$e^{-i\psi/2}\zeta = e^{-i\psi/2}B_1 + \overline{e^{-i\psi/2}B_1} = 2\Re(e^{-i\psi/2}B_1), \quad (207)$$

which says that the perpendicular projection of the bisector point onto the line  $\mathbb{R} e^{i\psi/2}$  carrying  $\zeta$  lands exactly on  $\frac{\zeta}{2}$ : the apex of the isosceles triangle sits over the midpoint of its base. This is `proj_eq_half_zeta`, and it is the dotted bisector line drawn in Figures 16 and 18.

## C.2 What is proved and what is hypothesized

Two analytic facts about  $\chi$  on the critical line enter as hypotheses:  $|\chi| = 1$ , used in the form  $\chi = e^{i \arg \chi}$  (`eq_exp_arg_of_norm_one`), and the functional equation  $\zeta = \chi \bar{\zeta}$ , which is  $\zeta(s) = \chi(s)\zeta(1-s)$  together with  $1-s = \bar{s}$  at  $\sigma = \frac{1}{2}$ . Everything else above is derived, including the reflection  $\Sigma_2 = \chi \bar{\Sigma}_1$  of Step 0, which is proved rather than assumed. The projection statement additionally takes  $\zeta = B_1 + B_2$ , which is Theorem 4.2, formalized in Appendix B.

The remainders are written here in the *ps* notation of §8.1, in the form  $R_{1ps} = d_1 e^{-i\omega}$ ,  $R_{2ps} = d_2 e^{i(\omega+\psi)}$  of (46) (where they are the plain  $R_1$ ,  $R_2$  of §4) rather than the  $[T]^{\mp iI(T)}$  form of (15)–(16); the bridge between them is `ceil_cpow` of Appendix B. As there,  $\zeta$ ,  $\chi$ ,  $\Sigma_1$  and  $\omega$  are free parameters, so what is checked is the geometry of the critical line and not the identity of the functions involved.

## C.3 Source listing

```

1  /-
2    Critical-line consequences of the decomposition R = R1ps + R2ps:
3      (1) e^{-i\psi/2} R is real, i.e. arg R = \psi/2 modulo \pi,
4      (2) d_1 = d_2,
5      (3) |R1ps| = |R2ps|, indeed R2ps = \chi \cdot conj R1ps,
6      (4) the two legs have equal length, and the bisector point projects onto \zeta/2.
7
8    Contains no 'axiom' and no 'sorry': every step is proved.
9  -/
10 import Mathlib.Analysis.SpecialFunctions.Complex.Circle
11 import Mathlib.Analysis.SpecialFunctions.Pow.Complex
12
13 open Complex Real ComplexConjugate
14
15 /-! ### Unit phases -/
16
17 /-- A purely imaginary exponent gives a unit phase. -/
18 lemma norm_exp_imag (x : ℝ) : ||Complex.exp (I * (x : ℂ))|| = 1 := by
19   rw [Complex.norm_exp]
20   simp
21
22 lemma norm_exp_neg_imag (x : ℝ) : ||Complex.exp (-(I * (x : ℂ)))|| = 1 := by
23   rw [Complex.norm_exp]
24   simp
25
26 /-- Conjugation flips the sign of a purely imaginary exponent. -/
27 lemma conj_exp_imag (x : ℝ) :
28   conj (Complex.exp (I * (x : ℂ))) = Complex.exp (-(I * (x : ℂ))) := by
29   rw [← Complex.exp_conj]
30   congr 1
31   simp [Complex.ext_iff]
32
33 lemma conj_exp_neg_imag (x : ℝ) :
34   conj (Complex.exp (-(I * (x : ℂ)))) = Complex.exp (I * (x : ℂ)) := by
35   rw [← Complex.exp_conj]
36   congr 1

```

```

37   simp [Complex.ext_iff]
38
39 /-- A number of unit modulus is the exponential of its argument: this is the
40 form in which ' $|\chi| = 1$ ' on the critical line will be used. -/
41 lemma eq_exp_arg_of_norm_one {χ : ℂ} (hχ : ‖χ‖ = 1) :
42   χ = Complex.exp (I * ((Complex.arg χ : ℝ) : ℂ)) := by
43   have h := Complex.norm_mul_exp_arg_mul_I χ
44   rw [hχ] at h
45   rw [mul_comm I _]
46   simpa using h.symm
47
48 /-! ### Step 0: the second partial sum is the reflection of the first
49
50 At ' $\sigma = 1/2$ ' we have ' $n^{\{s-1\}} = \text{conj } n^{\{-s\}}$ ' for every ' $n$ ', because
51 ' $\text{conj}(-s) = s - 1$ ' there. Summing, ' $\Sigma_2 = \chi \cdot \text{conj } \Sigma_1$ ', which is the hypothesis
52 the results below are stated with. -/
53
54 lemma conj_natCast_cpow (n : ℕ) (z : ℂ) :
55   conj ((n : ℂ) ^ z) = (n : ℂ) ^ (conj z) := by
56   have harg : ((n : ℂ)).arg  $\neq \pi$  := by
57     rw [show ((n : ℂ)) = ((n : ℝ)) : ℂ by push_cast; ring,
58         Complex.arg_ofReal_of_nonneg (by positivity)]
59     exact Ne.symm Real.pi_ne_zero
60   rw [Complex.cpow_conj _ _ harg, Complex.conj_natCast]
61
62 /-- On the critical line the two Dirichlet partial sums are conjugate. -/
63 theorem partial_sum_reflect (m : ℕ) (t : ℝ) :
64    $\sum n \in \text{Finset.Icc } 1 \ m, ((n : \mathbb{C})) ^ (((1 / 2 : \mathbb{C}) + (t : \mathbb{C}) * I) - 1)$ 
65   = conj ( $\sum n \in \text{Finset.Icc } 1 \ m, ((n : \mathbb{C})) ^ (-((1 / 2 : \mathbb{C}) + (t : \mathbb{C}) * I))$ ) := by
66   rw [map_sum]
67   refine Finset.sum_congr rfl fun n _ => ?_
68   rw [conj_natCast_cpow]
69   congr 1
70   simp [Complex.ext_iff]
71   norm_num
72
73 /-! ### Step 1: the rotated remainder is real
74
75 On the critical line the functional equation reads ' $\zeta = \chi \cdot \text{conj } \zeta$ ', and the
76 second partial sum is the reflection of the first, ' $S_2 = \chi \cdot \text{conj } S_1$ '. With
77 ' $u = e^{\{i\psi/2\}}$ ' a square root of ' $\chi$ ', the remainder ' $R = \zeta - S_1 - S_2$ ' satisfies
78 ' $\text{conj } (\text{conj } u * R) = \text{conj } u * R$ ', that is, ' $e^{\{-i\psi/2\}} R$ ' is real. -/
79
80 lemma rotated_remainder_self_conj (ζ S₁ χ u : ℂ)
81   (hu : u * conj u = 1) (hu2 : u ^ 2 = χ)
82   (hζ : ζ = χ * conj ζ) :
83   conj (conj u * (ζ - S₁ - χ * conj S₁)) = conj u * (ζ - S₁ - χ * conj S₁) := by
84   simp only [map_sub, map_mul, Complex.conj_conj]
85   rw [← hu2] at hζ
86   simp only [map_pow]
87   linear_combination (-conj u) * hζ
88   + (-u * conj ζ + u * conj S₁ - conj u * S₁) * hu
89
90 lemma rotated_remainder_im_eq_zero (ζ S₁ χ : ℂ) (p : ℝ)
91   (hp : χ = Complex.exp (I * (p : ℂ)))
92   (hζ : ζ = χ * conj ζ) :
93   (Complex.exp (-I * ((p / 2 : ℝ) : ℂ))) * (ζ - S₁ - χ * conj S₁).im = 0 := by
94   have hu : Complex.exp (I * ((p / 2 : ℝ) : ℂ)) * conj (Complex.exp (I * ((p / 2 : ℝ) : ℂ))) = 1
95     := by

```

```

95     rw [conj_exp_imag, ← Complex.exp_add]
96     simp
97 have hu2 : Complex.exp (I * ((p / 2 : ℝ) : ℂ)) ^ 2 = χ := by
98     rw [hp, pow_two, ← Complex.exp_add]
99     congr 1
100    push_cast
101    ring
102 have hc : conj (Complex.exp (I * ((p / 2 : ℝ) : ℂ)))
103     = Complex.exp (-(I * ((p / 2 : ℝ) : ℂ))) := conj_exp_imag _
104 rw [← hc]
105 exact Complex.conj_eq_iff_im.mp
106 (rotated_remainder_self_conj ζ S1 χ _ hu hu2 hζ)
107
108 /-! ### Step 2: reality of 'e^{-ip/2} R' says 'arg R = p/2' modulo 'π' -/
109
110 lemma sin_half_sub_arg_eq_zero (R : ℂ) (p : ℝ) (hR : R ≠ 0)
111   (him : (Complex.exp (-(I * ((p / 2 : ℝ) : ℂ))) * R).im = 0) :
112   Real.sin (p / 2 - Complex.arg R) = 0 := by
113 have hexp : Complex.exp (-(I * ((p / 2 : ℝ) : ℂ)))
114     = ((Real.cos (p / 2) : ℝ) : ℂ) - ((Real.sin (p / 2) : ℝ) : ℂ) * I := by
115 have h : -(I * ((p / 2 : ℝ) : ℂ)) = ((-p / 2) : ℝ) : ℂ * I := by
116     push_cast; ring
117 rw [h, Complex.exp_mul_I, ← Complex.ofReal_cos, ← Complex.ofReal_sin,
118     Real.cos_neg, Real.sin_neg]
119     push_cast
120     ring
121 have hnorm : ‖R‖ ≠ 0 := norm_ne_zero_iff.mpr hR
122 rw [hexp] at him
123 simp only [Complex.mul_im, Complex.sub_re, Complex.sub_im, Complex.mul_re,
124     Complex.ofReal_re, Complex.ofReal_im, Complex.I_re, Complex.I_im] at him
125 rw [Real.sin_sub, Complex.cos_arg hR, Complex.sin_arg]
126 field_simp
127 linear_combination -him
128
129 /-! ### The weights and the two fractional remainders -/
130
131 /- 'd1 = |R| sin(ω - arg R + ψ)/sin(2ω + ψ)' -/
132 noncomputable def d1 (R : ℂ) (w p : ℝ) : ℝ :=
133   ‖R‖ * (Real.sin (w - Complex.arg R + p) / Real.sin (2 * w + p))
134
135 /- 'd2 = |R| sin(ω + arg R)/sin(2ω + ψ)' -/
136 noncomputable def d2 (R : ℂ) (w p : ℝ) : ℝ :=
137   ‖R‖ * (Real.sin (w + Complex.arg R) / Real.sin (2 * w + p))
138
139 /- 'R1ps = d1 e^{-iω}' -/
140 noncomputable def R_1ps (R : ℂ) (w p : ℝ) : ℂ :=
141   ((d1 R w p : ℝ) : ℂ) * Complex.exp (-(I * ((w : ℝ) : ℂ)))
142
143 /- 'R2ps = d2 e^{i(ω+ψ)}' -/
144 noncomputable def R_2ps (R : ℂ) (w p : ℝ) : ℂ :=
145   ((d2 R w p : ℝ) : ℂ) * Complex.exp (I * ((w + p : ℝ) : ℂ))
146
147 /-! ### Result 1: the two weights are equal
148
149 Note that 'ω' is arbitrary here: the equality of the weights depends only on the
150 direction of 'R' relative to 'χ', not on the turn angle. -/
151
152 theorem d1_eq_d2_of_im_eq_zero (R : ℂ) (w p : ℝ)
153   (him : (Complex.exp (-(I * ((p / 2 : ℝ) : ℂ))) * R).im = 0) :

```

```

154     d1 R w p = d2 R w p := by
155 rcases eq_or_ne R 0 with h | h
156 · simp [d1, d2, h]
157 · have hs : Real.sin (p / 2 - Complex.arg R) = 0 :=
158   sin_half_sub_arg_eq_zero R p h hm
159   have hnum : Real.sin (w - Complex.arg R + p) = Real.sin (w + Complex.arg R) := by
160     have hd := Real.sin_sub_sin (w - Complex.arg R + p) (w + Complex.arg R)
161     have harg : (w - Complex.arg R + p - (w + Complex.arg R)) / 2
162       = p / 2 - Complex.arg R := by ring
163     rw [harg, hs, mul_zero, zero_mul] at hd
164     linarith
165     rw [d1, d2, hnum]
166
167 /-- **d1 = d2 on the critical line.** -/
168 theorem d1_eq_d2 (ζ S1 χ : ℂ) (w p : ℝ)
169   (hp : χ = Complex.exp (I * (p : ℂ)))
170   (hζ : ζ = χ * conj ζ) :
171   d1 (ζ - S1 - χ * conj S1) w p = d2 (ζ - S1 - χ * conj S1) w p :=
172   d1_eq_d2_of_im_eq_zero _ w p (rotated_remainder_im_eq_zero ζ S1 χ p hp hζ)
173
174 /-! ### Result 2: the two fractional remainders have equal length -/
175
176 /-- **|R1ps| = |R2ps|.** -/
177 theorem norm_R1ps_eq_norm_R2ps (R : ℂ) (w p : ℝ) (hd : d1 R w p = d2 R w p) :
178   ||R1ps R w p|| = ||R2ps R w p|| := by
179   rw [R1ps, R2ps, norm_mul, norm_mul, norm_exp_neg_imag, norm_exp_imag, hd]
180
181 /-- The sharper statement: 'R2ps' is the mirror image of 'R1ps' across the real
182 axis, followed by the rotation 'χ'. -/
183 theorem R2ps_eq_chi_conj_R1ps (R χ : ℂ) (w p : ℝ)
184   (hp : χ = Complex.exp (I * (p : ℂ)))
185   (hd : d1 R w p = d2 R w p) :
186   R2ps R w p = χ * conj (R1ps R w p) := by
187   have hexp : Complex.exp (I * ((w + p : ℝ) : ℂ))
188     = Complex.exp (I * (p : ℂ)) * Complex.exp (I * ((w : ℝ) : ℂ)) := by
189     rw [← Complex.exp_add]
190     congr 1
191     push_cast
192     ring
193   rw [R1ps, R2ps, map_mul, Complex.conj_ofReal, conj_exp_neg_imag, hd, hexp, hp]
194   ring
195
196 /-! ### Result 3: the two legs have equal length -/
197
198 /-- With 'B1 = S1 + R1ps' and 'B2 = S2 + R2ps = χ conj S1 + R2ps', the second leg
199 is the reflection of the first. -/
200 theorem leg2_eq_chi_conj_leg1 (S1 R χ : ℂ) (w p : ℝ)
201   (hp : χ = Complex.exp (I * (p : ℂ)))
202   (hd : d1 R w p = d2 R w p) :
203   χ * conj S1 + R2ps R w p = χ * conj (S1 + R1ps R w p) := by
204   rw [map_add, mul_add, R2ps_eq_chi_conj_R1ps R χ w p hp hd]
205
206 /-- **L1 = L2.** -/
207 theorem legs_norm_eq (S1 R χ : ℂ) (w p : ℝ) (hχ : ||χ|| = 1)
208   (hp : χ = Complex.exp (I * (p : ℂ)))
209   (hd : d1 R w p = d2 R w p) :
210   ||χ * conj S1 + R2ps R w p|| = ||S1 + R1ps R w p|| := by
211   rw [leg2_eq_chi_conj_leg1 S1 R χ w p hp hd, norm_mul, hχ, one_mul,
212     RCLike.norm_conj]

```

```

213
214 /-! ### Result 4: the bisector point projects onto  $\zeta/2$  -/
215
216 /-- If ' $B_2 = \chi \text{ conj } B_1$ ' and ' $\zeta = B_1 + B_2$ ', then the orthogonal projection of the
217 bisector point ' $B_1$ ' onto the line ' $\mathbb{R}e^{i\psi/2}$ ' through ' $\zeta$ ' is exactly ' $\zeta/2$ '. The
218 apex of the isosceles triangle sits over the midpoint of its base. -/
219 theorem proj_eq_half_zeta (B1  $\zeta$   $\chi$  :  $\mathbb{C}$ ) (p :  $\mathbb{R}$ )
220   (hp :  $\chi = \text{Complex.exp } (I * (p : \mathbb{C}))$ )
221   (hzeta :  $\zeta = B_1 + \chi * \text{conj } B_1$ ) :
222   (((Complex.exp  $-(I * ((p / 2 : \mathbb{R}) : \mathbb{C}))$ ) * B1).re :  $\mathbb{R}$ ) :  $\mathbb{C}$ ) *
223     Complex.exp  $(I * ((p / 2 : \mathbb{R}) : \mathbb{C}))$ )
224     =  $\zeta / 2$  := by
225   set em := Complex.exp  $-(I * ((p / 2 : \mathbb{R}) : \mathbb{C}))$  with hem
226   set ep := Complex.exp  $(I * ((p / 2 : \mathbb{R}) : \mathbb{C}))$  with hep
227   have hu : ep * em = 1 := by
228     rw [hep, hem, ← Complex.exp_add]
229     simp
230   have hu2 : ep ^ 2 =  $\chi$  := by
231     rw [hep, hp, pow_two, ← Complex.exp_add]
232     congr 1
233     push_cast
234     ring
235   have hconj : conj em = ep := by rw [hem, hep, conj_exp_neg_imag]
236   set z := em * B1 with hz
237   have key : ((z.re :  $\mathbb{R}$ ) :  $\mathbb{C}$ ) * 2 = z + conj z := by
238     rw [Complex.add_conj]
239     push_cast
240     ring
241   have hcz : conj z = ep * conj B1 := by rw [hz, map_mul, hconj]
242   rw [eq_div_iff (two_ne_zero : (2 :  $\mathbb{C}$ )  $\neq$  0), hzeta]
243   linear_combination ep * key + ep * hz + B1 * hu + ep * hcz + conj B1 * hu2

```

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